F039902182

【TRS231BSxxxx】

F039902183

【TRH231BSxxxx】

F039990144

【FRO232BSyyzz】

HOST CPU-B/E UART IF Specification Ver.1.0.1

OKAYA ELECTRONICS CORP.
M's STYLE TECHNOLOGY CORP.

Version	History	Date
0.01	1st Draft Version (Same as Frontier)	2021/07/22
0.02	4. DTV Status ・Add 「DTV OFF」、「DTV OFF with EPG SCAN」 5.1.1. HOST⇒B/E Command ・Add 「Get Number of Program info」 ・Add 「Get Audio ES info」 ・Add 「Set DTV Mode」 ・Add 「Set Video Resolution」 ・Add 「Get Video Resolution」 5.1.4. B/E⇒HOST Event ・Add 「Event relay」 ・Add 「Number of Program info」 5.2.3. Channel Up/Down ・Add New Parameters in Data 3. ・Add Comment (3) and (4) 5.2.4. Physical Channel Set ・Delete 「for debug」 5.3.1. Set auto tune Mode ・Add New Parameter in Data 2：03h Auto with Seamless 5.3.12. Set DTV Mode ・Add New Command 5.3.13. Set Video Resolution ・Add New Command 5.4.1. Get DTV Status ・ Add New Parameter in Data 4 ： 05h：DTV OFF / 06h：DTV OFF with EPG SCAN 5.4.3. Get Configure ・Add New Parameter in Data 6：03h：Auto with Seamless 5.4.15. Get Number of Program info ・Add New Command 5.4.16. Get Audio ES info ・Add New Command 5.4.17. Get Video Resolution ・Add New Command 5.7.1. DTV Status ・Add New Parameter in Data 2： 05h：DTV OFF / 06h：DTV OFF with EPG SCAN 5.7.8. Event relay ・Add New Command 5.7.9. Number of Program info ・Add New Command	2021/08/23
0.03	2.2. Communication Format ・Change Response Request number format. 5.1.1. HOST⇒B/E Command ・Add 「Set Banner Config」 ・Add 「Get Banner Config」 ・Add 「Set Character Set」 ・Add 「Get Character Set」 ・Add 「Get TS Name with UTF-8」 ・Add 「Get Program Information with UTF-8」 5.1.4. B/E ⇒ HOST Event	2021/09/10

	・Add 「Program Information with UTF-8」 5.2.1. Channel Scan ・Add New Parameter in Data 2 ・Add comment in Description. ・Change Error Condition in "Invalid Parameter" 5.2.8. Direct Channel Select (1) ・Add New Optional Parameter. ・Change Error Condition in "Invalid Parameter" 5.3.1. Set auto tune Mode ・Change Parameter in Data 2：Auto and Auto with Seamless 5.3.7. Set Baud rate ・Change default value in Data 2. 5.3.9. Set DTV Parameter ・Add New Parameter in Data 154 – 185 5.3.12. Set DTV Mode ・Add (3)：Description about scan channel. 5.3.13. Set Video Resolution ・Typographical correction (5) Sequence 5.3.14. Set Banner Config ・Add New Command 5.3.15. Set Character Set ・Add New Command 5.4.3. Get Configure (2) ・Change Parameter in Data 2：Auto and Auto with Seamless 5.4.7. Get Service Information ・Support UTF-8 Character set 5.4.8. Get DTV Parameter ・Add New Parameter in Data156 – 187 5.4.16. Get Audio ES info ・Change to New Format 5.4.18. Get Banner Config ・Add New Command 5.4.19. Get Character Set ・Add New Command 5.4.20. Get TS Name with UTF-8 ・Add New Command 5.4.21. Get Program Information with UTF-8 ・Add New Command 5.7.10. Program Information with UTF-8 ・Add New Event	
0.04	2.1. Communication Configuration ・Change Baud rate 2.2. Communication Format ・Data Length：Changed the definition of the unit of length ・Correct comment Request No. 5.1. Command List ・Delete Comment ・Delete Boot/Loader 5.1.1. HOST⇒B/E Command ・Add 「Set DTV Parameter 2」 ・Add 「Get DTV Parameter 2」	2021/09/28

	・Add 「Get Service Information with UTF-8」 ・Add 「Get Program Description with UTF-8」 ・Add 「Get Program Name with UTF-8」 ・Add 「Get Audio ES info with UTF-8」 ・Add 「Set User ID」 ・Add 「Get User ID」 ・Add 「Set EPG Config」 013Dh ・Add 「Get EPG Config」 013Eh ・Add 「Set Video Color」 013Fh ・Add 「Get Video Color」 0140h ・Add 「Set Event Mask」 0141h ・Add 「Get Event Mask」 0142h 5.1.4. B/E⇒HOST Event ・Add 「TS Name Update」 820E ・Add 「TS Name Update with UTF-8」802F 5.2.1. Channel Scan ・Add Result OK Send Timing. ・Data 3：Add Optional Function that Tune top channel 5.2.2. Channel Set ・Change No Service Comment. 5.2.4. Physical Channel Set ・Clerical error correction in (1) ・Antenna Number Condition Deleted in (4) ・No Service Error Deleted in (4). 5.2.7. ES Switch ・Correction Invalid Parameter Condition. 5.2.8. Direct Channel Select ・Update Description. ・Delete unsupported value from example. 5.2.11. Watch Dog ・Delete Error Handling 5.2.14. Display Banner ・Add Invalid Parameter Condition. 5.2.15. Service ID Channel Select ・Delete wrong comment. ・Delete Invalid Parameter Condition. 5.3.1. Set auto tune Mode ・Change Parameter in Data 2：Auto and Auto with Seamless Add Description additional function. 5.3.2. Set Closed Caption Mode ・Change to table in (1) Data2. 5.3.7. Set Baud rate (3) ・Add Comment 5.3.8. Set DTV System ・Add Fixed 3x1Diver System ・Add New Comment for F039902182. 5.3.12. Set DTV Mode ・Update Description. 5.3.14. Set Banner Config ・Add New Parameter in Data9-10. ・Add Comment.	

	5.3.16. Set DTV Parameter 2 ・Add New Command 5.3.17. Set User ID ・Add New Command 5.3.18. Set EPG Config (T.B.D.) 5.3.19. Set Video Color ・Add New Command. 5.3.20. Set Event Mask ・Add New Command 5.4.1. Get DTV Status ・Add Fixed 3x1Diver System in (2) Data 14. 5.4.3. Get Configure ・Change Parameter in Data 2：Auto and Auto with Seamless 5.4.7. Get Service Information ・Delete UTF-8 Support. 5.4.10. Get DTV System ・Add Fixed 3x1 Diver System 5.4.15. Get Number of Program info ・Add New Parameter in Data 2. 5.4.16. Get Audio ES info ・Delete UTF-8 Support 5.4.18. Get Banner Config ・Add New Parameter in Data11-13. 5.4.22. Get DTV Parameter 2 ・Add New Command 5.4.23. Get Service Information with UTF-8 ・Add New Command 5.4.24. Get Program Description with UTF-8 ・Add New Command 5.4.25. Get Program Name with UTF-8 ・Add New Command 5.4.26. Get Audio ES info with UTF-8 ・Add New Command 5.4.27. Get User ID ・Add New Command 5.4.28. Get EPG Config (T.B.D.) 5.4.29. Get Video Color ・Add New Command 5.4.30. Get Event Mask ・Add New Command 5.7.1. DTV Status ・Add Fixed 3x1 Diver System in (1) Data 12. 5.7.8. Event relay ・Change Command Type to Need Ack 5.7.9. Number of Program info ・Change Command Type to Need Ack 5.7.10. Program Information with UTF-8 ・Change Fragment Data Number 5.7.11. TS Name Update ・Add New Event 5.7.12. TS Name Update with UTF-8 ・Add New Event	

0.05	2.1. Communication Configuration ・Add Comment for Default Baud rate. 2.2. Communication Format ・Add Model Number 5.1.1. HOST⇒B/E Command ・Add 「Get Part Number」 ・Add 「Get Program Name by 3Digit Number」 ・Add 「Get Program Name with UTF-8 by 3Digit Number」 ・Add 「Get Logo Data by 3Digit Number」 5.1.3. B/E⇒HOST Command ・Boot Mode change Boot to DTV. 5.1.4. B/E⇒HOST Event ・Add 「Logo Update」 5.1.6. HOST⇒BE Test & Debug Command ・Del 「Write F/E Register」 ・Del 「Read F/E Register」 ・Hardware Test change to DTV. 5.2.4. Physical Channel Set (3) ・Add Description. 5.2.9. DTV Initialize ・Add New Parameter. 5.3.1. Set auto tune Mode (3) ・Add description about default value. 5.3.6. Set Auto Event Relay ・Add new parameter. 5.3.7. Set Baud rate (1) ・Delete default. 5.3.8. Set DTV System (1) ・Add new parameter. 5.3.9. Set DTV Parameter (1) ・Move new parameter to DTV Parameter 2. 5.3.13. Set Video Resolution (1) ・Change 720p/480p value. 5.3.14. Set Banner Config (3) ・Add New Description. 5.3.15. Set Character Set (3) ・Add New Description. 5.3.16. Set DTV Parameter (1) ・Video Resolution value changed. ・Move new parameter form DTV Parameter ・Add new parameter for EPG 5.3.17. Set User ID (1) ・Change User ID Max Value and Definition. 5.3.17. Set User ID (3) ・Delete reboot. 5.3.18. Set EPG Config ・Add Parameter 5.3.19 Set Video Color ・Add Parameter ・Update description 5.3.20. Set Event Mask (1) ・Add Logo Update Event Mask	2021/10/28

	5.4.1. Get DTV Status ・Add BML Status. ・Added new values for Diver Mode 5.4.3. Get Configure ・Add New Parameter. 5.4.7. Get Service Information ・Updated description 5.4.8. Get DTV Parameter (2) ・Move new parameter to DTV Parameter 2. 5.4.10. Get DTV System ・Added new values for DTV System 5.4.13. Get Logo Data (3) ・Add Type 7 Simple Logo with UTF-8 ・Add Simple Logo Examples 5.4.16. Get Audio ES info (3) ・Add language text length. 5.4.17. Get Video Resolution (2) ・Change 720p/480p value. 5.4.21. Get Program Information with UTF-8 (2) ・Change Length of Program Name fields. ・Corrected typo in description. 5.4.22. Get DTV Parameter 2 ・Add New Parameter. 5.4.23. Get Service Information with UTF-8 ・Updated description 5.4.24. Get Program Description with UTF-8 (2) ・Change Length of Program Description Length fields. 5.4.25. Get Program Name with UTF-8 (2) ・Change Length of Program Name fields. 5.4.26. Get Audio ES info with UTF-8 ・Add language text length. 5.4.27. Get User ID (2) ・Change User ID Max Value and Definition. 5.4.28. Get EPG Config (1) ・Add Parameter. 5.4.30. Get Event Mask (2) ・Add Logo Update Event Mask 5.4.31. Get Part Number ・Add New Command. 5.4.32. Get Program Name by 3Digit Number ・Add New Command 5.4.33. Get Program Name with UTF-8 by 3Digit Number ・Add New Command 5.4.34. Get Logo Data by 3Digit Number ・Add New Command 5.6.1. Wakeup (2) ・Add User ID. 5.7.1. DTV Status ・Add BML Status. ・Added new values for Diver Mode 5.7.4. DTV Event ・Add BML Errors 5.7.8. Event relay (1)	

	・Add Three Digit Number. 5.7.9. Number of Program info ・Change Home/Travel Both Data ・Updated description 5.7.10. Program Information with UTF-8 (1) ・Change Length of Program Name fields. 5.7.13. Logo Update ・Add New Command 5.9.1. Write F/E Register ・Not Support 5.9.2. Read F/E Register ・Not Support	
0.06	5.1.4. B/E ⇒ HOST Event ・Add 「EPG Button Status」(8211h) ・Add 「BML Error」(8212h) 5.2.4. Physical Channel Set (3) ・Modify description. 5.2.9. DTV Initialize (3) ・Add Initialize Data Table 5.2.12. Display EPG (3) ・Add New Description ・Deleted EPG update time/channel. 5.2.14. Display Banner (3) ・Change Automatically hide time. 5.3.6. Set Auto Event Relay (1)/(3) ・Change to number of notification and interval 5.3.8. Set DTV System (3) ・Delete change sequence. ・Change System configuration table 5.3.13. Set Video Resolution ・Delete reboot request ・Delete default ・Add New Parameter 「480i」 5.3.14. Set Banner Config ・Add New Parameter ・Modify Description 5.3.16. Set DTV Parameter 2 ・Regenerate data content 5.3.18. Set EPG Config ・Add New Data 5.3.19. Set Video Color ・Add Default Value 5.3.20. Set Event Mask ・Add New Event 5.4.1. Get DTV Status ・Modify Data：17 ・Delete Data：19 5.4.3. Get Configure ・Change Data25,26 5.4.12. Get Program Name (2) ・Correct data length 5.4.13. Get Logo Data ・Change PNG Data Format(include PLTE).	2021/11/13

	・Modify description. 5.4.17. Get Video Resolution ・Add New Parameter 「480i」 5.4.18. Get Banner Config (2) ・Add New Parameter 5.4.22. Get DTV Parameter 2 ・Corrected typo ・Regenerate data content 5.4.25. Get Program Name with UTF-8 (2) ・Correct data length 5.4.29. Get EPG Config ・Add New Data 5.4.29. Get Video Color (2) ・Add Gamma Parameter 5.4.30. Get Event Mask ・Add New Event 5.4.32. Get Program Name by 3Digit Number ・Correct data length 5.4.34. Get Logo Data by 3Digit Number ・Modify description. 5.6.1. Wakeup ・Add Current User 5.7.1. DTV Status ・Modify Data：15 ・Delete Data：17 ・Modify description. 5.7.4. DTV Event ・Delete BML Error 5.7.9. Number of Program Info (1) ・Modify Data number. 5.7.14. EPG Button Status ・Add New Event 5.7.15. BML Error ・Add New Event 5.8.1. SW Update (3) ・Delete NVRAM Data	
0.07	2.2. Communication Format ・Add New Format for 16bit Length Field 5.2.12. Display EPG ・Modify Description 5.3.6. Set Auto Event Relay ・Update description. 5.3.16. Set DTV Parameter 2 ・Add New Parameter 5.3.18. Set EPG Config ・Update description. 5.3.19. Set Video Color ・Corrected maximum band default values of gamma. 5.3.20. Set Event Mask ・Modify parameter description. 5.4.22. Get DTV Parameter 2 ・Add New Parameter	2021/11/22

	5.4.29. Get Video Color ・Corrected maximum values of gamma. 5.4.30. Get Event Mask ・Modify parameter description. 5.6.1. Wakeup ・Modify Description 5.7.14. EPG Button Status ・Update description.	
0.08	5.2.12. Display EPG ・Add Display Type Optional Data 5.3.16. Set DTV Parameter 2 ・Add New Parameter ・Delete duplicate definitions. 5.4.13. Get Logo Data (3) ・UTF-8 Length Support 4byte UTF-8 code 5.4.20. Get TS Name with UTF-8 (2) ・Support 4byte UTF-8 code 5.4.21. Get Program Information with UTF-8 (2) ・Support 4byte UTF-8 code 5.4.22. Get DTV Parameter 2 ・Add New Parameter ・Delete duplicate definitions. 5.4.23. Get Service Information with UTF-8 (3) ・Support 4byte UTF-8 code 5.4.26. Get Audio ES info with UTF-8 (3) ・Support 4byte UTF-8 code 5.7.10. Program Information with UTF-8 (1) ・Support 4byte UTF-8 code 5.7.12. TS Name Update with UTF-8 (1) ・Support 4byte UTF-8 code 5.9.6. HW Test ・New Add Operation.	2021/12/15
0.09	5.1.4. B/E⇒HOST Event ・Add New Event 「C/N Information」(8213h) 5.2.12. Display EPG ・Updated Description 5.3.16. Set DTV Parameter 2 ・Add comment in Receive Logo Type ・Add New Parameter 5.3.19. Set Video Color ・Updated Description 5.4.22. Get DTV Parameter 2 ・Add comment in Receive Logo Type ・Add New Parameter 5.7.3. Search Status ・Add New Parameter 5.7.15. BML Error ・Correct Data Length. 5.7.16. C/N Information ・Add New Command	
0.10	5.2.5. Search Up/Down ・Add New Description 5.2.6. Broadcast Search	2022/03/11

	・Add New Description 5.2.9 DTV initialize ・Change Description content 5.2.12. Display EPG ・Add New Description 5.3.9. Set DTV Parameter ・Add New Description 5.3.13. Set Video ・Add New Description 5.3.16. Set DTV Parameter2 ・Add New Parameter ・Add New Description to some data 5.3.18. Set EPG Config ・Data length correct ・Add New Parameter 5.3.20. Set Event Mask ・Add New Description in Data0,1 and 2. ・Change Data 2 parameter. 5.4.7. Get Service Information ・Delete Sentence in Description 5.4.13. Get Logo Data ・The value inside of Description are changed. ・Change a sentence in Description 5.7.1. DTV Status ・Data 14, 15 are changed. 5.7.4. DTV Event ・Add New Description 5.7.16. C/N Information ・Data description change 5.9.3. Test Mode ・Added additional Audio Modes for pink noise ・Updated video color table ・Replaced color bar image with actual one.	
0.11	2.2 Communication Format ・Add new description. 5.2.9. DTV Initialize ・Add Description content. 5.2.12. Display EPG ・Add description on Data 3 ・Add new content on Description. 5.3.12. Set DTV Mode ・Change Data length. ・Add new Data 4. ・Add new contents on Description. 5.3.14. Set Banner Config ・Banner Width Minimum / Maximum changed. ・Add new data(13), change Data length. ・Add and changed description. 5.3.16 Set DTV parameter 2 ・Add description on data 25~27,46,47. ・Add new data (107), (108) ・Add new contents on Error Handling. 5.3.18. Set EPG Config	

	・EPG Type Image Update. ・Add description about figure. ・Add new description on Data 2. ・Add new data (52). 5.4.1. Get DTV status ・Add new parameter on Data 17. 5.4.11. Get Program List ・Delete Sentence in Description. 5.4.13. Get Logo Data ・Correct Data Length in (2). ・Delete Sentence in Description. 5.4.18 Get Banner config ・Add new data(15) ・Change data length 5.4.22. Get DTV Parameter 2 ・Add new data (109),(110). 5.4.23. Get Service Information with UTF-8 ・Delete Sentence in Description. 5.4.28 Get EPG Config ・Add new data (51~54). 5.7.1. DTV status ・Add new parameter on Data 15. 5.7.3 Search Status ・Add new description on Data 2, 4 ・Add new contents on Description. 5.7.4 DTV Event ・Add new data(20h) in DTV event. ・Changed description in DTV event code(code 82, 84). 5.7.15 BML Status ・Change chapter. ・Add new BML status(B3h,B4h,B5h,B6h). 5.9.3 Test mode ・Correct missing data in description.	
0.12	3 Error Code ・Add new error code on the table. 5.2.10 Shutdown ・Add new description on Sequence. 5.2.12 Display EPG ・Add new description about broadcast search behaver. 5.3.5 Set Residential Area ・Change of description method of Data 3. 5.3.14 Set Banner Config ・Add new Data 14. 5.3.16 Set DTV Parameter 2 ・Change of description method of Data 8. ・Change the data contents on Data 46. ・Change the description of Data name. ・Change of description method of Data 47. ・Add new Data 109. 5.3.18 Set EPG Config ・Change of description method of Data 2. ・Change the description of Data name.	2022/04/08

	・EPG Type Image Update. 5.4.18 Get Banner Config ・Add new Data 16. 5.4.22 Get DTV Parameter 2 ・Change of description method of Data 10. ・Change the description of Data name. ・Change of description method of Data 49. ・Add new Data 111. 5.4.28 Get EPG config ・Change of description method of Data 4. ・Change the description of Data name. 5.7.4 DTV Event ・Add new description on Description. ・Add supplement to "20h". ・Add new flow. 5.9.3 Test mode ・Change contents on Data 3 5.9.5 Monitor Status ・Change of description method of Data 3. 5.9.7 2nd TS Test ・Change of description method of Data 4. 5.9.8 2nd TS Test Status ・Change of description method of Data 2.	
0.13	5.2.8. Direct Channel Select ・Add Comment in (1) Data 5. 5.2.13. Update Preset Table ・Change Data length in (1). 5.2.14. Display Banner ・Add new description in (3) 5.3.5. Set Residential Area ・Add comment in (1) Data 3. 5.3.6. Set Auto Event Relay ・Change description in (3) 5.3.7. Set Baud rate ・Delete parameter 0Bh in (1) Data 2. 5.3.12. Set DTV Mode ・Add condition in (1) Data 4. 5.3.15. Set Character Set ・Change Data length in (1) 5.4.12. Get Program Name ・Change Max Data length in (2) 5.4.13. Get Logo Data ・Change Data length in (1) 5.4.25. Get Program Name with UTF-8 ・Change description in (3) 5.4.32. Get Program Name by 3Digit Number ・Change Max Data length in (2) 5.4.33. Get Program Name with UTF-8 by 3Digit Number ・Change Max Data length in (2) 5.4.34. Get Logo Data by 3Digit Number ・Change Data length in (1)/(2) 5.7.2 Scan Status ・Change description of Data4 in (1)/(2)	2022/4/22

	5.7.4. DTV Event ・Add figure Title. 5.7.8. Event relay ・Change Sequence 5.7.9. Number of Program info ・Change Sequence 5.7.11. TS Name Update ・Change Data length in (1) 5.7.12. TS Name Update with UTF-8 ・Change Data length in (1)	
0.14	5.3.9. Set DTV Parameter ・Add range value to each data 5.3.14. Set Banner Config ・Add range value to each data 5.3.16. Set DTV Parameter 2 ・Add range value to each data 5.3.17. Set User ID ・Add New description on Description 5.3.18. Set EPG Config ・Add range value to each data ・Change data range of Data 46,47 5.4.22. Get DTV Parameter 2 ・Change data range of Data 91,92 5.7.4. DTV Event ・Change the description of chart. ・Add new event in state diagram about Event Code 5.9.3. Test Mode ・Add new description. 6.DTV information ・Update the chapter content.	2022/6/1
0.15	5.7.4. DTV Event ・Add new event in state diagram about Event Code 5.9.3. Test Mode ・Change the description 6.1. Channel List ・Add note in the table 6.2. Last Channel ・Add note in the table 6.3. DTV Configuration ・Add note in the table	2022/6/8
0.16	5.2.1. Channel Scan ・Remove sentence in Description. 5.3.6. Set Auto Event Relay ・Add new description on data 2 5.3.14. Set Banner Config ・Remove sentence in Description. ・Change sentence in Description. 5.3.16. Set DTV Parameter 2 ・Change data name on EPG related data. ・Change data range of data 33, 87 ・Add new description on data 32,33,34 ・Change data name on data 109 5.3.18. Set EPG config	2022/7/6

	・Change data name. ・Change image on Description 5.4.22. Get DTV Parameter 2 ・Change data name on EPG related data ・Change data name on data 111 5.4.28. Get EPG Config ・Change data name 6.4. DTV System ・Removed note	
0.16.1	5.3.9. Set DTV parameter ・Correct the BER related data range ・Correct the description of time-related calculation method	2022/8/4
0.17	This document is integrated with New Frontier project F039990144【FRO232BSyyzz】. 2. UART Communication Specification ・Add New Frontier Baud rate data 5.2.1 Channel Scan ・Add a new comment in Description 5.2.12 Display EPG ・Add EPG scale information in Description 5.3.5 Set Residential Area ・Change the sentence in Description 5.3.16 Set DTV Parameter 2 ・Change the description on Data 32,33,34 ・Change the description on Description 5.3.18 Set EPG Config ・Add New Frontier project image or EPG type 5.3.19 Set Video Color ・Change the contents of the data 4 to 6 5.7.15 BML Status ・Change the sentence of B5h, B4h in Description. 5.8.1 SW Update ・Add (6) Appendix (SW update flow) 5.9.3 Test Mode ・Add New Frontier project test mode image. 6.4 DTV System ・Change contents in Note	2022/9/6
0.18	5.1.1. HOST ⇒ B/E Command ・Add New Command 「0147h Start Background Search」 ・Add New Command 「0148h Start Background Scan」 5.1.4. B/E ⇒ HOST Event ・Add New Command 「8214h Manual EPG Scan Result」 5.2.16. Start Background Search ・Add New Command 5.2.17. Start Background Scan ・Add New Command 5.3.6. Set Auto Event Relay ・Updated Description 5.3.16. Set DTV Parameter 2 ・Modified the explanation of Data32 ・Removed the table showing Data format 5.3.19. Set Video Color ・Removed Gamma related line from Description	

	5.3.20. Set Event Mask (1) Data4 ・Add New Parameter 5.4.1. Get DTV Status (2) Data16 ・Add New Parameter 5.4.6. Get Program Information ・Corrected Data indices in the response table 5.4.7. Get Service Information ・Added new table for response with Header Type 2 ・Updated Description 5.4.11. Get Program List ・Corrected Data Length in the response table ・Added new table for response with Header Type 2 ・Corrected the Number of Events in Description ・Updated Description 5.4.13. Get Logo Data ・Added new table for response with Header Type 2 ・Updated Description 5.4.16. Get Audio ES info ・Added new table for response with Header Type 2 ・Updated Description 5.4.20. Get TS Name with UTF-8 ・Corrected Data Length and indices in the response table ・Added new table for response with Header Type 2 ・Updated Description 5.4.21. Get Program Information with UTF-8 ・Corrected Data indices in the response table ・Corrected Data11 value in the response table ・Added new table for response with Header Type 2 ・Updated Description 5.4.23. Get Service Information with UTF-8 ・Added new table for response with Header Type 2 ・Updated Description ・Corrected upper limit of Service Name Length in Description 5.4.24. Get Program Description with UTF-8 ・Added new table for response with Header Type 2 ・Updated Description 5.4.25. Get Program Name with UTF-8 ・Added new table for response with Header Type 2 ・Updated Description 5.4.26. Get Audio ES info with UTF-8 ・Added new table for response with Header Type 2 ・Updated Description 5.4.30. Get Event Mask (2) Data6 ・Add New Parameter 5.4.33. Get Program Name with UTF-8 by 3Digit Number ・Added new table for response with Header Type 2 ・Updated Description 5.4.34. Get Logo Data by 3Digit Number ・Added new table for response with Header Type 2 ・Updated Description 5.7.1. DTV Status (1) Data14	

	・Add New Parameter 5.7.10. Program Information with UTF-8 ・Corrected Data indices in the event table ・Corrected Data9 value in the event table ・Added new table for event with Header Type 2 ・Updated Description 5.7.11. TS Name Update ・Added new table for event with Header Type 2 ・Updated Description 5.7.12. TS Name Update with UTF-8 ・Corrected Data indices in the event table ・Added new table for event with Header Type 2 ・Updated Description	
0.19	2.2. Communication Format ・Added title to the data format ・Updated Description 5.2.3. Channel Up/Down ・Updated Description 5.3.6. Set Auto Event Relay ・Updated Description 5.3.13. Set Video Resolution ・Added New Parameter 「1080p@30Hz」 5.3.16. Set DTV Parameter 2 ・Added 1080p@30Hz resolution to Data 5 5.3.20. Set Event Mask ・Changed BML Error to BML Status 5.4.5. Get Audio ES ・Changed the contents of Data 4 5.4.7. Get Service Information ・For Header Type 2, Data Length in 2rows changed to one. ・Updated Description 5.4.11. Get Program List ・For Header Type 2, Data Length in 2rows changed to one. ・For Header Type 2, Added Data Length value if Result is NG ・Updated Description 5.4.13. Get Logo Data ・For Header Type 2, Data Length in 2rows changed to one. 5.4.16. Get Audio ES Info ・For Header Type 2, Data Length in 2rows changed to one. 5.4.17. Get Video Resolution ・Added New Parameter 「1080p@30Hz」 5.4.20. Get TS Name with UTF-8 ・For Header Type 2, Data Length in 2rows changed to one. 5.4.21. Get Program Information with UTF-8 ・For Header Type 2, Data Length in 2rows changed to one. 5.4.22. Get DTV Parameter 2 ・Added 1080p@30Hz resolution to Data 7 5.4.23. Get Service Information with UTF-8 ・For Header Type 2, Data Length in 2rows changed to one. ・For Header Type 2, Added Data Length upper limit ・Updated Description 5.4.24. Get Program Description with UTF-8	2022/11/25

	・For Header Type 2, Data Length in 2rows changed to one. 5.4.25. Get Program Name with UTF-8 ・For Header Type 2, Data Length in 2rows changed to one. 5.4.26. Get Audio ES Info with UTF-8 ・Corrected table title ・For Header Type 2, Data Length in 2rows changed to one. 5.4.33. Get Program Name with UTF-8 by 3Digit Number ・For Header Type 2, Data Length in 2rows changed to one. ・For Header Type 2, Added Data Length value if Result is NG ・For Header Type 2, Start Sync[5Ah] to [59h] changed. 5.4.34. Get Logo Data by 3Digit Number ・For Header Type 2, Data Length in 2rows changed to one. ・For Header Type 2, Corrected Data length format 5.7.10. Program Information with UTF-8 ・For Header Type 2, Data Length in 2rows changed to one. ・For Header Type 2, Added Data Length maximum. 5.7.11. TS Name Update ・For Header Type 2, Data Length in 2rows changed to one. 5.7.12. TS Name Update with UTF-8 ・For Header Type 2, Data Length in 2rows changed to one.	
0.20	5.3.16. Set DTV Parameter 2 ・Updated Description 5.3.18 Set EPG Config ・Updated Description ・Removed unnecessary text 5.4.22. Get DTV Parameter 2 ・Updated Description 5.7.1. DTV Status ・Added new parameter in data 2. 5.8.3. Background SW Update Start ・Added new command 5.8.4. Background SW Update Cancel ・Added new command 5.8.5. Background SW Update ・Added new command 5.8.6. Background SW Update Status ・Added new command 6.1. Channel List ・Updated store condition 6.2. Last Channel ・Updated store condition 6.3. DTV Configuration ・Updated store condition 6.4. DTV System ・Updated store condition 6.5. TRMP Information ・Updated store condition 6.6. Logo ・Updated store condition 6.7. BML Information ・Updated store condition 6.8. Channel List	

	・Updated store condition 7.Background SW Update Sequence ・Add chapters 7.1, 7.2, 7.3.	
0.21	5.1.5. SW Update Command ・Title changed(BE⇒HOST SW Update Command -> SW Update Command) ・Add new command. 5.3.13. Set Video Resolution ・Updated Description 5.3.19. Set Video Color ・Brightness range changed. 5.4.29. Get Video Color ・Brightness range changed.	
0.22	5.1.1. HOST⇒B/E Command ・Added new command "DTV Reset". 5.1.4. B/E ⇒ HOST Event ・Changed BML Error to BML Status. ・Added new event "DTV Critical Error" 5.2.9. DTV Initialize ・Correction of typos in Description. 5.2.18. DTV Reset ・Added new chapter. 5.3.8. Set DTV System ・Correction of typos in Description. ・Removed invalid statement. ・Corrected Invalid Parameter condition in Error Handling. 5.3.9. Set DTV Parameter ・Correction of typos in Description. 5.4.1. Get DTV Status ・Added value 07h to Data4 in Get DTV Status Response table. ・Updated Description. 5.4.8. Get DTV Parameter ・Correction of typos in Description. 5.4.11. Get Program List ・Correction of typos in Description. 5.7.1. DTV Status ・Updated Description. 5.7.4. DTV Event ・Removed references to Frontier 5.7.18. DTV Critical Error ・Added new chapter 5.9.4. Set Debug Parameter ・Removed references to Frontier 6.4. DTV System ・Correction of typos	2023/05/17
0.23	5.1.1. HOST⇒B/E Command ・Changed command "DTV Reset" to "DTV Restart". 5.2.18. DTV Restart ・Updated DTV Restart Command format. ・Updated DTV Restart Response format. ・Updated Description.	2023/05/26

	・Updated Error Handling ・Updated Sequence. 5.7.18. DTV Critical Error ・Updated Event format. ・Updated Description. ・Updated Sequence.	
0.24	2.3.6. B/E ⇒ Host Sequence [Command(Need ACK) – NAK] ・Updated Description. 2.3.8. B/E ⇒ Host Sequence [Command(No Need ACK)] ・Updated Description. 5.3.17. Set User ID ・Corrected the length of Response. 5.4.1. Get DTV Status ・Updated Description. 5.4.21. Get Program Information with UTF-8 ・Corrected the length of Response. 5.4.24. Get Program Description with UTF-8 ・Corrected the length of Response. 5.4.25. Get Program Name with UTF-8 ・Corrected the length of Response. 5.4.31. Get Part Number ・Corrected the length of Response. 5.4.33. Get Program Name with UTF-8 by 3Digit Number ・Corrected the length of Response. 5.4.34. Get Logo Data by 3Digit Number ・Corrected the length of Response. 5.6.1. Wakeup ・Corrected the length of Response. 5.7.1. DTV Status ・Updated Description. 5.7.9. Number of Program info ・Updated Description. ・Corrected Data Length. 5.8.1. SW Update ・Updated Error Handling. 5.8.3. Background SW Update Start ・Updated Error Handling. 5.9. Test & Debug Command ・Removed chapter "Write F/E Register (Not Support)". ・Removed chapter "Read F/E Register (Not Support)".	2023/08/09
0.25	5.3.7. Set Baud rate. ・Updated (3)Description. 5.3.8. Set DTV System. ・Updated (3)Description. 5.3.9. Set DTV Parameter. ・Remove parameter range information from each data. *These information moves to Chapter 8. ・Update (3)Description. 5.3.14. Set Banner Config. ・Remove parameter range information from each data. *These information refers to Chapter 8.2 or 8.3. ・Update (3)Description. 5.3.16. Set DTV Parameter 2.	2023/12/08

	・Update data 3 information. ・Remove parameter range information from each data. *These information moves to Chapter 8. ・Update (3)Description. 5.3.18. Set EPG Config. ・Remove parameter range information from each data. *These information refers to Chapter 8.2 or 8.3 ・Update (3)Description. 5.4.21. Get Program Information with UTF-8. ・Correct Data Length. 5.4.24. Get Program Description with UTF-8. ・Correct Data Length. 5.4.25. Get Program Name with UTF-8. ・Correct Data Length. 5.4.33. Get Program Name with UTF-8 by 3Digit Number. ・Correct Data Length. 5.4.34. Get Logo Data by 3Digit Number. ・Correct Data Length. 5.9.1 Test Mode. ・Update (3)Description. 5.9.3. Monitor Status. ・Updated Data 2：Diver Mode(05h~08h). 8. DTV parameter and DTV parameter 2 data range. ・Add chapters 8.1 to 8.3.	
0.26	2.2 Communication Format ・Add applicable range of Header Type 2 5.3.9 Set DTV Parameter ・Correct data supplements 5.4.7 Get Service Information ・Change Data Length range ・Change Description 5.4.23 Get Service Information with UTF-8 ・Change Data Length range ・Change Description 5.7.5 EWS Event ・Add a Description 5.9.1 Test & Debug command ・Change the color value for New Frontier 8.1 DTV Parameter ・Change data range description ・Add Description 8.2 DTV Parameter 2 data range for Thrush ・Change data range description ・Add Description 8.3. DTV Parameter 2 data range for New Frontier ・Change data range description ・Add Description ・Changed data that cannot be configure to gray color	2024/4/24
0.27	5.3.9 Set DTV Parameter ・Correct description of data range information ・Add Time data range information 5.3.16 Set DTV Parameter 2 ・Correct description of data range information	2024/5/29

	5.4.23 Get Service Information with UTF-8 ・Correct description content 5.7.16 C/N Information ・Correct description of data range information 5.9.3 Monitor Status ・Correct description of data range information 8.1 DTV Parameter data range ・Correct description of data range information ・Add Time data range information 8.2 DTV Parameter 2 data range for Thrush ・Correct description of data range information 8.3. DTV Parameter 2 data range for New Frontier ・Correct description of data range information	
V0.28	5.2.11 Watch Dog ・Add error handling 5.3.6 Set Auto Event Relay ・Add the unit information to some data	2024/7/25
V1.0	The official release version 5.2.1 Channel Scan ・Revised description about Data 3. ・Add error handling about Data 3 content. 5.8.5 Background SW Update ・Revised data length	2024/8/10
V1.0.1	5.2.1 Channel Scan ・Changed description	2024/9/5

Contents

1. Introduction 28
2. UART Communication Specification 28
2.1. Communication Configuration 28
2.2. Communication Format 29
2.3. Communication Sequence 30
2.3.1. Host ⇒ B/E Sequence [Command(Need ACK) – ACK] 30
2.3.2. Host ⇒ B/E Sequence [Command(Need ACK) – NAK] 31
2.3.3. Host ⇒ B/E Sequence [Command(Need ACK) – Timeout] 32
2.3.4. Host ⇒ B/E Sequence [Command(No Need ACK)] 33
2.3.5. B/E ⇒ Host Sequence [Command(Need ACK) – ACK] 34
2.3.6. B/E ⇒ Host Sequence [Command(Need ACK) – NAK] 35
2.3.7. B/E ⇒ Host Sequence [Command(Need ACK) – Timeout] 36
2.3.8. B/E ⇒ Host Sequence [Command(No Need ACK)] 37
2.4. Communication Specification 38
2.5. Timeout Value 38
3. Error Code 39
4. DTV Status 40
5. Data Detail 40
5.1. Command List 40
5.1.1. HOST⇒B/E Command 40
5.1.2. HOST⇒B/E Event 41
5.1.3. B/E⇒HOST Command 42
5.1.4. B/E⇒HOST Event 42
5.1.5. SW Update Command 42
5.1.6. HOST⇒BE Test & Debug Command 42
5.1.7. B/E⇒HOST Test & Debug Command 43
5.2. HOST⇒B/E Command 44
5.2.1. Channel Scan 44
5.2.2. Channel Set 46
5.2.3. Channel Up/Down 48
5.2.4. Physical Channel Set 50
5.2.5. Search Up/Down 52
5.2.6. Broadcast Search 54
5.2.7. ES Switch 56
5.2.8. Direct Channel Select 58
5.2.9. DTV Initialize 60
5.2.10. Shutdown 63
5.2.11. Watch Dog 65

5.2.12. Display EPG 67
5.2.13. Update Preset Table 71
5.2.14. Display Banner 73
5.2.15. Service ID Channel Select 75
5.2.16. Start Background Search 77
5.2.17. Start Background Scan 79
5.2.18. DTV Restart 81
5.3. HOST⇒B/E Set Command 83
5.3.1. Set auto tune Mode 83
5.3.2. Set Closed Caption Mode 85
5.3.3. Set Auto Broadcast Search Mode 87
5.3.4. Set Audio Mode 89
5.3.5. Set Residential Area 91
5.3.6. Set Auto Event Relay 95
5.3.7. Set Baud rate 97
5.3.8. Set DTV System 99
5.3.9. Set DTV Parameter 102
5.3.10. Set Low Power Mode 106
5.3.11. BML Forced Exit 108
5.3.12. Set DTV Mode 110
5.3.13. Set Video Resolution 112
5.3.14. Set Banner Config 114
5.3.15. Set Character Set 118
5.3.16. Set DTV Parameter 2 120
5.3.17. Set User ID 125
5.3.18. Set EPG Config 127
5.3.19. Set Video Color 133
5.3.20. Set Event Mask 135
5.4. HOST⇒B/E Get Command 137
5.4.1. Get DTV Status 137
5.4.2. Get TS Name 141
5.4.3. Get Configure 144
5.4.4. Get Version 147
5.4.5. Get Audio ES 149
5.4.6. Get Program Information 151
5.4.7. Get Service Information 153
5.4.8. Get DTV Parameter 156
5.4.9. Get Program Description 159
5.4.10. Get DTV System 161

5.4.11. Get Program List 163
5.4.12. Get Program Name 167
5.4.13. Get Logo Data 170
5.4.14. Get RMP Maker ID 174
5.4.15. Get Number of Program info 177
5.4.16. Get Audio ES info 180
5.4.17. Get Video Resolution 183
5.4.18. Get Banner Config 185
5.4.19. Get Character Set 187
5.4.20. Get TS Name with UTF-8 189
5.4.21. Get Program Information with UTF-8 192
5.4.22. Get DTV Parameter 2 195
5.4.23. Get Service Information with UTF-8 199
5.4.24. Get Program Description with UTF-8 203
5.4.25. Get Program Name with UTF-8 206
5.4.26. Get Audio ES info with UTF-8 210
5.4.27. Get User ID 213
5.4.28. Get EPG Config 215
5.4.29. Get Video Color 218
5.4.30. Get Event Mask 220
5.4.31. Get Part Number 222
5.4.32. Get Program Name by 3Digit Number 224
5.4.33. Get Program Name with UTF-8 by 3Digit Number 226
5.4.34. Get Logo Data by 3Digit Number 229
5.5. HOST⇒B/E Event 232
5.5.1. Key Event 232
5.6. B/E⇒HOST Command 234
5.6.1. Wakeup 234
5.6.2. Boot Mode 237
5.7. B/E⇒HOST Event 239
5.7.1. DTV Status 239
5.7.2. Scan Status 243
5.7.3. Search Status 244
5.7.4. DTV Event 246
5.7.5. EWS Event 250
5.7.6. Program Information 251
5.7.7. EPG Status 253
5.7.8. Event relay 254
5.7.9. Number of Program info 255

5.7.10. Program Information with UTF-8 257
5.7.11. TS Name Update 260
5.7.12. TS Name Update with UTF-8 261
5.7.13. Logo Update 262
5.7.14. EPG Button Status 263
5.7.15. BML Status 264
5.7.16. C/N Information 265
5.7.17. EPG Scan Result 266
5.7.18. DTV Critical Error 268
5.8. SW Update Command 270
5.8.1. SW Update 270
5.8.2. SW Update Status 275
5.8.3. Background SW Update Start 276
5.8.4. Background SW Update Cancel 278
5.8.5. Background SW Update 280
5.8.6. Background SW Update Status 281
5.9. Test & Debug Command 283
5.9.1. Test Mode 283
5.9.2. Set Debug Parameter 287
5.9.3. Monitor Status 289
5.9.4. HW Test 292
5.9.5. 2nd TS Test 294
5.9.6. 2nd TS Test Status 296
6. DTV Information 298
6.1. Channel List 298
6.2. Last Channel 298
6.3. DTV Configuration 298
6.4. DTV System 299
6.5. TRMP Information 299
6.6. Logo 299
6.7. BML Information 299
6.8. DTV Parameter2 299
7. Background SW Update Sequence (Examples) 300
7.1. Update Success 300
7.2. Update Failed – CRC Error 301
7.3. Update Failed – Invalid Fragment 302
8. DTV parameter and DTV parameter 2 data range 303
8.1. DTV Parameter data range 303
8.2. DTV Parameter 2 data range for Thrush 305

8.3. DTV Parameter 2 data range for New Frontier 311

1. Introduction

This document is described the communication specification of HOST CPU and B/E.

2. UART Communication Specification

2.1. Communication Configuration

Baud rate	921600bits ※1
Start Bit	1
Stop Bit	1
Parity	None
H/W Flow Control	None

Baud rate : Default Baud rate is Below

- TRS231BSxxxx : 921600bits
- TRH231BSxxxx : 921600bits
- FRO232BSyyzz : 19200bits

2.2. Communication Format

Header Type 1

Header Part					Data Part	
Start Flag	Type	Request No	Data Length	Header CRC8	Data	Data CRC8
← 16bit → A5 5A	← 8bit →	← 8bit →	←8bit→	← 8bit →	← *X*byte →	← 8bit →

Header Type 2 (Using for TRS231BSxxxx and TRH231BSxxxx)

Header Part					Data Part	
Start Flag	Type	Request No	Data Length	Header CRC8	Data	Data CRC8
← 16bit → A5 59	← 8bit →	← 8bit →	←16bit→	← 8bit →	← *X*byte →	← 8bit →

Start Sync :	A5h,5Ah : Data Length Field is 8 bit A5h,59h : Data Length Field is 16bit
Type :	Type 00h : Command(Need Ack) 01h : Command(No Need Ack) 80h : ACK 90h : NAK(Header CRC Error) 91h : NAK(Data CRC Error)
Request No. :	Sequential Number 00h ~ FFh
Data Length :	Length includes the following fields. Unit : Model dependent - Data - Data CRC In case of ACK/NAK, this field is always 0.
Header CRC8 :	CRC include the following fields. - Start Flag - Type - Request No - Data Length 8bit CRC Value(Refer to the following Note)
Data :	Refer to each command.
Data CRC8 :	CRC include following field. - Data 8bit CRC Value(Refer to the following Note)

Note: CRC8: g(x) =x8+x5+x4+1

Request No. :

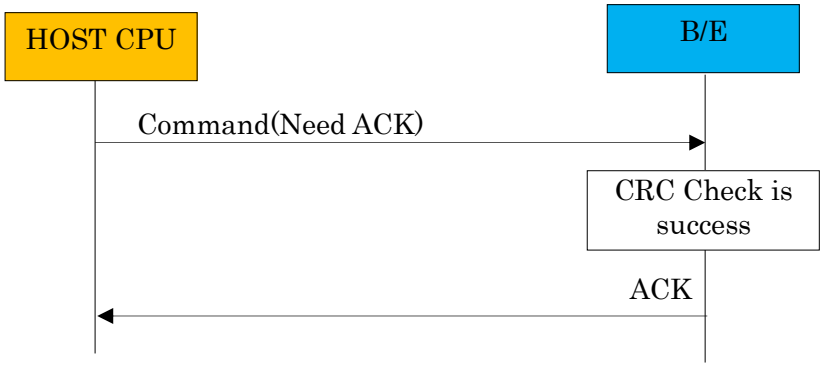
- Response from B/E shall use the same request No. from Host so that Host can distinguish the correct response.
- Response from HOST can be incremental request number which is different than Request from Module.
- For Events, B/E uses an incrementing Request No.

Header Type.

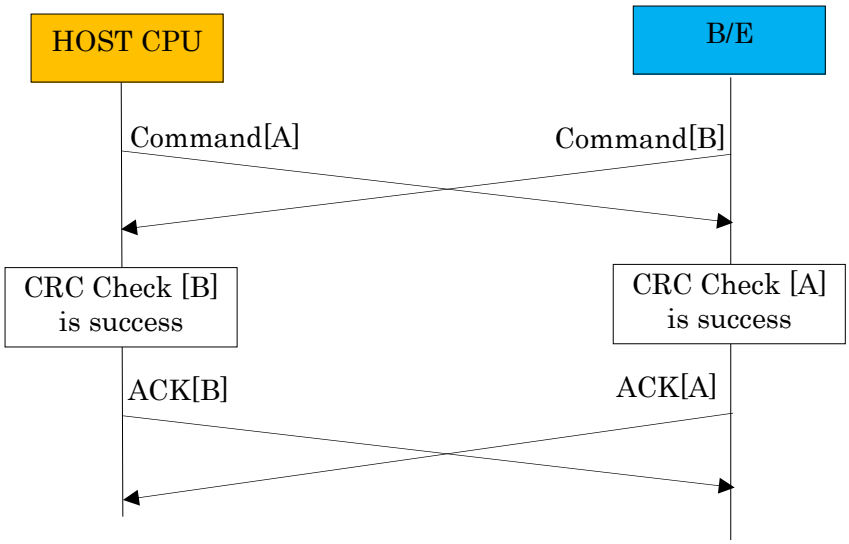
- Host should always use Header Type 1 for sending commands, responses, ACKs.

2.3. Communication Sequence

2.3.1. Host ⇒ B/E Sequence [Command(Need ACK) – ACK]

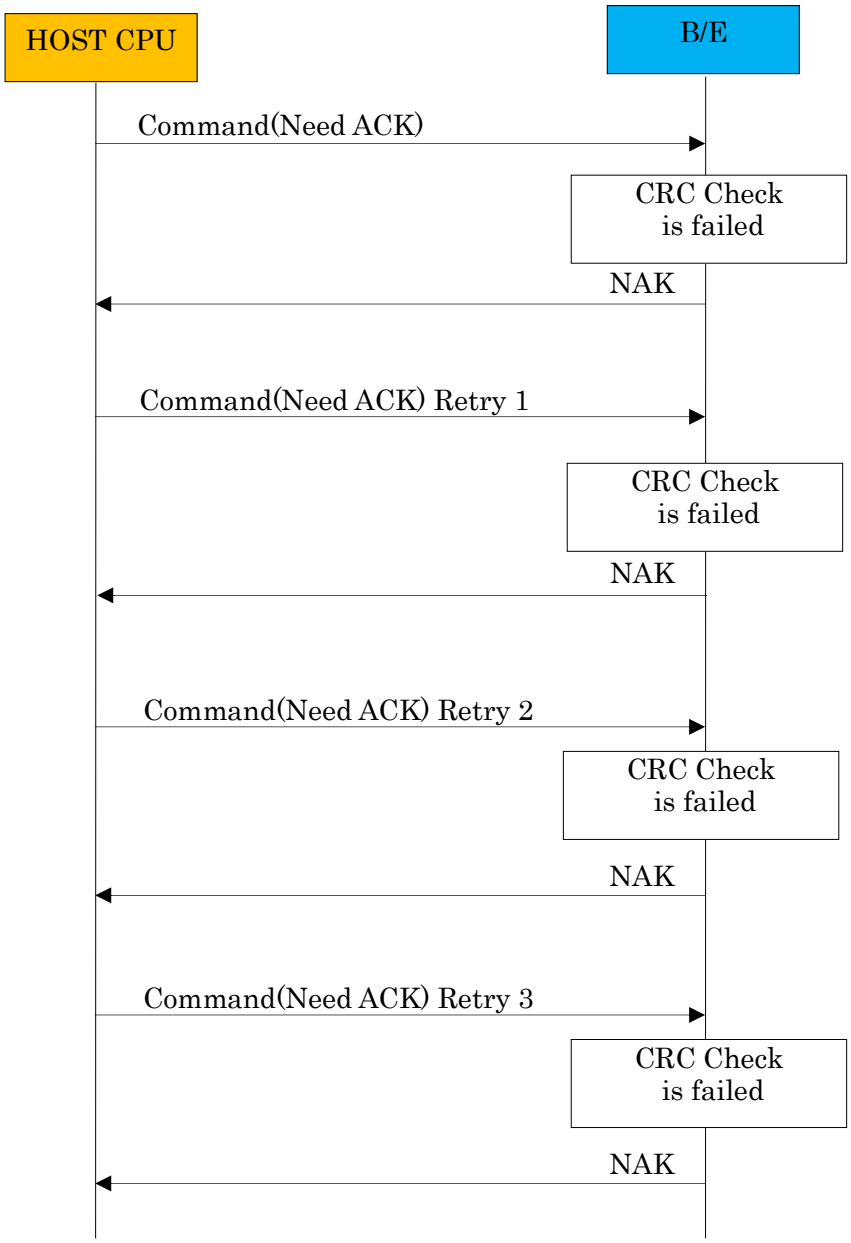


- If HOST CPU need ACK for Command, HOST CPU should transmit Command (Need ACK).
- If B/E receive Command (Need ACK), B/E should transmit ACK as the same Request No of command.
- If HOST CPU receive ACK as the same Request No for command, it means that the command communication is completed.
- If HOST CPU receive ACK as the different Request No for command, the ACK should be discarded.
- HOST CPU shouldn't transmit the next command until receiving the ACK of the same Request No (except retry out).



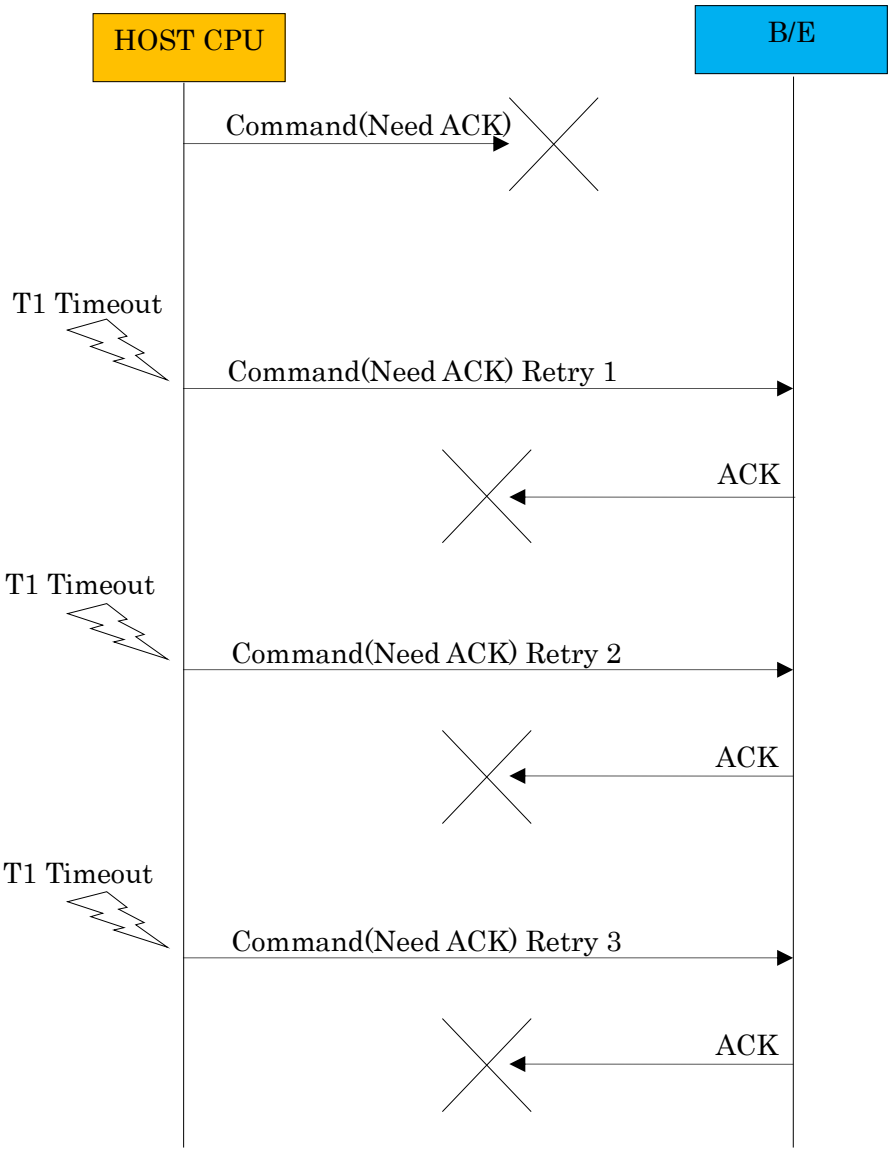
- If HOST CPU receive the command ([B]) from B/E before receiving the ACK ([A]) for the command ([A]), HOST CPU should process the command ([B]) from B/E.

2.3.2. Host ⇒ B/E Sequence [Command(Need ACK) – NAK]



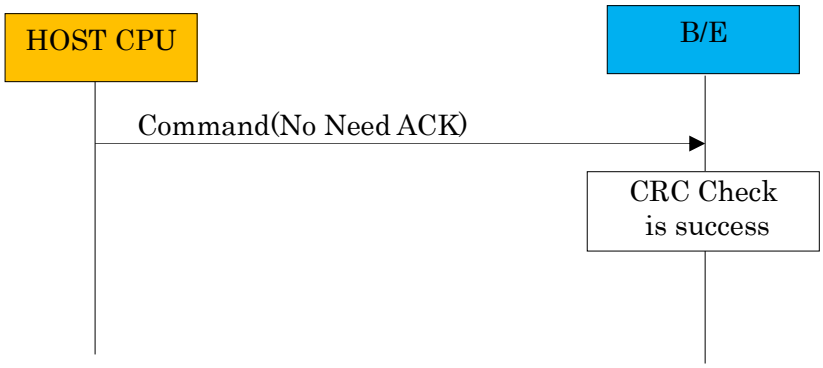
- If Command (Need ACK) is failed, B/E should return NAK.
- If HOST CPU receive NAK as the same Request No for command, HOST CPU retry to transmit the same command.
- Max of retry count is 3.
- 1st command and retry command should be same Request No.
- If HOST CPU receive NAK as the different Request No for command, the NAK should be discarded.

2.3.3. Host ⇒ B/E Sequence [Command(Need ACK) – Timeout]



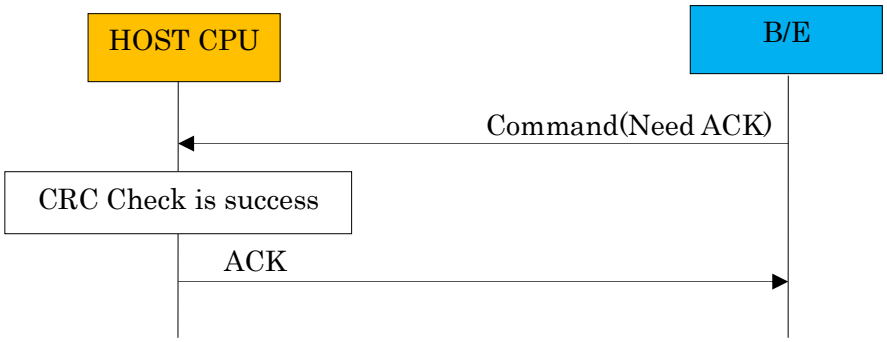
- If HOST CPU transmit Command (Need ACK), HOST CPU should start timer.
- If HOST CPU don't receive ACK as the same Request No for command within T1 sec, HOST CPU retry to transmit the same command. (Refer to "2.5.Timeout Value")
- Max of retry count is 3.
- 1st command and retry command should be same Request No.

2.3.4. Host ⇒ B/E Sequence [Command(No Need ACK)]

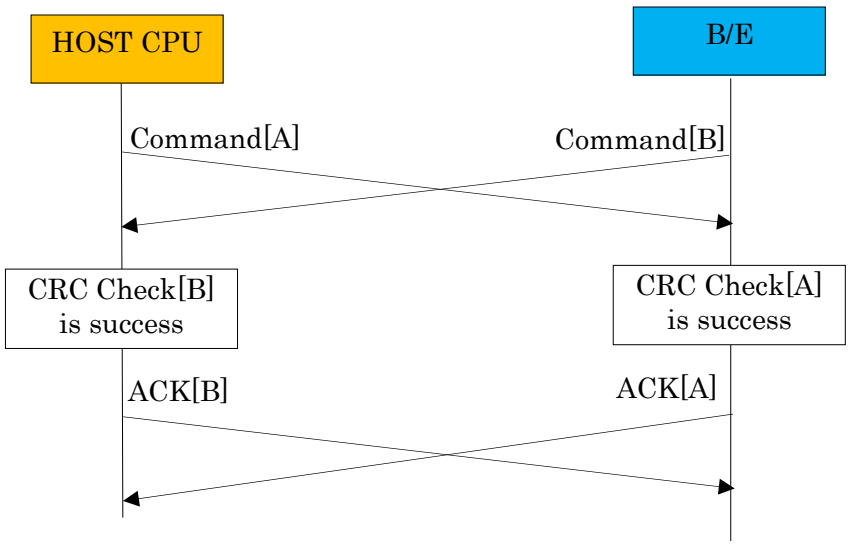


- If HOST CPU doesn't need ACK for Command, HOST CPU should transmit Command (No Need ACK).
- If B/E receive Command (No Need ACK), B/E shouldn't transmit ACK as the same Request No of command.

2.3.5. B/E ⇒ Host Sequence [Command(Need ACK) – ACK]

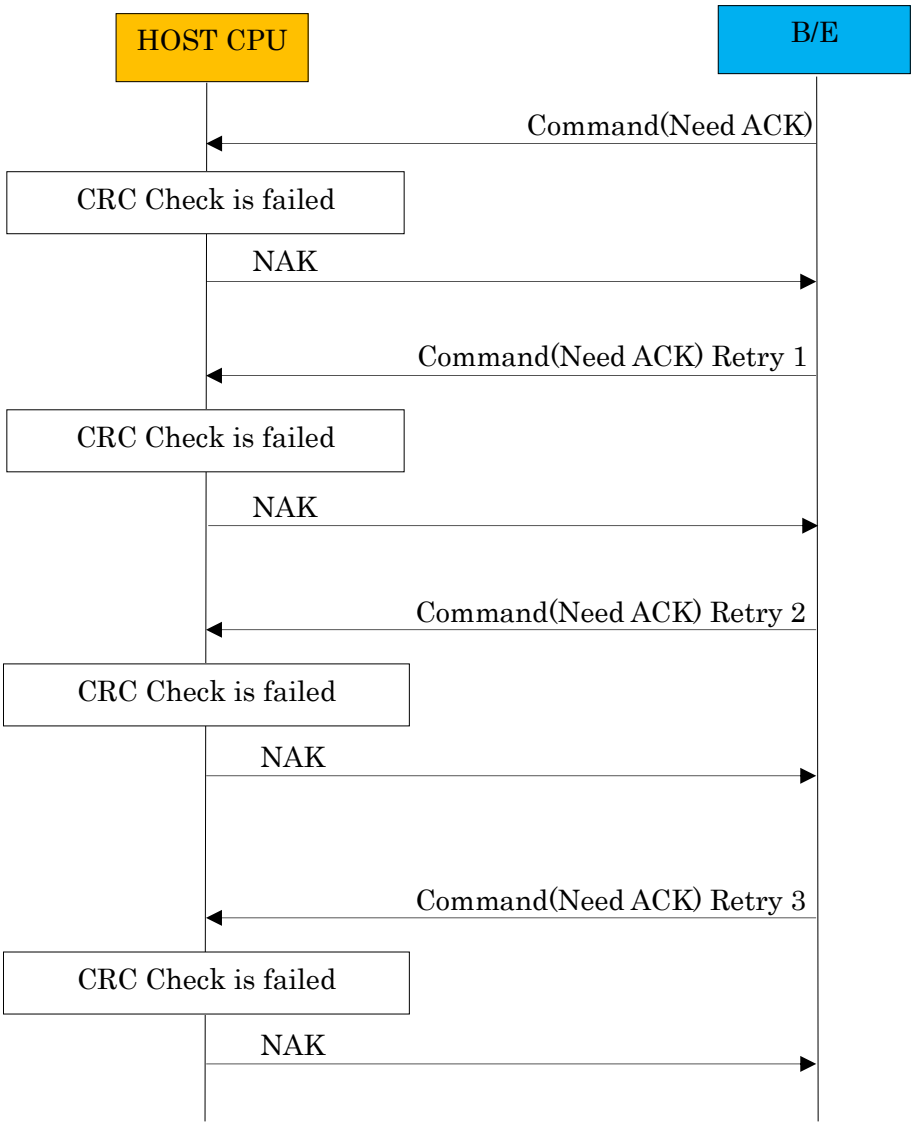


- If B/E need ACK for Command, B/E should transmit Command (Need ACK).
- If HOST CPU receive Command (Need ACK), HOST CPU should transmit ACK as the same Request No of command.
- If B/E receive ACK as the same Request No for command, it means that the command communication is completed.
- If B/E receive ACK as the different Request No for command, the ACK should be discarded.
- B/E shouldn't transmit the next command until receiving the ACK of the same Request No (except retry out).



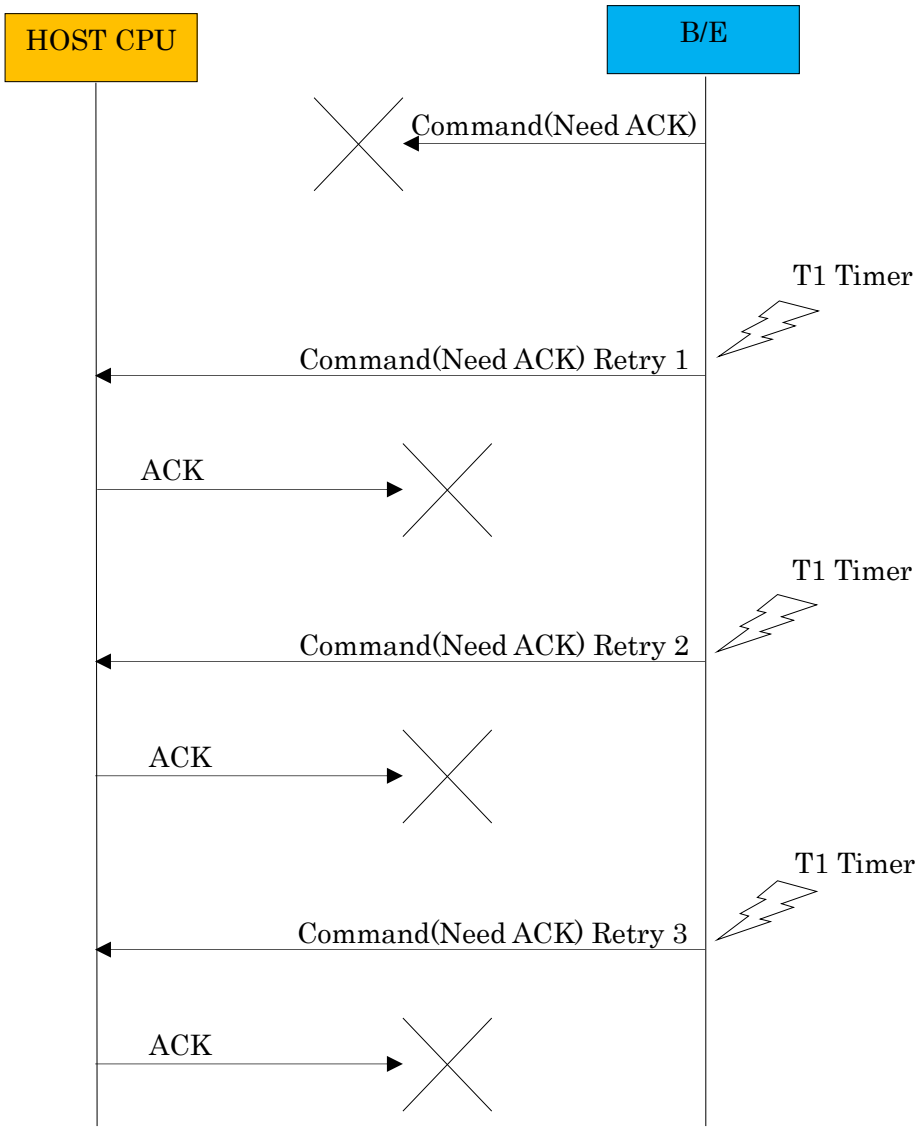
- If B/E receive the command ([A]) from HOST CPU before receiving the ACK ([B]) for the command ([B]), B/E should process the command ([A]) from HOST CPU, after receiving the ACK ([B]) for the command ([B]).

2.3.6. B/E ⇒ Host Sequence [Command(Need ACK) – NAK]



- If Command (Need ACK) is failed, HOST CPU should ignore the data and return NAK.
- If B/E receive NAK as the same Request No for command, B/E retry to transmit the same command.
- Max of retry count is 3.
- 1st command and retry command should be same Request No.
- If B/E receive NAK as the different Request No for command, the NAK should be discarded.

2.3.7. B/E ⇒ Host Sequence [Command(Need ACK) – Timeout]



- If B/E transmit Command (Need ACK), B/E should start timer.
- If B/E don't receive ACK as the same Request No for command within T1 sec, B/E retry to transmit the same command. (Refer to "2.5.Timeout Value")
- Max of retry count is 3.
- 1st command and retry command should be same Request No.
- HOST CPU should execute only once with same Request No.

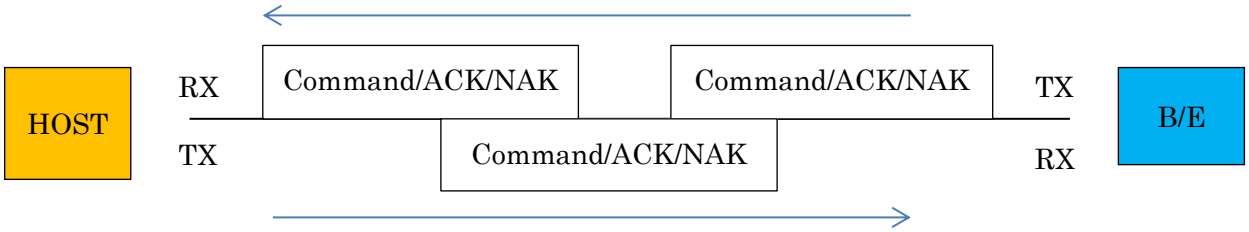
2.3.8. B/E ⇒ Host Sequence [Command(No Need ACK)]



- If B/E doesn't need ACK for Command, B/E should transmit Command (No Need ACK).
- If HOST CPU receive Command (No Need ACK), HOST CPU shouldn't transmit ACK as the same Request No of command.
- If CRC check is failed Host should ignore the data completely.

2.4. Communication Specification

・ The UART communication support full duplex.



2.5. Timeout Value

Item	Value	Comment
T1	1sec	Command Retry Timer

3. Error Code

Error Code	Description
01h	Invalid Parameter
02h	Invalid Data
03h	Invalid Length
04h	No Service
05h	Not Support
06h	Can't receive this channel, because weak signal.
07h	Can't receive this channel.
08h	This channel is not being broadcast.
09h	TRMP Information is invalid
0Ah	Invalid Status
0Bh	RMP Error EC21
0Ch	RMP Error EC22
0Dh	RMP Error EC23
20h	Tuner Error(I2C)
FFh	Unknown Error

RMP Error EC21：

● Error details:

This error is displayed when a work key for a transmitted ECM does not exist (when an EMM with a valid work key has not yet been received and when the time taken after selecting a station is less than twice the maximum value of an EMM resending cycle).

● Error message example:

The reception control data is not set. Please wait for a moment. (It may take a maximum of 30 seconds.)

● Code: EC21

● Others:

This error can be displayed as the error message above. It is also effective to display the error such that viewers can easily understand the current status using the progress bar, remaining time display, etc.

RMP Error EC22：

● Error details:

This error is displayed when a work key for a transmitted ECM does not exist (when an EMM with a valid work key has not yet been received and when the time taken after selecting a station exceeds twice the maximum value of an EMM resending cycle).

● Error message example:

This channel cannot be viewed. Please contact the customer center of the channel.

● Code: EC22

RMP Error EC23：

● Error details:

This error occurs when an access control descriptor included in the PMT is not consistent with CA_system_id as specified in Part 2 of this volume. (See 5.4 in Part 2 of this volume.)

● Error message example:

This program cannot be viewed with this receiver. Please contact the customer center of the channel.

● Code: EC23

4. DTV Status

DTV Status	Description
01h	Booting(Power OFF→ON) (Not support)
02h	Shutdown(Power ON→OFF)
03h	In Service
04h	Low Power
05h	DTV OFF
06h	DTV OFF with EPG SCAN

5. Data Detail

5.1. Command List

5.1.1. HOST⇒B/E Command

Code (Data0/1)	Command		DTV	
0001h	Channel Scan		X	
0002h	Channel Set		X	
0003h	Channel Up/Down		X	
0004h	Physical Channel Set		X	
0005h	Search Up/Down		X	
0006h	Broadcast Search		X	
			X	
			X	
0009h	ES Switch		X	
000Ah	Direct Channel Select		X	
0101h	DTV Initialize		X	
			X	
			X	
0104h	Get DTV Status		X	
0105h	Shutdown		X	
0106h	Watch Dog		X	
0107h	Get TS Name		X	
0108h	Set auto tune Mode		X	
010Ah	Set Closed Caption Mode		X	
010Bh	Set Auto BC Search Mode		X	
010Ch	Get Configure		X	
010Dh	Set Audio mode		X	
0110h	Set Residential Area		X	
0111h	Set Auto Event Relay		X	
0112h	Display EPG		X	
0113h	Get Version		X	
0114h	Get Audio ES		X	
0117h	Update Preset Table		X	
0118h	Get Program Information		X	
0119h	Set Baud rate		X	
011Ch	Get Service Information		X	
011Dh	Set DTV Parameter		X	
011Eh	Get DTV Parameter		X	
011Fh	Display Banner		X	

0120h	Set DTV System		X	
0121h	Service ID Channel Select		X	
0122h	Get Program Description		X	
0123h	Get DTV System		X	
0124h	Get Program List		X	
0125h	Get Program Name		X	
0126h	Get Logo Data		X	
0127h	Set Low Power Mode		X	
0128h	BML Forced Exit		X	
0129h	Get RMP Maker ID		X	
012Ah	Get Number of Program info		X	
012Bh	Get Audio ES info		X	
012Ch	Set DTV Mode		X	
012Dh	Set Video Resolution		X	
012Eh	Get Video Resolution		X	
012Fh	Set Banner Config		X	
0130h	Get Banner Config		X	
0131h	Set Character Set		X	
0132h	Get Character Set		X	
0133h	Get TS Name with UTF-8		X	
0134h	Get Program Information with UTF-8		X	
0135h	Set DTV Parameter 2		X	
0136h	Get DTV Parameter 2		X	
0137h	Get Service Information with UTF-8		X	
0138h	Get Program Description with UTF-8		X	
0139h	Get Program Name with UTF-8		X	
013Ah	Get Audio ES info with UTF-8		X	
013Bh	Set User ID		X	
013Ch	Get User ID		X	
013Dh	Set EPG Config		X	
013Eh	Get EPG Config		X	
013Fh	Set Video Color		X	
0140h	Get Video Color		X	
0141h	Set Event Mask		X	
0142h	Get Event Mask		X	
0143h	Get Part Number		X	
0144h	Get Program Name by 3Digit Number		X	
0145h	Get Program Name with UTF-8 by 3Digit Number		X	
0146h	Get Logo Data by 3Digit Number		X	
0147h	Start Background Search		X	
0148h	Start Background Scan		X	
01F0h	DTV Restart		X	

5.1.2. HOST⇒B/E Event

Code (Data0/1)	Command		DTV	
0201h	Key Event		X	

5.1.3. B/E⇒HOST Command

Code (Data0/1)	Command		DTV	
8001h	Wakeup		X	
8003h	Boot Mode		X	

5.1.4. B/E⇒HOST Event

Code (Data0/1)	Command		DTV	
8201h	DTV Status		X	
8203h	Scan Status		X	
8204h	Search Status		X	
8206h	DTV Event		X	
8208h	EWS Event		X	
8209h	Program Information		X	
820Ah	EPG Status		X	
820Bh	Event relay		X	
820Ch	Number of Program info		X	
820Dh	Program Information with UTF-8		X	
820Eh	TS Name Update		X	
820Fh	TS Name Update with UTF-8		X	
8210h	Logo Update		X	
8211h	EPG Button Status		X	
8212h	BML Status		X	
8213h	C/N Information		X	
8214h	Manual EPG Scan Result		X	
8301h	DTV Critical Error		X	

5.1.5. SW Update Command

Code (Data0/1)	Command		DTV	
A001h	SW Update		X	
A002h	SW Update Status		X	
A101h	Background SW Update Start		X	
A102h	Background SW Update Cancel		X	
A103h	Background SW Update		X	
A104h	Background SW Update Status		X	

5.1.6. HOST⇒BE Test & Debug Command

Can use these commands for testing purposes only.

Code (Data0/1)	Command		DTV	
F005h	Test Mode		X	
F006h	Set Debug Parameter		X	
F008h	Hardware Test		X	

F009h	2nd TS Test		X	

5.1.7. B/E⇒HOST Test & Debug Command

Code (Data0/1)	Command		DTV	
F007h	Monitor Status		X	
F00Ah	2nd TS Test Status		X	

5.2. HOST⇒B/E Command

5.2.1. Channel Scan

(1) Channel Scan Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	04h or 05h
Header CRC8 :	--
Data 0 :	00h Channel Scan(Upper)
Data 1 :	01h Channel Scan(Lower)
Data 2 :	Operation 00h : Start 10h : Start with Update Home Preset 11h : Start with Update Drive Preset FFh : Stop
Data 3 :	Update Operation (optional) 00h : Update without Tune top channel 01h : Update with Tune top channel
Data CRC8 :	--

(2) Channel Scan Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	00h Channel Scan(Upper)
Data 1 :	01h Channel Scan(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- If Operation is 00h, B/E start Channel Scan.
- If Operation is FFh, B/E stop Channel Scan.
- When finishing and stopping Channel Scan, B/E doesn't update automatically, if Data2 is 00h. For updating the preset table, HOST have to send "5.2.13.Update Preset Table".

  Operation in Data 2 with Update Home/Drive Preset case, no need "5.2.13. Update Preset Table".
- If you start channel scan during back search or back scan in Dynamic 4Diver System, back search or back scan will be canceled immediately.
- When BE received the scan command, response is sent to Host immediately after starting the scan.

- Data 3 : Optional Function

Data 2	Data 3	Operation
10h or 11h	None / 00h	Update without Tune top channel (Idle status)
10h or 11h	01h	Update with Tune top channel
Other than the above		Scan without update *If optional data(Data3)is there, it will be invalid error.

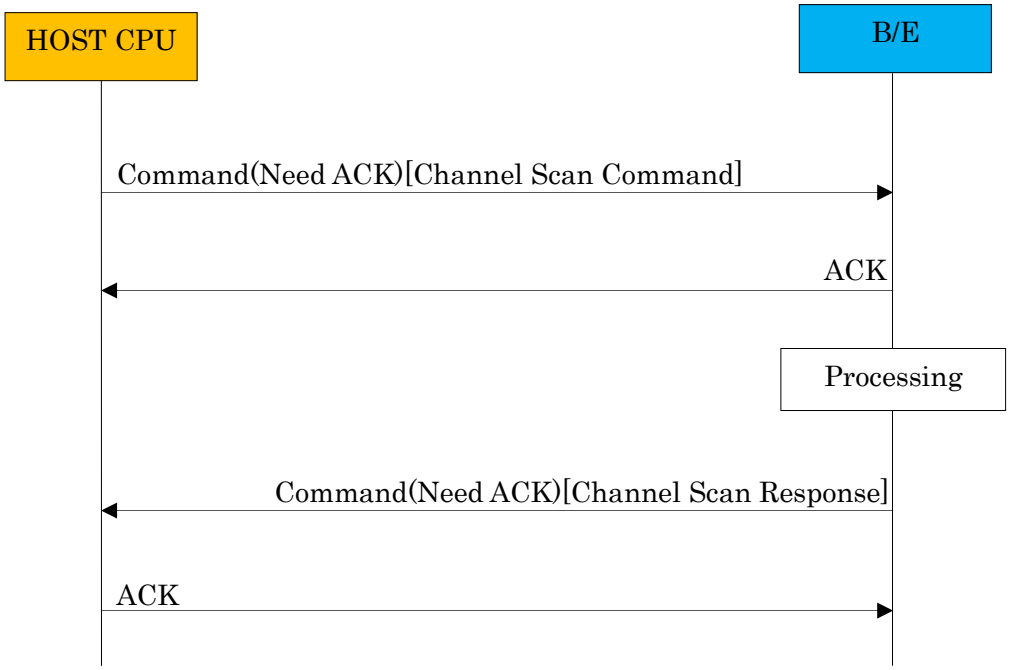
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.2.2. Channel Set

(1) Channel Set Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	00h Channel Set(Upper)
Data 1 :	02h Channel Set(Lower)
Data 2 :	Preset Number 00h : Home Preset 01h : Drive Preset FFh : Last Preset
Data 3 :	Preset Channel Number 01h ~ 0Ch
Data CRC8 :	--

(2) Channel Set Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	00h Channel Set(Upper)
Data 1 :	02h Channel Set(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- B/E start tuning Preset Channel Number.
- Start playback of the first service.

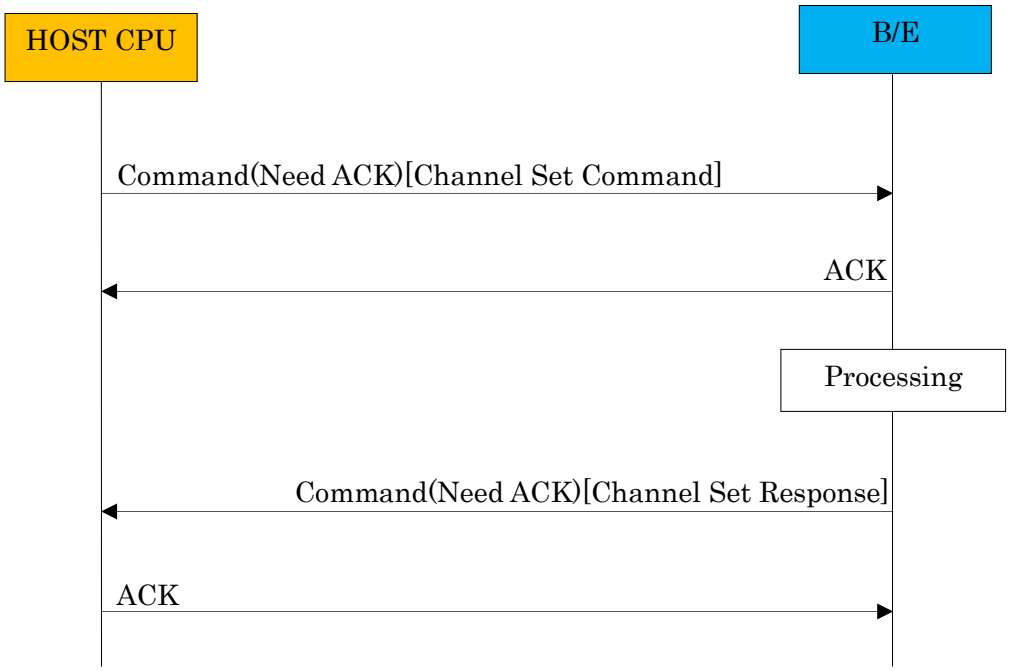
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid. - Preset Number is not 00h or 01h or FFh. - Preset Channel Number is not 01h~0Ch.
03h(Invalid Length)	Data Length is invalid.

04h(No Service)	Desired Service is not found in Preset Table. Eg: Auto Tune Mode is Oneseg and the set channel doesn't have an Oneseg service
FFh(Unknown Error)	Unknown error

(5) Sequence



5.2.3. Channel Up/Down

(1) Channel Up/Down Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	00h Channel Up/Down(Upper)
Data 1 :	03h Channel Up/Down(Lower)
Data 2 :	Preset Number FFh : Last Preset
Data 3 :	Direction 00h : Up 01h : Down 10h : Up (First Service Select) 11h : Down (First Service Select) 20h : Up without Channel Change 21h : Down without Channel Change
Data CRC8 :	--

(2) Channel Up/Down Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	00h Channel Up/Down(Upper)
Data 1 :	03h Channel Up/Down(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- B/E start tuning by Up/Down in Preset Table.
- If different service is there, switch service.
- Suspend service is skipped.
- Data 3 : 20h/21h Same program is skipped.

  e.g., 011->up->012->up->013->up->011.
- Select the last channel in the idle state.
- If there is no last channel, result error and do not select channel.

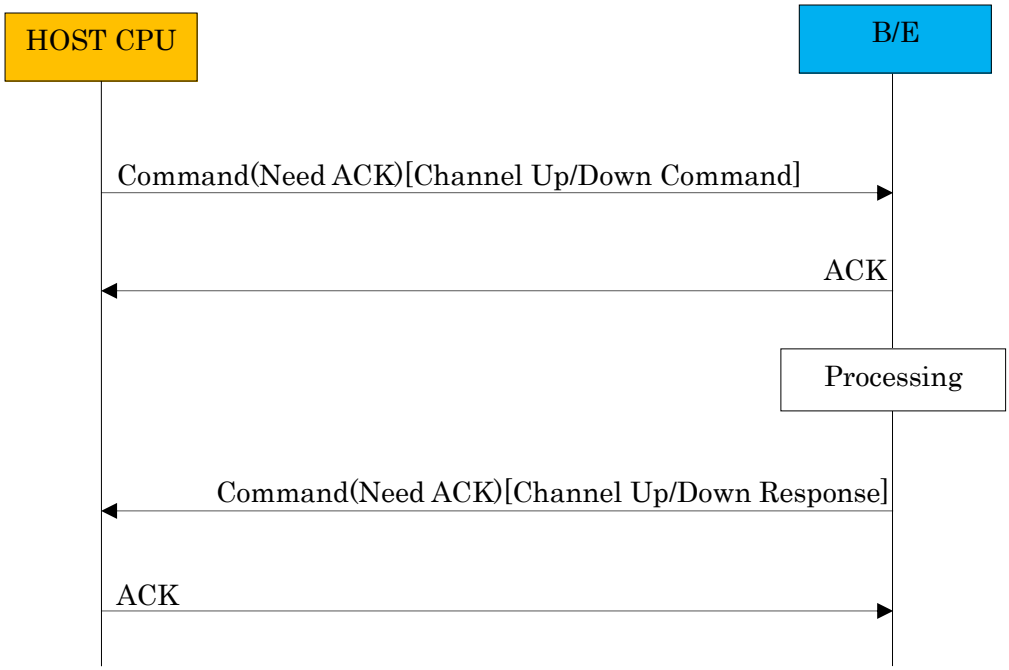
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid. - Preset Number is not FFh. - Direction is other than the defined value.
03h(Invalid Length)	Data Length is invalid.
04h(No Service)	Preset table is empty.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.2.4. Physical Channel Set

(1) Physical Channel Set Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	00h Physical Channel Set(Upper)
Data 1 :	04h Physical Channel Set(Lower)
Data 2 :	Preset Number FFh : Last Preset
Data 3 :	Physical Channel Number 0Dh ~ 34h
Data CRC8 :	--

(2) Physical Channel Set Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	00h Physical Channel Set(Upper)
Data 1 :	04h Physical Channel Set(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- B/E start tuning by Physical Channel Number.
- If the tuning is successfully completed, the process of the physical channel set command has ended normally, so the search stop command is not necessary.
- If the tuning is not completed (e.g., when there is no signal input), the channel selecting operation is continued. Host is free to issue other commands like scan or tune.

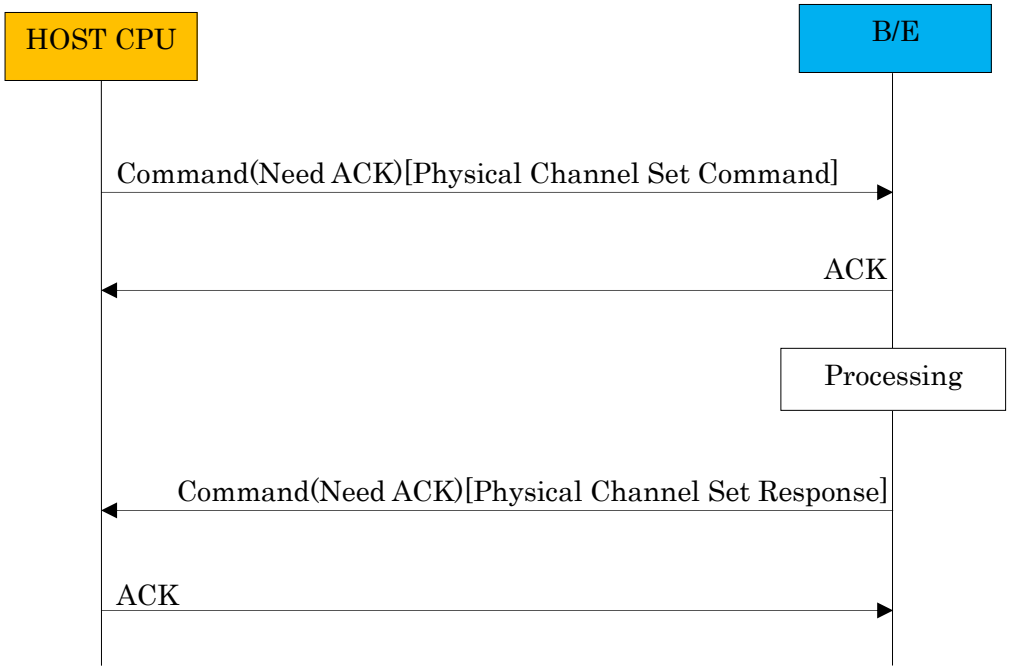
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid. - Preset Number is not FFh. - Physical Channel Number is not 0Dh~34h
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.2.5. Search Up/Down

(1) Search Up/Down Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	00h Search Up/Down(Upper)
Data 1 :	05h Search Up/Down(Lower)
Data 2 :	Preset Number FFh : Last Preset
Data 3 :	Operation 00h : Start to search up 01h : Start to search down FFh : Stop to search
Data CRC8 :	--

(2) Search Up/Down Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	00h Search Up/Down(Upper)
Data 1 :	05h Search Up/Down(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- B/E starts to search.
- If Operation is 00h, B/E start to search up.
- If Operation is 01h, B/E start to search down.
- After starting search up/down, "Search Status" starts to notify.
- If Operation is FFh, B/E stop to search.
- After finishing or stopping search, "Search Status" stops to notify.
- When stopping search, B/E tune the last channel. But if B/E doesn't have valid last channel, B/E become idle status.
- Start next physical channel, and increment (decrement) physical channel number.
- If channel detected, start playback of the first service.
- If any channel is not detected, tune the last channel.
- This will abort background search if it was running.

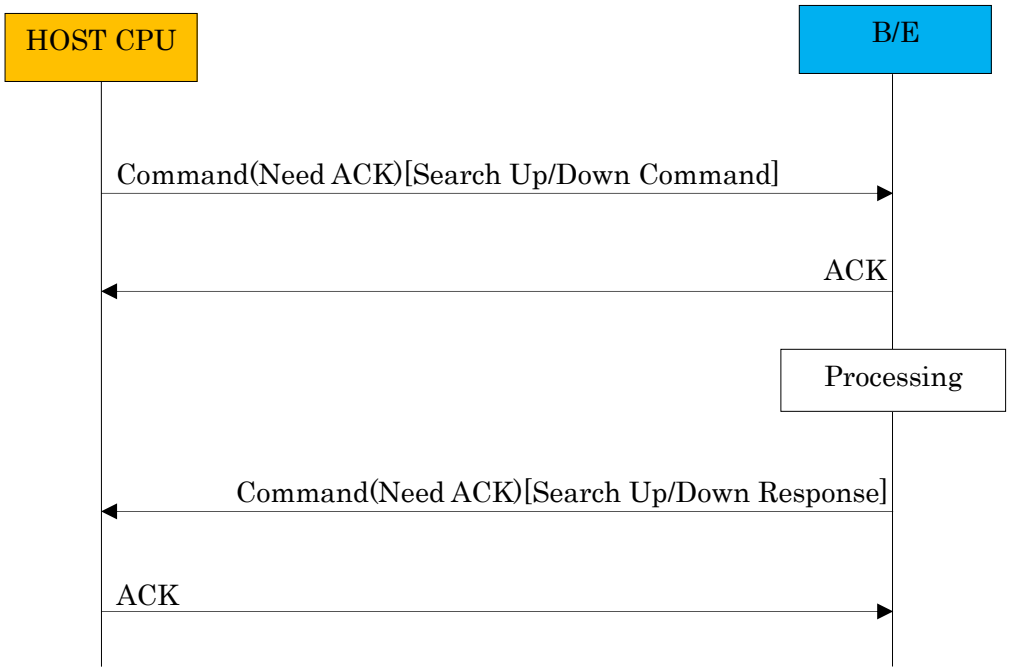
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid. - Preset Number is not FFh. - Operation is not 00h or 01h or FFh.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.2.6. Broadcast Search

(1) Broadcast Search Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	04h
Header CRC8 :	--
Data 0 :	00h Broadcast Search(Upper)
Data 1 :	06h Broadcast Search (Lower)
Data 2:	Search Mode 01h : Relay Search Start 02h : Affiliated Search Start 03h : Relay + Affiliated Search Start FFh : Search Stop
Data CRC8 :	--

(2) Broadcast Search Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	00h Broadcast Search(Upper)
Data 1 :	06h Broadcast Search (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- B/E starts to search with foreground.
- If Search mode is 01h, B/E start relay station search.
- If Search mode is 02h, B/E start affiliated station search.
- If Search mode is 03h, B/E start relay and affiliated station search.
- After starting Broadcast search, "Search Status" starts to notify.
- If search mode is FFh, B/E stop relay/affiliated station search. Then B/E return last channel.
- After finishing or stopping Broadcast search, "Search Status" stops to notify.
- This will abort background search if it was running.

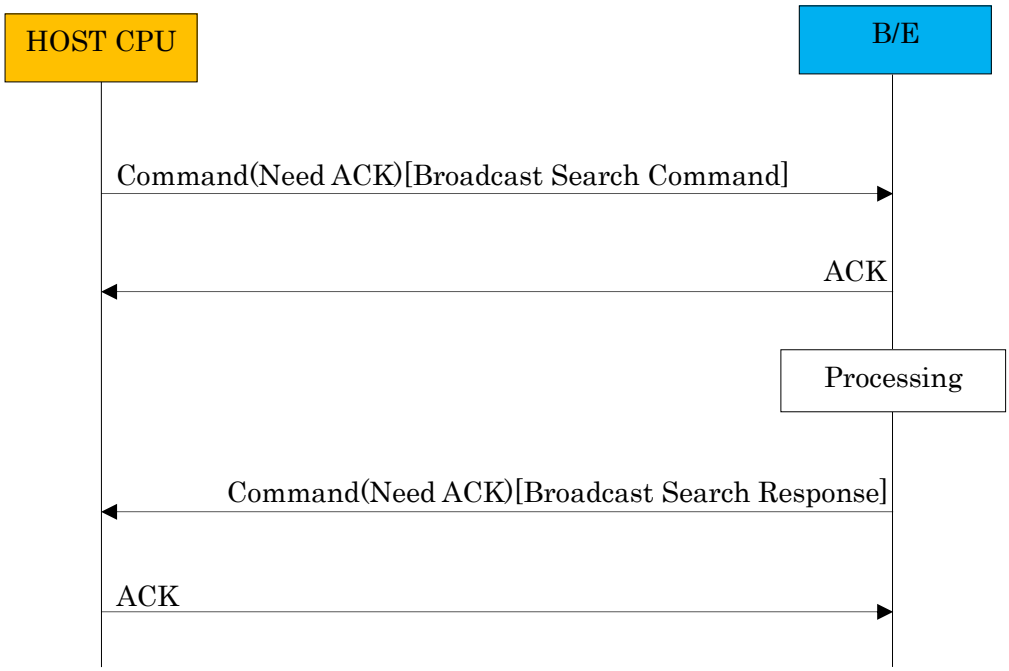
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid. - Search Mode is not 01h or 02h or 03h or FFh.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.2.7. ES Switch

(1) ES Switch Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	00h ES Switch(Upper)
Data 1 :	09h ES Switch(Lower)
Data 2 :	ES Switch Type 01h : Video ES Switch(Not support) 02h : Audio ES Switch
Data 3 :	Component Tag
Data CRC8 :	--

(2) ES Switch Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	00h ES Switch(Upper)
Data 1 :	09h ES Switch(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- Switch Audio ES.
- Not support to switching Video ES.
- Component Tag of Audio ES should be set Component Tag of "5.4.5.Get Audio ES".

(4) Error Handling

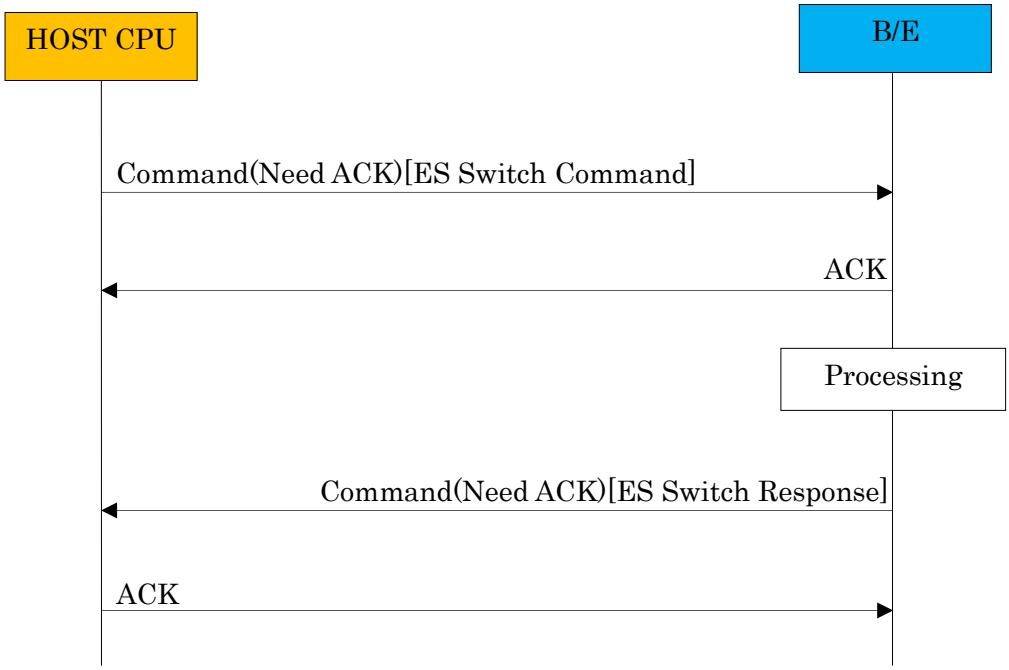
- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid. - ES Switch Type is not 01h or 02h.
03h(Invalid Length)	Data Length is invalid.
05h(Not Support)	The parameter is not supported.

	- ES Switch Type is 01h.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.2.8. Direct Channel Select

(1) Direct Channel Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	06h 07h (Optional)
Header CRC8 :	--
Data 0 :	00h Direct Service(Upper)
Data 1 :	0Ah Direct Service(Lower)
Data 2 :	Three Digit Number(Upper) Refer to "(3) Description"
Data 3 :	Three Digit Number(Lower) Refer to "(3) Description"
Data 4 :	Branch Identifier Number(0~9)
Data 5 :	Preset number (Optional) 00h : Home Preset 01h : Drive Preset FFh : Last Preset
Data CRC8 :	--

(2) Direct Channel Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	00h Direct Service(Upper)
Data 1 :	0Ah Direct Service(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- B/E start tuning by the three-digit number and the branch identifier number.
- Branch Identifier for not-stored channel should be set to 00h.
- If Data Length is 06h, B/E considers it as Last Preset.
- Refer to the following about the formant of Three Digit Number.

three-digit number	Data 2	Data 3
011	00h	0Bh
712	02h	C8h

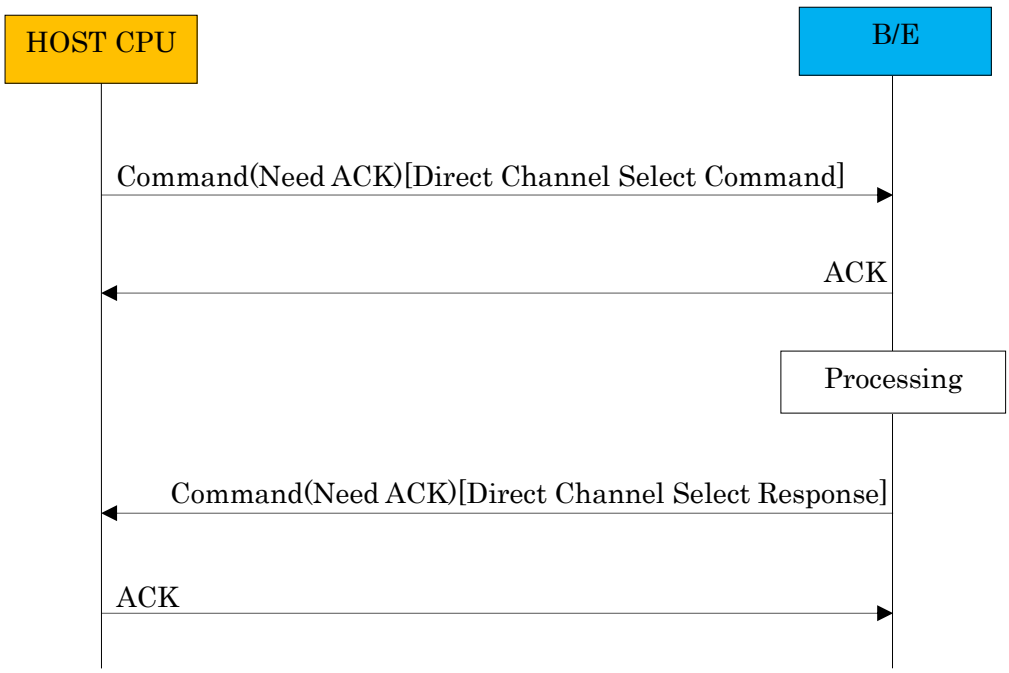
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid. - Three-digit number is not 000~199, 600~799. - Branch Identifier Number is not 00h ~ 09h. - Preset number is not 00h/01h/FFh.
03h(Invalid Length)	Data Length is invalid.
04h(No Service)	The specified service is not there in Preset table.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.2.9. DTV Initialize

(1) DTV Initialize Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	04h
Header CRC8 :	--
Data 0 :	01h DTV Initialize(Upper)
Data 1 :	01h DTV Initialize (Lower)
Data 2 :	Target Data 00h : All Data Initialize 01h : Configuration Data Initialize 02h : Home Preset Initialize 03h : Drive Preset Initialize 8Xh : User Data Initialize (X=User ID)
Data CRC8 :	--

(2) DTV Initialize Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h DTV Initialize(Upper)
Data 1 :	01h DTV Initialize (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- Initialize DTV (Channel List/DTV Config/TRMP Information/Logo/BML Information)
- Refer to the followings about the initial values of DTV Setting. Initialization of each item is depending on "Data 2"

Items	Initial Value	Data 2
Audio Mode	00h : Main	00h or 01h
Closed Caption Mode	00h : Off	00h or 01h
Auto Tune Mode	＊1	00h or 01h
Auto Broadcast Search Mode	＊2	00h or 01h
Auto Event Relay	＊3	00h or 01h
Region ID(Home Preset)	FFh	00h or 02h
Zip code(Home Preset)	All data are 00h	00h or 02h
Region ID(Drive Preset)	FFh	00h or 03h
Zip code(Drive Preset)	All data are 00h	00h or 03h
EPG Scale	00h(Type 1) ＊4	00h or 01h
＊1: The value is DTV Parameter 2 Auto tune mode default		

＊2: The value is DTV Parameter 2 Default Auto Broadcast Search mode
＊3: The value is DTV Parameter 2 Event relay Auto Event rely On/Off
＊4: If Type 1 is disabled, next available Type will work.

- B/E stop to playback and become the black screen during Initialize DTV (Preset Table, Last Channel, DTV Setting). If B/E has valid last channel after initialization, B/E tune the last channel.
  - Initialize Data Table

Content Data2	Config BML	Home LastCh	Drive LastCh	Logo RMP	Remark
00h	All User				All Initialize/Factory Reset
01h	Current User	—	—	—	Current User Configuration Data Initialize
02h	—	Current User	—	—	Current User Home Preset Initialize
03h	—	—	Current User	—	Current User Drive Preset Initialize
81h	User 1			—	User 1 All Initialize
82h	User 2			—	User 2 All Initialize
83h	User 3			—	User 3 All Initialize
84h	User 4			—	User 4 All Initialize
85h	User 5			—	User 5 All Initialize
86h	User 6			—	User 6 All Initialize

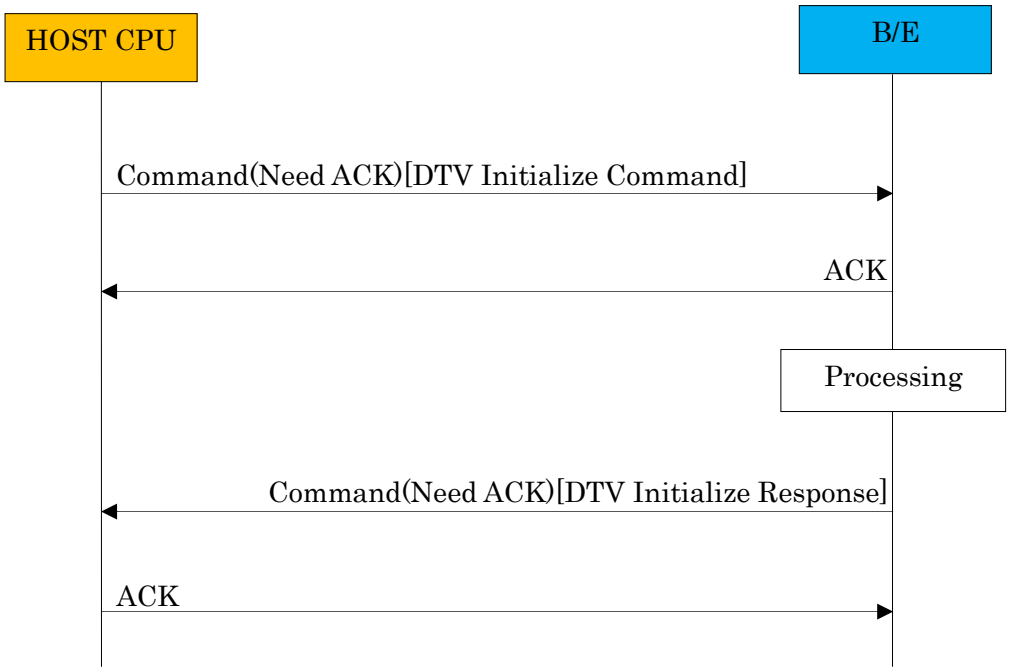
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 22.5sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid. - Target Data is not 00h ~ 03h.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence


HOST CPU
B/E
Command(Need ACK)[DTV Initialize Command]
ACK
Processing
Command(Need ACK)[DTV Initialize Response]
ACK

5.2.10. Shutdown

(1) Shutdown Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h~04h
Header CRC8 :	--
Data 0 :	01h Shutdown(Upper)
Data 1 :	05h Shutdown(Lower)
Data 2 :	Shutdown Mode(Option) 00h : immediately shutdown 01h : shutdown after storing data
Data CRC8 :	--

(2) Shutdown Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Shutdown(Upper)
Data 1 :	05h Shutdown(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- Shutdown B/E
- If Shutdown Mode is 00h, B/E stop to playback. Then B/E become to idle status.
- If Shutdown Mode is 01h, B/E stop to playback. Then B/E store DTV Information to NVRAM and become to idle status.
- If Shutdown Mode is omitted, B/E behavior same as Shutdown Mode=00h.

(4) Error Handling

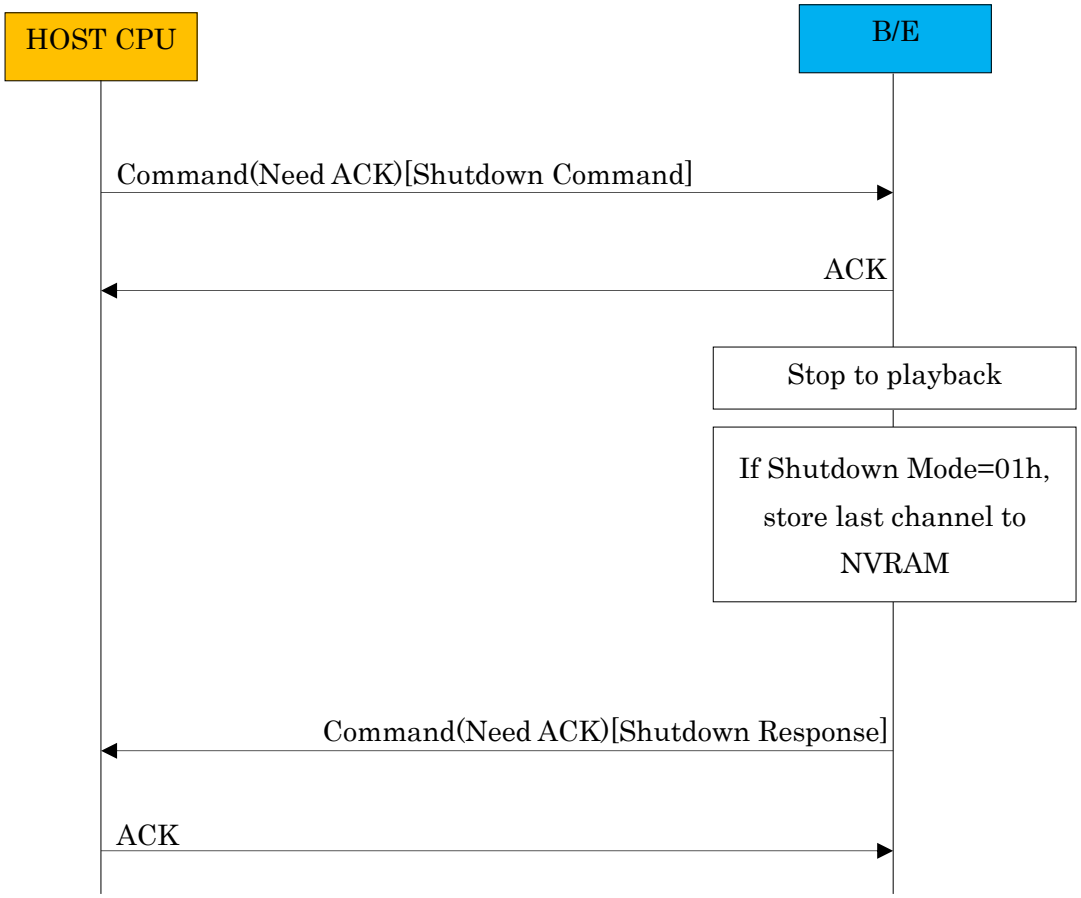
- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command (Shutdown Mode=00h).
  - ✧ Doesn't receive the response within 10sec after HOST transmit the command (Shutdown Mode=01h).
  - ✧ The result of response is NG.

・ Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid. - Shutdown Mode is not 00h ~ 01h.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



・ If Shutdown Mode=00h of Shutdown Command, Timeout of Shutdown Response is 1sec.

・ If Shutdown Mode=01h of Shutdown Command, Timeout of Shutdown Response is 10sec.

・ After shutdown B/E, issuing UART control command is prohibited, and any information output from the UART communication should be ignored except Get DTV Status command.

5.2.11. Watch Dog

(1) Watch Dog Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	01h : Command(No Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h
Header CRC8 :	--
Data 0 :	01h Watch Dog(Upper)
Data 1 :	06h Watch Dog(Lower)
Data CRC8 :	--

(2) Watch Dog Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	01h : Command(No Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Watch Dog(Upper)
Data 1 :	06h Watch Dog(Lower)
Data 2 :	00h : Reserved
Data 3 :	00h : Reserved
Data CRC8 :	--

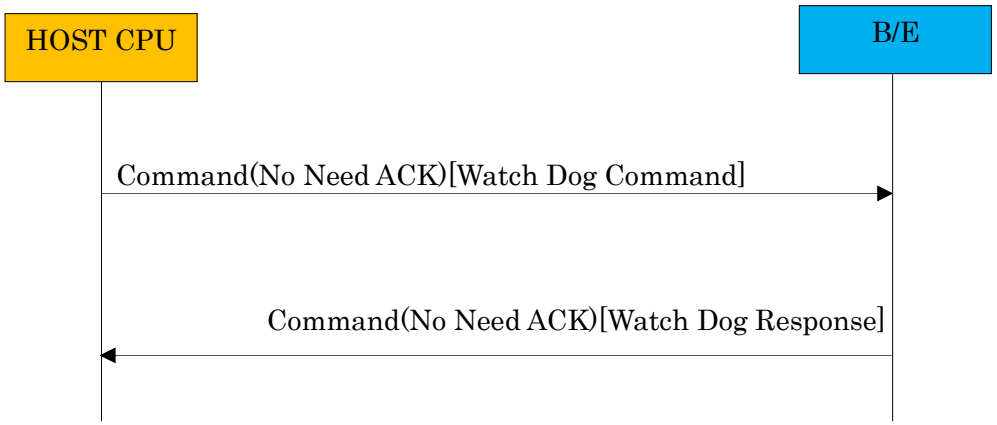
(3) Description

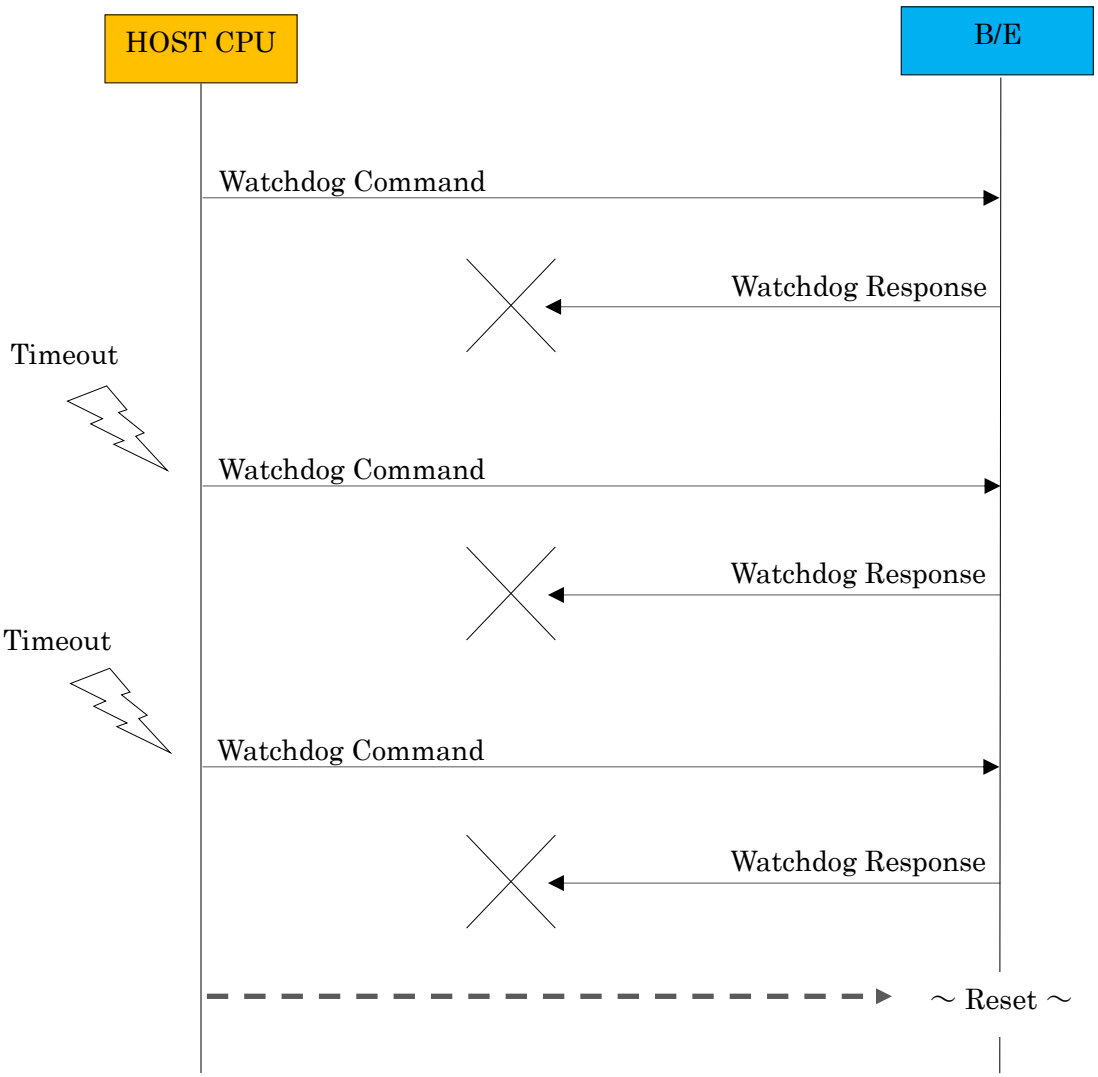
- Watch Dog
- During SW Update, HOST shouldn't send this command.

(4) Error Handling

- Reset module (Refer to sequence)

(5) Sequence


HOST CPU
B/E
Command(No Need ACK)[Watch Dog Command]
Command(No Need ACK)[Watch Dog Response]


HOST CPU
B/E
Watchdog Command
Watchdog Response
Timeout
Watchdog Command
Watchdog Response
Timeout
Watchdog Command
Watchdog Response
～ Reset ～

5.2.12. Display EPG

(1) Display EPG Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	04h or 05h(option)
Header CRC8 :	--
Data 0 :	01h Display EPG (Upper)
Data 1 :	12h Display EPG (Lower)
Data 2 :	Operation 00h : Display EPG List(Current segment) 01h : Display EPG Information 02h : Up Arrow 03h : Down Arrow 04h : Left Arrow 05h : Right Arrow 06h : Back 07h : Enter 08h : Switch segment 09h : Update 0Ah : Previous Day(Blue Button) 0Bh : Next Day(Red Button) 0Ch : Multi service(Green Button) 0Dh : Change Scale(Yellow Button) 0Eh : Cancel update 0Fh : Close All
Data 3 :	Display EPG Type (Option) Data2 = 00h : 00h : Last display type 01h~04h :EPG Type 1~4 Data2 != 00h : Fix 00h
Data CRC8 :	--

(2) Display EPG Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : 14h Result=NG : 05h
Data 0 :	01h Display EPG (Upper)
Data 1 :	12h Display EPG (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Screen Status 00h : No display 01h : EPG List(Fullseg) 02h : EPG Information(Fullseg) 03h : EPG List(Oneseg) 04h : EPG Information(Oneseg)
Data 5 :	Display EPG List 0: invalid 1:valid
Data 6 :	Display EPG Information 0: invalid 1:valid

Data 7 :	Up Arrow 0: invalid 1:valid
Data 8 :	Down Arrow 0: invalid 1:valid
Data 9 :	Left Arrow 0: invalid 1:valid
Data 10 :	Right Arrow 0: invalid 1:valid
Data 11 :	Back 0: invalid 1:valid
Data 12 :	Enter 0: invalid 1:valid
Data 13 :	Switch segment 0: invalid 1:valid
Data 14 :	Update 0: invalid 1:valid
Data 15 :	Previous Day(Blue Button) 0: invalid 1:valid
Data 16 :	Next Day(Red Button) 0: invalid 1:valid
Data 17 :	Multi service(Green Button) 0: invalid 1:valid
Data 18 :	Change Scale(Yellow Button) 0: invalid 1:valid
Data CRC8 :	--

(3) Description

- Display and control EPG.
- Refer to the following table about each operation

Screen / Operation	No EPG ①	EPG List ②	EPG Information ③
Display EPG List	⇒②	②	⇒②
Display EPG Information	⇒③	⇒③	③
Up Arrow	-(*1)	-(*2)	-(*3)
Down Arrow	-(*1)	-(*2)	-(*3)
Left Arrow	-(*1)	-(*2)	-(*1)
Right Arrow	-(*1)	-(*2)	-(*1)
Back	-(*1)	⇒①	⇒① ⇒② (*9)
Enter	-(*1)	⇒③	-(*1)
Switch segment	-(*1)	-(*4)	-(*1)
Update	-(*1)	-(*5)	-(*1)
Previous Day	-(*1)	-(*6)	-(*1)
Next Day	-(*1)	-(*6)	-(*1)
Multi service	-(*1)	-(*7)	-(*1)
Change Scale	-(*1)	-(*8)	-(*1)
Close All	-(*1)	⇒①	⇒①

*1: This operation is ignored.

*2: This operation moves the focus.

*3: This operation scrolls the screen.

*4: This operation switch Fullseg EPG List/Oneseg EPG List.

*5: This operation update EPG List.

*6: This operation moves previous day or next day.

*7: This operation switches all service and the top service.

*8: This operation changes the scale (4Level).

*9: If moving from EPG List, return to EPG List. If directly display EPG Information, close EPG Information.

- If B/E receive "Cancel update" during updating, B/E display updated EPG Information.
- During update, it is not possible to playback AV.

(EPG Scan use foreground antennas.)

- HOST can know screen status by Data4 of Response.
- HOST can know valid button by Data5~18 of Response.
- Notifies HOST about the real time button status via EPG Button Status event.
- If changing scale (Data2=0Dh), B/E store the last scale to NVRAM. When displaying at next, EPG is displayed with last scale.
- When receiving ack of "Display EPG Response", B/E update the screen.
- Change Scale: It changes as follows each time a command is sent.
  Level1 -> Level2 -> Level3 -> Level4 -> Level1 ・・・
  - Those Levels (Types) disabled with Data 2 of Set EPG Config will be skipped from the change scale sequence.
  - When Data Length = 04h and Data 2 = 00h : Display EPG List(Current segment), last type of EPG list display.
  - When Data Length = 05h and Data 2 = 00h : Display EPG List(Current segment), The EPG Type can be select directly by Data 3.
  - EPG Scale :
    - TRS231BSxxxx / TRH231BSxxxx

EPG Scale	Number of Broadcast	Character Size
Level1	4	Small
Level2	5	Small
Level3	4	Middle
Level4	4	Large

    - FRO232Bsyyzz

EPG Scale	Number of Broadcast	Character Size
Level1	3	Small
Level2	4	Small
Level3	3	Middle
Level4	3	Large

- EPG update time per channel can be configured with DTV Parameter 2.
- The close command cannot be used during updating.
- When Update start then background scan will abort.
- During EPG display, automatic broadcast search does not work.

(4) Error Handling

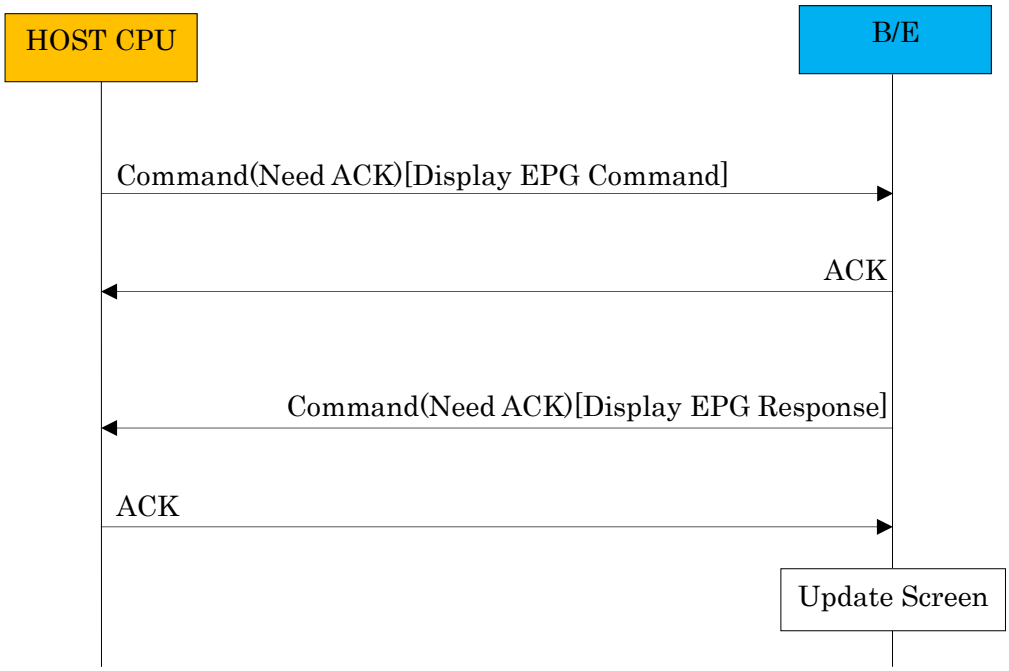
- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.

- Regarding the condition that Result of the response is NG, refer the followings:

  Display EPG Response supports the following error codes.

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid. -Data Length less than 2. - Operation is not 00h ~ 0Eh.
03h(Invalid Length)	Data Length is invalid.
0Ah(Invalid Status)	DTV cannot execute. Ex) Send when not show EPG.
FFh(Unknown Error)	Unknown error

(5) Sequence



5.2.13. Update Preset Table

(1) Update Preset Table Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	06h
Header CRC8 :	--
Data 0 :	01h Update Preset Table (Upper)
Data 1 :	17h Update Preset Table (Lower)
Data 2 :	Preset Number 00h : Home Preset 01h : Drive Preset
Data 3 :	Operation 00h : Update the result of channel scan 01h : Entry current channel
Data 4 :	Operation=00h 00h : Cancel 01h : Update Operation=01h 01h~0Ch : One Touch Number
Data CRC8 :	--

(2) Update Preset Table Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Update Preset Table (Upper)
Data 1 :	17h Update Preset Table (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- Update Preset Table.
- If updating the result of channel scan, HOST should send Data3=00h & Data4=01h.
- If cancel the result of channel scan, HOST should send Data3=00h & Data4=00h. B/E discard the result of channel scan.
- If entry current playback channel to specified one touch number, HOST should send Data3=01h & Data4=01h~0Ch.

(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:

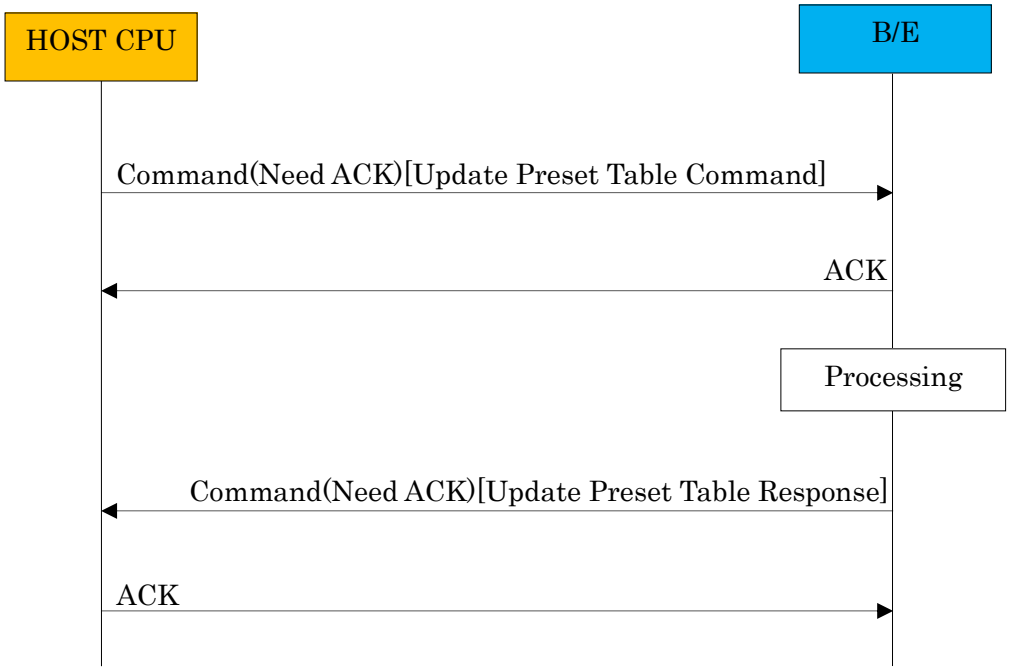
- ✧ Doesn't receive the response within 1sec after HOST transmit the command.
- ✧ The result of response is NG.

・ Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	the parameter is invalid. - Data2 is not 00h or 01h. - Data3 is not 00h or 01h. - If Data3 is 00h, Data4 is not 00h or 01h. - If Data3 is 01h, Data4 is not 01h~0Ch.
03h(Invalid Length)	Data Length is invalid.
0Ah(Invalid Status)	- HOST send Data3=00h without channel scan. - HOST send Data3=01h~0Ch without tuning.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.2.14. Display Banner

(1) Display Banner Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Display Banner(Upper)
Data 1 :	1Fh Display Banner(Lower)
Data 2 :	Operation 00h : Display Banner 01h : Hide Banner
Data 3 :	Data2=00h(Display Banner) 00h : No timeout (Continue to display until Hide Banner) 01h~FFh : Display Time(Second) Data2=01h(Hide Banner) 00h : Fixed Value
Data CRC8 :	--

(2) Display Banner Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Display Banner(Upper)
Data 1 :	1Fh Display Banner(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- This command controls the display of banner.
- If Data2=00h (Display Banner) & Data3=00h (No timeout), B/E continue to display the banner until receiving Hide Banner.
- If Data2=00h (Display Banner) & Data3=01h~FFh, Data3 means the timeout value. B/E display during the time set by Data3.
- If Auto Banner Display is enabled, B/E automatically display the banner without this command when the channel selecting.
  In this case, B/E automatically hide after Seconds. (X : DTV Parameter 2)
- When receiving ack of "Display Banner Response", B/E update the screen.

(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to

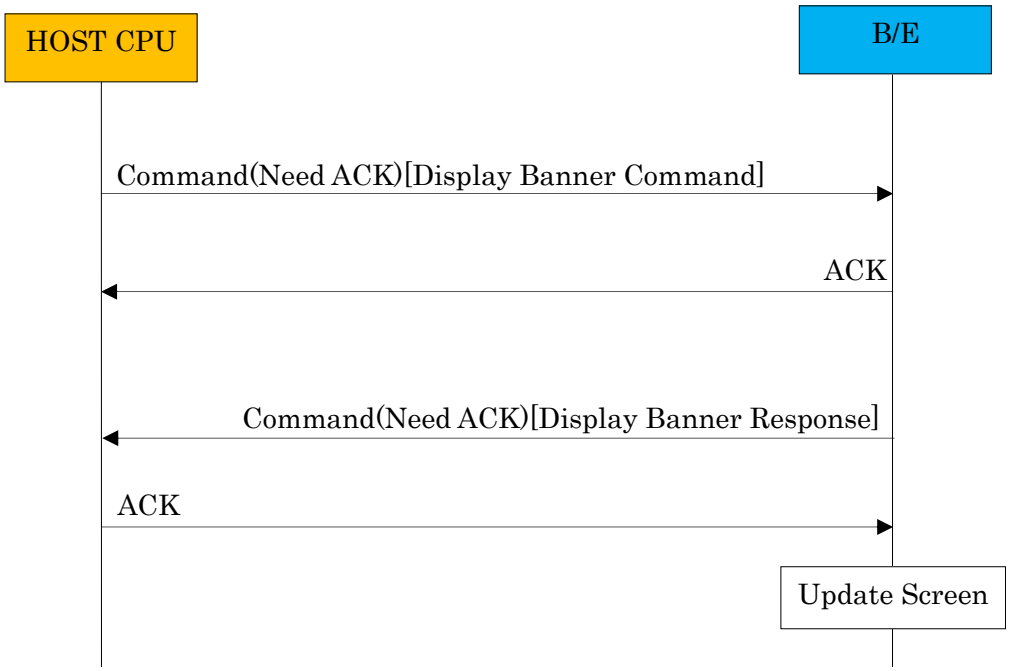
handle the failure of response.

- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 10**sec** after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	the parameter is invalid. - Data2 is not 00h or 01h.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.2.15. Service ID Channel Select

(1) Service ID Channel Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Service ID Channel Select(Upper)
Data 1 :	21h Service ID Channel Select(Lower)
Data 2 :	Service ID (Upper)
Data 3 :	Service ID(Lower)
Data CRC8 :	--

(2) Service ID Channel Select Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Service ID Channel Select(Upper)
Data 1 :	21h Service ID Channel Select(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- B/E start tuning by Service ID.

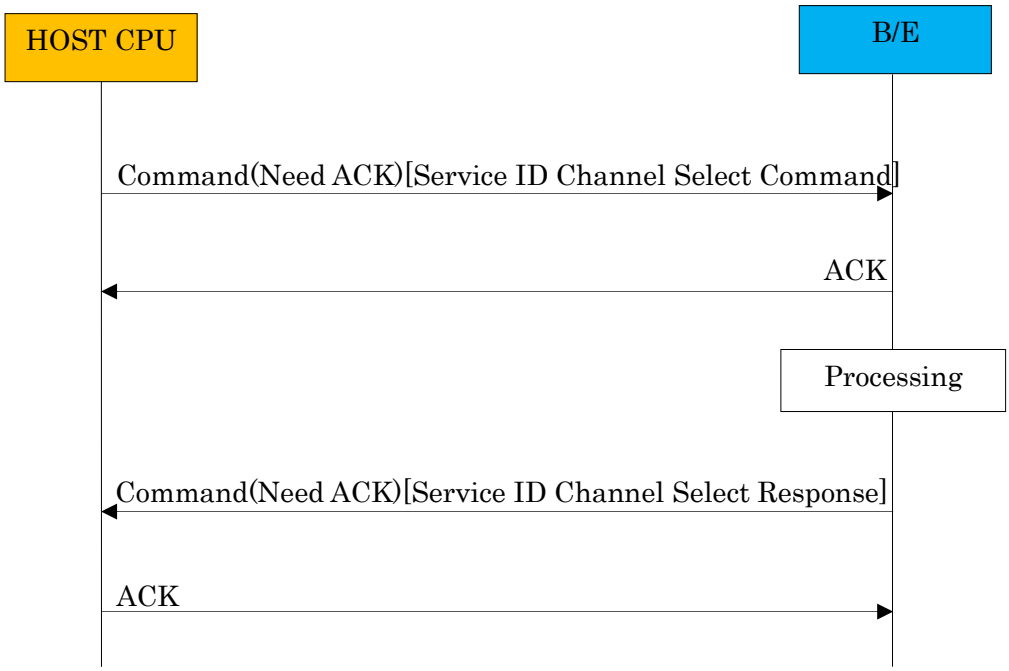
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
03h(Invalid Length)	Data Length is invalid.
04h(No Service)	The specified service is not there in Preset table.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence


HOST CPU
B/E
Command(Need ACK)[Service ID Channel Select Command]
ACK
Processing
Command(Need ACK)[Service ID Channel Select Response]
ACK

5.2.16. Start Background Search

(1) Start Background Search Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	04h
Header CRC8 :	--
Data 0 :	01h Start Background Search (Upper)
Data 1 :	47h Start Background Search (Lower)
Data 2:	Search Mode 01h : Relay Search Start 02h : Affiliated Search Start 03h : Relay + Affiliated Search Start FFh : Search Stop
Data CRC8 :	--

(2) Start Background Search Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Start Background Search (Upper)
Data 1 :	47h Start Background Search (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- B/E starts to search with background.
- If Search mode is 01h, B/E start relay station search.
- If Search mode is 02h, B/E start affiliated station search.
- If Search mode is 03h, B/E start relay and affiliated station search.
- If search mode is FFh, B/E stop relay/affiliated station search.

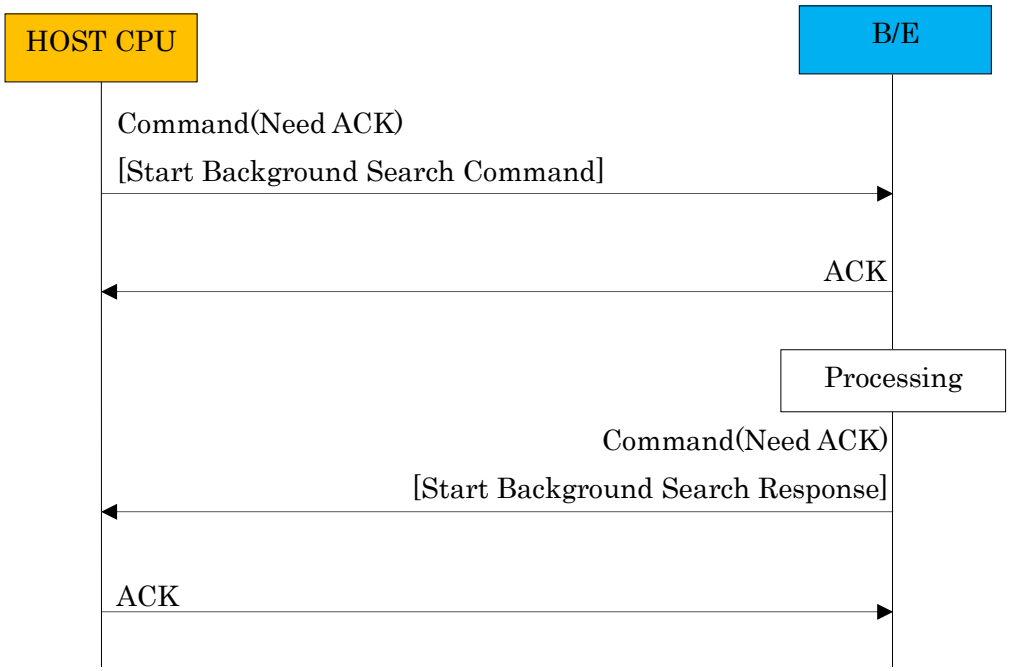
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid. - Search Mode is not 01h or 02h or 03h or FFh.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.2.17. Start Background Scan

(1) Start Background Scan Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	07h
Header CRC8 :	--
Data 0 :	01h Start Background Scan (Upper)
Data 1 :	48h Start Background Scan (Lower)
Data 2 :	DTV Mode 00h : Start Background Scan FFh : Stop Background Scan
Data 3 :	Preset Number 00h : Home 01h : Drive 02h : Home & Drive
Data 4:	EPG Scan time per channel 01h – FFh
Data 5 :	Number of days 1~8days
Data CRC8 :	--

(2) Start Background Scan Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Start Background Scan (Upper)
Data 1 :	48h Start Background Scan (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- Scans the preset registered channels for EPG.
  - EPG Scan channels: Registered channels in Home and Drive Preset. (Excludes duplicate channels.)
  - EPG Scan time per Channel : Data 3 : 1[sec] ～ 255[sec]
  - Start Channel: Last Preset and last channel.

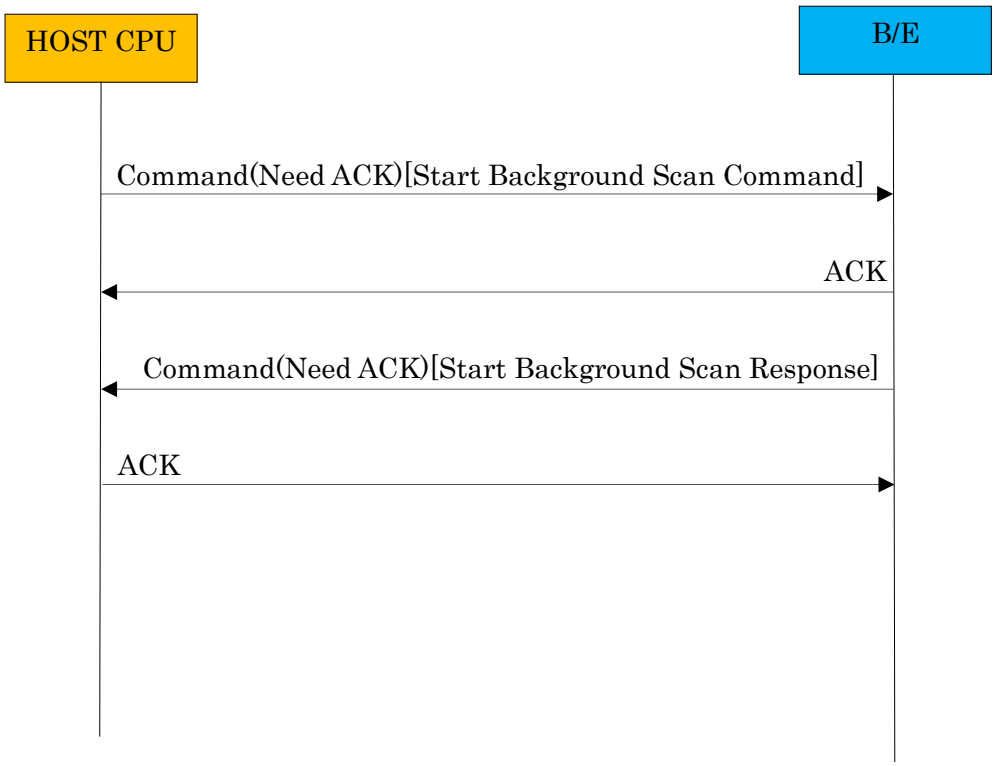
(4) Error Handling

- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.2.18. DTV Restart

(1) DTV Restart Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h
Header CRC8 :	--
Data 0 :	01h DTV Restart (Upper)
Data 1 :	F0h DTV Restart (Lower)
Data CRC8 :	--

(2) DTV Restart Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h DTV Restart (Upper)
Data 1 :	F0h DTV Restart (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

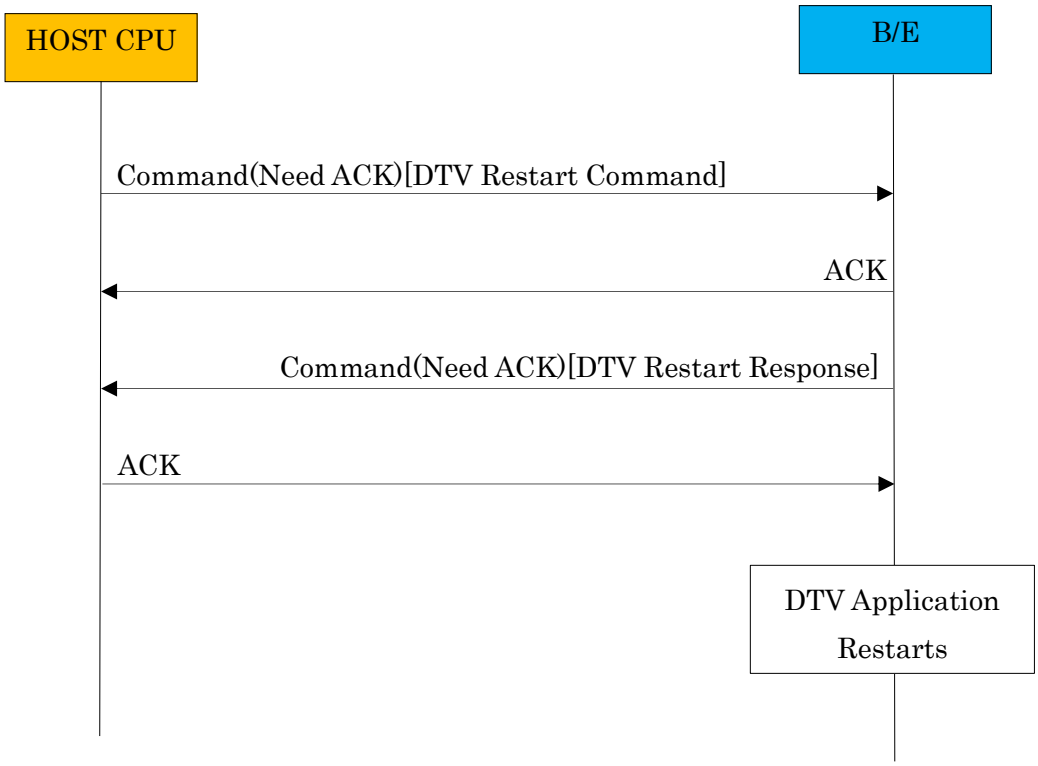
(3) Description

- The host may use this command to restart the DTV application.
- This command is valid only after the DTV Critical Error Event (Refer 5.7.18 DTV Critical Error).
- The BE follows the normal startup sequence for a restart with this command.
- After this command and response, the host should be ready for receiving Wakeup Command from the BE.

(4) Error Handling

- If this command is sent without the DTV Critical Error event, BE responds with error code 0Ah(Invalid Status).

(5) Sequence


HOST CPU
B/E
Command(Need ACK)[DTV Restart Command]
ACK
Command(Need ACK)[DTV Restart Response]
ACK
DTV Application Restarts

5.3. HOST⇒B/E Set Command

5.3.1. Set auto tune Mode

(1) Set auto tune Mode Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	04h
Header CRC8 :	--
Data 0 :	01h Set auto tune Mode (Upper)
Data 1 :	08h Set auto tune Mode (Lower)
Data 2 :	00h : Auto 01h : Fullseg 02h : Oneseg 03h : Auto with Seamless
Data CRC8 :	--

(2) Set auto tune Mode Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Set auto tune Mode (Upper)
Data 1 :	08h Set auto tune Mode (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- Set auto tune Mode.
- This setting is saved by NVRAM.
- It is possible to configure the default value using DTV Parameter 2.

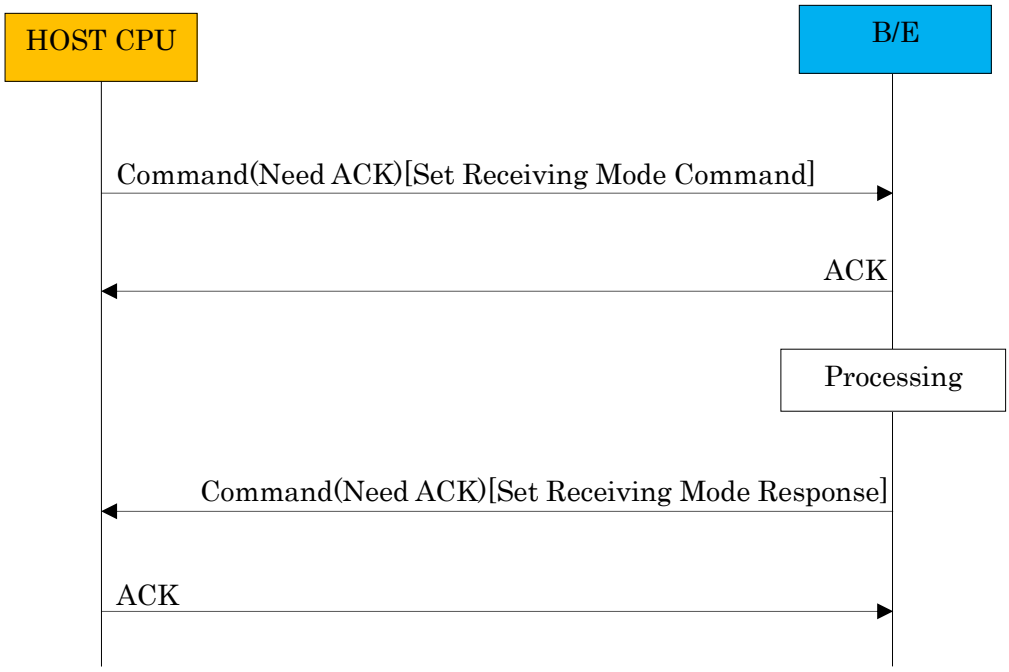
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.3.2. Set Closed Caption Mode

(1) Set Closed Caption Mode Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	04h
Header CRC8 :	--
Data 0 :	01h Set Closed Caption Mode (Upper)
Data 1 :	0Ah Set Closed Caption Mode (Lower)
Data 2 :	

Value	Closed Caption	Superimpose
00h	OFF	OFF
01h	ON - 1st	ON - 1st
02h	ON - 2nd	ON - 2nd
03h	OFF	ON - 1st
04h	ON - 1st	OFF
05h	ON - 2nd	OFF

Initial value : 00h

Data CRC8 : --

(2) Set Closed Caption Mode Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Set Closed Caption Mode (Upper)
Data 1 :	0Ah Set Closed Caption Mode (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- Set Closed Caption Mode.
- This setting is saved by NVRAM.
- Refer to the following table regarding to the specification of Closed Caption display.

Closed Caption	Description
OFF	Don't display Closed Caption
1st	If the program has 1st Closed Caption, B/E displays 1st Closed Caption.
2nd	If the program has 2nd Closed Caption, B/E displays 2nd Closed Caption. If the program has only 1st Closed Caption, B/E displays 1st Closed Caption. If the program doesn't have Closed Caption, B/E doesn't display Closed Caption.

- Refer to the following table regarding to the specification of Superimpose display.

Superimpose	Description
OFF	Don't display Superimpose.
1st	If the program has 1st Superimpose, B/E displays 1st Superimpose.
2nd	If the program has 2nd Superimpose, B/E displays 2nd Superimpose. If the program has only 1st Superimpose, B/E displays 1st Superimpose. If the program doesn't has Superimpose, B/E doesn't display Superimpose.

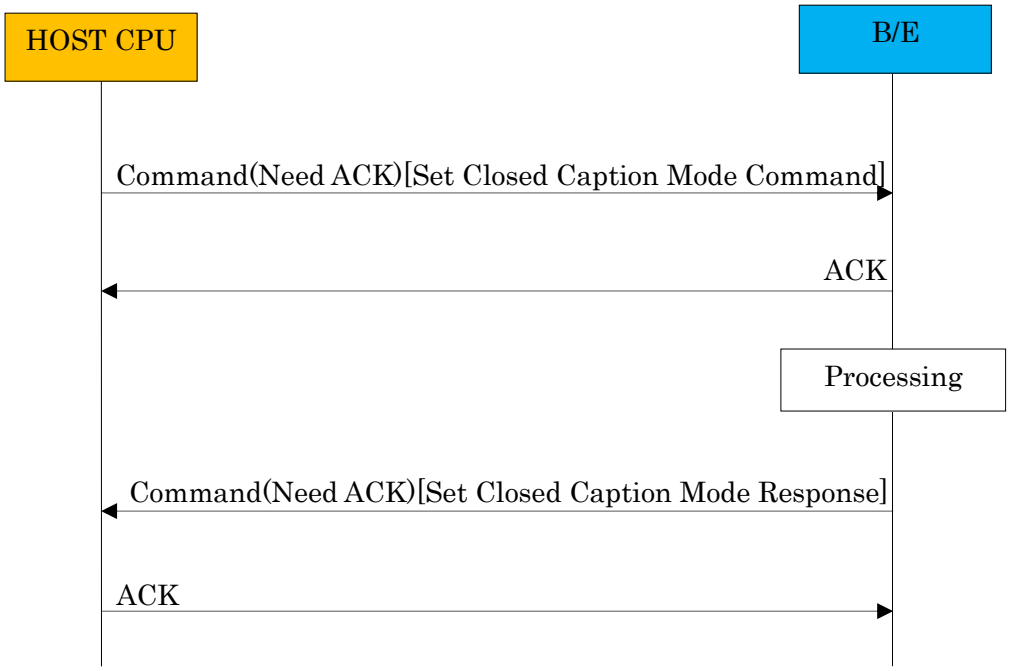
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.3.3. Set Auto Broadcast Search Mode

(1) Set Auto Broadcast Search Mode Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	04h
Header CRC8 :	--
Data 0 :	01h Set Auto Broadcast Search Mode (Upper)
Data 1 :	0Bh Set Auto Broadcast Search Mode (Lower)
Data 2 :	00h : Automatic Broadcast Search Off 01h : Automatic Relay Station Search On 02h : Automatic Affiliated Station Search On 03h : Automatic Relay & Affiliated Station Search On
Data CRC8 :	--

(2) Set Auto Broadcast Search Mode Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Set Auto Broadcast Search Mode (Upper)
Data 1 :	0Bh Set Auto Broadcast Search Mode (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- Set Automatic Broadcast Search Mode.
- If Diver Mode is Dynamic 4Diver Mode or Fixed 2x2 Diver Mode, Automatic Broadcast Search run via background. If Diver Mode is Fixed 2Diver Mode, Automatic Broadcast Search run via foreground (Refer to "5.3.8.Set DTV System").
- This setting is saved by NVRAM.

(4) Error Handling

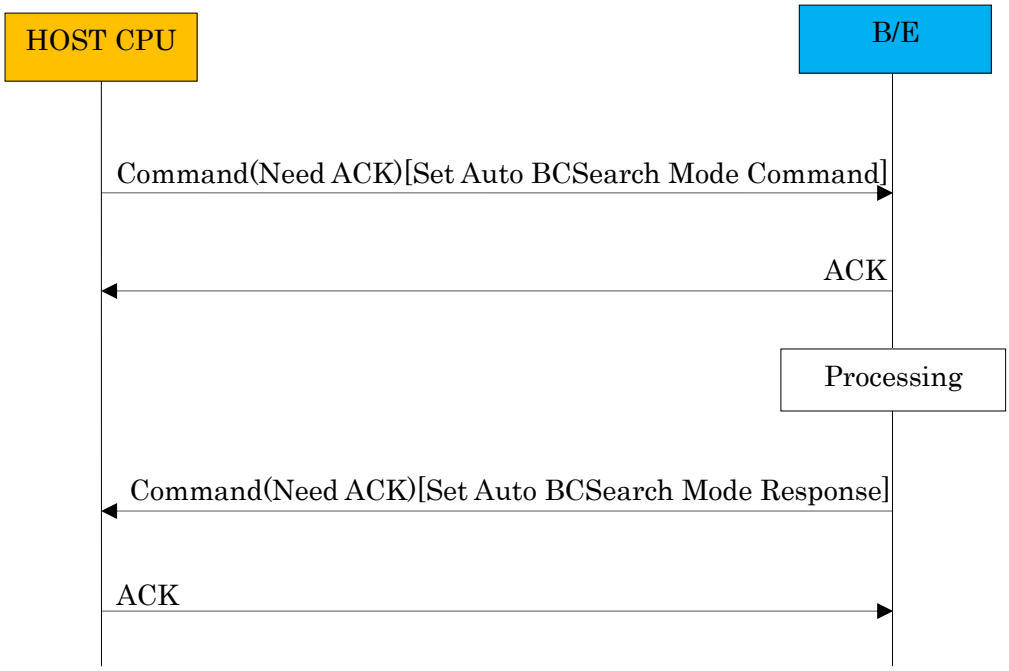
- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.

03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.3.4. Set Audio Mode

(1) Set Audio Mode Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	04h
Header CRC8 :	--
Data 0 :	01h Set Audio Mode (Upper)
Data 1 :	0Dh Set Audio Mode (Lower)
Data 2 :	00h : Main 01h : Sub 02h : Main(Lch)+Sub(Rch)
Data CRC8 :	--

(2) Set Audio Mode Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Set Audio Mode (Upper)
Data 1 :	0Dh Set Audio Mode (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- Set Audio Mode.
- This setting is saved by NVRAM.

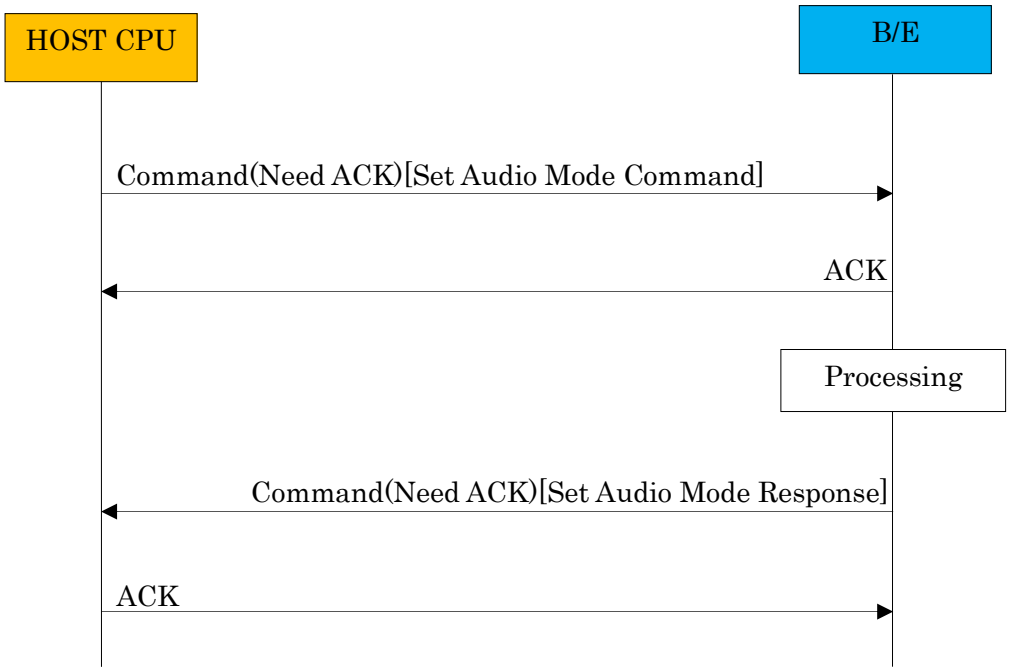
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence


HOST CPU
B/E
Command(Need ACK)[Set Audio Mode Command]
ACK
Processing
Command(Need ACK)[Set Audio Mode Response]
ACK

5.3.5. Set Residential Area

(1) Set Residential Area Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	0Dh
Header CRC8 :	--
Data 0 :	01h Set Residential Area(Upper)
Data 1 :	10h Set Residential Area(Lower)
Data 2 :	00h : Home Preset 01h : Drive Preset FFh : Last Preset
Data 3 :	Target Bit0 : Set Region ID Bit1 : Set Zip-Code Bit2~Bit7 : Reserved (Fix 0)
Data 4 :	Region ID
Data 5 :	Zip-Code[0] Ascii Code (In case of "1234567", '1')
Data 6 :	Zip-Code[1] Ascii Code (In case of "1234567", '2')
Data 7 :	Zip-Code[2] Ascii Code (In case of "1234567", '3')
Data 8 :	Zip-Code[3] Ascii Code (In case of "1234567", '4')
Data 9 :	Zip-Code[4] Ascii Code (In case of "1234567", '5')
Data 10 :	Zip-Code[5] Ascii Code (In case of "1234567", '6')
Data 11 :	Zip-Code[6] Ascii Code (In case of "1234567", '7')
Data CRC8 :	--

(2) Set Residential Area Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Set Residential Area(Upper)
Data 1 :	10h Set Residential Area(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- Set Residential Area (Region ID and Zip code).
- In case of setting only Region ID, set Data3=01h, Data4=Input Value, Data5~11=00h (B/E doesn't care Data5~11).
- If case of setting only Zip code, set Data3=02h, Data4=00h (B/E doesn't care Data4), Data5~11=Input Value.
- If case of Setting Region ID and Zip code, set Data3=03h, Data4=Input Value, Data5~11=Input Value.
- This setting is saved by NVRAM.
- BML browser will not use zip code anymore if 0000000 used.

・ Refer the following about Region ID.

ID	Region (English)	Region (Japanese)	Wide area	ID	Region (English)	Region (Japanese)	Wide area
1	Hokkaido (Sapporo)	北海道（札幌）	Hokkaido	28	Shizuoka	静岡	―
2	Hokkaido (Hakodate)	北海道（函館）	Hokkaido	29	Aichi	愛知	Chukyo
3	Hokkaido (Asahikawa)	北海道（旭川）	Hokkaido	30	Mie	三重	Chukyo
4	Hokkaido (Obihiro)	北海道（帯広）	Hokkaido	31	Shiga	滋賀	Kinki
5	Hokkaido (Kushiro)	北海道（釧路）	Hokkaido	32	Kyoto	京都	Kinki
6	Hokkaido (Kitami)	北海道（北見）	Hokkaido	33	Osaka	大阪	Kinki
7	Hokkaido (Muroran)	北海道（室蘭）	Hokkaido	34	Hyogo	兵庫	Kinki
8	Aomori	青森	―	35	Nara	奈良	Kinki
9	Iwate	岩手	―	36	Wakayama	和歌山	Kinki
10	Miyagi	宮城	―	37	Tottori	鳥取	Shimane, Tottori
11	Akita	秋田	―	38	Shimane	島根	Shimane, Tottori
12	Yamagata	山形	―	39	Okayama	岡山	Okayama, Kagawa
13	Fukushima	福島	―	40	Hiroshima	広島	―
14	Ibaraki	茨城	Kanto	41	Yamaguchi	山口	―
15	Tochigi	栃木	Kanto	42	Tokushima	徳島	―
16	Gunma	群馬	Kanto	43	Kagawa	香川	Okayama, Kagawa
17	Saitama	埼玉	Kanto	44	Ehime	愛媛	―
18	Chiba	千葉	Kanto	45	Kochi	高知	―
19	Tokyo	東京	Kanto	46	Fukuoka	福岡	―
20	Kanagawa	神奈川	Kanto	47	Saga	佐賀	―
21	Niigata	新潟	―	48	Nagasaki	長崎	―
22	Toyama	富山	―	49	Kumamoto	熊本	―
23	Ishikawa	石川	―	50	Oita	大分	―
24	Fukui	福井	―	51	Miyazaki	宮崎	―
25	Yamanashi	山梨	―	52	Kagoshima	鹿児島	―
26	Nagano	長野	―	53	Okinawa	沖縄	―
27	Gifu	岐阜	Chukyo				

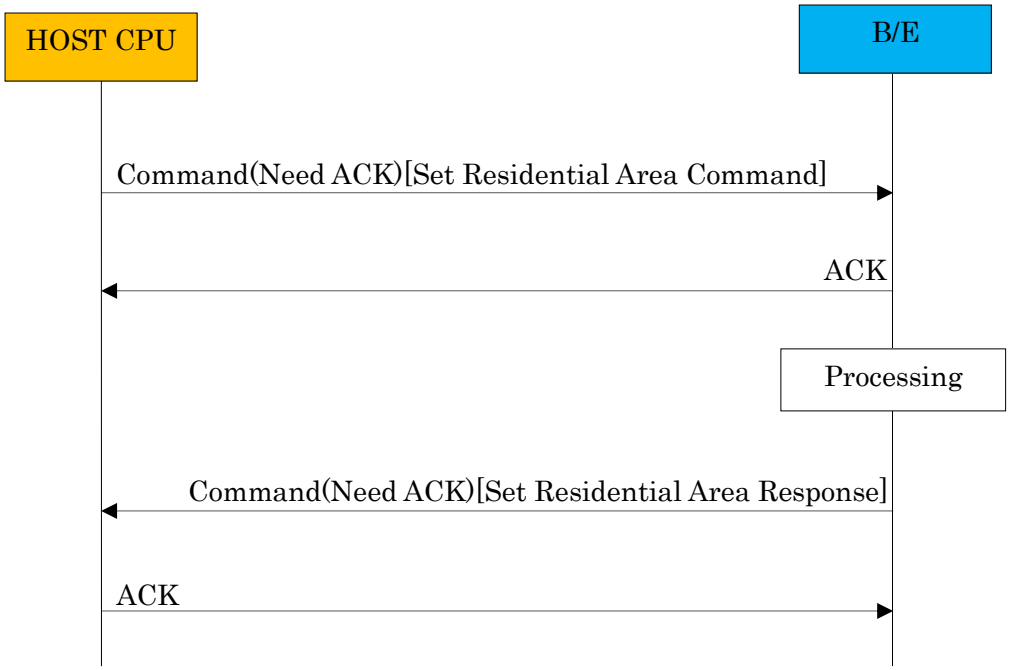
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid. - Both Target.b0=0 and Target.b1=0. - Target.b0=1 & Region ID is invalid. - Target.b1=1 & Zip code is not Ascii Number
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.3.6. Set Auto Event Relay

(1) Set Auto Event Relay Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	04h or 06h(option)
Header CRC8 :	--
Data 0 :	01h Set Auto Event Relay (Upper)
Data 1 :	11h Set Auto Event Relay (Lower)
Data 2 :	00h : Auto Event Relay Off 01h : Auto Event Relay On *When this configured as off, notification will be skipped.
Data 3 :	Notification interval before switching 5 to 30 [s]
Data 4 :	Number of notifications before switching 0 to 6 [times]
Data CRC8 :	--

(2) Set Auto Event Relay Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Set Auto Event Relay (Upper)
Data 1 :	11h Set Auto Event Relay (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

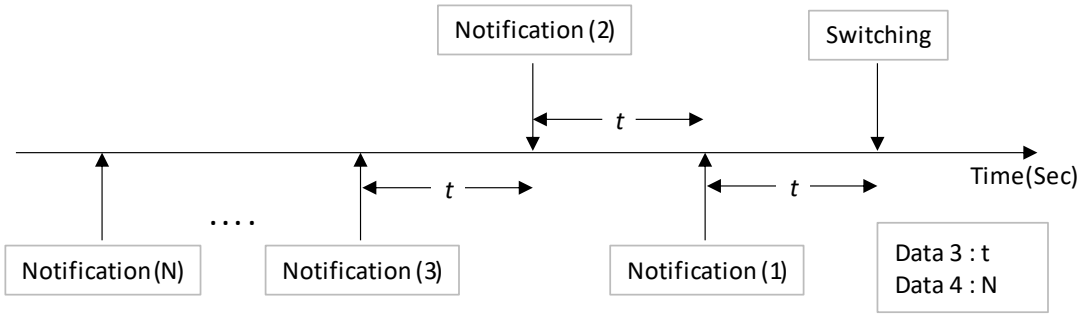
(3) Description

- Set Auto Event Relay.
- This setting is saved by NVRAM.
- Data 3 / 4 : Event timing
- Data 3 is the interval between notifications and the time of switching.
- Data 4 is the number of notifications before switching.

  Data3 × Data4 : Be equal to or less than 30

  Ex) Data3 = 12 / Data4 = 2

  Host will be notified 2 times – 12[s] and 24[s] before switching.
- The host can read the current set values of Data 3, Data 4 using Get Configure command.
- "Event Relay done" notification will be sent irrespective of the notification settings made by Data3, Data4 and DTV Parameter 2 Data 33, Data 34.



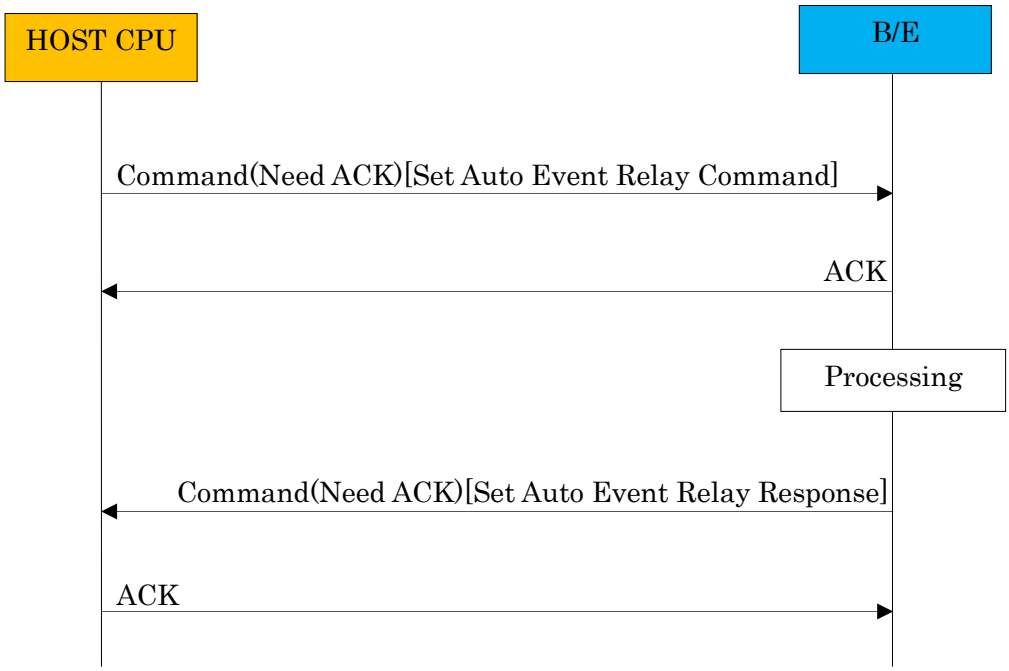
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.3.7. Set Baud rate

(1) Set Baud rate Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	04h
Header CRC8 :	--
Data 0 :	01h Set Baud rate (Upper)
Data 1 :	19h Set Baud rate (Lower)
Data 2 :	Baud rate 03h : 9600 04h : 19200 05h : 38400 06h : 57600 07h : 115200 08h : 230400 09h : 460800 0Ah : 921600
Data CRC8 :	--

(2) Set Baud rate Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Set Baud rate (Upper)
Data 1 :	19h Set Baud rate (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- Set Baud rate.
- Regarding to the timing of changing baud rate, refer to "(5) Sequence".
- Data2 (03h~0Ah) don't guarantee the communication.
- This command setting is not stored to NVRAM.

(4) Error Handling

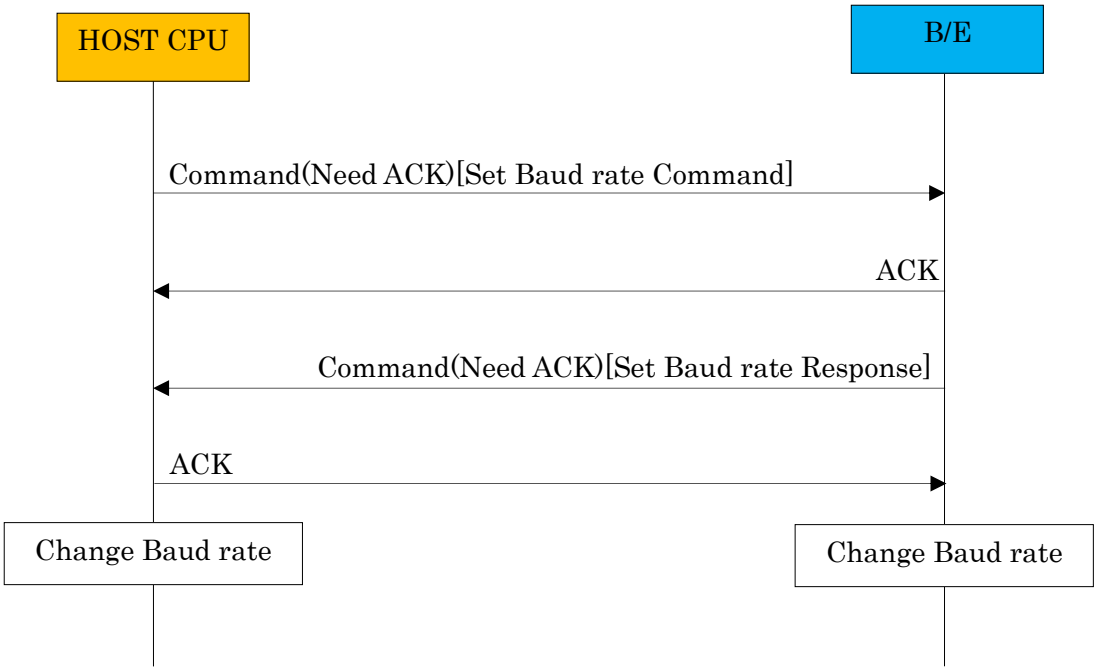
- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.

・ Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.3.8. Set DTV System

(1) Set DTV System Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	04h
Header CRC8 :	--
Data 0 :	01h Set DTV System (Upper)
Data 1 :	20h Set DTV System (Lower)
Data 2 :	DTV System 00h : Dynamic 4Diver System (Fore Br0/1) 01h : Fixed 2x2Diver System (Fore Br0/1) 02h : Fixed 2Diver System(Antenna Br2/3) 03h : Fixed 4Diver System 04h : Fixed 2Diver System(Antenna Br0/1) 05h : Fixed 3x1Diver System (Fore Br0/1/2) 06h : Dynamic 4Diver System (Fore Br2/3) 07h : Fixed 2x2Diver System (Fore Br2/3) 08h : Fixed 3x1Diver System (Fore Br1/2/3)
Data CRC8 :	--

(2) Set DTV System Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Set DTV System (Upper)
Data 1 :	20h Set DTV System (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- Set DTV System.
- DTV System shows the following table.

DTV System	Content
Dynamic 4Diver System	Automatically switch 4Diver Mode and 2x2Diver Mode
Fixed 2x2Diver System	Always 2x2Diver Mode
Fixed 2Diver System	Always 2Diver Mode
Fixed 4Diver System	Always 4Diver Mode
Fixed 3x1Diver System	Always 3x1Diver Mode

*: 4Diver Mode is 4 antennas with foreground (Not use background).

*: 2x2Diver Mode is 2 antennas with foreground and 2 antennas with background.

*: 2Diver Mode is 2 antennas with foreground (Not use background).

*: 3x1Diver Mode is 3 antenna with foreground and 1 antenna with background.

- DTV Program load DTV System (Data2) only once at the booting. If HOST send "Set DTV System", HOST has to reset B/E.
- If DTV System Read Error happened, Selected Fixed 4Diver System.
- Set DTV System Response is performed after data of Set DTV System Command is written to NVRAM.

・Antenna input pin assignment

Depending on the DTV System mode setting and background processing status, the function assignment of the antenna Br0-3 is switched as shown in below Table. Here, the DTV system mode setting switches the valid antenna input pins and the effectiveness of background processing.

Data 2	System	Background processing status	Br0	Br1	Br2	Br3
00h	Dynamic 4Diver System	Active	F	F	B	B
		Inactive	F	F	F	F
01h	Fixed 2x2Diver System	Active	F	F	B	B
02h	Fixed 2Diver System	Inactive	—	—	F	F
03h	Fixed 4Diver System	Inactive	F	F	F	F
04h	Fixed 2Diver System	Inactive	F	F	—	—
05h	Fixed 3x1Diver System	Active	F	F	F	B
06h	Dynamic 4Diver System	Active	B	B	F	F
		Inactive	F	F	F	F
07h	Fixed 2x2Diver System	Active	B	B	F	F
08h	Fixed 3x1Diver System	Active	B	F	F	F

F : used for Foreground processing.

B : used for Background processing.

— : not used for either Foreground processing or Background processing.

- For F039902182, this setting is irrelevant and so the response will be OK for any values. 04h is the only effective setting.
- After change any setting on DTV system, a power cycle on/off is required to adapt the setting.

(4) Error Handling

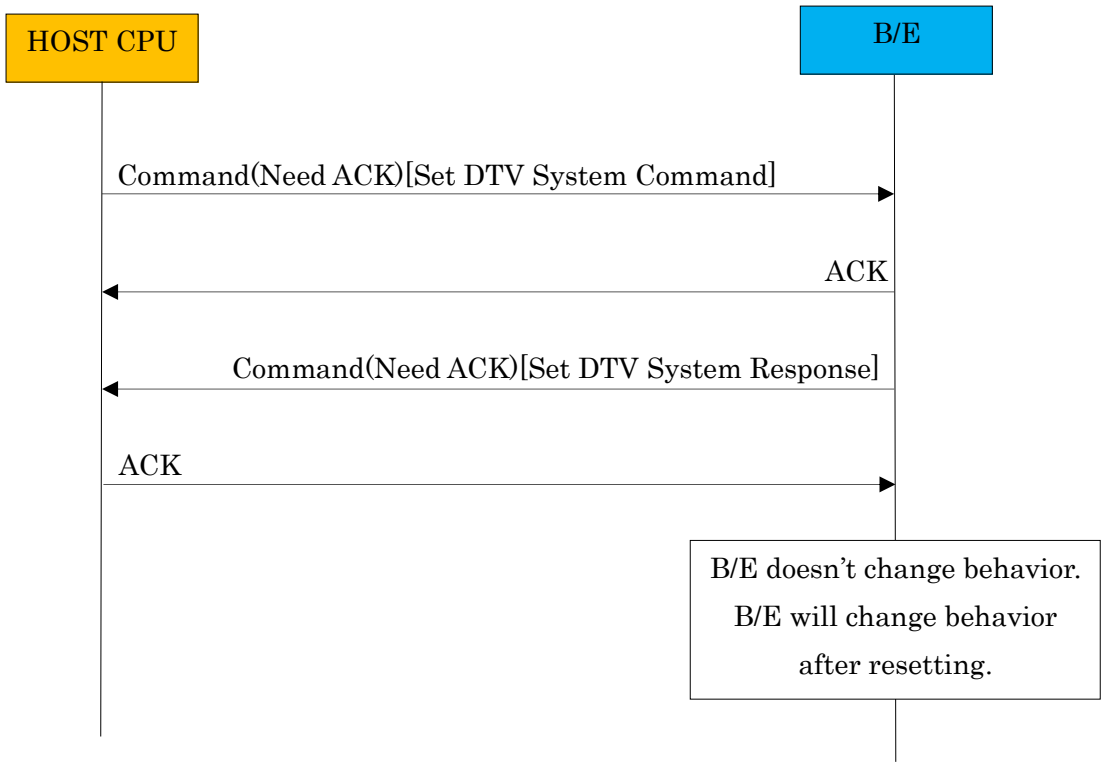
- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 10sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid. - DTV System is not 00h ~ 08h.

03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.3.9. Set DTV Parameter

(1) Set DTV Parameter Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	E3h
Header CRC8 :	--
Data 0 :	01h Set DTV Parameter (Upper)
Data 1 :	1Dh Set DTV Parameter (Lower)
Data 2~5 :	Enable/Disable Data2 : 01h , Data3~5 : 00h : Enable Parameter Data2:00h, Data3~5: 00h: Disable Parameter (In case of Disable, use Default Parameter)
Data 6~9 :	C/N for Tuner Unlock Detection.
Data 10~13 :	BER for Tuner Unlock Detection.
Data 14~17 :	PER for Tuner Unlock Detection.
Data 18~21 :	Time for Tuner Unlock Detection.
Data 22~25 :	C/N for change to Fullseg.
Data 26~29 :	BER for change to Fullseg.
Data 30~33 :	PER for change to Fullseg.
Data 34~37 :	Time for change to Fullseg.
Data 38~41 :	C/N for change to Oneseg.
Data 42~45 :	BER for change to Oneseg.
Data 46~49 :	PER for change to Oneseg.
Data 50~53 :	Time for change to Oneseg.
Data 54~57 :	C/N for start Broadcast Search.
Data 58~61 :	BER for start Broadcast Search.
Data 62~65 :	PER for start Broadcast Search.
Data 66~69 :	Time for start Broadcast Search.
Data 70~73 :	Time (1) for disable Broadcast Search.
Data 74~77 :	Time (2) for disable Broadcast Search.
Data 78~81 :	Time for Background Tuner Unlock Detection.
Data 82~85 :	C/N for start Background Search.
Data 86~89 :	Time for start Background Search.
Data 90~93 :	C/N for start Background Scan.
Data 94~97 :	Time for start Background Scan.
Data 98~101 :	Time for change to 4Div mode.
Data 102~105 :	C/N difference for switching to background channel.
Data 106~109 :	BER for switching to background channel.
Data 110~113 :	PER for switching to background channel.
Data 114~117 :	Time for switching to background channel.
Data 118~121 :	C/N for Receive Error Detection.
Data 122~125 :	BER for Receive Error Detection.
Data 126~129 :	PER for Receive Error Detection.
Data 130~133 :	Time for Receive Error Detection.
Data 134~137 :	C/N for Fullseg Detection at fixed Fullseg search.
Data 138~141 :	C/N for Oneseg Detection at fixed or auto switch mode search.
Data 142~145 :	Detect Time (count) for other broadcast.
Data 146~149 :	Detect During Time for Another broadcast.
Data 150~153 :	C/N Offset for Fullseg Detection at fixed Fullseg broadcast search.
Data 154~157 :	Reserved
Data 158~161 :	Reserved
Data 162~165 :	Reserved
Data 166~169 :	Reserved
Data 170~173 :	Reserved
Data 174~177 :	Reserved
Data 178~181 :	Reserved

Data 182~185 :	Reserved
Data 186~189 :	Reserved
Data 190~193 :	Reserved
Data 194~197 :	Reserved
Data 198~201 :	Reserved
Data 202~205 :	Reserved
Data 206~209 :	Reserved
Data 210~213 :	Reserved
Data 214~217 :	Reserved
Data 218~221 :	Reserved
Data 222~225 :	Reserved
Data CRC8 :	--

・C/N: signed 4byte data (Divide by 100)

・BER : unsigned 4byte data (Divide by 1000000)

・PER : unsigned 4byte data (Divide by 10000)

・Time : unsigned 4byte data (Multiply by 0.2)

・C/N range 0dB to 40.0 dB

・BER range 0.0000 (No error) to 0.0441 (Error 4.41%)

・PER range 0.0(No error) to 1.0 (Error100%)

・Time range 200ms to 2000s (Excluding Data 142 to 145)

(2) Set DTV Parameter Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Set DTV Parameter (Upper)
Data 1 :	1Dh Set DTV Parameter (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

・ Set DTV Parameter.

・ Each parameter is 32bit. Refer to the following table about the format.

Data		Each Parameter
Data N		Upper
Data N+1		↑
Data N+2		↓
Data N+3		Lower

・ Determination method

- Tuner Unlock Detection
- Change to Oneseg
- Start Broadcast Search
- Receive Error

  When the following OR condition continues for Time.

  a. C/N < C/N Threshold
  b. BER < BER Threshold
  c. PER < PER Threshold

- Change to Fullseg
- Switching to background channel

  When the following AND condition continues for Time.

  a. C/N > C/N Threshold
  b. BER > BER Threshold
  c. PER > PER Threshold

・If DTV Parameter Read Error happened, The value implemented in the program is used.

・The SET DTV Parameter command can be used during idle status or during tuning status.

・This command stores data in FLASH asynchronously with command response. So, if reset of power off immediately after the response, need the shutdown (shutdown after storing data) command.

・When changing DTV System / DTV Parameter continuously,
Execute the command in the following procedure.

① Execute Set DTV Parameter
② Execute Set DTV System
③ Execute Shutdown (shutdown after storage data)

Otherwise, DTV System will not be changed.

・Data118~121 is for one/auto segment only, for full segment, it sets with DTV Parameter2 Data10-11.

・Data 122-125, 126-129, 130-133 :common for all segment.

・Data 134-137, 138-141 are used for Broadcast search and seek together.

・Even if Data 2-5 is set to Disable, the parameter read value will read the set value. (Value of parameters set as default cannot be read.)

(4) Error Handling

・ In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.

・ The failure of response means that the followings:

✧ Doesn't receive the response within 1sec after HOST transmit the command.

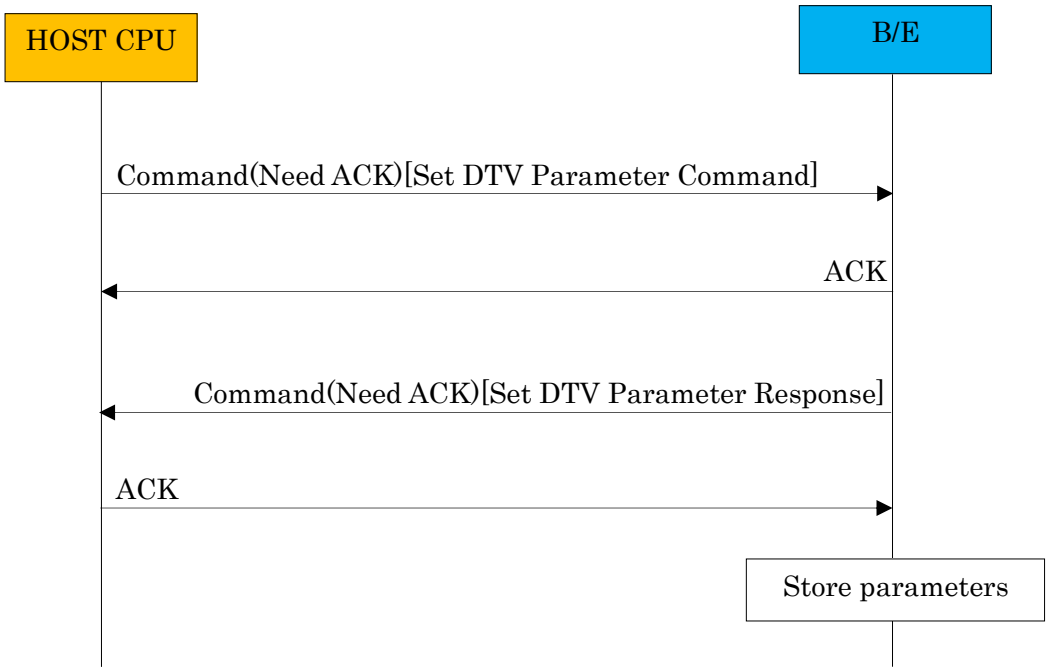
✧ The result of response is NG.

・ Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.3.10. Set Low Power Mode

(1) Set Low Power Mode Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	04h
Header CRC8 :	--
Data 0 :	01h Set Low Power Mode(Upper)
Data 1 :	27h Set Low Power Mode(Lower)
Data 2 :	Low Power Mode 00h : Low Power Mode OFF 01h : Low Power Mode ON
Data CRC8 :	--

(2) Set Low Power Mode Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Set Low Power Mode(Upper)
Data 1 :	27h Set Low Power Mode(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- Set Low Power Mode.

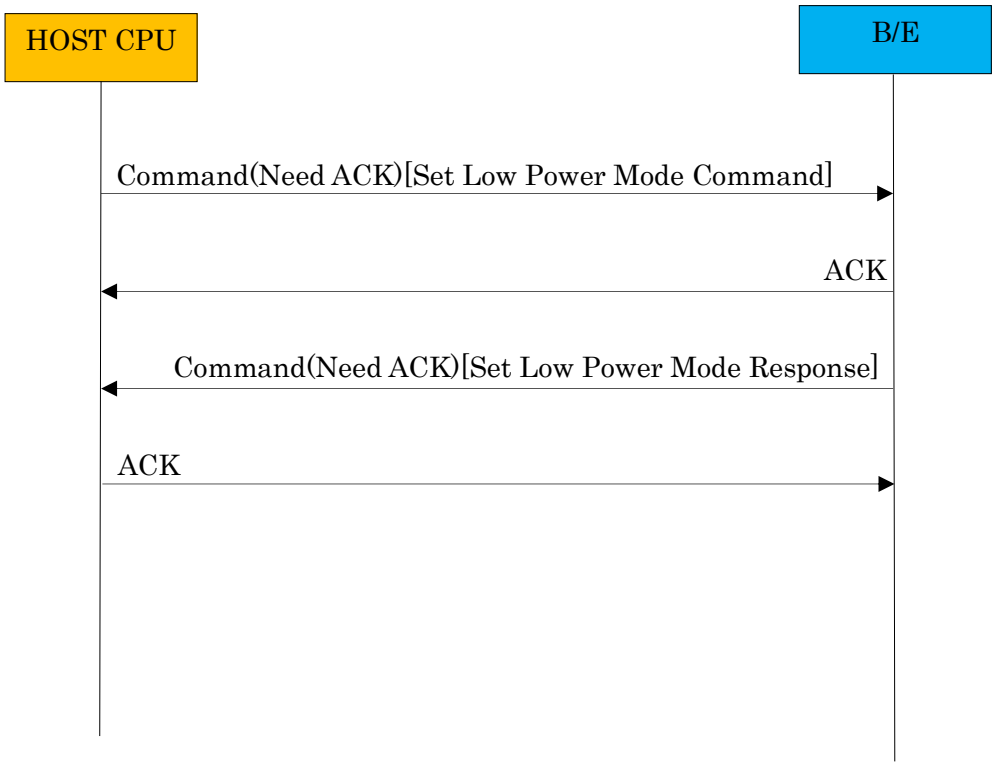
(4) Error Handling

- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence


HOST CPU
B/E
Command(Need ACK)[Set Low Power Mode Command]
ACK
Command(Need ACK)[Set Low Power Mode Response]
ACK

5.3.11. BML Forced Exit

(1) BML Forced Exit Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h
Header CRC8 :	--
Data 0 :	01h BML Forced Exit(Upper)
Data 1 :	28h BML Forced Exit(Lower)
Data CRC8 :	--

(2) BML Forced Exit Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h BML Forced Exit(Upper)
Data 1 :	28h BML Forced Exit(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- Forced Exit BML.
- Depending on the content, it may be displayed again. (Auto Start BML content.)

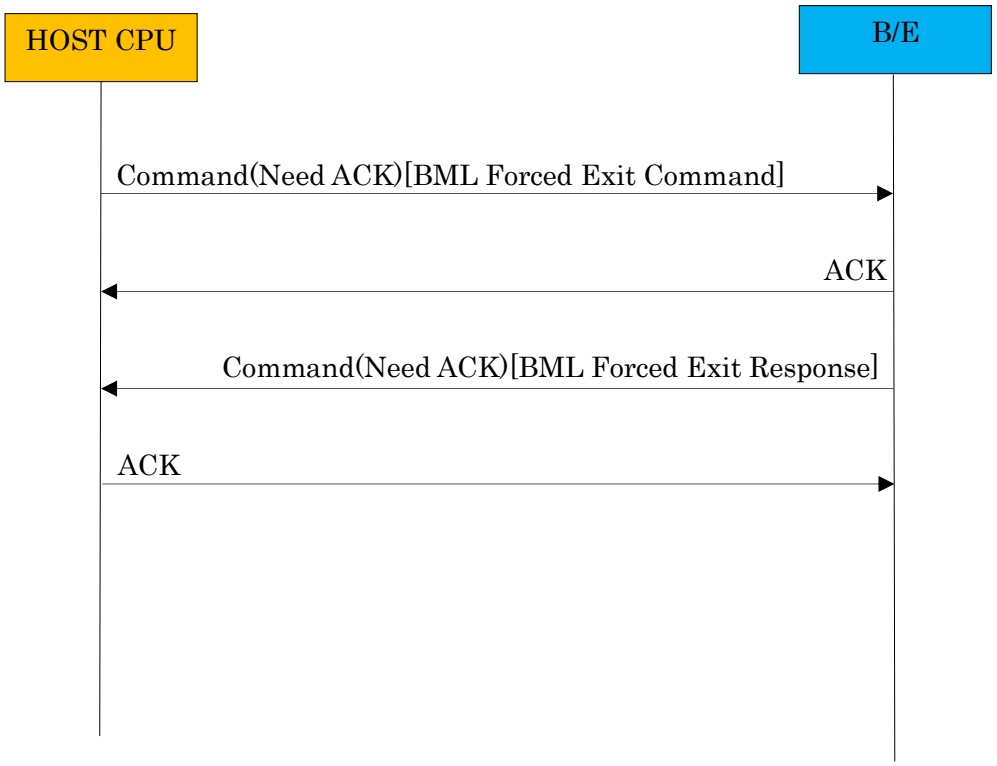
(4) Error Handling

- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence


HOST CPU
B/E
Command(Need ACK)[BML Forced Exit Command]
ACK
Command(Need ACK)[BML Forced Exit Response]
ACK

5.3.12. Set DTV Mode

(1) Set DTV Mode Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	06h
Header CRC8 :	--
Data 0 :	01h Set DTV Mode(Upper)
Data 1 :	2Ch Set DTV Mode(Lower)
Data 2 :	DTV Mode 00h : DTV Mode ON without last channel tune 01h : DTV Mode ON with last channel tune 10h : DTV Mode OFF without EPG Scan 11h : DTV Mode OFF with EPG Scan
Data 3 :	EPG Scan time per channel only Data2=11h 01h – FFh Data 2 != 11h : 00h
Data 4:	Number of days only Data2=11h 1~8days Data 2 != 11h : 00h
Data CRC8 :	--

(2) Set DTV Mode Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Set DTV Mode(Upper)
Data 1 :	2Ch Set DTV Mode(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- DTV Mode ON without/with last channel tune: Sets DTV Mode to ON and remains idle or tune to the last channel.
- When the DTV mode is OFF, power consumption will be reduced / similar to lower power mode but slightly more power consumed.
- DTV Mode OFF with Scan: Scans the preset registered channels for EPG.
  - EPG Scan channels: Registered channels in Home and Drive Preset. (Excludes duplicate channels.)
  - EPG Scan time per Channel : Data 3 : 1[sec] ～ 255[sec]
  - Start Channel: Last Preset and last channel.
  - EPG scan completed status refers to DTV status, Data 2.

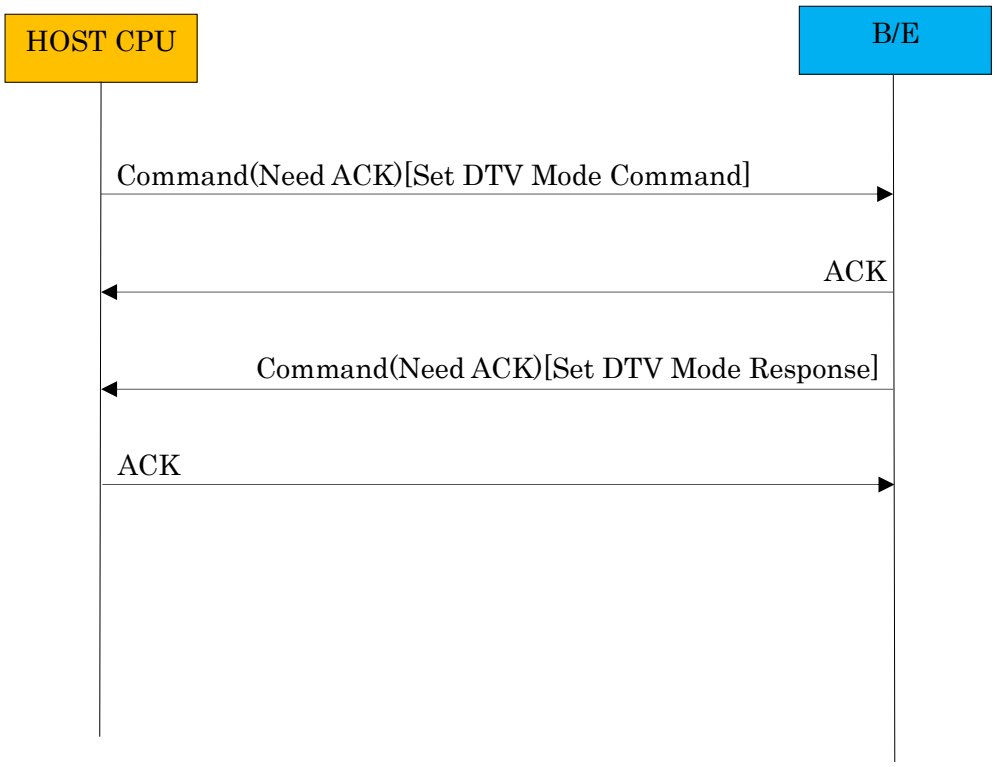
(4) Error Handling

・ Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.3.13. Set Video Resolution

(1) Set Video Resolution Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	04h
Header CRC8 :	--
Data 0 :	01h Set Video Resolution(Upper)
Data 1 :	2Dh Set Video Resolution(Lower)
Data 2 :	Resolution 00h : 720p 01h : 480p 02h : 1080p 03h : 480i (NTSC) 04h : 1080p@30Hz
Data CRC8 :	--

(2) Set Video Resolution Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : 06h Result=NG : 05h
Header CRC8 :	--
Data 0 :	01h Set Video Resolution(Upper)
Data 1 :	2Dh Set Video Resolution(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Reserved Fixx 00h
Data CRC8 :	--

(3) Description

- 720p is the default for F03990218x
- 480i : BT656 System only.
- This Setting change is not stored in NVRAM.
- Playback will be stopped when changing the resolution and host may start again by any of the tune commands.

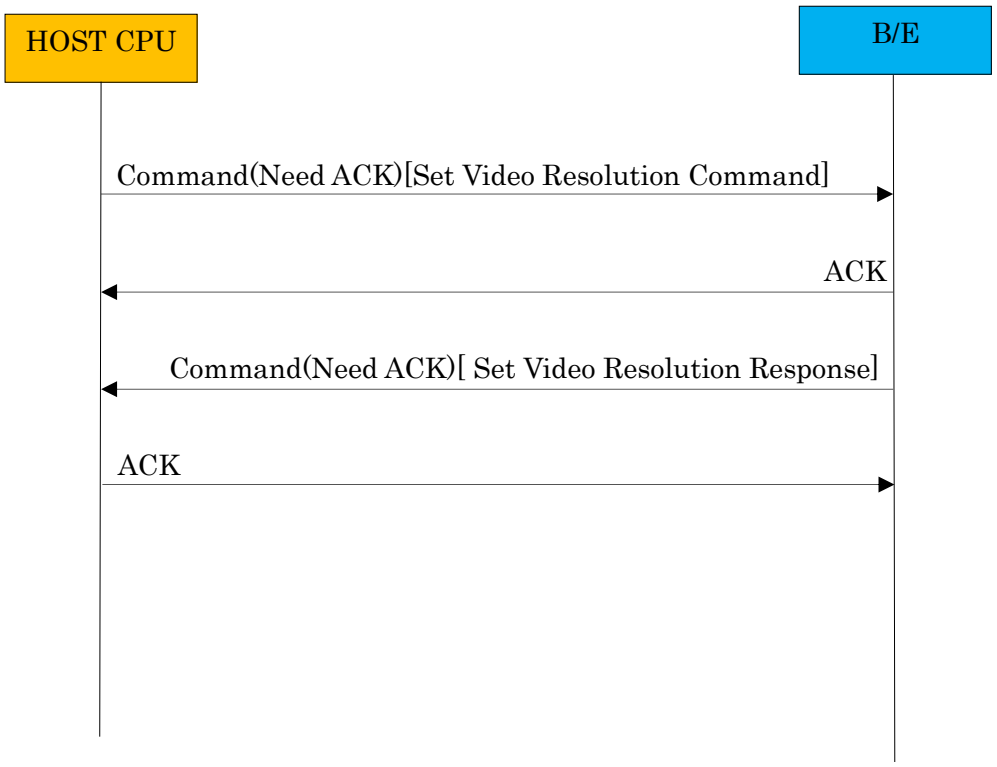
(4) Error Handling

・ Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.3.14. Set Banner Config

(1) Set Banner Config Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	10h
Header CRC8 :	--
Data 0 :	01h Set Banner Config(Upper)
Data 1 :	2Fh Set Banner Config(Lower)
Data 2 :	Banner Position X (Upper)
Data 3 :	Banner Position X (Lower)
Data 4 :	Banner Position Y (Upper)
Data 5 :	Banner Position Y (Lower)
Data 6 :	Font Size [px]
Data 7 :	Transparency [%]
Data 8 :	Logo Position Offset [px]
Data 9 :	Margin of Top and Bottom [px]
Data 10 :	Margin of Left and Right [px]
Data 11 :	Line spacing [px]
Data 12 :	Max number of characters
Data 13 :	Segment icon 00h:enable 01h:disable
Data 14 :	Automatic banner display enable 00h:enable 01h:disable
Data CRC8 :	--

(2) Set Banner Config Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Set Banner Config(Upper)
Data 1 :	2Fh Set Banner Config(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- Banner Configuration Parameter.



- Broadcast logo size will be change according to the font size, irrespective of the logotype set by host.
- Banner Height is Font Size × 2 + Top and Bottom Margin × 2 + Line Spacing.
- Banner Width Maximum Font Size × [Data : 12] + Left and Right Margin × 2 + Font Size × 6(Time information)
- If any value is invalid, returns error and the setting is completely ignored. Banner will be displayed using last valid settings.
- This command setting is not stored to NVRAM.
- All coordinate resolutions are set at 1280x720.
- Segment icon size will be changed according to the font size.
- The actual parameter range refer to chapter 8.2 or 8.3, data 37 to 46 ,data 107 and data 109.
- The Maximum Number of Characters of Banner is following content.
  - Max number of characters is the maximum number of characters of the program name.
  - Banner width is calculated based on the set value, Font Size × Max number of characters + Left and Right Margin × 2 + Font Size × 6 (showing the program time)
  - The above calculation is based on the second row of the banner.
  - The first row contains 3Digit number, Branch ID, Logo and TS Name.
  - The effective banner width will be the "Maximum of these 2 row-widths"
  - In doing so, if the effective width is more than the calculated second row width, more characters of the program name will be displayed.

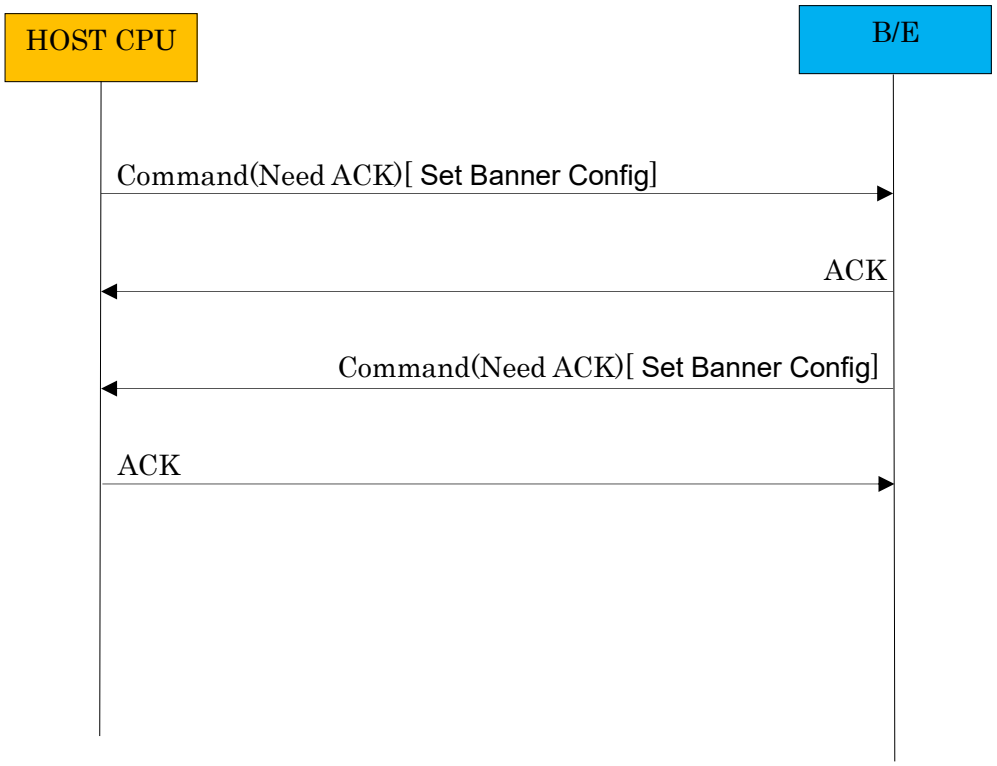
(4) Error Handling

・ Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence


HOST CPU
B/E
Command(Need ACK)[Set Banner Config]
ACK
Command(Need ACK)[Set Banner Config]
ACK

5.3.15. Set Character Set

(1) Set Character Set Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	04h
Header CRC8 :	--
Data 0 :	01h Set Character Set(Upper)
Data 1 :	31h Set Character Set(Lower)
Data 2 :	Character Set 00h : ARIB 01h : UTF-8
Data CRC8 :	--

(2) Set Character Set Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Set Character Set(Upper)
Data 1 :	31h Set Character Set(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- Change Character set
- Affects the following commands.

T.B.D.

- This command setting is not stored to NVRAM.

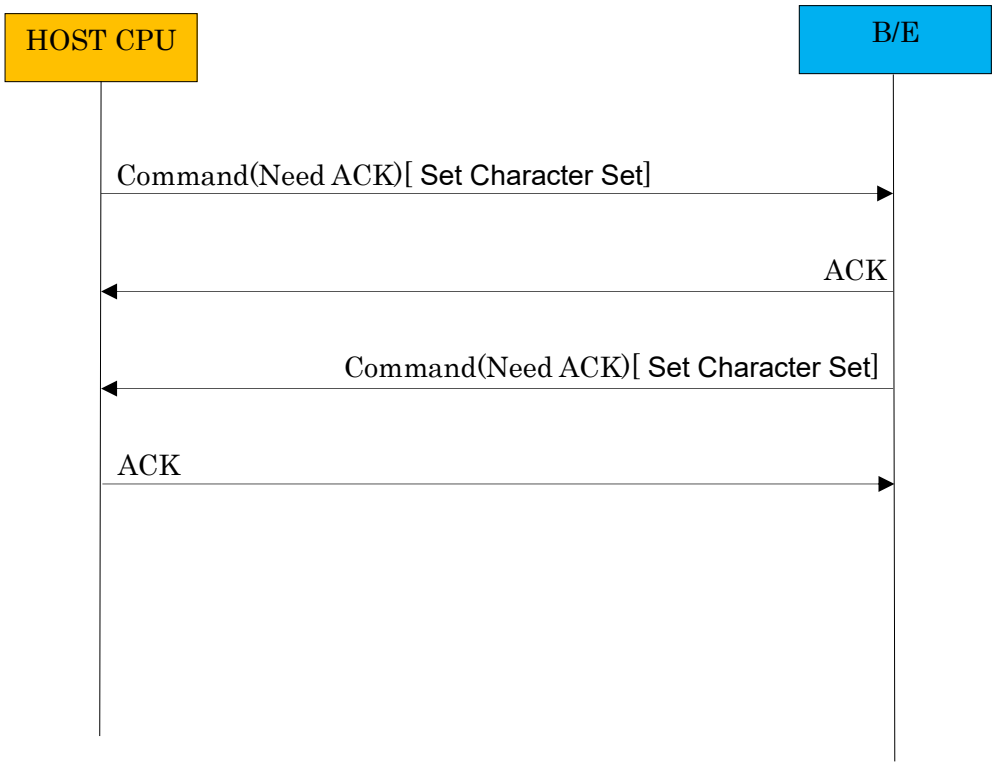
(4) Error Handling

- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence


HOST CPU
B/E
Command(Need ACK)[Set Character Set]
ACK
Command(Need ACK)[Set Character Set]
ACK

5.3.16. Set DTV Parameter 2

(1) Set DTV Parameter 2 Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	E3h
Header CRC8 :	--
Data 0 :	01h Set DTV Parameter 2 (Upper)
Data 1 :	35h Set DTV Parameter 2 (Lower)
Data 2 :	Enable / Disable 01h : Enable Parameter 00h : Disable Parameter (Use Default Parameter)
Data 3 :	Auto tune mode default 00h:Auto 01h:Fullseg 02h:Oneseg 03h:Auto with Seamless
Data 4:	Default Character Set 00h : ARIB 01h : UTF-8
Data 5 :	Default Video Resolution 00h : 720p 01h : 480p 02h : 1080p 03h : 480i 04h : 1080p@30Hz
Data 6 :	Closed Caption display over BML browser 00h : Enable Display 01h : Disable Display
Data 7 :	BML Error Message 00h : Enable Display 01h : Disable Display
Data 8 :	Receive Logo Type Bit0 : Logo Type 0 Bit1 : Logo Type 1 Bit2 : Logo Type 2 Bit3 : Logo Type 3 Bit4 : Logo Type 4 Bit5 : Logo Type 5 Bit6/Bit7 : Fix 0 0 : Receive 1 : Not Receive
Data 9 :	Search Detection Timeout *This is using for Broadcast Search and Search up/down.
Data 10 ~ 11:	C/N for Receive Error Detection for Fix Fullseg mode
Data 12 ~ 13:	C/N for Preset Scan Detection for Fix Oneseg mode and Auto mode
Data 14 ~ 15:	C/N for Preset Scan Detection for Fix Fullseg mode
Data 16 :	Timeout for Preset Scan per channel
Data 17 ~ 18 :	C/N Level 3 threshold for full seg
Data 19 ~ 20 :	C/N Level 2 threshold for full seg
Data 21 ~ 22 :	C/N Level 1 threshold for full seg
Data 23 ~ 24 :	EPG Update scan time / channel
Data 25 :	EPG Scan Number of Days (Only foreground EPG Scan)
Data 26 ~ 27 :	EPG Scan Signal Threshold (Only foreground EPG Scan)
Data 28 ~ 29 :	Audio Const offset for Fullseg

Data 30 ~ 31 :	Audio Const offset for Oneseg
Data 32 :	Event relay : Auto Event relay On/Off *When this configured as off, notification will be skipped.
Data 33 :	Event relay : Notification interval before switching [s]
Data 34 :	Event relay : Number of notifications before switching [times]
Data 35 ~ 36	Banner: Banner Position X
Data 37 ~ 38	Banner: Banner Position Y
Data 39 :	Banner: Font Size [px]
Data 40 :	Banner: Transparency [%]
Data 41 :	Banner: Logo Position Offset [px]
Data 42 :	Banner: Margin of Top and Bottom [px]
Data 43 :	Banner: Margin of Left and Right [px]
Data 44 :	Banner: Line spacing [px]
Data 45 :	Banner: Max number of characters
Data 46 :	Banner: Default Timeout [s]
Data 47 :	EPG List Type Enable Bit0 : List Type 1 Enable Bit1 : List Type 2 Enable Bit2 : List Type 3 Enable Bit3 : List Type 4 Enable
Data 48 :	EPG List: Type1 Program name Font Size [px]
Data 49 :	EPG List: Type1 TS Name Font Size [px]
Data 50 :	EPG List: Type1 EPG Content Font Size [px]
Data 51 :	EPG List: Type2 Program name Font Size [px]
Data 52 :	EPG List: Type2 TS Name Font Size [px]
Data 53 :	EPG List: Type2 EPG Content Font Size [px]
Data 54 :	EPG List: Type3 Program name Font Size [px]
Data 55 :	EPG List: Type3 TS Name Font Size [px]
Data 56 :	EPG List: Type3 EPG Content Font Size [px]
Data 57 :	EPG List: Type4 Program name Font Size [px]
Data 58 :	EPG List: Type4 TS Name Font Size [px]
Data 59 :	EPG List: Type4 EPG Content Font Size [px]
Data 60 :	EPG List: Table Title Color Red
Data 61 :	EPG List: Table Title Color Green
Data 62 :	EPG List: Table Title Color Blue
Data 63 :	EPG List: Table Title Font Color Red
Data 64 :	EPG List: Table Title Font Color Green
Data 65 :	EPG List: Table Title Font Color Blue
Data 66 :	EPG List: Cursor Color Red
Data 67 :	EPG List: Cursor Color Green
Data 68 :	EPG List: Cursor Color Blue
Data 69 :	EPG List: Cursor Font Color Red
Data 70 :	EPG List: Cursor Font Color Green
Data 71 :	EPG List: Cursor Font Color Blue
Data 72 :	EPG List: Current Program Color Red
Data 73 :	EPG List: Current Program Color Green
Data 74 :	EPG List: Current Program Color Blue
Data 75 :	EPG List: Current Program Font Color Red
Data 76 :	EPG List: Current Program Font Color Green
Data 77 :	EPG List: Current Program Font Color Blue
Data 78 :	EPG List: Balloon color Red
Data 79 :	EPG List: Balloon color Green
Data 80 :	EPG List: Balloon color Blue
Data 81 :	EPG List: Balloon Font color Red
Data 82 :	EPG List: Balloon Font color Green
Data 83 :	EPG List: Balloon Font color Blue
Data 84 :	EPG List: Border line color Red
Data 85 :	EPG List: Border line color Green

Data 86 :	EPG List: Border line color Blue
Data 87 :	EPG Information: Font Size [px]
Data 88 :	EPG Information: Title Font Color Red
Data 89 :	EPG Information: Title Font Color Green
Data 90 :	EPG Information: Title Font Color Blue
Data 91 :	EPG Information: Line Spacing for Title
Data 92 :	EPG Information: Line Spacing for Contents
Data 93 :	EPG Information: Margin
Data 94 :	Default Auto Broadcast Search mode
Data 95~96 :	C/N Level 3 threshold for one seg
Data 97~98 :	C/N Level 2 threshold for one seg
Data 99~100 :	C/N Level 1 threshold for one seg
Data 101 :	Default event mask 0 *Refer to chapter "5.3.20. Set Event Mask"
Data 102 :	Default event mask 1 *Refer to chapter "5.3.20. Set Event Mask"
Data 103 :	Default event mask 2 *Refer to chapter "5.3.20. Set Event Mask"
Data 104 :	EPG List: Short program border line color Red
Data 105 :	EPG List: Short program border line color Green
Data 106 :	EPG List: Short program border line color Blue
Data 107 :	Banner: segment icon
Data 108 :	Number of scrolling line on EPG detail
Data 109 :	Automatic banner display enable. 00h:Enable 01h:Disable
Data 110~225 :	Reserved
Data CRC8 :	--

・C/N: signed 4byte data (Divide by 100)

・C/N range 0dB to 40.0 dB

(2) Set DTV Parameter 2 Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Set DTV Parameter 2 (Upper)
Data 1 :	35h Set DTV Parameter 2 (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- Set DTV Parameter.
- Each parameter is big endian.
- Each parameters data range move to Chapter 8.2 or 8.3, please refer to them.
- For New Frontier(F039990144) case, Data 5 is fixed, cannot be changed.

- For New Frontier(F039990144) case, changing to the following data ate no supported, so the display will not be guaranteed if changed.
  *Data 6 to 7 and 48 to 93, 104 to 105.
- C/N: signed 4byte data (Divide by 100)

(4) Error Handling

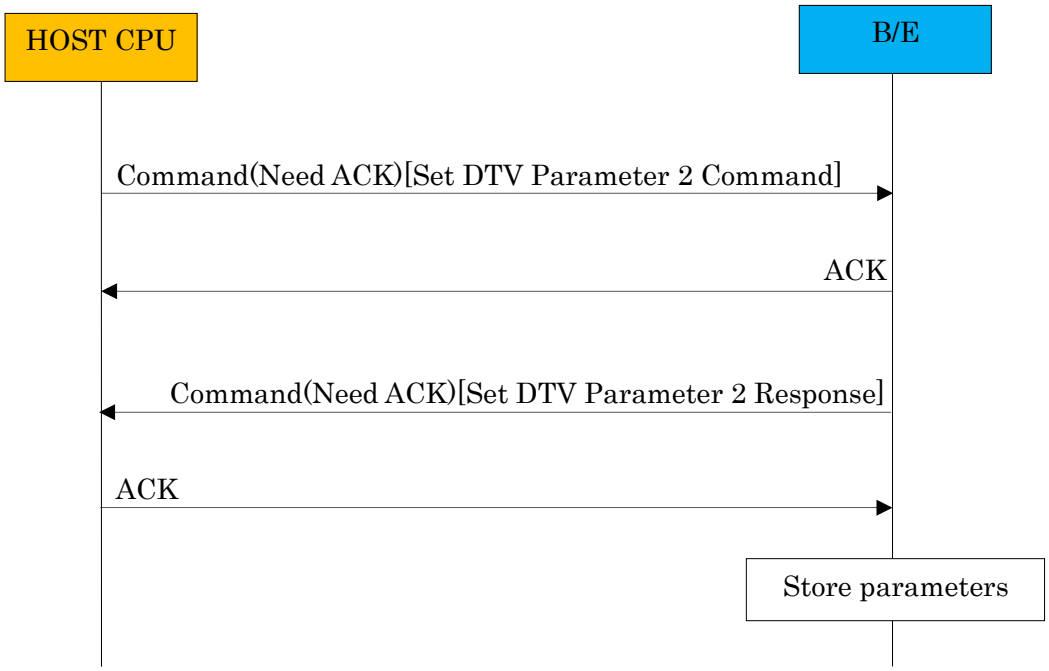
- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	unknown error

For other errors, refer to Chapter [3. Error Code]

When 01h (invalid parameter) is happened, all parameter is not effective.

(5) Sequence



5.3.17. Set User ID

(1) Set User ID Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	04h
Header CRC8 :	--
Data 0 :	01h Set User ID (Upper)
Data 1 :	3Bh Set User ID (Lower)
Data 2 :	User ID 01h to 06h FFh : Demo mode User
Data CRC8 :	--

(2) Set User ID Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Set User ID (Upper)
Data 1 :	3Bh Set User ID (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- During Demo mode User, below Commands are supported.

Code (Data0/1)	Command
0104h	Get DTV Status
0105h	Shutdown
0106h	Watch dog
0107h	Get TS Name
0113h	Get Version
0118h	Get Program information
0119h	Set Baud rate
012Ch	Set Banner Config
0130h	Get Banner Config
0133h	Get TS Name with UTF-8
0134h	Get Program Information with UTF-8
0135h	Set DTV Parameter 2
0136h	Get DTV Parameter 2
013Bh	Set User ID
013Ch	Get User ID
0141h	Set Event Mask
0142h	Get Event Mask
0143h	Get Part Number

0145h	Get Program Name with UTF-8 by 3Digit Number

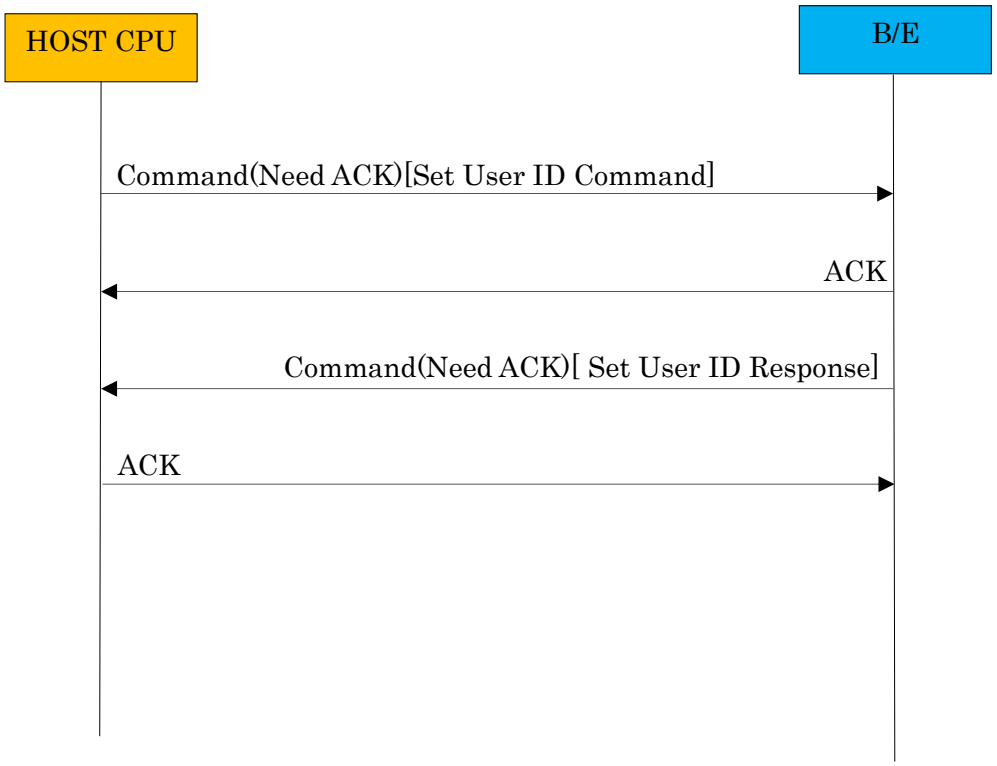
(4) Error Handling

・ Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.3.18. Set EPG Config

(1) Set EPG Config Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	36h
Header CRC8 :	--
Data 0 :	01h Set EPG Config(Upper)
Data 1 :	3Dh Set EPG Config(Lower)
Data 2 :	EPG List Type Enable Bit0 : Type 1 Enable Bit1 : Type 2 Enable Bit2 : Type 3 Enable Bit3 : Type 4 Enable 0 : Enable 1 : Disable
Data 3 :	EPG List: Type1 Program name Font Size [px]
Data 4 :	EPG List: Type1 TS Name Font Size [px]
Data 5 :	EPG List: Type1 EPG Content Font Size [px]
Data 6 :	EPG List: Type2 Program name Font Size [px]
Data 7 :	EPG List: Type2 TS Name Font Size [px]
Data 8 :	EPG List: Type2 EPG Content Font Size [px]
Data 9 :	EPG List: Type3 Program name Font Size [px]
Data 10 :	EPG List: Type3 TS Name Font Size [px]
Data 11 :	EPG List: Type3 EPG Content Font Size [px]
Data 12 :	EPG List: Type4 Program name Font Size [px]
Data 13 :	EPG List: Type4 TS Name Font Size [px]
Data 14 :	EPG List: Type4 EPG Content Font Size [px]
Data 15 :	EPG List: Table Title Color Red
Data 16 :	EPG List: Table Title Color Green
Data 17 :	EPG List: Table Title Color Blue
Data 18 :	EPG List: Table Title Font Color Red
Data 19 :	EPG List: Table Title Font Color Green
Data 20 :	EPG List: Table Title Font Color Blue
Data 21 :	EPG List: Cursor Color Red
Data 22 :	EPG List: Cursor Color Green
Data 23 :	EPG List: Cursor Color Blue
Data 24 :	EPG List: Cursor Font Color Red
Data 25 :	EPG List: Cursor Font Color Green
Data 26 :	EPG List: Cursor Font Color Blue
Data 27 :	EPG List: Current Program Color Red
Data 28 :	EPG List: Current Program Color Green
Data 29 :	EPG List: Current Program Color Blue
Data 30 :	EPG List: Current Program Font Color Red
Data 31 :	EPG List: Current Program Font Color Green
Data 32 :	EPG List: Current Program Font Color Blue
Data 33 :	EPG List: Balloon color Red
Data 34 :	EPG List: Balloon color Green
Data 35 :	EPG List: Balloon color Blue
Data 36 :	EPG List: Balloon Font color Red
Data 37 :	EPG List: Balloon Font color Green
Data 38 :	EPG List: Balloon Font color Blue
Data 39 :	EPG List: Border line color Red
Data 40 :	EPG List: Border line color Green
Data 41 :	EPG List: Border line color Blue
Data 42 :	EPG information: Font Size [px]

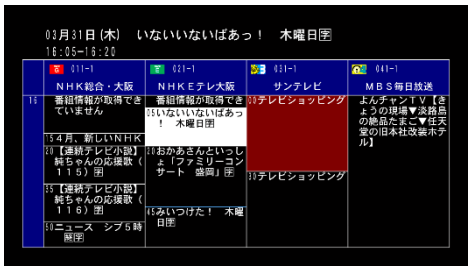
Data 43 :	EPG information: Title Font Color Red
Data 44 :	EPG information: Title Font Color Green
Data 45 :	EPG information: Title Font Color Blue
Data 46 :	EPG information: Line Spacing for Title
Data 47 :	EPG information: Line Spacing for Contents
Data 48 :	EPG information: Margin(Applied to all four sides)
Data 49 :	EPG List: Short program border line color Red
Data 50 :	EPG List: Short program border line color Green
Data 51 :	EPG List: Short program border line color Blue
Data 52 :	Number of scrolling line on EPG detail
Data CRC8 :	--

(2) Set EPG Config Response (BE ⇒ HOST)

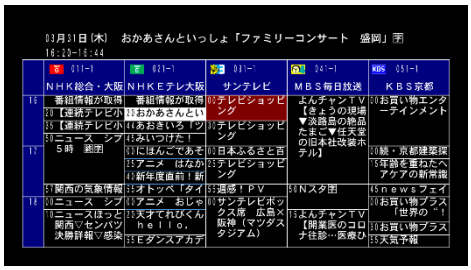
Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Set EPG Config(Upper)
Data 1 :	3Dh Set EPG Config(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

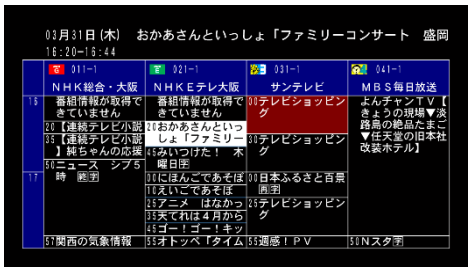
- EPG Type for F039902182【TRS231BSxxxx】,F03990218【TRH231BSxxxx】
- For New Frontier(F039990144) case, Data 3 to 51 should not be changed. Changing these parameters may corrupt the display.



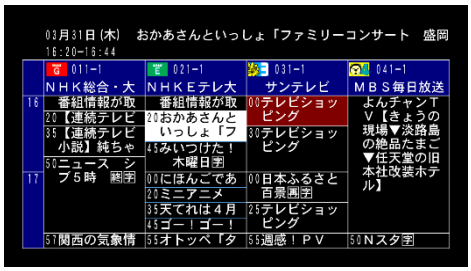
Type1



Type2



Type3



Type4

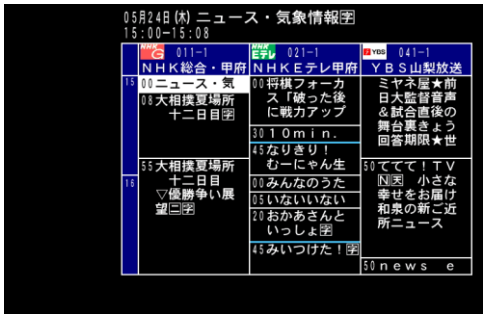
・ EPG Type for F039990144【FRO232BSyyzz】



Type1



Type2

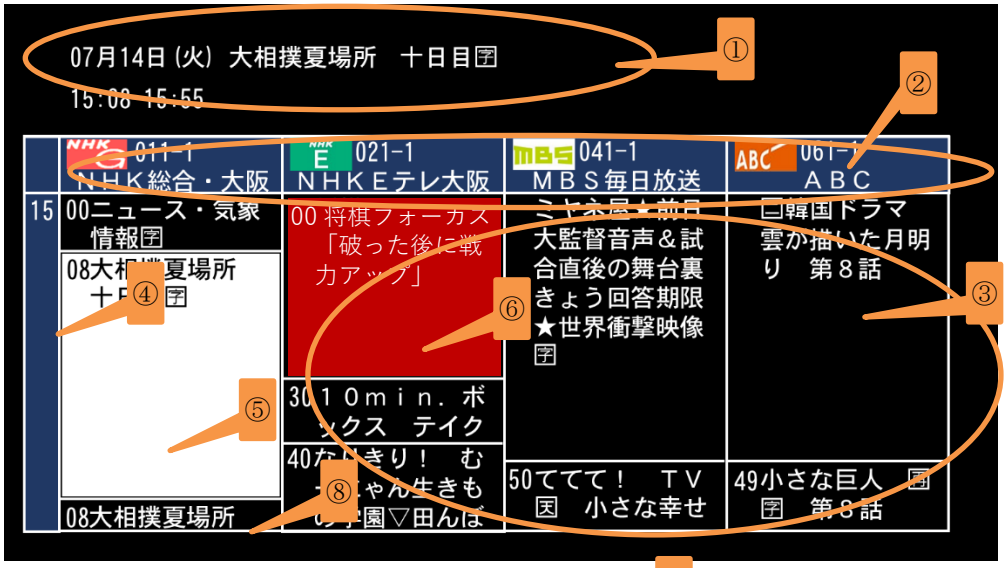


Type3



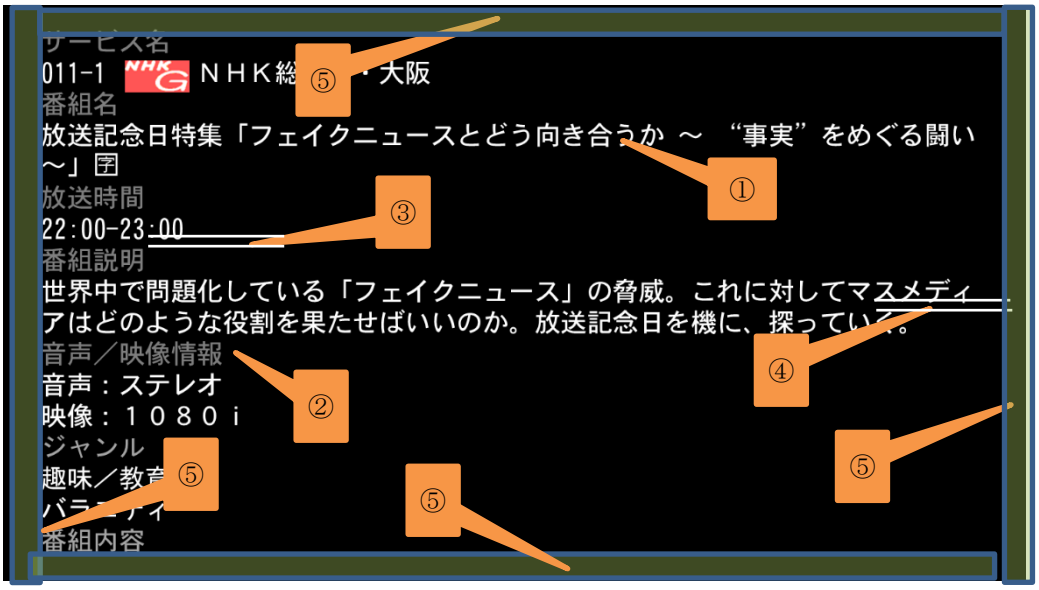
Type4

EPG Configuration Parameter.





① Program Name Font Size
② TS Name Font Size
③ EPG Content Font Size
④ Table Title Color & Font Color
⑤ Cursor Color & Font Color
⑥ Current Program Color & Font Color
⑦ Balloon Color & Font Color
⑧ Border Line Color
⑨ Short program border line color



① Font size (Title and Content)
② Title Font Color
③ Title Line Spacing
④ Content Line Spacing
⑤ EPG Detail Margin

- The Title Line Spacing will not be applied to first title on the page, to avoid the title being displayed too low due to Margin and Line Spacing.
- This command setting is not stored to NVRAM.
- The actual parameter range refers to chapter 8.2 and 8.3, data 48 to 93, data 108.
- For F039990144【FRO232BSyyzz】, the EPG list design is adjusted to meet your requests as much as possible, so it is not guaranteed that the display will be correct when parameters are changed.

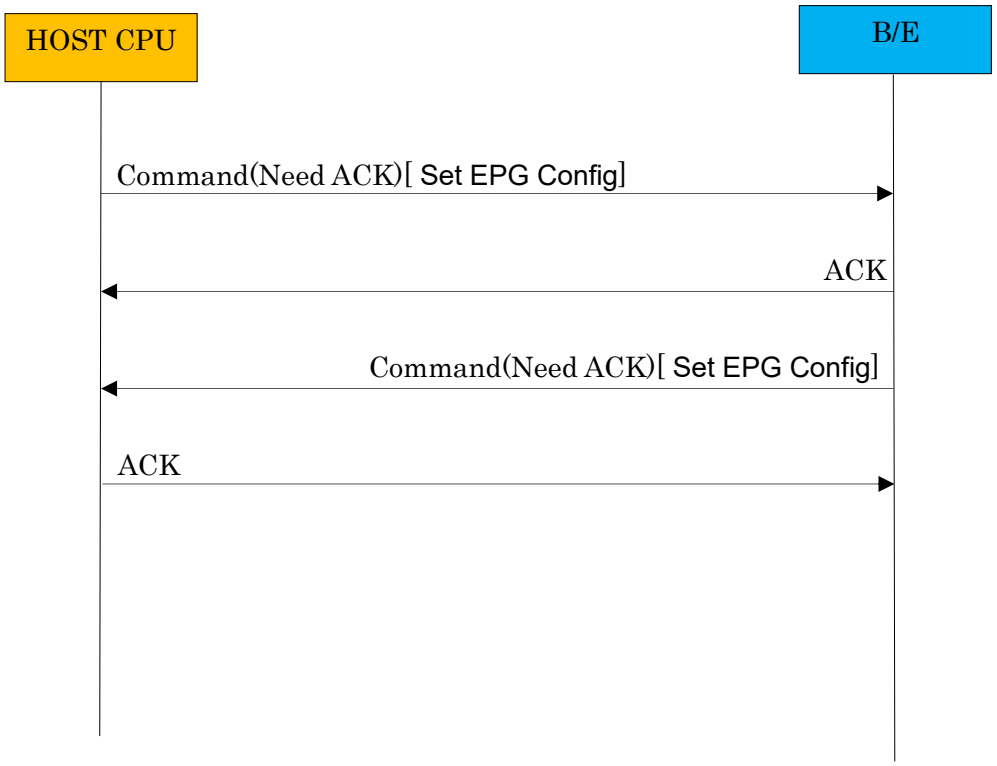
(4) Error Handling

- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence


HOST CPU
B/E
Command(Need ACK)[Set EPG Config]
ACK
Command(Need ACK)[Set EPG Config]
ACK

5.3.19. Set Video Color

(1) Set Video Color Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	08h
Header CRC8 :	--
Data 0 :	01h Set Video Color (Upper)
Data 1 :	3Fh Set Video Color (Lower)
Data 2 :	Brightness -127(81h) – 127(7Fh) [Default is 0]
Data 3 :	Contrast 0 – 7 [Default is 0]
Data 4～6 :	Reserved
Data CRC8 :	--

(2) Set Video Color Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Set Video Color (Upper)
Data 1 :	3Fh Set Video Color (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- Set Video Color
- Brightness: Brightness of the original image can be decreased or increased and consists of 255 levels. These values are 2's complements and can be set between -127 and +127.
- Contrast: Contrast can be adjusted in 8 levels, 0 to 7. 0 represents the contrast of original image. The contrast of the original image cannot be reduced.
- Set User ID does not affect last color setting.

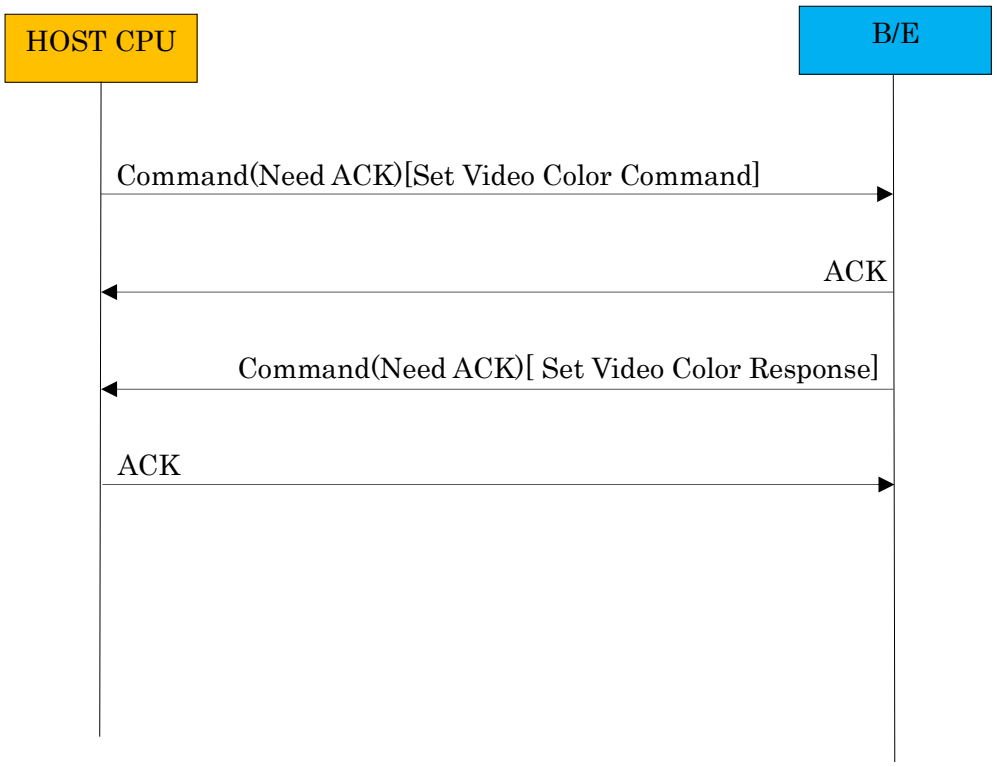
(4) Error Handling

・ Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.3.20. Set Event Mask

(1) Set Event Mask Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	06h
Header CRC8 :	--
Data 0 :	01h Set Event Mask (Upper)
Data 1 :	41h Set Event Mask (Lower)
Data 2 :	Event mask 0 bit0 : DTV Status bit1 : Scan Status bit2 : Search Status bit3 : DTV Event bit4 : EWS Event bit5 : EPG Status bit6 : Event relay bit7 : Number of Program info 0 : Not Mask 1 : Mask - Disables the event notification.
Data 3 :	Event mask 1 bit0 : Program Information bit1 : Program Information with UTF-8 bit2 : TS Name Update bit3 : TS Name Update with UTF-8 bit4 : Logo Update bit5 : EGP Button Status bit6 : BML Status bit7 : C/N Information 0 : Not Mask 1 : Mask - Disables the event notification.
Data 4 :	Event mask 2 bit0 : EPG Scan Result bit1 – 7 Reserve 0 : Not Mask 1 : Mask - Disables the event notification.
Data CRC8 :	--

(2) Set Event Mask Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Set Event Mask (Upper)
Data 1 :	41h Set Event Mask (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")

Data CRC8 : --

(3) Description

- Set Event Enable / Disable

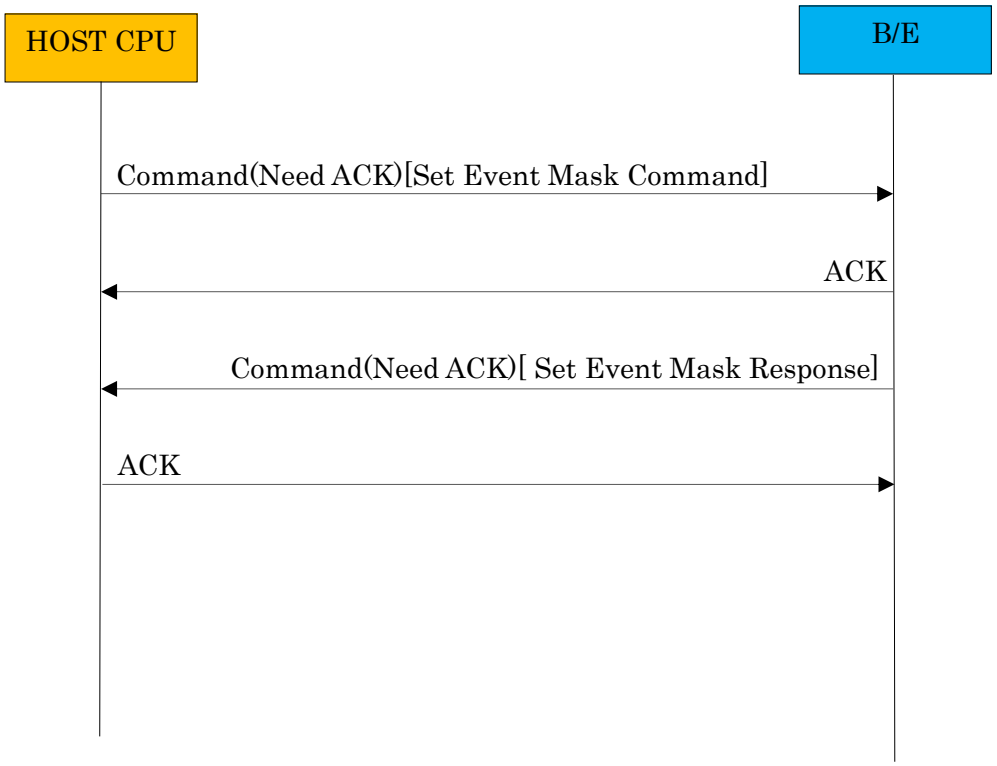
(4) Error Handling

- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.4. HOST⇒B/E Get Command

5.4.1. Get DTV Status

(1) Get DTV Status Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h
Header CRC8 :	--
Data 0 :	01h Get DTV Status(Upper)
Data 1 :	04h Get DTV Status(Lower)
Data CRC8 :	--

(2) Get DTV Status Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	14h
Header CRC8 :	--
Data 0 :	01h Get DTV Status(Upper)
Data 1 :	04h Get DTV Status(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	DTV Status 00h : Power OFF (Not Support) 01h : Booting (Not Support) 02h : Shutdown (After Shutdown B/E) 03h : In Service (After Booting B/E) 04h : Low Power (Low Power mode) 05h : DTV OFF 06h : DTV OFF with EPG SCAN 07h : Demo mode
Data 5 :	Tuning Status 00h : Idle 01h : Tuning
Data 6 :	Service Type (Now Playing Service Type) 00h : ISDB-T Fullseg 01h : ISDB-T Oneseg FFh : None (Not Tuning)
Data 7 :	Service Status 00h : Invalid 01h : Normal Service 02h : Special Service 03h : Emergency Service 04h : Suspend Service
Data 8 :	Video Output 00h : No output (From Tuning to Video Output) 01h : Outputted
Data 9 :	Preset Number 00h : Preset No.1(Home) 01h : Preset No.2(Drive)

	FFh : None(Not Tuning / Not Stored channel)
Data 10 :	Preset Channel Number 00h : tuned by Search/BC Search/Physical Channel Set 01h～0Ch : Preset Channel Number FFh : None (Not Tuning)
Data 11 :	Copy Control (Refer to "5.7.1 DTV Status (2) Description")
Data 12 :	Copy Control (Refer to "5.7.1 DTV Status (2) Description")
Data 13 :	Event Code Refer to "5.7.4.DTV Event" ※Event code of "0x80" or more will not occur in this data.
Data 14 :	Diver Mode 00h : Dynamic 4Diver System (Fore Br0/1) 01h : Fixed 2x2Diver System (Fore Br0/1) 02h : Fixed 2Diver System(Antenna Br2/3) 03h : Fixed 4Diver System 04h : Fixed 2Diver System(Antenna Br0/1) 05h : Fixed 3x1Diver System (Fore Br0/1/2) 06h : Dynamic 4Diver System (Fore Br2/3) 07h : Fixed 2x2Diver System (Fore Br2/3) 08h : Fixed 3x1Diver System (Fore Br1/2/3)
Data 15 :	Antenna Level 00h : Antenna Level 0 01h : Antenna Level 1 02h : Antenna Level 2 03h : Antenna Level 3
Data 16 :	Background Status 00h : idle 01h : Background Search 02h : Background Scan 03h : Manual Background Search 04h : Manual Background Scan
Data 17 :	EPG/BML Status 00h : No EPG/BML displayed 01h : Updating 02h : EPG List (full seg) 03h : EPG List (one seg) 04h : EPG Information (full seg) 05h : EPG Information (one seg) 06h : BML Display
Data18	Tuner0 Physical Channel Number
Data CRC8	--

(3) Description

- Get DTV Status
- DTV Status information in Data2 will be set to In Service (03h) immediately after sending "Wakeup" from B/E. B/E will not wait for Wakeup Response for setting this.
- If Data4 = "04h: Low Power mode", tune commands, Display EPG command, etc., are not supported.
- If Data4 = "07h: Demo mode ", Only a few commands are supported. See "5.3.17. Set User ID" for the list of supported commands.
- "Service Status(Data 7)" is :

Service Status	Description
Invalid	If Tuning Status (Data5) isn't "Tuning", Service Status is invalid.
Normal Service	
Special Service	This service is defined by "ARIB TR-B14 Volume4 23. Special Services".
Emergency Service	This service is defined by "ARIB TR-B14 Volume4 26. Use of Emergency Warning Broadcasting (Emergency Warning System (EWS))".
Suspend Service	This service is "Off service" that defined by "ARIB TR-B14 Volume4 15. Definition of Service "On service" / "Off service"".

- If "Diver Mode" is 00h or 01h, B/E use both tuner0/tuner1.
- Tuner0 is used for Foreground processing, and Tuner1 is used for Background processing.
- "Antenna Level" indicates the number of antenna bars on the display. Antenna Level 0 is bad signal. Antenna Level 3 is good signal. If "Tuning Status (Data5)" is idle, Antenna Level is 0.
- If "EPG Update Status" is 00h, B/E isn't updating EPG of specified channel. But B/E may be updating EPG with back scan. If 01h, B/E is updating forcibly by "5.2.12.Display EPG" command. If finishing to update EPG, "EPG Update Status" change to 00h.

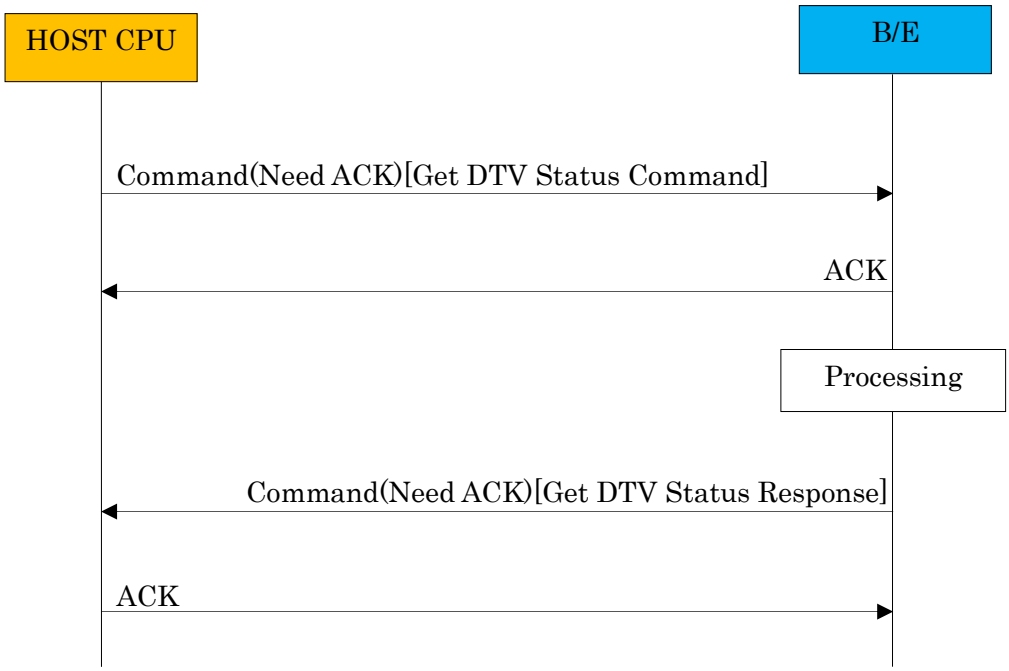
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence


HOST CPU
B/E
Command(Need ACK)[Get DTV Status Command]
ACK
Processing
Command(Need ACK)[Get DTV Status Response]
ACK

5.4.2. Get TS Name

(1) Get TS Name Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Get TS Name (Upper)
Data 1 :	07h Get TS Name (Lower)
Data 2 :	01h~0Ch : Preset Channel Number FFh : All Preset Channel Number 00h : Not Stored Channel.
Data 3 :	00h : Home Preset 01h : Drive Preset FFh : Last Preset
Data CRC8 :	--

(2) Get TS Name Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : Preset Channel Number =00h～0Ch: 1Ah Preset Channel Number = FFh : F6h Result=NG : 05h
Header CRC8 :	--
Data 0 :	01h Get TS Name (Upper)
Data 1 :	07h Get TS Name (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	01h~0Ch : Preset Channel Number FFh : All Channels in Preset 00h : Not Stored Channel.
Data 5~24 :	TS Name(8bit character code set) Data 4 is 00h~0Ch:Specified Preset Channel Number Data 4 is FFh : Preset Channel Number = 1
Data 25~44 :	TS Name(8bit character code set) Data 4 is 00h ~0Ch : Don't use Data 4 is FFh : Preset Channel Number = 2
Data 45~64 :	TS Name(8bit character code set) Data 4 is 00h ~0Ch : Don't use Data 4 is FFh : Preset Channel Number = 3
Data 65~84 :	TS Name(8bit character code set) Data 4 is 00h ~0Ch : Don't use Data 4 is FFh : Preset Channel Number = 4
Data 85~104 :	TS Name(8bit character code set) Data 4 is 00h ~0Ch : Don't use Data 4 is FFh : Preset Channel Number = 5
Data 105~124 :	TS Name(8bit character code set) Data 4 is 00h ~0Ch : Don't use Data 4 is FFh : Preset Channel Number = 6

Data 125~144 :	TS Name(8bit character code set) Data 4 is 00h ~0Ch : Don't use Data 4 is FFh : Preset Channel Number = 7
Data 145~164 :	TS Name(8bit character code set) Data 4 is 00h ~0Ch : Don't use Data 4 is FFh : Preset Channel Number = 8
Data 165~184 :	TS Name(8bit character code set) Data 4 is 00h ~0Ch : Don't use Data 4 is FFh : Preset Channel Number = 9
Data 185~204 :	TS Name(8bit character code set) Data 4 is 00h ~0Ch : Don't use Data 4 is FFh : Preset Channel Number = 10
Data 205~224 :	TS Name(8bit character code set) Data 4 is 00h ~0Ch : Don't use Data 4 is FFh : Preset Channel Number = 11
Data 225~244 :	TS Name(8bit character code set) Data 4 is 00h ~0Ch : Don't use Data 4 is FFh : Preset Channel Number = 12
Data CRC8 :	--

(3) Description

- Get TS Name.
- If Preset Channel Number is 01h~0Ch, get TS Name of specified Preset Channel and specified Preset Table.
- If Preset Channel Number is FFh, get All TS Name in specified Preset Table.
- If Preset Channel Number is 00h, get TS Name of Not Stored Channel.

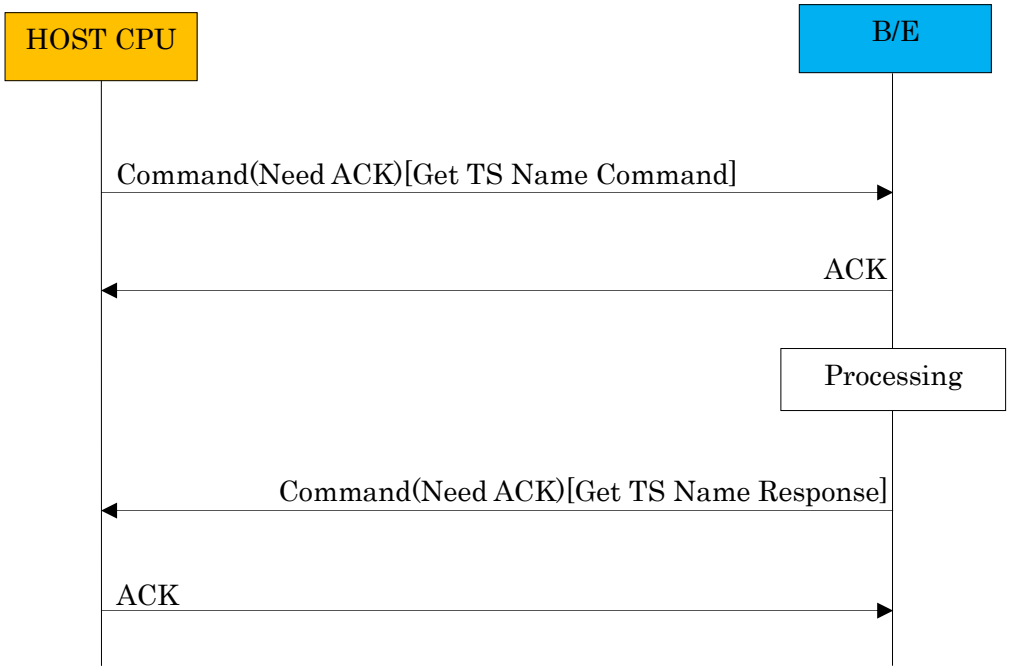
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence


HOST CPU
B/E
Command(Need ACK)[Get TS Name Command]
ACK
Processing
Command(Need ACK)[Get TS Name Response]
ACK

5.4.3. Get Configure

(1) Get Configure Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h
Header CRC8 :	--
Data 0 :	01h Get Configure(Upper)
Data 1 :	0Ch Get Configure(Lower)
Data CRC8 :	--

(2) Get Configure Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : 1Ch Result=NG : 05h
Header CRC8 :	--
Data 0 :	01h Get Configure(Upper)
Data 1 :	0Ch Get Configure(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Audio Mode 00h : Main 01h : Sub 02h : Main + Sub
Data 5 :	Closed Caption Mode 00h : closed caption off but superimpose off 01h : 1st closed caption on and superimpose 1st on 02h : 2nd closed caption on and superimpose 2nd on 03h : closed caption off and superimpose 1st on 04h : 1st closed caption on and superimpose off 05h : 2nd closed caption on and superimpose off Initial value : 00h
Data 6 :	Auto Tune Mode 00h : Auto 01h : Only Fullseg 02h : Only Oneseg 03h : Auto with Seamless
Data 7 :	Auto Broadcast Search Mode 00h : Automatic Broadcast Search Off 01h : Automatic Relay Station Search On 02h : Automatic Affiliated Station Search On 03h : Automatic Relay & Affiliated Station Search On
Data 8 :	Auto Event Relay 00h : Off 01h : On
Data 9 :	Region ID (Refer to "5.3.5.Set Residential Area") (Home Preset)

Data 10 :	Zip-Code[0](Home Preset) Ascii Code In case of "1234567", '1'
Data 11 :	Zip-Code[1](Home Preset) Ascii Code In case of "1234567", '2'
Data 12 :	Zip-Code[2](Home Preset) Ascii Code In case of "1234567", '3'
Data 13 :	Zip-Code[3](Home Preset) Ascii Code In case of "1234567", '4'
Data 14 :	Zip-Code[4](Home Preset) Ascii Code In case of "1234567", '5'
Data 15 :	Zip-Code[5](Home Preset) Ascii Code In case of "1234567", '6'
Data 16 :	Zip-Code[6](Home Preset) Ascii Code In case of "1234567", '7'
Data 17 :	Region ID (Refer to "5.3.5.Set Residential Area") (Drive Preset)
Data 18 :	Zip-Code[0](Drive Preset) Ascii Code In case of "1234567", '1'
Data 19 :	Zip-Code[1](Drive Preset) Ascii Code In case of "1234567", '2'
Data 20 :	Zip-Code[2](Drive Preset) Ascii Code In case of "1234567", '3'
Data 21 :	Zip-Code[3](Drive Preset) Ascii Code In case of "1234567", '4'
Data 22 :	Zip-Code[4](Drive Preset) Ascii Code In case of "1234567", '5'
Data 23 :	Zip-Code[5](Drive Preset) Ascii Code In case of "1234567", '6'
Data 24 :	Zip-Code[6](Drive Preset) Ascii Code In case of "1234567", '7'
Data 25 :	Event Relay Notification Interval before switching [s]
Data 26 :	Event Relay Number of Notification before switching [times]
Data CRC8 :	--

(3) Description

- Get Configure

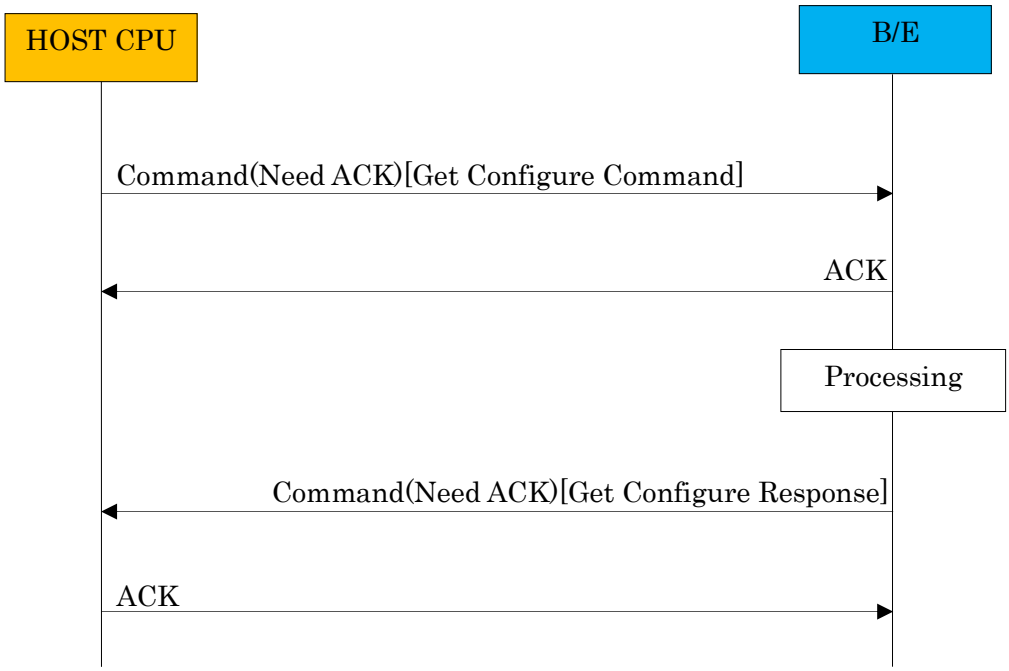
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence


HOST CPU
B/E
Command(Need ACK)[Get Configure Command]
ACK
Processing
Command(Need ACK)[Get Configure Response]
ACK

5.4.4. Get Version

(1) Get Version Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h
Header CRC8 :	--
Data 0 :	01h Get Version (Upper)
Data 1 :	13h Get Version (Lower)
Data CRC8 :	--

(2) Get Version Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : 15h Result=NG : 05h
Header CRC8 :	--
Data 0 :	01h Get Version (Upper)
Data 1 :	13h Get Version (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Hardware Version 0(Upper)
Data 5 :	Hardware Version 1
Data 6 :	Hardware Version 2
Data 7 :	Hardware Version 3(Lower)
Data 8 :	Boot Version 0(Upper)
Data 9 :	Boot Version 1
Data 10 :	Boot Version 2
Data 11 :	Boot Version 3(Lower)
Data 12 :	Software Version 0(Upper)
Data 13 :	Software Version 1
Data 14 :	Software Version 2
Data 15 :	Software Version 3
Data 16 :	Software Version 4
Data 17 :	Software Version 5
Data 18 :	Software Version 6
Data 19 :	Software Version 7(Lower)
Data CRC8 :	--

(3) Description

- Get Software Version.
- The format of Hardware Version is ASCII.(Ex. "004A")
- The format of Boot Version is ASCII. (Ex. "0102")
- The format of Software Version is ASCII. (Ex. "AABB1234")

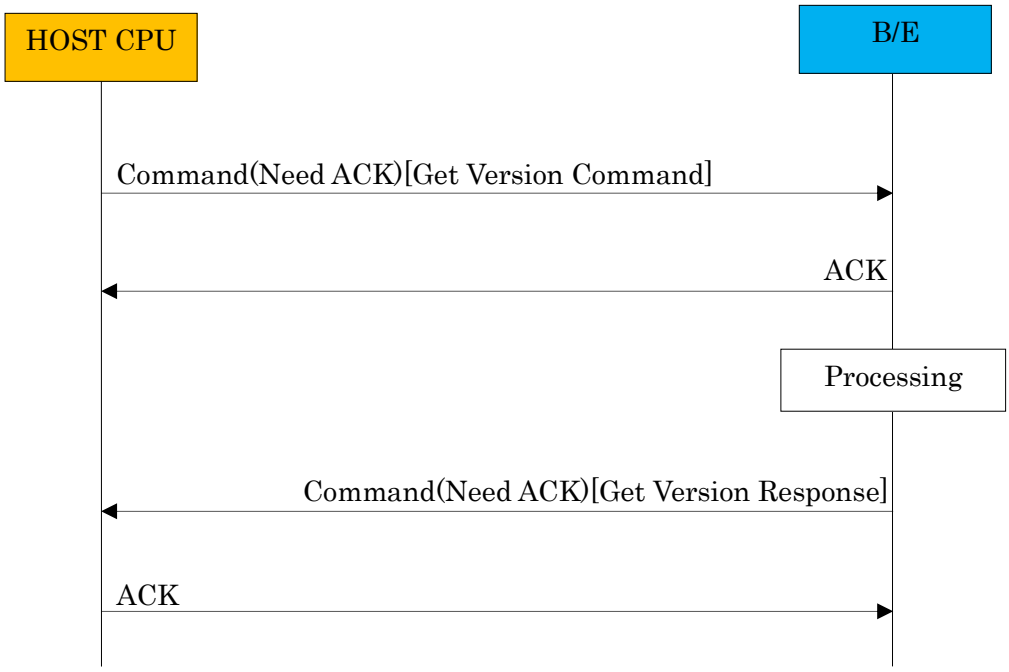
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.4.5. Get Audio ES

(1) Get Audio ES Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h
Header CRC8 :	--
Data 0 :	01h Get Audio ES (Upper)
Data 1 :	14h Get Audio ES (Lower)
Data CRC8 :	--

(2) Get Audio ES Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : 06h~26h Result=NG : 05h
Header CRC8 :	--
Data 0 :	01h Get Audio ES (Upper)
Data 1 :	14h Get Audio ES (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	ES Num 0x01~0x0F : Number of Audio ES 0xFF : not available yet
Data 5 :	Component Tag 1 (Current playing Audio ES component Tag)
Data 6 :	Component Tag 2 (Lowest Component Tag(except during playback) …
Data n :	Component Tag N (Highest Component Tag(except during playback)
Data CRC8 :	--

(3) Description

- Get Audio ES.
- Component Tag is 010h~2Fh (Fullseg), 83h~86h (Oneseg).
- Component Tag is used by "5.2.7.ES Switch".
- If B/E doesn't receive Audio ES Information, Data 4(ES Num) is set to 0xFF.

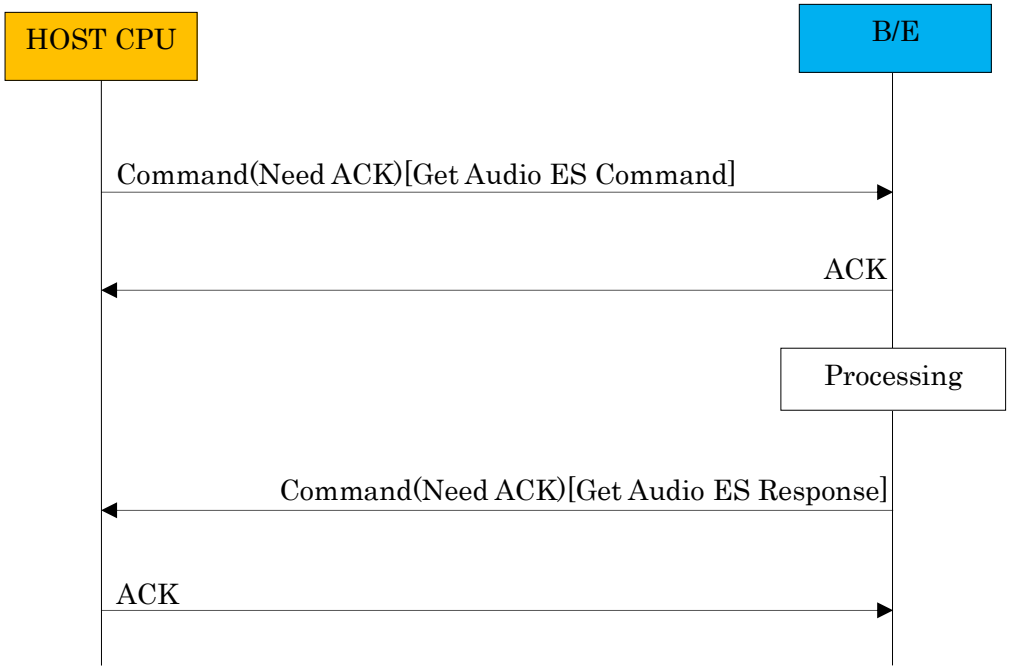
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.4.6. Get Program Information

(1) Get Program Information Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h
Header CRC8 :	--
Data 0 :	01h Get Program Information (Upper)
Data 1 :	18h Get Program Information (Lower)
Data CRC8 :	--

(2) Get Program Information Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h~80h
Header CRC8 :	--
Data 0 :	01h Get Program Information (Upper)
Data 1 :	18h Get Program Information (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Preset Number 00h : Preset No.1(Home) 01h : Preset No.2(Drive) FFh：None
Data 5 :	00h : Not stored service 01h~0Ch : Preset Channel Number FFh：None
Data 6 :	Three Digit Number(Upper)
Data 7 :	Three Digit Number(Lower)
Data 8 :	Branch Identifier Number
Data 9 :	Length of Station Name(N1) : 00h~14h
Data 10~9+N1 :	Station Name(TS Name)
Data 10+N1 :	Length of Program Name(N2) : 00h~60h
Data 11+N1~10+N1+N2:	Program Name
Data CRC8 :	--

(3) Description

- Get the program information in current service.
- The program information contains Preset Channel Number Station Name / Program name).
- Preset Channel Number (Data5) of Response is current playback channel. If current playback channel is not stored, Data5 is 00h and Data4 is FFh.
- Three Digit Number & Branch Identifier Number of Response are current playback service. If current playback channel is not stored, these data are set.

- The length of Station Name is 00h~14h.
- The format of Station Name is 8bit character code set.
- The length of Program Name is 00h~60h.
- The format of Program name is 8bit character code set.
- If B/E doesn't have Program name, Length of Program Name is set to 0.

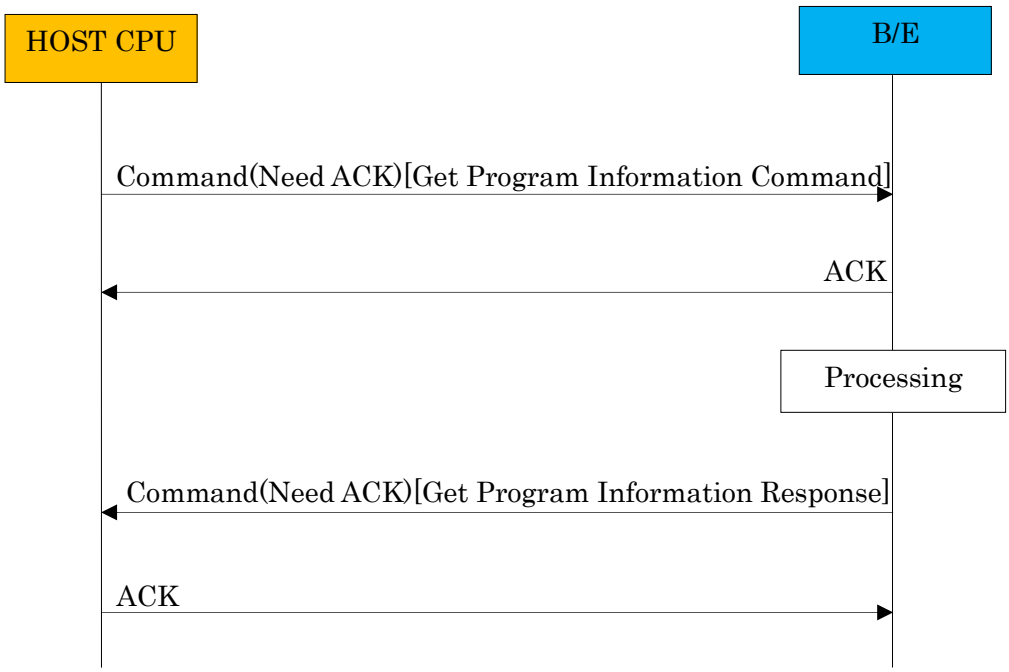
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.4.7. Get Service Information

(1) Get Service Information Command (HOST ⇒ BE)

Start Sync :	A5h,5ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Get Service Information (Upper)
Data 1 :	1Ch Get Service Information (Lower)
Data 2 :	Preset Table 00h : Home Preset 01h : Drive Preset FFh : Last Preset
Data 3 :	One Touch Number 01h ~ 0Ch : Service Information of Specified Channel 00h : Service Information of Not Stored Channel. FFh : Service Information of All Channel.
Data CRC8 :	--

(2) Get Service Information Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : 08h~ Result=NG : 05h
Header CRC8 :	--
Data 0 :	01h Get Service Information (Upper)
Data 1 :	1Ch Get Service Information (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Fragment Number
Data 5 :	Last Fragment Number
Data 6~ :	Service_Information_Data Refer to "(4) Description"
Data CRC8 :	--

(3) Get Service Information Response (BE ⇒ HOST with Header Type 2)

Start Sync :	A5h,59h
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length(2 Bytes) :	Result=OK : 0008h~0FFFh Result=NG : 0005h
Header CRC8 :	--
Data 0 :	01h Get Service Information (Upper)
Data 1 :	1Ch Get Service Information (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Fragment Number
Data 5 :	Last Fragment Number
Data 6~ :	Service_Information_Data Refer to "(4) Description"
Data CRC8 :	--

(4) Description

- Get Service Information.
- If One Touch Number (Data3) of Command is FFh, B/E response the service information of all channels. If One Touch Number (Data3) of Command is 00h, B/E response the service information of not-stored channel. If One Touch Number (Data3) is 01h~0Ch, B/E response the service information of specified channel.
- "Fragment Number" is the order of the fragmented response.
- For Header Type 1, If the length of Data Part is 255byte and under, "Fragment Number" is 0. If the length of Data Part is over 255byte, it is the order number. The B/E sends the data in actual order.
- For Header Type 2, If the length of Data Part is 4095byte and under, "Fragment Number" is 0. If the length of Data Part is over 4095byte, it is the order number. The B/E sends the data in actual order.
- "Last Fragment Number" is the max number of "Fragment Number".
- "Service Information Data" shows the following table.

Number_of_TS_Information(N1)	1byte
for(i = 0; i < N1 ; i++){	
OneTouchNumber	1byte
Number_of_Service_Information(N2)	1byte
for(j = 0; j < N2; j++){	
Service_Information()	n byte
}	
}	

- "Number_of_TS_Information" is the number of the followings TS Information. The range is 00h~0Ch.
- "OneTouchNumber" is valid one touch number. The range is 00h~0Ch

- "Number_of_Service_Information" is the number of the followings Service_Information(). The range of "Number_of_Service_Information" is 00h~17h.
- The format of "Service_Information()" shows the following table.

Field	Size	Value
Three Digit Number	2Byte	011~
Service ID	2Byte	0x0000~0xFFFF
Branch Identifier Number	1Byte	0~9
Media Type	1Byte	0 : Full Seg Service 1 : Data Service(Not Support) 2 : Data Service(Not Support) 3 : One Seg Service
Service Name Length	1Byte	00h ~ 14h
Service Name	0~20Bytes	8bit character code set

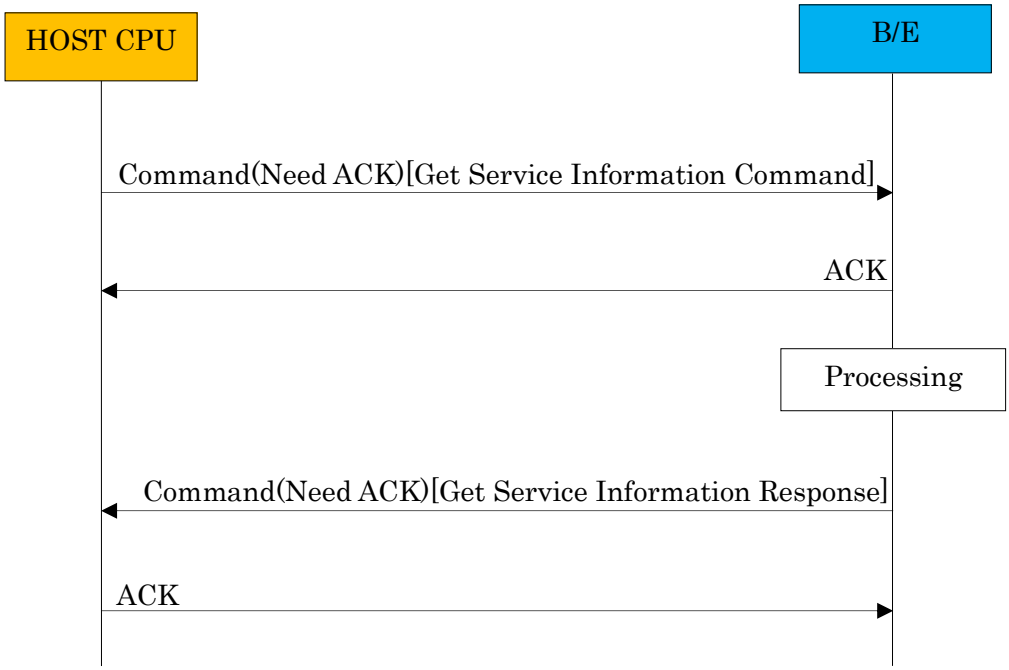
(5) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(6) Sequence



5.4.8. Get DTV Parameter

(1) Get DTV Parameter Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h
Header CRC8 :	--
Data 0 :	01h Get DTV Parameter (Upper)
Data 1 :	1Eh Get DTV Parameter (Lower)
Data CRC8 :	--

(2) Get DTV Parameter Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : E5h Result=NG : 05h
Header CRC8 :	--
Data 0 :	01h Get DTV Parameter (Upper)
Data 1 :	1Eh Get DTV Parameter (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4~7 :	Enable/Disable Data4 : 01h , Data5~7 : 00h : Enable Parameter Data4 : 00h , Data5~7 : 00h : Disable Parameter (In case of Disable, use Default Parameter
Data 8~11 :	C/N for Tuner Unlock Detection.
Data 12~15 :	BER for Tuner Unlock Detection.
Data 16~19 :	PER for Tuner Unlock Detection.
Data 20~23 :	Time for Tuner Unlock Detection.
Data 24~27 :	C/N for change to Fullseg.
Data 28~31 :	BER for change to Fullseg.
Data 32~35 :	PER for change to Fullseg.
Data 36~39 :	Time for change to Fullseg.
Data 40~43 :	C/N for change to Oneseg.
Data 44~47 :	BER for change to Oneseg.
Data 48~51 :	PER for change to Oneseg.
Data 52~55 :	Time for change to Oneseg.
Data 56~59 :	C/N for start Broadcast Search.
Data 60~63 :	BER for start Broadcast Search.
Data 64~67 :	PER for start Broadcast Search.
Data 68~71 :	Time for start Broadcast Search.
Data 72~75 :	Time (1) for disable Broadcast Search.
Data 76~79 :	Time (2) for disable Broadcast Search.
Data 80~83 :	Time for Background Tuner Unlock Detection.
Data 84~87 :	C/N for start Background Search.
Data 88~91 :	Time for start Background Search.
Data 92~95 :	C/N for start Background Scan.
Data 96~99 :	Time for start Background Scan.

Data 100~103 :	Time for change to 4Div mode.
Data 104~107 :	C/N difference for switching to background channel.
Data 108~111 :	BER for switching to background channel.
Data 112~115 :	PER for switching to background channel.
Data 116~119 :	Time for switching to background channel.
Data 120~123 :	C/N for Receive Error Detection.
Data 124~127 :	BER for Receive Error Detection.
Data 128~131 :	PER for Receive Error Detection.
Data 132~135 :	Time for Receive Error Detection.
Data 136~139 :	C/N for Fullseg Detection at fixed Fullseg search.
Data 140~143 :	C/N for Oneseg Detection at fixed or auto switch mode search.
Data 144~147 :	Detect Time (count) for other broadcast.
Data 148~151 :	Detect During Time for other broadcast.
Data 152~155 :	C/N Offset for Fullseg Detection at fixed Fullseg broadcast search.
Data 156~159 :	Reserved
Data 160~163 :	Reserved
Data 164~167 :	Reserved
Data 168~171 :	Reserved
Data 172~175 :	Reserved
Data 176~179 :	Reserved
Data 180~183 :	Reserved
Data 184~187 :	Reserved
Data 188~191 :	Reserved
Data 192~195 :	Reserved
Data 196~199 :	Reserved
Data 200~203 :	Reserved
Data 204~207 :	Reserved
Data 208~211 :	Reserved
Data 212~215 :	Reserved
Data 216~219 :	Reserved
Data 220~223 :	Reserved
Data 224~227 :	Reserved
Data CRC8 :	--

※ Parameters and range: Refer to 5.3.9 Set DTV Parameter.

(3) Description

- Get DTV Parameter.
- Even if Data 2-5 is set to Disable, the parameter read value will read the set value. (Value of parameters set as default cannot be read.)
- Each parameter is 32bit. Refer to the following table about the format.

・ Data	Each Parameter
Data N	Upper
Data N+1	↑
Data N+2	↓
Data N+3	Lower

(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.

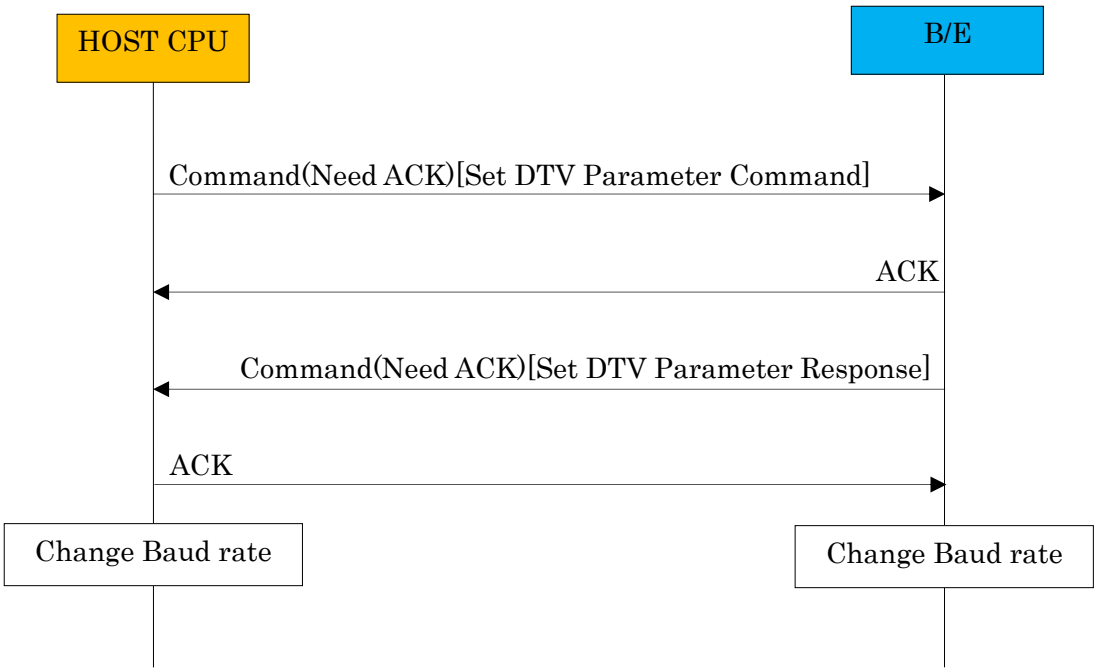
✧ The result of response is NG.

・ Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.4.9. Get Program Description

(1) Get Program Description Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	07h
Header CRC8 :	--
Data 0 :	01h Get Program Description (Upper)
Data 1 :	22h Get Program Description (Lower)
Data 2 :	Service ID(Upper)
Data 3 :	Service ID (Lower)
Data 4 :	Event ID(Upper)
Data 5 :	Event ID (Lower)
Data CRC8 :	--

(2) Get Program Description Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h~CAh
Header CRC8 :	--
Data 0 :	01h Get Program Description (Upper)
Data 1 :	22h Get Program Description (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Service ID(Upper)
Data 5 :	Service ID (Lower)
Data 6 :	Event ID(Upper)
Data 7 :	Event ID (Lower)
Data 8 :	Program Description Length
Data 9 ~:	Program Description
Data CRC8 :	--

(3) Description

- Get the program description.
- Service ID identify the service in Broadcaster (0x0000~0xFFFF).Event ID identify the program (0x0000~0xFFFF).
- If Service ID and Event ID are 0xFFFF then Get Current Data.
  (Only [Tuning Status = Tuning])
- Program Description Length is the length of Program Description (0~192Byte).
- Program Description is 8bit character code set Format.

(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.

- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
02h(Invalid Data)	Event ID is invalid.
03h(Invalid Length)	Data Length is invalid.
04h(No Service)	Service ID is invalid
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.4.10. Get DTV System

(1) Get DTV System Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h
Header CRC8 :	--
Data 0 :	01h Get DTV System (Upper)
Data 1 :	23h Get DTV System (Lower)
Data CRC8 :	--

(2) Get DTV System Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	06h
Header CRC8 :	--
Data 0 :	01h Get DTV System (Upper)
Data 1 :	23h Get DTV System (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	DTV System 00h : Dynamic 4Diver System (Fore Br0/1) 01h : Fixed 2x2Diver System (Fore Br0/1) 02h : Fixed 2Diver System(Antenna Br2/3) 03h : Fixed 4Diver System 04h : Fixed 2Diver System(Antenna Br0/1) 05h : Fixed 3x1Diver System (Fore Br0/1/2) 06h : Dynamic 4Diver System (Fore Br2/3) 07h : Fixed 2x2Diver System (Fore Br2/3) 08h : Fixed 3x1Diver System (Fore Br1/2/3)
Data CRC8 :	--

(3) Description

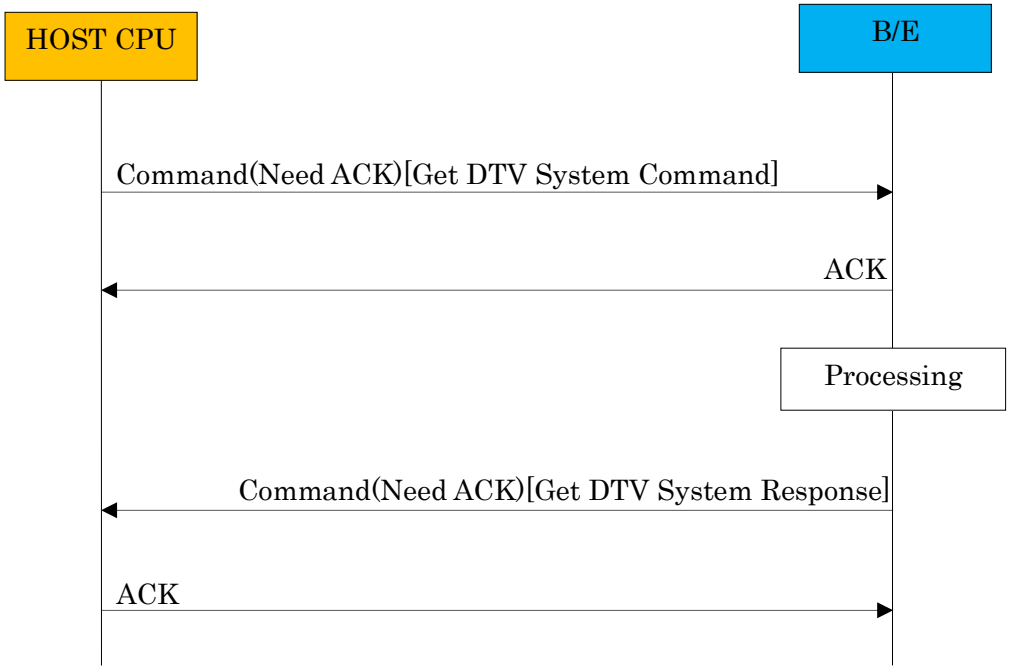
- Get DTV System Parameter.
- F039902182 : response Data 4 : Always return 04h.

(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

(5) Sequence



5.4.11. Get Program List

(1) Get Program List Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	08h
Header CRC8 :	--
Data 0 :	01h Get Program List (Upper)
Data 1 :	24h Get Program List (Lower)
Data 2 :	Year
Data 3 :	Month
Data 4 :	Day
Data 5 :	Service ID(Upper)
Data 6 :	Service ID (Lower)
Data CRC8 :	--

(2) Get Program List Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h~ FFh
Header CRC8 :	--
Data 0 :	01h Get Program List (Upper)
Data 1 :	24h Get Program List (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Fragment Number
Data 5 :	Last Fragment Number
Data 6 :	Year
Data 7 :	Month
Data 8 :	Day
Data 9 :	Service ID(Upper)
Data 10 :	Service ID (Lower)
Data 11 :	Number of Event
Data 12 :	Event ID(0) Information (Event ID (Upper))
Data 13 :	Event ID(0) Information (Event ID (Lower))
Data 14 :	Event ID(0) Service ID(Upper)
Data 15 :	Event ID(0) Service ID (Lower)
Data 16 :	Event ID(0) Information (Hour)
Data 17 :	Event ID(0) Information (Minute)
Data 18 :	Event ID(1) Information (Event ID (Upper))
Data 19:	Event ID(1) Information (Event ID (Lower))
Data 20 :	Event ID(1) Service ID(Upper)
Data 21:	Event ID(1) Service ID (Lower)
Data 22 :	Event ID(1) Information (Hour)
Data 23:	Event ID(1) Information (Minute) ~
Data N*6+12 :	Event ID(N) Information (Event ID (Upper))
Data N*6+13 :	Event ID(N) Information (Event ID (Lower))
Data N*6+14 :	Event ID(N) Service ID(Upper)
Data N*6+15 :	Event ID(N) Service ID (Lower)

Data N*6+16 :	Event ID(N) Information (Hour)
Data N*6+17 :	Event ID(N) Information (Minute)
Data CRC8 :	--

(3) Get Program List Response (BE ⇒ HOST with Header Type 2)

Start Sync :	A5h,59h
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length(2 Bytes) :	Result=OK : 000Dh~ 0175h Result=NG : 0005h
Header CRC8 :	--
Data 0 :	01h Get Program List (Upper)
Data 1 :	24h Get Program List (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Fragment Number (00h : Fixed Value)
Data 5 :	Last Fragment Number (00h : Fixed Value)
Data 6 :	Year
Data 7 :	Month
Data 8 :	Day
Data 9 :	Service ID(Upper)
Data 10 :	Service ID (Lower)
Data 11 :	Number of Event
Data 12 :	Event ID(0) Information (Event ID (Upper))
Data 13 :	Event ID(0) Information (Event ID (Lower))
Data 14 :	Event ID(0) Service ID(Upper)
Data 15 :	Event ID(0) Service ID (Lower)
Data 16 :	Event ID(0) Information (Hour)
Data 17 :	Event ID(0) Information (Minute)
Data 18 :	Event ID(1) Information (Event ID (Upper))
Data 19:	Event ID(1) Information (Event ID (Lower))
Data 20 :	Event ID(1) Service ID(Upper)
Data 21:	Event ID(1) Service ID (Lower)
Data 22 :	Event ID(1) Information (Hour)
Data 23:	Event ID(1) Information (Minute) ~
Data N*6+12 :	Event ID(N) Information (Event ID (Upper))
Data N*6+13 :	Event ID(N) Information (Event ID (Lower))
Data N*6+14 :	Event ID(N) Service ID(Upper)
Data N*6+15 :	Event ID(N) Service ID (Lower)
Data N*6+16 :	Event ID(N) Information (Hour)
Data N*6+17 :	Event ID(N) Information (Minute)
Data CRC8 :	--

(4) Description

- Get the program list.
- Header Type 1:
  - "Fragment Number" is the order of the fragmented response. If the length of Data Part is 255byte and under, "Fragment Number" is 0. If the length of Data Part is over 255byte, it is the order number. The B/E sends the data in actual order.

- "Last Fragment Number" is the max number of "Fragment Number" (0~2).

- For Header Type 2, Fragment Numbers - Data 4, Data 5 are set to 00h.
- "Year" means the year of Start Time for all events.(Only low 2-digits ex.2017 -> 17)
- "Month" means the month of Start Time for all events.(Ex. December -> 12)
- "Day" means the date of Start Time for all events.(Ex. 13th -> 13)
- "Service ID" identify the service in Broadcaster (0x0000~0xFFFF).
- "Number of Event" means the number of events.(0~96)
- "Event ID" identify the program (0x0000~0xFFFF).
- "Hour" means the hour of Start Time.(0~23)
- "Minute" means the hour of Start Time.(0~59)
- B/E doesn't save EPG Information to NVRAM. If B/E power off, B/E doesn't have EPG Information.

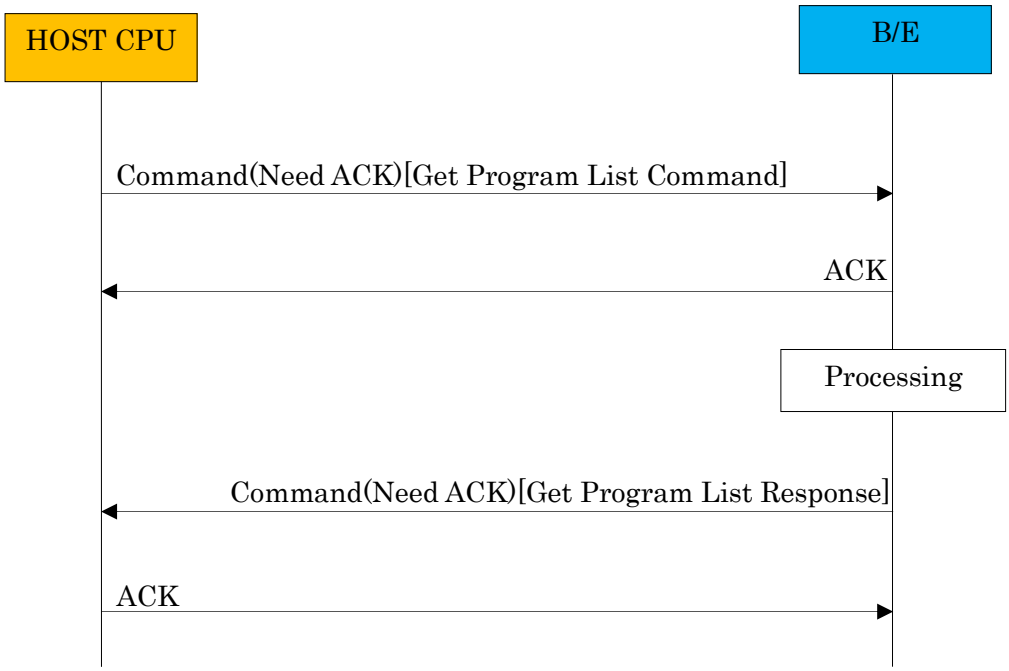
(5) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
03h(Invalid Length)	Data Length is invalid.
04h(No Service)	Service ID is invalid
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(6) Sequence


HOST CPU
B/E
Command(Need ACK)[Get Program List Command]
ACK
Processing
Command(Need ACK)[Get Program List Response]
ACK

5.4.12. Get Program Name

(1) Get Program Name Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	07h
Header CRC8 :	--
Data 0 :	01h Get Program Name (Upper)
Data 1 :	25h Get Program Name (Lower)
Data 2 :	Service ID(Upper)
Data 3 :	Service ID (Lower)
Data 4 :	Event ID(Upper)
Data 5 :	Event ID (Lower)
Data CRC8 :	--

(2) Get Program Name Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h~ 74h
Header CRC8 :	--
Data 0 :	01h Get Program Name (Upper)
Data 1 :	25h Get Program Name (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Service ID(Upper)
Data 5 :	Service ID (Lower)
Data 6 :	Event ID(Upper)
Data 7 :	Event ID (Lower)
Data 8 :	Year of Start Time
Data 9 :	Month of Start Time
Data 10 :	Day of Start Time
Data 11 :	Hour of Start Time
Data 12 :	Minute of Start Time
Data 13 :	Year of End Time
Data 14 :	Month of End Time
Data 15 :	Day of End Time
Data 16 :	Hour of End Time
Data 17 :	Minute of End Time
Data 18 :	Program Name Length
Data 19 ~:	Program Name
Data CRC8 :	--

(3) Description

- Get the program Name
- "Service ID" identify the service in Broadcaster (0x0000~0xFFFF).
- "Event ID" identify the program (0x0000~0xFFFF).
- If Service ID and Event ID are 0xFFFF then Get Current Data.
  (Only [Tuning Status = Tuning])
- "Year of Start Time" means the year of Start Time.(Only low 2-digit ex.2017->17)
- "Month of Start Time" means the month of Start Time.(Ex. December -> 12)
- "Day of Start Time" means the date of Start Time.(Ex. 13th -> 13)
- "Hour of Start Time" means the hour of Start Time.(0~23)
- "Minute of Start Time" means the hour of Start Time.(0~59)
- "Year of End Time" means the year of End Time.(Only low 2-digit ex.2017->17)
- "Month of End Time" means the month of End Time.(Ex. December -> 12)
- "Day of End Time" means the date of End Time.(Ex. 13th -> 13)
- "Hour of End Time" means the hour of End Time.(0~23)
- "Minute of End Time" means the hour of End Time.(0~59)
- If End Time is not yet fixed, "Year"/"Month"/"Day"/"Hour"/"Minute" are 0xFF.
- "Program Name Length" is the length of Program Name (0~96Byte).
- "Program Name" is 8bit character code set Format.
- B/E doesn't save EPG Information to NVRAM. If B/E power off, B/E doesn't have EPG Information.

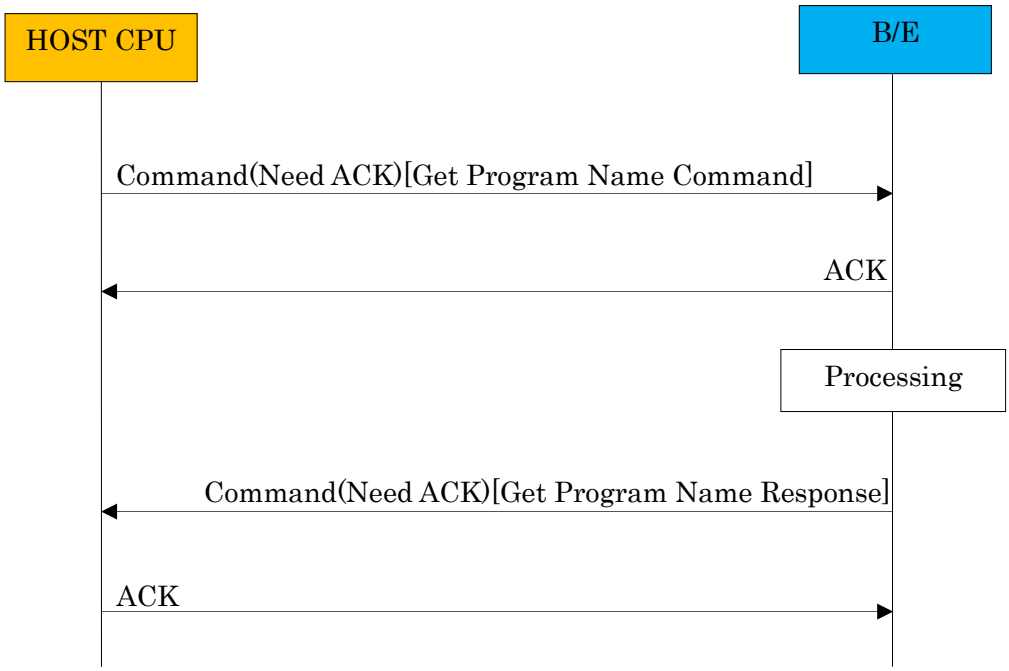
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
02h(Invalid Data)	Event ID is invalid.
03h(Invalid Length)	Data Length is invalid.
04h(No Service)	Service ID is invalid
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence


HOST CPU
B/E
Command(Need ACK)[Get Program Name Command]
ACK
Processing
Command(Need ACK)[Get Program Name Response]
ACK

5.4.13. Get Logo Data

(1) Get Logo Data Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	06h
Header CRC8 :	--
Data 0 :	01h Get Logo Data (Upper)
Data 1 :	26h Get Logo Data (Lower)
Data 2 :	Service ID(Upper)
Data 3 :	Service ID (Lower)
Data 4 :	Logo Type
Data CRC8 :	--

(2) Get Logo Data Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : 08h~ Result=NG : 05h
Header CRC8 :	--
Data 0 :	01h Get Logo Data (Upper)
Data 1 :	26h Get Logo Data (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Fragment Number
Data 5 :	Last Fragment Number
Data 6 :	Service ID(Upper)
Data 7 :	Service ID (Lower)
Data 8 ~:	Logo Information Refer to "(4)Description"
Data CRC8 :	--

(3) Get Logo Data Response (BE ⇒ HOST with Header Type 2)

Start Sync :	A5h,59h
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length(2 Bytes) :	Result=OK : 0008h~0734h Result=NG : 0005h
Header CRC8 :	--
Data 0 :	01h Get Logo Data (Upper)
Data 1 :	26h Get Logo Data (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Fragment Number (00h : Fixed Value)
Data 5 :	Last Fragment Number (00h : Fixed Value)
Data 6 :	Service ID(Upper)
Data 7 :	Service ID (Lower)
Data 8 ~:	Logo Information Refer to "(4)Description"
Data CRC8 :	--

(4) Description

- Get Logo Data.
- "Logo Type" is :

Logo Type	Kind	Height [dot]	Width [dot]	Max Size [byte]	Format
0	SD 4:3 Small	24	48	1400	PNG
1	SD 16:9 Small	24	36	1184	PNG
2	HD Small	27	48	1508	PNG
3	SD 4:3 Large	36	72	1832	PNG
4	SD 16:9 Large	36	54	1508	PNG
5	HD Large	36	64	1688	PNG
6	Simple Logo	-	-	10byte (5characters)	8bit character code set
7	Simple Logo	-	-	20byte (5characters)	UTF-8

- Only the Logo Types set in DTV Parameter 2 will be available.
- This will be used for all screens where a logo is displayed.
- Result (Data 3) will be NG if host request Logo Type not set in DTV Parameter 2.
- For Header Type 1
  - "Fragment Number" is the order of the fragmented response. If the length of Data Part is 255byte and under, "Fragment Number" is 0. If the length of Data Part is over 255byte, it is the order number.
  - "Last Fragment Number" is the max number of "Fragment Number" (0~6).
- For Header Type 2, Fragment Numbers - Data 4, Data 5 are set to 00h.

- "Logo Information Data" shows the following table.

Logo Type	1byte
Length of Logo Data	2byte
Logo Data	n byte

- If B/E doesn't receive Logo Data yet, B/E response that "Length of Logo Data" is zero and "Logo Data" is empty.
- Logo PNG Data will be complete including the PLTE Chunk data(536 Byte) .
- Simple Logo examples

Broadcast	Simple Logo
NHK 総合大阪	ＮＨＫ－Ｇ
NHK 教育大阪	ＮＨＫ－Ｅ
毎日放送	ＭＢＳ
朝日放送	ＡＢＣ
テレビ大阪	ＴＶＯ
関西テレビ	ＫＴＶ
読売テレビ	ＹＴＶ

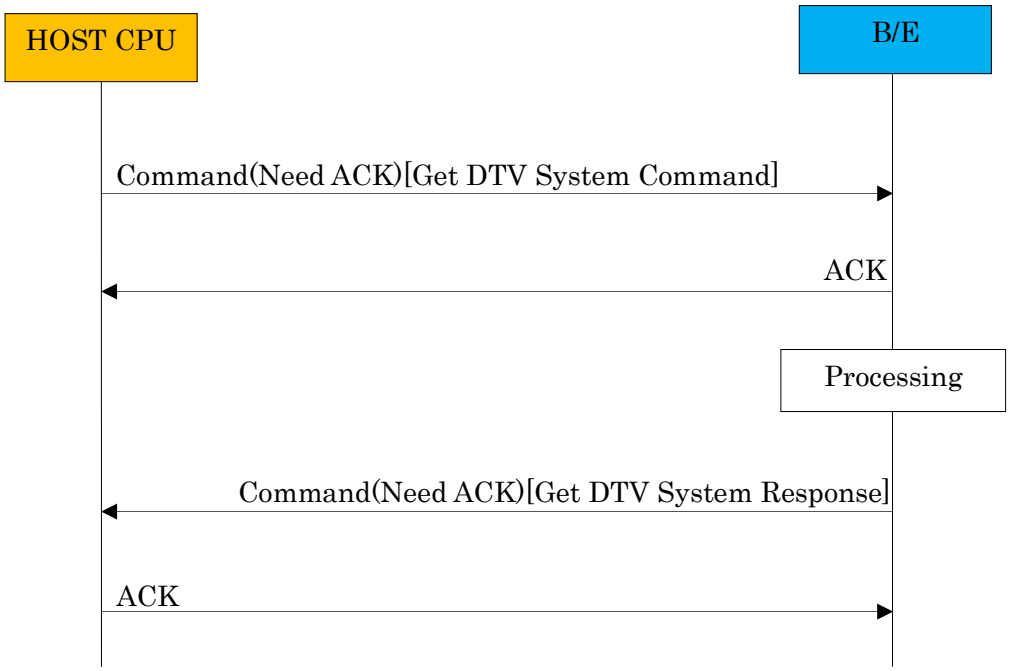
(5) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid. - Logo Type is not 00h ~ 06h.
03h(Invalid Length)	Data Length is invalid.
04h(No Service)	Service ID is invalid
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(6) Sequence


HOST CPU
B/E
Command(Need ACK)[Get DTV System Command]
ACK
Processing
Command(Need ACK)[Get DTV System Response]
ACK

5.4.14. Get RMP Maker ID

(1) Get RMP Maker ID Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Get RMP Maker ID (Upper)
Data 1 :	29h Get RMP Maker ID (Lower)
Data 2 :	Preset Number 00h : Preset No.1(Home) 01h : Preset No.2(Drive) FFh : Original RMP Maker ID
Data 3 :	One touch Number 00h : Not stored service 01h~0Ch : One touch Number FFh : Original RMP Maker ID
Data CRC8 :	--

(2) Get RMP Maker ID Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	13h
Header CRC8 :	--
Data 0 :	01h Get RMP Maker ID (Upper)
Data 1 :	29h Get RMP Maker ID (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Display0 (Upper)
Data 5 :	Display0 (Lower)
Data 6 :	Display1 (Upper)
Data 7 :	Display1 (Lower)
Data 8 :	Display2 (Upper)
Data 9 :	Display2 (Lower)
Data 10 :	Display3 (Upper)
Data 11 :	Display3 (Lower)
Data 12 :	Display4 (Upper)
Data 13 :	Display4 (Lower)
Data 14 :	Display5 (Upper)
Data 15 :	Display5 (Lower)
Data 16 :	Display6 (Upper)
Data 17 :	Display6 (Lower)
Data CRC8 :	--

(3) Description

- Get RMP Maker ID

  If "Preset No." = FFh and "One touch No." = FFh, Return Original RMP Maker ID.

- How to Display

  If Response values are below.

  Display0 = 1

  Display1 = 2345

  Display2 = 6789

  Display3 = 0123

  Display4 = 4567

  Display5 = 8901

  Display6 = 2345

  「1 – 2345 – 6789 – 0123 – 4567 – 8901 – 2345」

  If Response values are All 00h: It indicates that it has not been acquired yet.

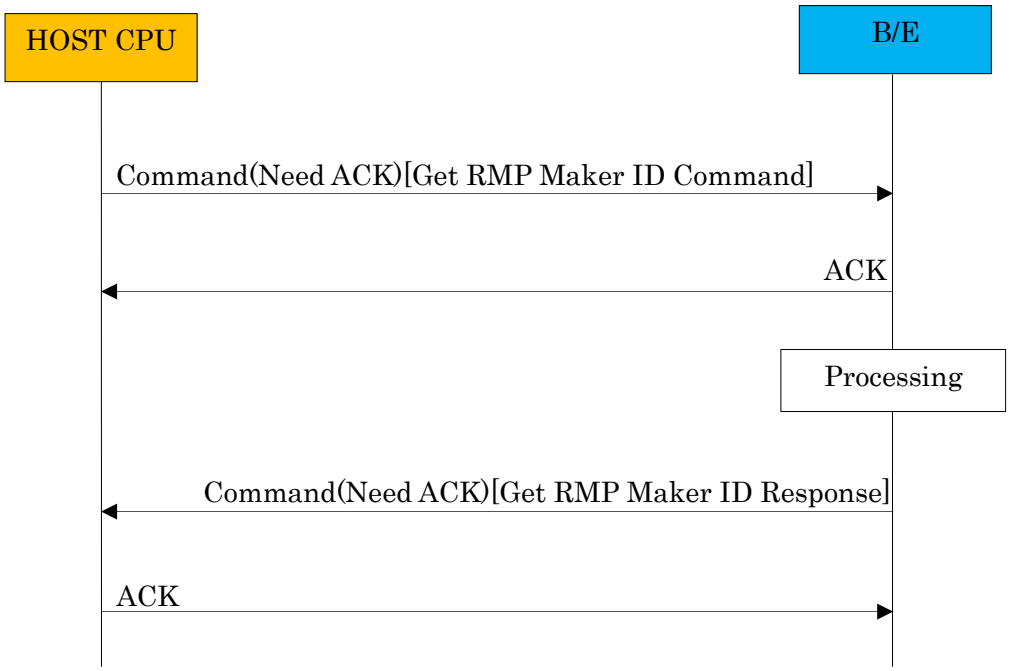
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
04h(No Service)	One touch Number is invalid (Not registered channel)
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence


HOST CPU
B/E
Command(Need ACK)[Get RMP Maker ID Command]
ACK
Processing
Command(Need ACK)[Get RMP Maker ID Response]
ACK

5.4.15. Get Number of Program info

(1) Get Number of Program info Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	04h
Header CRC8 :	--
Data 0 :	01h Get Number of Program info (Upper)
Data 1 :	2Ah Get Number of Program info (Lower)
Data 2 :	00h : Home Preset 01h : Drive Preset FEh: Last Channel FFh : Last Preset
Data CRC8 :	--

(2) Get Number of Program info Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : 1Fh or 07h Result=NG : 05h
Header CRC8 :	--
Data 0 :	01h Get Number of Program info (Upper)
Data 1 :	2Ah Get Number of Program info (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Preset no.0(Not Stored Channel) (Fullseg) : Number of Program info 00h : Not Active 01h – 08h : Number of Program info FFh : unknown Data2=FEh Last channel Last Channel Fullseg : Number of Program info 00h : Not Active 01h – 08h : Number of Program info FFh : unknown
Data 5 :	Preset no.1(Fullseg) : Number of Program info 00h : Not Active (Not Stored) 01h – 08h : Number of Program info FFh : unknown Data2=FEh Last channel Last Channel Oneseg : Number of Program info 00h : Not Active 01h – 08h : Number of Program info FFh : unknown
Data 6 :	Preset no.2(Fullseg) : Number of Program info
Data 7 :	Preset no.3(Fullseg) : Number of Program info
Data 8 :	Preset no.4(Fullseg) : Number of Program info
Data 9 :	Preset no.5(Fullseg) : Number of Program info

Data 10 :	Preset no.6(Fullseg) : Number of Program info
Data 11 :	Preset no.7(Fullseg) : Number of Program info
Data 12 :	Preset no.8(Fullseg) : Number of Program info
Data 13 :	Preset no.9(Fullseg) : Number of Program info
Data 14 :	Preset no.10(Fullseg) : Number of Program info
Data 15 :	Preset no.11(Fullseg) : Number of Program info
Data 16 :	Preset no.12(Fullseg) : Number of Program info
Data 17 :	Preset no.0(Not Stored Channel) (Oneseg) : Number of Program info 00h : Not Active 01h – 08h : Number of Program info FFh : unknown
Data 18 :	Preset no.1(Oneseg) : Number of Program info 00h : Not Active (Not Stored) 01h – 08h : Number of Program info FFh : unknown
Data 19 :	Preset no.2(Oneseg) : Number of Program info
Data 20 :	Preset no.3(Oneseg) : Number of Program info
Data 21 :	Preset no.4(Oneseg) : Number of Program info
Data 22 :	Preset no.5(Oneseg) : Number of Program info
Data 23 :	Preset no.6(Oneseg) : Number of Program info
Data 24 :	Preset no.7(Oneseg) : Number of Program info
Data 25 :	Preset no.8(Oneseg) : Number of Program info
Data 26 :	Preset no.9(Oneseg) : Number of Program info
Data 27 :	Preset no.10(Oneseg) : Number of Program info
Data 28 :	Preset no.11(Oneseg) : Number of Program info
Data 29 :	Preset no.12(Oneseg) : Number of Program info
Data CRC8 :	--

(3) Description

- Get current time Number of programs per Channel.
- Number of Program info = FFh : Haven't got current program info yet

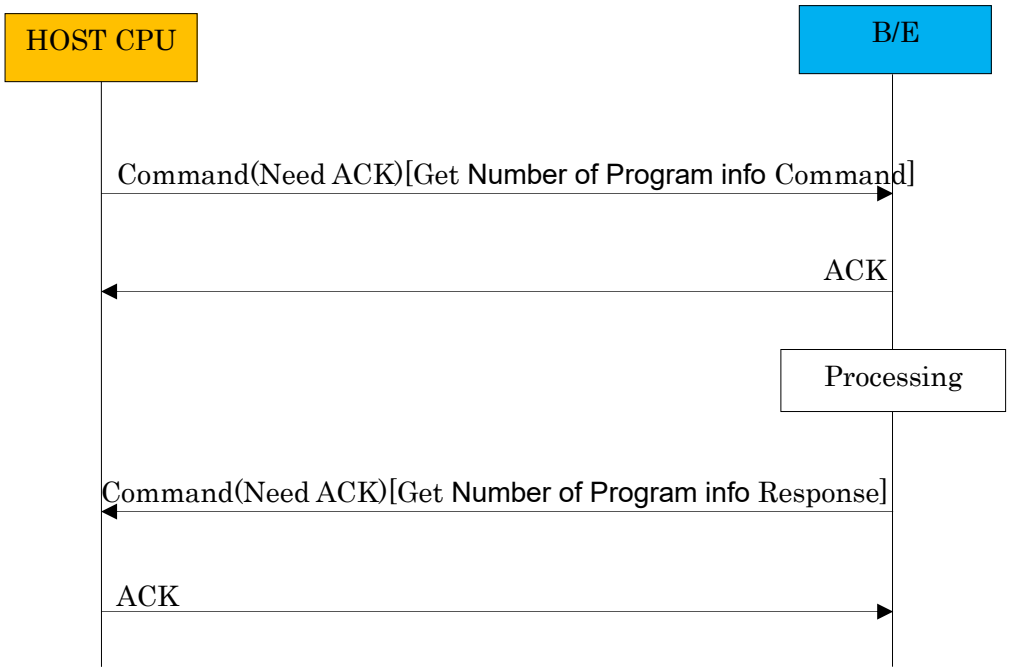
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence


HOST CPU
B/E
Command(Need ACK)[Get Number of Program info Command]
ACK
Processing
Command(Need ACK)[Get Number of Program info Response]
ACK

5.4.16. Get Audio ES info

(1) Get Audio ES info Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h
Header CRC8 :	--
Data 0 :	01h Get Audio ES info(Upper)
Data 1 :	2Bh Get Audio ES info(Lower)
Data CRC8 :	--

(2) Get Audio ES info Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : 07h ~ FFh Result=NG : 05h
Header CRC8 :	--
Data 0 :	01h Get Audio ES info(Upper)
Data 1 :	2Bh Get Audio ES info(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Fragment Number
Data 5 :	Last Fragment Number
Data 6 ~	Audio ES info Data
Data CRC8 :	--

(3) Get Audio ES info Response (BE ⇒ HOST with Header Type 2)

Start Sync :	A5h,59h
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length(2Bytes) :	Result=OK : 0007h ~ 0242h Result=NG : 0005h
Header CRC8 :	--
Data 0 :	01h Get Audio ES info(Upper)
Data 1 :	2Bh Get Audio ES info(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Fragment Number (00h : Fixed Value)
Data 5 :	Last Fragment Number (00h : Fixed Value)
Data 6 ~	Audio ES info Data
Data CRC8 :	--

(4) Description

- Get Audio ES information.
- For Header Type 1
  - "Fragment Number" is the order of the fragmented response. If the length of Data Part is 255byte and under, "Fragment Number" is 0. If the length of Data Part is over 255byte, it is the order number.
  - "Last Fragment Number" is the max number of "Fragment Number" (0~2).
- For Header Type 2, Fragment Numbers - Data 4, Data 5 are set to 00h.
- "Audio ES Info Data" shows the following table.

Number_of_AudioES_Information(N1) for(i = 0; i < N1 ; i++){	1byte
CompornetTag	1byte
Number_of_Language(N2)	1byte
for(j = 0; j < N2; j++){	1byte
language code	1byte
language text length (N3)	1byte
language_text[N3]}	N3 byte
}	

- N1 max = 15 / N2 max = 2 / N3 max = 16
- Language code :

Value	Language
0	Japanese
1	English
2	German
3	French
4	Italian
5	Russian
6	Chinese
7	Korean
8	Spanish
9	Etc Foreign language : Language than the above Unknown : Cannot determine one language because multiple language is mixed

- language Text : Maximum 16 bytes for 8bit character code set.

(5) Error Handling

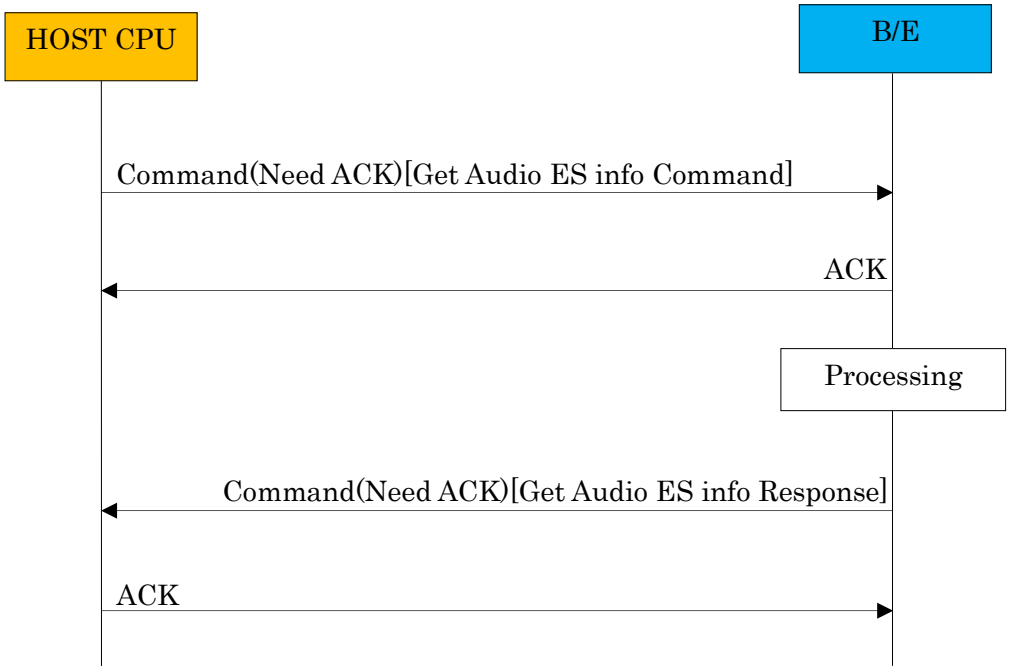
- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.

・ Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
05h(Not Support)	The parameter is not supported.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(6) Sequence



5.4.17. Get Video Resolution

(1) Get Video Resolution Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h
Header CRC8 :	--
Data 0 :	01h Get Video Resolution(Upper)
Data 1 :	2Eh Get Video Resolution(Lower)
Data CRC8 :	--

(2) Get Video Resolution Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : 06h Result=NG : 05h
Header CRC8 :	--
Data 0 :	01h Get Video Resolution(Upper)
Data 1 :	2Eh Get Video Resolution(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Resolution 00h : 720p 01h : 480p 02h : 1080p 03h : 480i 04h : 1080p@30Hz
Data CRC8 :	--

(3) Description

- Get Video Output Resolution

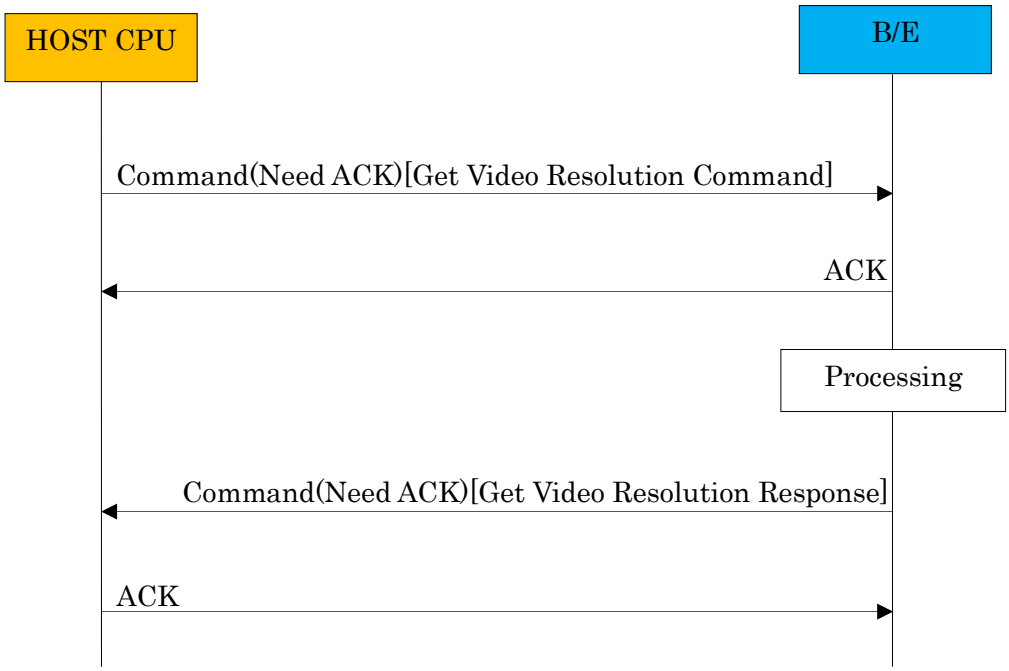
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence


HOST CPU
B/E
Command(Need ACK)[Get Video Resolution Command]
ACK
Processing
Command(Need ACK)[Get Video Resolution Response]
ACK

5.4.18. Get Banner Config

(1) Get Banner Config Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h
Header CRC8 :	--
Data 0 :	01h Get Banner Config(Upper)
Data 1 :	30h Get Banner Config(Lower)
Data CRC8 :	--

(2) Get Banner Config Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : 12h Result=NG : 05h
Header CRC8 :	--
Data 0 :	01h Get Banner Config(Upper)
Data 1 :	30h Get Banner Config(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Banner Position X (Upper)
Data 5 :	Banner Position X (Lower)
Data 6 :	Banner Position Y (Upper)
Data 7 :	Banner Position Y (Lower)
Data 8 :	Font Size [px]
Data 9 :	Transparency [%]
Data 10 :	Logo Position Offset [px]
Data 11 :	Margin of Top and Bottom [px]
Data 12 :	Margin of Left and Right [px]
Data 13 :	Line Spacing [px]
Data 14 :	Max number of characters
Data 15 :	Banner segment icon enable 00h /disable 01h
Data 16 :	Automatic banner display disables enable 00h /disable 01h
Data CRC8 :	--

(3) Description

- Get Banner Configuration

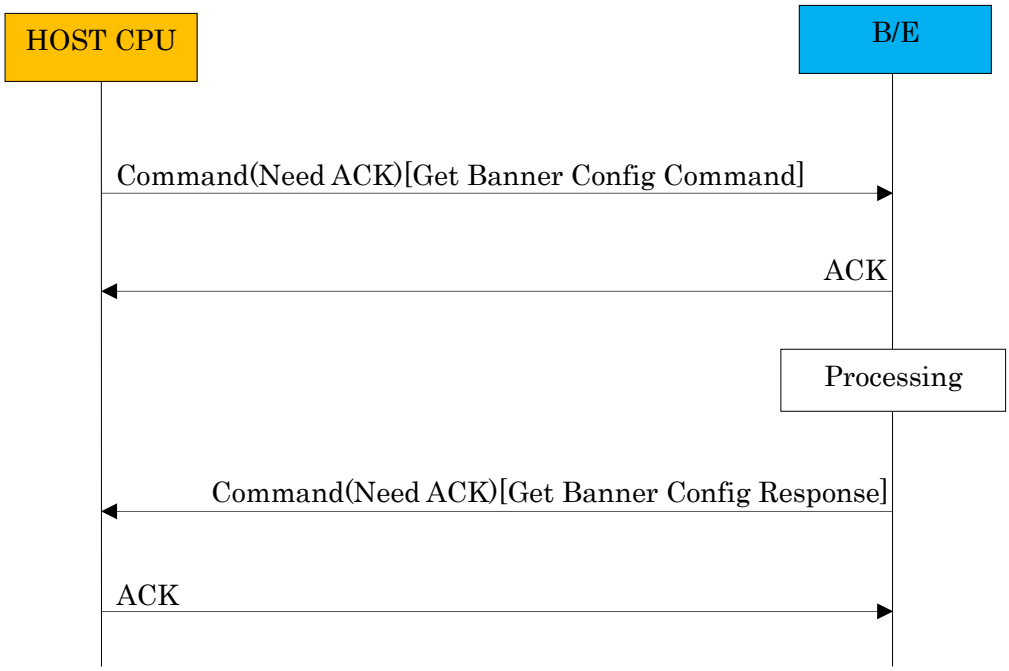
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.4.19. Get Character Set

(1) Get Character Set Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h
Header CRC8 :	--
Data 0 :	01h Get Character Set(Upper)
Data 1 :	32h Get Character Set(Lower)
Data CRC8 :	--

(2) Get Character Set Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : 06h Result=NG : 05h
Header CRC8 :	--
Data 0 :	01h Get Character Set(Upper)
Data 1 :	32h Get Character Set(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Character Set 00h : ARIB 01h : UTF-8
Data CRC8 :	--

(3) Description

- Get Character Set

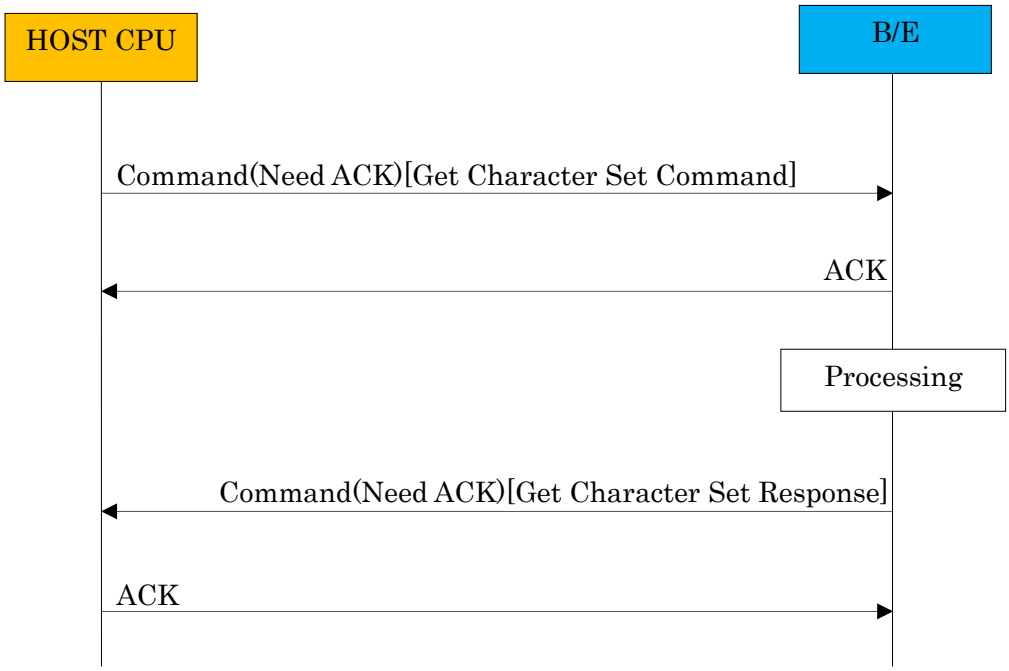
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.4.20. Get TS Name with UTF-8

(1) Get TS Name with UTF-8 Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Get TS Name with UTF-8 (Upper)
Data 1 :	33h Get TS Name with UTF-8(Lower)
Data 2 :	01h~0Ch : Preset Channel Number FFh : All Preset Channel Number 00h : Not Stored Channel.
Data 3 :	00h : Home Preset 01h : Drive Preset FFh : Last Preset
Data CRC8 :	--

(2) Get TS Name with UTF-8 Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : 58h ~ Result=NG : 05h
Header CRC8 :	--
Data 0 :	01h Get TS Name with UTF-8 (Upper)
Data 1 :	33h Get TS Name with UTF-8(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Fragment Number
Data 5 :	Last Fragment Number
Data 6 :	01h~0Ch : Preset Channel Number FFh : All Preset Channel Number 00h : Not Stored Channel.
Data 7~86 :	TS Name (UTF-8) Data 6 is 00h~0Ch:Specified Preset Channel Number Data 6 is FFh : Preset Channel Number = 1
Data 87~ :	TS Name (UTF-8) 80bytesper Preset Channel Number
Data CRC8 :	--

(3) Get TS Name with UTF-8 Response (BE ⇒ HOST with Header Type 2)

Start Sync :	A5h,59h
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length(2 Bytes) :	Result=OK : 0056h ~ 03C8h Result=NG : 0005h
Header CRC8 :	--
Data 0 :	01h Get TS Name with UTF-8 (Upper)
Data 1 :	33h Get TS Name with UTF-8(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Fragment Number (00h : Fixed value)
Data 5 :	Last Fragment Number (00h : Fixed value)
Data 6 :	01h~0Ch : Preset Channel Number FFh : All Preset Channel Number 00h : Not Stored Channel.
Data 7~86 :	TS Name (UTF-8) Data 6 is 00h~0Ch:Specified Preset Channel Number Data 6 is FFh : Preset Channel Number = 1
Data 87~ :	TS Name (UTF-8) 80bytesper Preset Channel Number
Data CRC8 :	--

(4) Description

- Get TS Name with UTF-8.
- If Preset Channel Number is 01h~0Ch, get TS Name of specified Preset Channel and specified Preset Table.
- If Preset Channel Number is FFh, get All TS Name in specified Preset Table.
- If Preset Channel Number is 00h, get TS Name of Not Stored Channel.
- For Header Type 2, Fragment Numbers - Data4, Data5 are set to 00h.

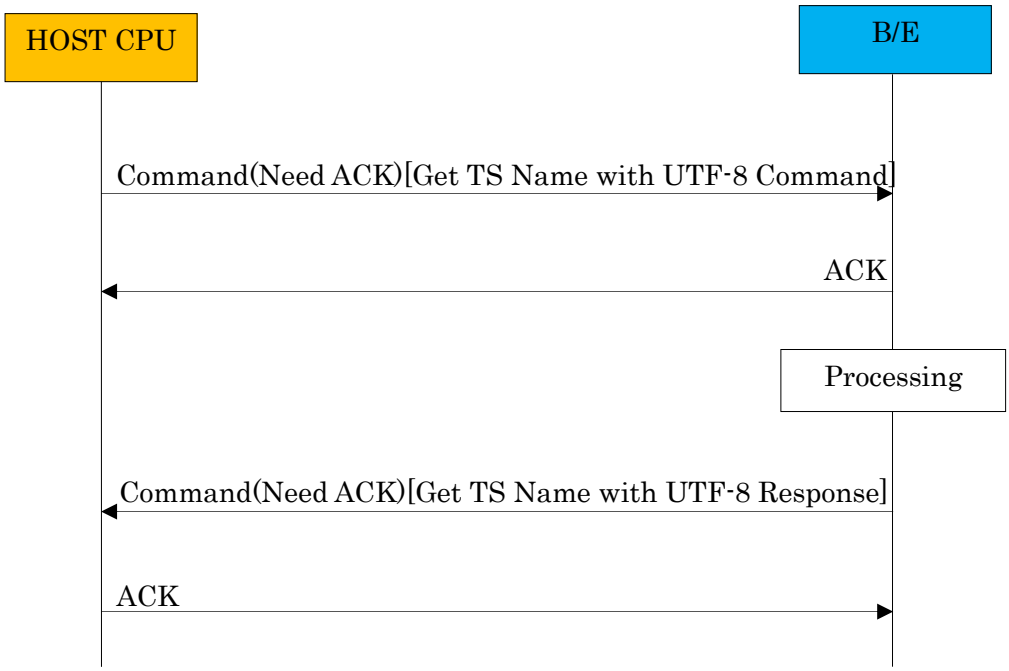
(5) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(6) Sequence


HOST CPU
B/E
Command(Need ACK)[Get TS Name with UTF-8 Command]
ACK
Processing
Command(Need ACK)[Get TS Name with UTF-8 Response]
ACK

5.4.21. Get Program Information with UTF-8

(1) Get Program Information with UTF-8 Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h
Header CRC8 :	--
Data 0 :	01h Get Program Information with UTF-8 (Upper)
Data 1 :	34h Get Program Information with UTF-8 (Lower)
Data CRC8 :	--

(2) Get Program Information Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result = OK: 08h~FFh Result = NG: 05h
Header CRC8 :	--
Data 0 :	01h Get Program Information with UTF-8 (Upper)
Data 1 :	34h Get Program Information with UTF-8 (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Fragment Number
Data 5 :	Last Fragment Number
Data 6 :	Preset Number 00h : Preset No.1(Home) 01h : Preset No.2(Drive) FFh：None
Data 7 :	00h : Not stored service 01h~0Ch : Preset Channel Number FFh：None
Data 8 :	Three Digit Number(Upper)
Data 9 :	Three Digit Number(Lower)
Data 10 :	Branch Identifier Number
Data 11 :	Length of Station Name(N1) : 00h~50h
Data 12~(11+N1) :	Station Name(TS Name with UTF-8)
Data 12+N1 :	Length of Program Name(N2)(Upper) : 00h~180h
Data 13+N1	Length of Program Name(N2)(Lower) : 00h~180h
Data 14+N1~13+N1+N2:	Program Name with UTF-8
Data CRC8 :	--

(3) Get Program Information Response (BE ⇒ HOST with Header Type 2)

Start Sync :	A5h,59h
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length(2 Bytes) :	Result = OK: ~01D4h Result = NG: 0005h
Header CRC8 :	--
Data 0 :	01h Get Program Information with UTF-8 (Upper)
Data 1 :	34h Get Program Information with UTF-8 (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Fragment Number (00h : Fixed Value)
Data 5 :	Last Fragment Number (00h : Fixed Value)
Data 6 :	Preset Number 00h : Preset No.1(Home) 01h : Preset No.2(Drive) FFh : None
Data 7 :	00h : Not stored service 01h~0Ch : Preset Channel Number FFh : None
Data 8 :	Three Digit Number(Upper)
Data 9 :	Three Digit Number(Lower)
Data 10 :	Branch Identifier Number
Data 11 :	Length of Station Name(N1) : 00h~50h
Data 12~(11+N1) :	Station Name(TS Name with UTF-8)
Data 12+N1 :	Length of Program Name(N2)(Upper) : 00h~180h
Data 13+N1	Length of Program Name(N2)(Lower) : 00h~180h
Data 14+N1~13+N1+N2:	Program Name with UTF-8
Data CRC8 :	--

(4) Description

- Get the program information in current service.
- The program information contains Preset Channel Number Station Name / Program name).
- Preset Channel Number (Data7) of Response is current playback channel. If current playback channel is not stored, Data7 is 00h and Data6 is FFh.
- Three Digit Number & Branch Identifier Number of Response are current playback service. If current playback channel is not stored, these data are set.
- If B/E doesn't have Program name, Length of Program Name is set to 0.
- For Header Type 2, Fragment Numbers - Data4, Data5 are set to 00h.

(5) Error Handling

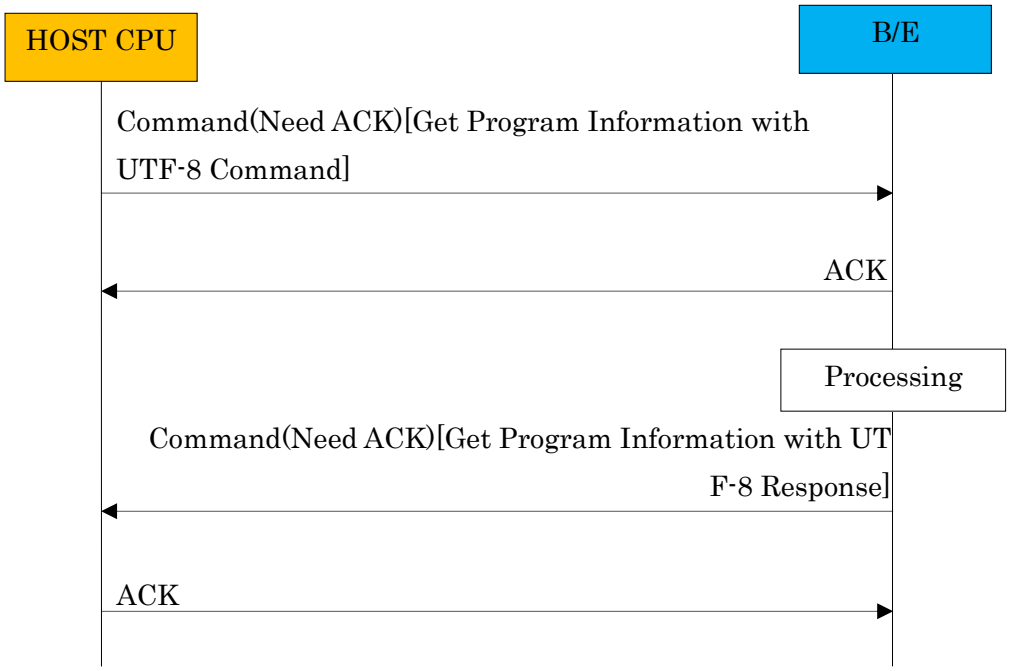
- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.

- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(6) Sequence



5.4.22. Get DTV Parameter 2

(1) Get DTV Parameter 2 Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h
Header CRC8 :	--
Data 0 :	01h Get DTV Parameter 2 (Upper)
Data 1 :	36h Get DTV Parameter 2 (Lower)
Data CRC8 :	--

(2) Get DTV Parameter 2 Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : E5h Result=NG : 05h
Header CRC8 :	--
Data 0 :	01h Get DTV Parameter 2 (Upper)
Data 1 :	36h Get DTV Parameter 2 (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Enable / Disable 01h : Enable Parameter 00h : Disable Parameter (Use Default Parameter)
Data 5 :	Auto tune mode default 00h:Auto 01h:Fullseg 02h:Oneseg 03h:Auto with Seamless
Data 6:	Default Character Set 00h : ARIB 01h : UTF-8
Data 7 :	Default Video Resolution 00h :720p 01h : 480p 02h : 1080p 03h : 480i 04h : 1080p@30Hz
Data 8 :	Closed Caption display over BML browser 00h : Enable Display 01h : Disable Display
Data 9 :	BML Error Message 00h : Enable Display 01h : Disable Display
Data 10 :	Receive Logo Type Bit0 : Logo Type 0 Bit1 : Logo Type 1 Bit2 : Logo Type 2

	Bit3 : Logo Type 3 Bit4 : Logo Type 4 Bit5 : Logo Type 5 Bit6/Bit7 : Fix 0 0 : Receive 1 : Not Receive
Data 11 :	Search Detection Timeout
Data 12 ~ 13:	C/N for Receive Error Detection for Fix Fullseg mode
Data 14 ~ 15:	C/N for Preset Scan Detection for Fix Oneseg mode and Auto mode
Data 16 ~ 17:	C/N for Preset Scan Detection for Fix Fullseg mode
Data 18 :	Timeout for Preset Scan per channel
Data 19 ~ 20 :	C/N Level 3 threshold
Data 21 ~ 22 :	C/N Level 2 threshold
Data 23 ~ 24 :	C/N Level 1 threshold
Data 25 ~ 26 :	EPG Update scan time / channel 1 to 180[s]
Data 27 :	EPG Scan Number of Days
Data 28 ~ 29 :	EPG Scan Signal Threshold
Data 30 ~ 31 :	Audio Const offset for Fullseg
Data 32 ~ 33 :	Audio Const offset for Oneseg
Data 34 :	Event relay : Auto Event relay On/Off
Data 35 :	Event relay : Notification interval before switching [s]
Data 36 :	Event relay : Number of notifications before switching [times]
Data 37 ~ 38	Banner: Banner Position X
Data 39 ~ 40	Banner: Banner Position Y
Data 41 :	Banner: Font Size [px]
Data 42 :	Banner: Transparency [%]
Data 43 :	Banner: Logo Position Offset [px]
Data 44 :	Banner: Margin of Top and Bottom [px]
Data 45 :	Banner: Margin of Left and Right [px]
Data 46 :	Banner: Line spacing [px]
Data 47 :	Banner: Max number of characters
Data 48 :	Banner: Default Banner Timeout [s]
Data 49 :	EPG List Type Enable Bit0 : List Type 1 Enable Bit1 : List Type 2 Enable Bit2 : List Type 3 Enable Bit3 : List Type 4 Enable
Data 50 :	EPG List: Type1 Program name Font Size [px]
Data 51 :	EPG List: Type1 TS Name Font Size [px]
Data 52 :	EPG List: Type1 EPG Content Font Size [px]
Data 53 :	EPG List: Type2 Program name Font Size [px]
Data 54 :	EPG List: Type2 TS Name Font Size [px]
Data 55 :	EPG List: Type2 EPG Content Font Size [px]
Data 56 :	EPG List: Type3 Program name Font Size [px]
Data 57 :	EPG List: Type3 TS Name Font Size [px]
Data 58 :	EPG List: Type3 EPG Content Font Size [px]
Data 59 :	EPG List: Type4 Program name Font Size [px]
Data 60 :	EPG List: Type4 TS Name Font Size [px]
Data 61 :	EPG List: Type4 EPG Content Font Size [px]
Data 62 :	EPG List: Table Title Color Red
Data 63 :	EPG List: Table Title Color Green
Data 64 :	EPG List: Table Title Color Blue
Data 65 :	EPG List: Table Title Font Color Red
Data 66 :	EPG List: Table Title Font Color Green
Data 67 :	EPG List: Table Title Font Color Blue
Data 68 :	EPG List: Cursor Color Red

Data 69 :	EPG List: Cursor Color Green
Data 70 :	EPG List: Cursor Color Blue
Data 71 :	EPG List: Cursor Font Color Red
Data 72 :	EPG List: Cursor Font Color Green
Data 73 :	EPG List: Cursor Font Color Blue
Data 74 :	EPG List: Current Program Color Red
Data 75 :	EPG List: Current Program Color Green
Data 76 :	EPG List: Current Program Color Blue
Data 77 :	EPG List: Current Program Font Color Red
Data 78 :	EPG List: Current Program Font Color Green
Data 79 :	EPG List: Current Program Font Color Blue
Data 80 :	EPG List: Balloon color Red
Data 81 :	EPG List: Balloon color Green
Data 82 :	EPG List: Balloon color Blue
Data 83 :	EPG List: Balloon Font color Red
Data 84 :	EPG List: Balloon Font color Green
Data 85 :	EPG List: Balloon Font color Blue
Data 86 :	EPG List: Border line color Red
Data 87 :	EPG List: Border line color Green
Data 88 :	EPG List: Border line color Blue
Data 89 :	EPG Information: Font Size [px]
Data 90 :	EPG Information: Title Font Color Red
Data 91 :	EPG Information: Title Font Color Green
Data 92 :	EPG Information: Title Font Color Blue
Data 93 :	EPG Information: Line Spacing for Title
Data 94 :	EPG Information: Line Spacing for Contents
Data 95 :	EPG Information: Margin
Data 96 :	Default Auto Broadcast Search Mode
Data 97~98 :	C/N Level 3 threshold for one seg
Data 99~100 :	C/N Level 2 threshold for one seg
Data 101~102 : :	C/N Level 1 threshold for one seg
Data 103:	Default event mask 0
Data 104:	Default event mask 1
Data 105:	Default event mask 2
Data 106:	EPG List: Short program border line color Red
Data 107:	EPG List: Short program border line color Green
Data 108:	EPG List: Short program border line color Blue
Data 109:	Banner: Banner segment icon 00h:Enable 01h:Disable
Data 110:	Number of scrolling line on EPG detail (01h~
Data 111:	Automatic banner display enable enable 00h /disable 01h
Data 112~ 227:	Reserved
Data CRC8 :	--

※ Parameters and range: Refer to 5.3.16 Set DTV Parameter 2.

(3) Description

- Get DTV Parameter.
- Each parameter is big endian.

(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:

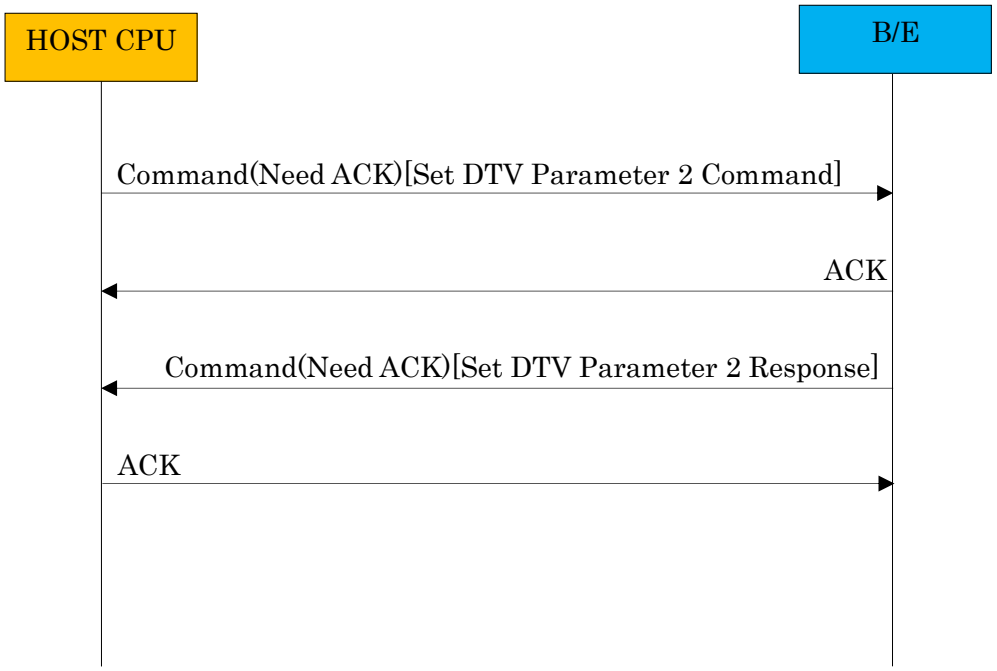
- ✧ Doesn't receive the response within 1sec after HOST transmit the command.
- ✧ The result of response is NG.

Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.4.23. Get Service Information with UTF-8

(1) Get Service Information with UTF-8 Command (HOST ⇒ BE)

Start Sync :	A5h,5ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	01h Get Service Information with UTF-8 (Upper)
Data 1 :	37h Get Service Information with UTF-8 (Lower)
Data 2 :	Preset Table 00h : Home Preset 01h : Drive Preset FFh : Last Preset
Data 3 :	One Touch Number 01h ~ 0Ch : Service Information of Specified Channel 00h : Service Information of Not Stored Channel. FFh : Service Information of All Channel.
Data CRC8 :	--

(2) Get Service Information with UTF-8 Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : 08h~ Result=NG : 05h
Header CRC8 :	--
Data 0 :	01h Get Service Information with UTF-8 (Upper)
Data 1 :	37h Get Service Information with UTF-8 (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Fragment Number
Data 5 :	Last Fragment Number
Data 6~ :	Service_Information_Data Refer to "(4) Description"
Data CRC8 :	--

(3) Get Service Information with UTF-8 Response (BE ⇒ HOST with Header Type 2)

Start Sync :	A5h,59h
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length(2 Bytes) :	Result=OK : 0008h~ 0FFFh Result=NG : 0005h
Header CRC8 :	--
Data 0 :	01h Get Service Information with UTF-8 (Upper)
Data 1 :	37h Get Service Information with UTF-8 (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Fragment Number
Data 5 :	Last Fragment Number
Data 6~ :	Service_Information_Data Refer to "(4) Description"
Data CRC8 :	--

(4) Description

- Get Service Information.
- If One Touch Number (Data3) of Command is FFh, B/E response the service information of all channels. If One Touch Number (Data3) of Command is 00h, B/E response the service information of not-stored channel. If One Touch Number (Data3) is 01h~0Ch, B/E response the service information of specified channel.
- For Header Type 1
  - "Fragment Number" is the order of the fragmented response. If the length of Data Part is 255byte and under, "Fragment Number" is 0. If the length of Data Part is over 255byte, it is the order number. The B/E sends the data in actual order.
  - "Last Fragment Number" is the max number of "Fragment Number".
- For Header Type 2
  - "Fragment Number" is the order of the fragmented response. If the length of Data Part is 4095byte and under, "Fragment Number" is 0. If the length of Data Part is over 4095byte, it is the order number. The B/E sends the data in actual order.
  - "Last Fragment Number" is the max number of "Fragment Number".
- "Service Information Data" shows the following table.

Number_of_TS_Information(N1)	1byte
for(i = 0; i < N1 ; i++){	
OneTouchNumber	1byte
Number_of_Service_Information(N2)	1byte
for(j = 0; j < N2; j++){	
Service_Information()	n byte
}	
}	

- "Number_of_TS_Information" is the number of the followings TS Information. The

range is 00h~0Ch.

- "OneTouchNumber" is valid one touch number. The range is 00h~0Ch
- "Number_of_Service_Information" is the number of the followings Service_Information(). The range of "Number_of_Service_Information" is 00h~17h.
- The format of "Service_Information()" shows the following table.

Field	Size	Value
Three Digit Number	2Byte	011~
Service ID	2Byte	0x0000~0xFFFF
Branch Identifier Number	1Byte	0~9
Media Type	1Byte	0 : Full Seg Service 1 : Data Service(Not Support) 2 : Data Service(Not Support) 3 : One Seg Service
Service Name Length	1Byte	00h ~ 50h
Service Name	0~80Bytes	UTF-8

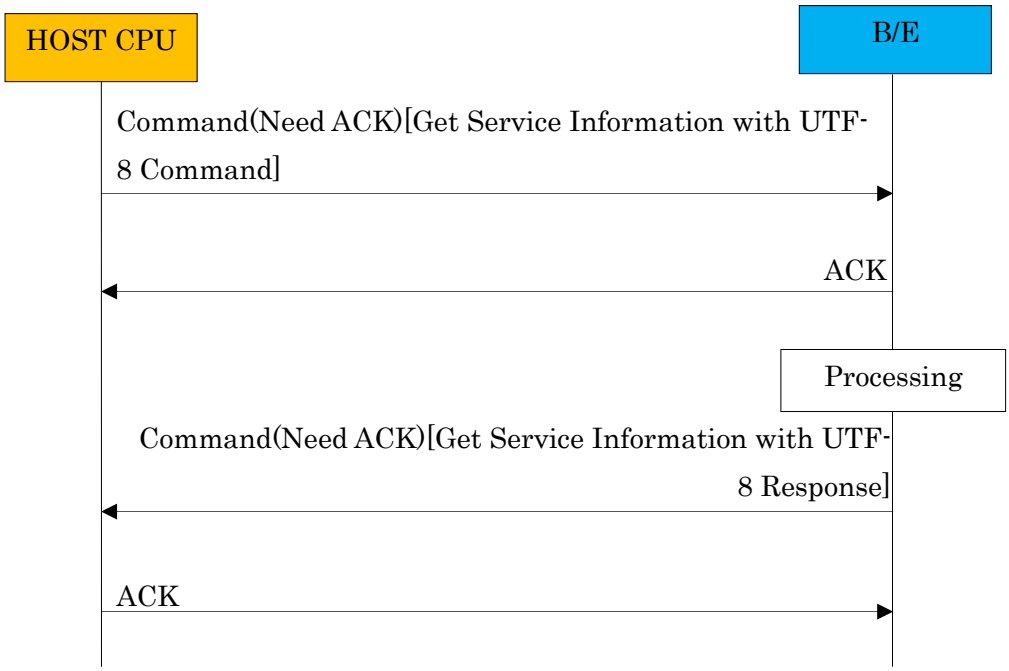
(5) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(6) Sequence


HOST CPU
B/E
Command(Need ACK)[Get Service Information with UTF-8 Command]
ACK
Processing
Command(Need ACK)[Get Service Information with UTF-8 Response]
ACK

5.4.24. Get Program Description with UTF-8

(1) Get Program Description with UTF-8 Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	07h
Header CRC8 :	--
Data 0 :	01h Get Program Description with UTF-8 (Upper)
Data 1 :	38h Get Program Description with UTF-8 (Lower)
Data 2 :	Service ID(Upper)
Data 3 :	Service ID (Lower)
Data 4 :	Event ID(Upper)
Data 5 :	Event ID (Lower)
Data CRC8 :	--

(2) Get Program Description with UTF-8 Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result OK : 08h~FFh Result NG : 05h
Header CRC8 :	--
Data 0 :	01h Get Program Description with UTF-8 (Upper)
Data 1 :	38h Get Program Description with UTF-8 (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Fragment Number
Data 5 :	Last Fragment Number
Data 6 :	Service ID(Upper)
Data 7 :	Service ID (Lower)
Data 8 :	Event ID(Upper)
Data 9 :	Event ID (Lower)
Data 10 :	Program Description Length (Upper)
Data 11 :	Program Description Length (Lower)
Data 12 ~:	Program Description
Data CRC8 :	--

(3) Get Program Description with UTF-8 Response (BE ⇒ HOST with Header Type 2)

Start Sync :	A5h,59h
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length(2 Bytes) :	Result OK : ~030Dh Result NG : 0005h
Header CRC8 :	--
Data 0 :	01h Get Program Description with UTF-8 (Upper)
Data 1 :	38h Get Program Description with UTF-8 (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Fragment Number (00h : Fixed Value)
Data 5 :	Last Fragment Number (00h : Fixed Value)
Data 6 :	Service ID(Upper)
Data 7 :	Service ID (Lower)
Data 8 :	Event ID(Upper)
Data 9 :	Event ID (Lower)
Data 10 :	Program Description Length (Upper)
Data 11 :	Program Description Length (Lower)
Data 12 ~:	Program Description
Data CRC8 :	--

(4) Description

- Get the program description.
- Service ID identify the service in Broadcaster (0x0000~0xFFFF).Event ID identify the program (0x0000~0xFFFF).
- If Service ID and Event ID are 0xFFFF then Get Current Data. (Only [Tuning Status = Tuning])
- Program Description Length is the length of Program Description (Max : 192Bytes × 4).
- Program Description is UTF-8 character code set Format.
- For Header Type 2, Fragment Numbers - Data4, Data5 are set to 00h.

(5) Error Handling

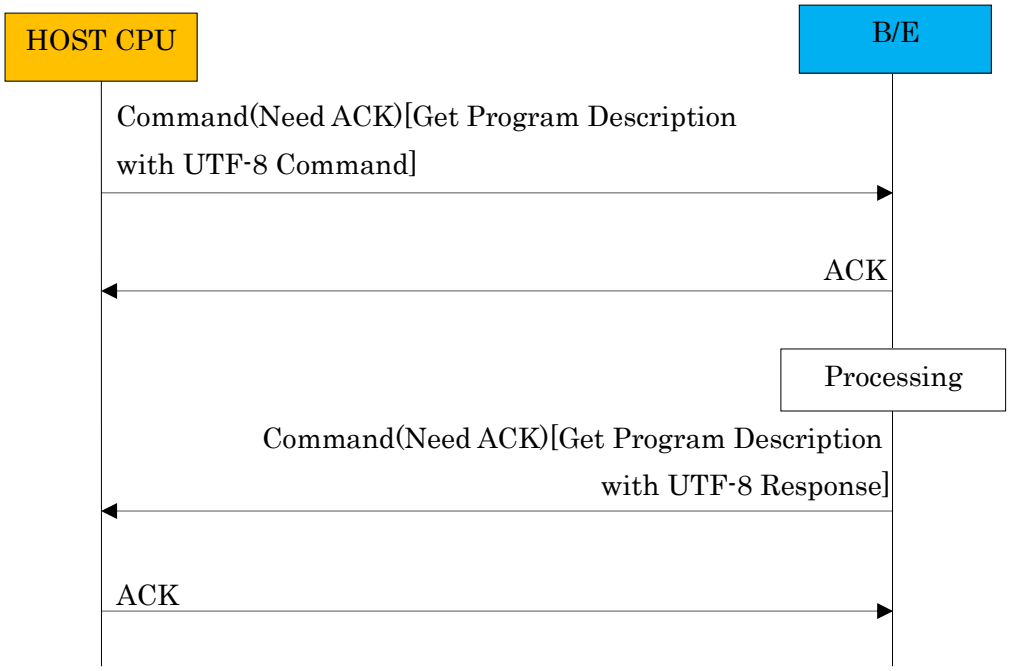
- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
02h(Invalid Data)	Event ID is invalid.
03h(Invalid Length)	Data Length is invalid.

04h(No Service)	Service ID is invalid
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(6) Sequence



5.4.25. Get Program Name with UTF-8

(1) Get Program Name with UTF-8 Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	07h
Header CRC8 :	--
Data 0 :	01h Get Program Name with UTF-8 (Upper)
Data 1 :	39h Get Program Name with UTF-8 (Lower)
Data 2 :	Service ID(Upper)
Data 3 :	Service ID (Lower)
Data 4 :	Event ID(Upper)
Data 5 :	Event ID (Lower)
Data CRC8 :	--

(2) Get Program Name with UTF-8 Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result = OK : 08h~FFh Result = NG : 05h
Header CRC8 :	--
Data 0 :	01h Get Program Name with UTF-8 (Upper)
Data 1 :	39h Get Program Name with UTF-8 (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Fragment Number
Data 5 :	Last Fragment Number
Data 6 :	Service ID(Upper)
Data 7 :	Service ID (Lower)
Data 8 :	Event ID(Upper)
Data 9 :	Event ID (Lower)
Data 10 :	Year of Start Time
Data 11 :	Month of Start Time
Data 12 :	Day of Start Time
Data 13 :	Hour of Start Time
Data 14 :	Minute of Start Time
Data 15 :	Year of End Time
Data 16 :	Month of End Time
Data 17 :	Day of End Time
Data 18 :	Hour of End Time
Data 19 :	Minute of End Time
Data 20 :	Program Name Length (Upper)
Data 21 :	Program Name Length (Lower)
Data 22 ~:	Program Name
Data CRC8 :	--

(3) Get Program Name with UTF-8 Response (BE ⇒ HOST with Header Type 2)

Start Sync :	A5h,59h
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length(2 Bytes) :	Result = OK : ~ 0197h Result = NG : 0005h
Header CRC8 :	--
Data 0 :	01h Get Program Name with UTF-8 (Upper)
Data 1 :	39h Get Program Name with UTF-8 (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Fragment Number (00h : Fixed Value)
Data 5 :	Last Fragment Number (00h : Fixed Value)
Data 6 :	Service ID(Upper)
Data 7 :	Service ID (Lower)
Data 8 :	Event ID(Upper)
Data 9 :	Event ID (Lower)
Data 10 :	Year of Start Time
Data 11 :	Month of Start Time
Data 12 :	Day of Start Time
Data 13 :	Hour of Start Time
Data 14 :	Minute of Start Time
Data 15 :	Year of End Time
Data 16 :	Month of End Time
Data 17 :	Day of End Time
Data 18 :	Hour of End Time
Data 19 :	Minute of End Time
Data 20 :	Program Name Length (Upper)
Data 21 :	Program Name Length (Lower)
Data 22 ~:	Program Name
Data CRC8 :	--

(4) Description

- Get the program description.
- "Service ID" identify the service in Broadcaster (0x0000~0xFFFF).
- "Event ID" identify the program (0x0000~0xFFFF).
- If Service ID and Event ID are 0xFFFF then Get Current Data.
  (Only [Tuning Status = Tuning])
- "Year of Start Time" means the year of Start Time.(Only low 2-digit ex.2017->17)
- "Month of Start Time" means the month of Start Time.(Ex. December -> 12)
- "Day of Start Time" means the date of Start Time.(Ex. 13th -> 13)
- "Hour of Start Time" means the hour of Start Time.(0~23)
- "Minute of Start Time" means the hour of Start Time.(0~59)
- "Year of End Time" means the year of End Time.(Only low 2-digit ex.2017->17)
- "Month of End Time" means the month of End Time.(Ex. December -> 12)
- "Day of End Time" means the date of End Time.(Ex. 13th -> 13)
- "Hour of End Time" means the hour of End Time.(0~23)
- "Minute of End Time" means the hour of End Time.(0~59)
- If End Time is not yet fixed, "Year"/"Month"/"Day"/"Hour"/"Minute" are 0xFF.
- "Program Name Length" is the length of Program Name (0~96Byte ~~× 3~~ × 4).
- "Program Name" is UTF-8 character code set Format.
- B/E doesn't save EPG Information to NVRAM. If B/E power off, B/E doesn't have EPG Information.
- For Header Type 2, Fragment Numbers - Data 4, Data5 are set to 00h.

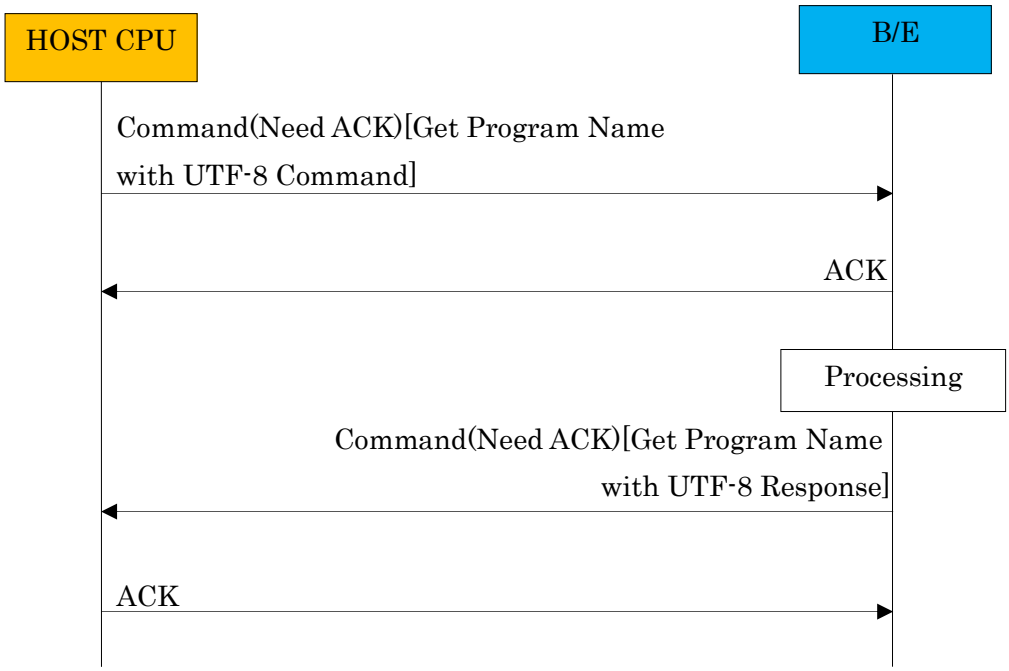
(5) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
02h(Invalid Data)	Event ID is invalid.
03h(Invalid Length)	Data Length is invalid.
04h(No Service)	Service ID is invalid
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(6) Sequence


HOST CPU
B/E
Command(Need ACK)[Get Program Name with UTF-8 Command]
ACK
Processing
Command(Need ACK)[Get Program Name with UTF-8 Response]
ACK

5.4.26. Get Audio ES info with UTF-8

(1) Get Audio ES info with UTF-8 Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h
Header CRC8 :	--
Data 0 :	01h Get Audio ES info with UTF-8 (Upper)
Data 1 :	3Ah Get Audio ES info with UTF-8 (Lower)
Data CRC8 :	--

(2) Get Audio ES info with UTF-8 Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : 07h ~ FFh Result=NG : 05h
Header CRC8 :	--
Data 0 :	01h Get Audio ES info with UTF-8 (Upper)
Data 1 :	3Ah Get Audio ES info with UTF-8 (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Fragment Number
Data 5 :	Last Fragment Number
Data 6 ~	Audio ES info Data
Data CRC8 :	--

(3) Get Audio ES info with UTF-8 Response (BE ⇒ HOST with Header Type 2)

Start Sync :	A5h,59h
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length(2 Bytes) :	Result=OK : 0007h ~ 07E2h Result=NG : 0005h
Header CRC8 :	--
Data 0 :	01h Get Audio ES info with UTF-8 (Upper)
Data 1 :	3Ah Get Audio ES info with UTF-8 (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Fragment Number (00h : Fixed Value)
Data 5 :	Last Fragment Number (00h : Fixed Value)
Data 6 ~	Audio ES info Data
Data CRC8 :	--

(4) Description

- Get Audio ES information.
- "Audio ES Info Data" shows the following table.

Number_of_AudioES_Information(N1)
for(i = 0; i < N1 ; i++){
CompornetTag
Number_of_Language(N2)
for(j = 0; j < N2; j++){
language code
language text length (N3)
language text[N3]
}
}
	1byte
	
	1byte
	1byte
	1byte
	1byte
	1byte
	N3 byte

- N1 max = 15 / N2 max = 2 / N3 max = 64
- Language code :

Value	Language
0	Japanese
1	English
2	German
3	French
4	Italian
5	Russian
6	Chinese
7	Korean
8	Spanish
9	Etc Foreign language : Language than the above Unknown : Cannot determine one language because multiple languages are mixed

- language Text : Maximum 16*4 bytes for UTF-8
- For Header Type 2, Fragment Numbers - Data4, Data5 are set to 00h.

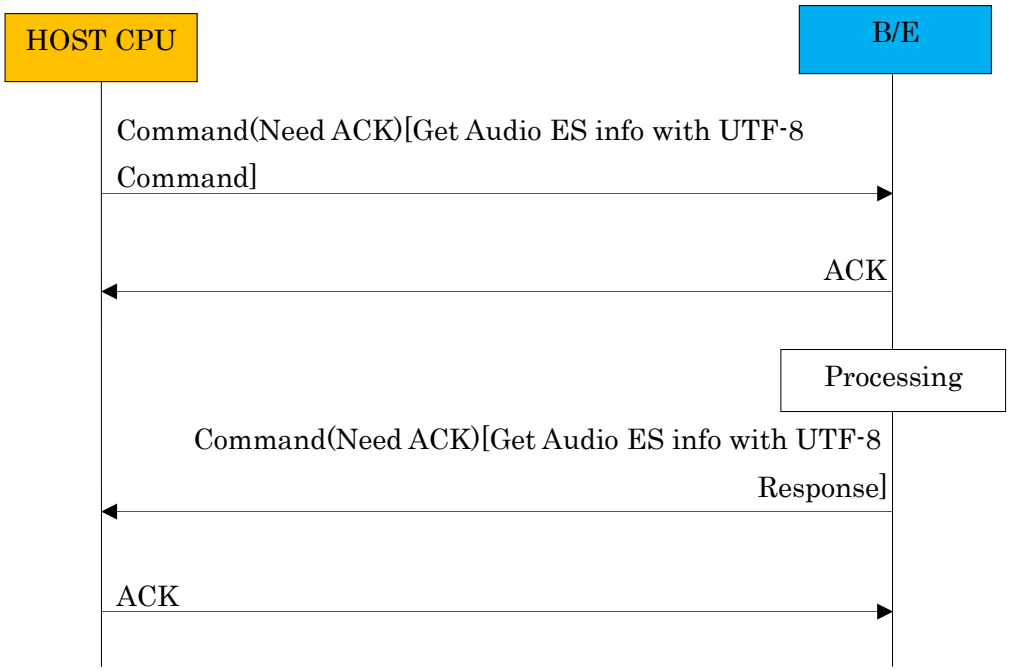
(5) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
05h(Not Support)	The parameter is not supported.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(6) Sequence


HOST CPU
B/E
Command(Need ACK)[Get Audio ES info with UTF-8 Command]
ACK
Processing
Command(Need ACK)[Get Audio ES info with UTF-8 Response]
ACK

5.4.27. Get User ID

(1) Get User ID Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h
Header CRC8 :	--
Data 0 :	01h Get User ID (Upper)
Data 1 :	3Ch Get User ID (Lower)
Data CRC8 :	--

(2) Get User ID Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : 06h Result=NG : 05h
Header CRC8 :	--
Data 0 :	01h Get User ID (Upper)
Data 1 :	3Ch Get User ID (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	User ID 01h to 06h or FFh
Data CRC8 :	--

(3) Description

- Get Current User ID

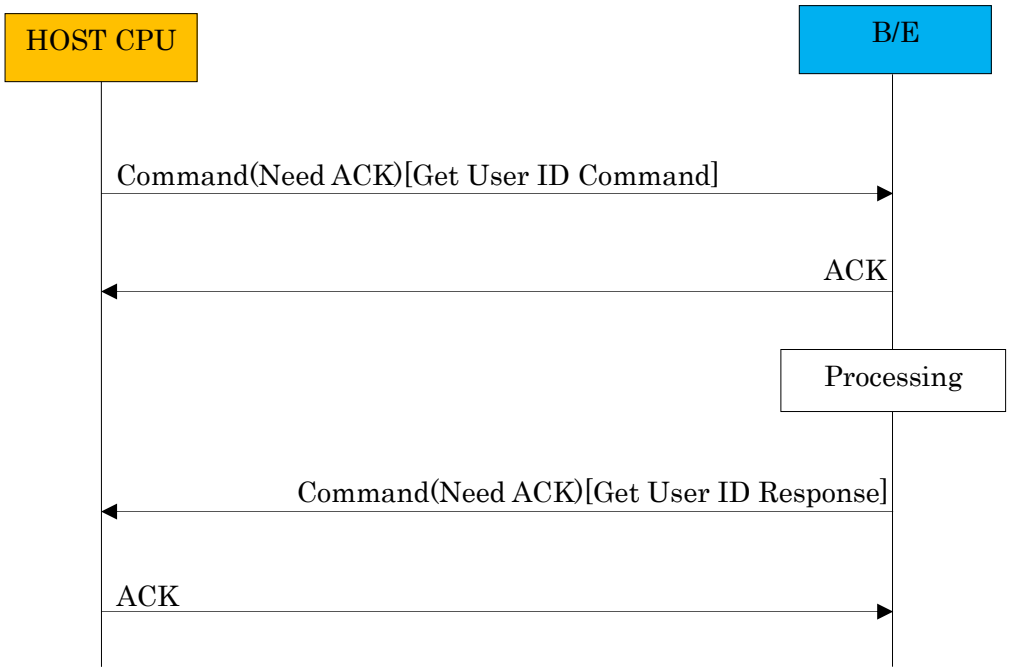
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence


HOST CPU
B/E
Command(Need ACK)[Get User ID Command]
ACK
Processing
Command(Need ACK)[Get User ID Response]
ACK

5.4.28. Get EPG Config

(1) Get EPG Config Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h
Header CRC8 :	--
Data 0 :	01h Get EPG Config(Upper)
Data 1 :	3Eh Get EPG Config(Lower)
Data CRC8 :	--

(2) Get EPG Config Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : 38h Result=NG : 05h
Header CRC8 :	--
Data 0 :	01h Get EPG Config(Upper)
Data 1 :	3Eh Get EPG Config(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	EPG List Type Enable Bit0 : Type 1 Enable Bit1 : Type 2 Enable Bit2 : Type 3 Enable Bit3 : Type 4 Enable
Data 5 :	EPG List: Type1 Program name Font Size [px]
Data 6 :	EPG List: Type1 TS Name Font Size [px]
Data 7 :	EPG List: Type1 EPG Content Font Size [px]
Data 8 :	EPG List: Type2 Program name Font Size [px]
Data 9 :	EPG List: Type2 TS Name Font Size [px]
Data 10 :	EPG List: Type2 EPG Content Font Size [px]
Data 11 :	EPG List: Type3 Program name Font Size [px]
Data 12 :	EPG List: Type3 TS Name Font Size [px]
Data 13 :	EPG List: Type3 EPG Content Font Size [px]
Data 14 :	EPG List: Type4 Program name Font Size [px]
Data 15 :	EPG List: Type4 TS Name Font Size [px]
Data 16 :	EPG List: Type4 EPG Content Font Size [px]
Data 17 :	EPG List: Table Title Color Red
Data 18 :	EPG List: Table Title Color Green
Data 19 :	EPG List: Table Title Color Blue
Data 20 :	EPG List: Table Title Font Color Red
Data 21 :	EPG List: Table Title Font Color Green
Data 22 :	EPG List: Table Title Font Color Blue
Data 23 :	EPG List: Cursor Color Red
Data 24 :	EPG List: Cursor Color Green
Data 25 :	EPG List: Cursor Color Blue
Data 26 :	EPG List: Cursor Font Color Red
Data 27 :	EPG List: Cursor Font Color Green
Data 28 :	EPG List: Cursor Font Color Blue

Data 29 :	EPG List: Current Program Color Red
Data 30 :	EPG List: Current Program Color Green
Data 31 :	EPG List: Current Program Color Blue
Data 32 :	EPG List: Current Program Font Color Red
Data 33 :	EPG List: Current Program Font Color Green
Data 34 :	EPG List: Current Program Font Color Blue
Data 35 :	EPG List: Balloon color Red
Data 36 :	EPG List: Balloon color Green
Data 37 :	EPG List: Balloon color Blue
Data 38 :	EPG List: Balloon Font color Red
Data 39 :	EPG List: Balloon Font color Green
Data 40 :	EPG List: Balloon Font color Blue
Data 41 :	EPG List: Border line color Red
Data 42 :	EPG List: Border line color Green
Data 43 :	EPG List: Border line color Blue
Data 44 :	EPG Information: Font Size [px]
Data 45 :	EPG Information: Title Font Color Red
Data 46 :	EPG Information: Title Font Color Green
Data 47 :	EPG Information: Title Font Color Blue
Data 48 :	EPG Information: Line Spacing for Title
Data 49 :	EPG Information: Line Spacing for Contents
Data 50 :	EPG Information: Margin(Applied to all four sides)
Data 51 :	EPG List: Short program border line color Red
Data 52 :	EPG List: Short program border line color Green
Data 53 :	EPG List: Short program border line color Blue
Data 54 :	Number of scrolling line on EPG detail (01h~
Data CRC8 :	--

(3) Description

- Get EPG Configuration

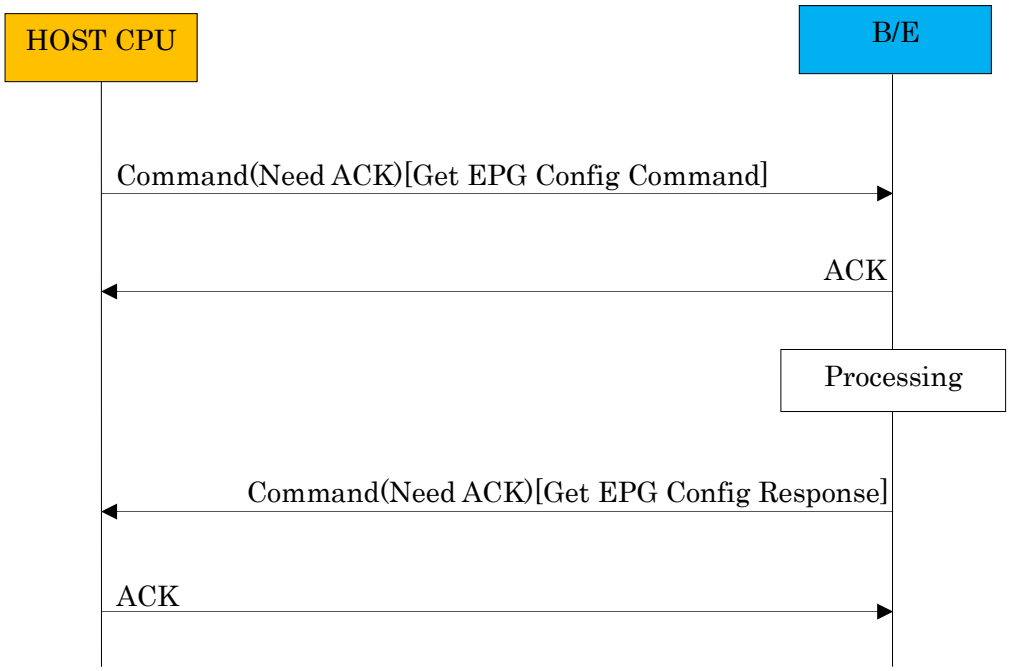
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence


HOST CPU
B/E
Command(Need ACK)[Get EPG Config Command]
ACK
Processing
Command(Need ACK)[Get EPG Config Response]
ACK

5.4.29. Get Video Color

(1) Get Video Color Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h
Header CRC8 :	--
Data 0 :	01h Get Video Color (Upper)
Data 1 :	40h Get Video Color (Lower)
Data CRC8 :	--

(2) Get Video Color Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : 0Ah Result=NG : 05h
Header CRC8 :	--
Data 0 :	01h Get Video Color (Upper)
Data 1 :	40h Get Video Color (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Brightness -127(81h) – 127(7Fh)
Data 5 :	Contrast 0 – 7
Data 6～8:	Reserved
Data CRC8 :	--

(3) Description

- Get Video Color

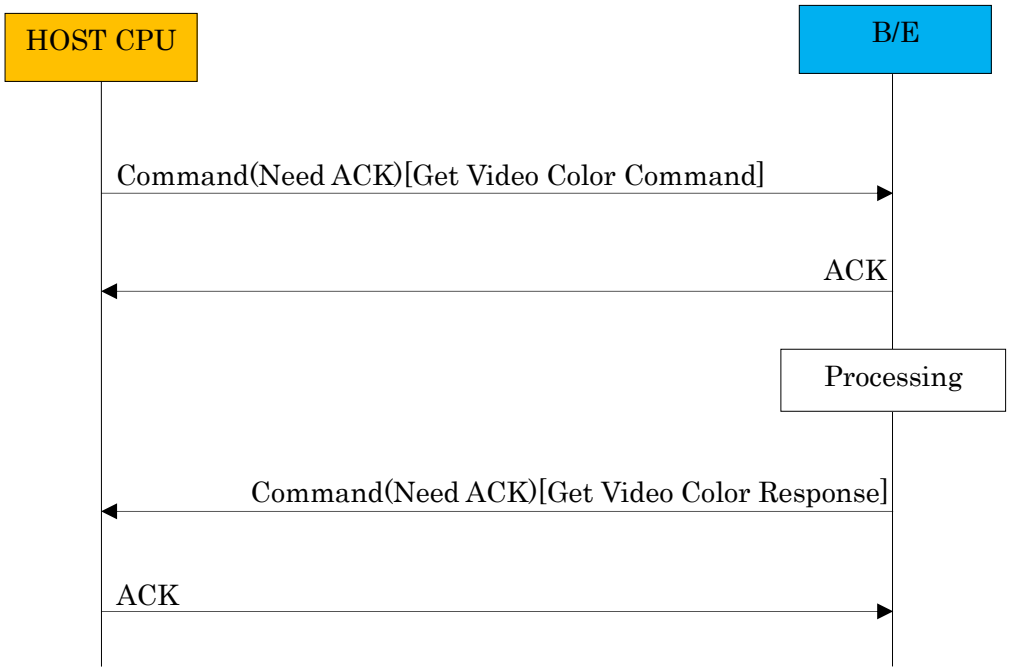
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.4.30. Get Event Mask

(1) Get Event Mask Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h
Header CRC8 :	--
Data 0 :	01h Get Event Mask (Upper)
Data 1 :	42h Get Event Mask (Lower)
Data CRC8 :	--

(2) Get Event Mask Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : 08h Result=NG : 05h
Header CRC8 :	--
Data 0 :	01h Get Event Mask (Upper)
Data 1 :	42h Get Event Mask (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	bit0 : DTV Status bit1 : Scan Status bit2 : Search Status bit3 : DTV Event bit4 : EWS Event bit5 : EPG Status bit6 : Event relay bit7 : Number of Program info 0 : Not Mask 1 : Mask - Disables the event notification.
Data 5 :	bit0 : Program Information bit1 : Program Information with UTF-8 bit2 : TS Name Update bit3 : TS Name Update with UTF-8 bit4 : Logo Update bit5 : EGP Button Status bit6 : BML Error bit7 : C/N Information 0 : Not Mask 1 : Mask - Disables the event notification.
Data 6 :	bit0 : EPG Scan Result bit1 – 7 Reserve 0 : Not Mask 1 : Mask - Disables the event notification.
Data CRC8 :	--

(3) Description

- Get Event Mask

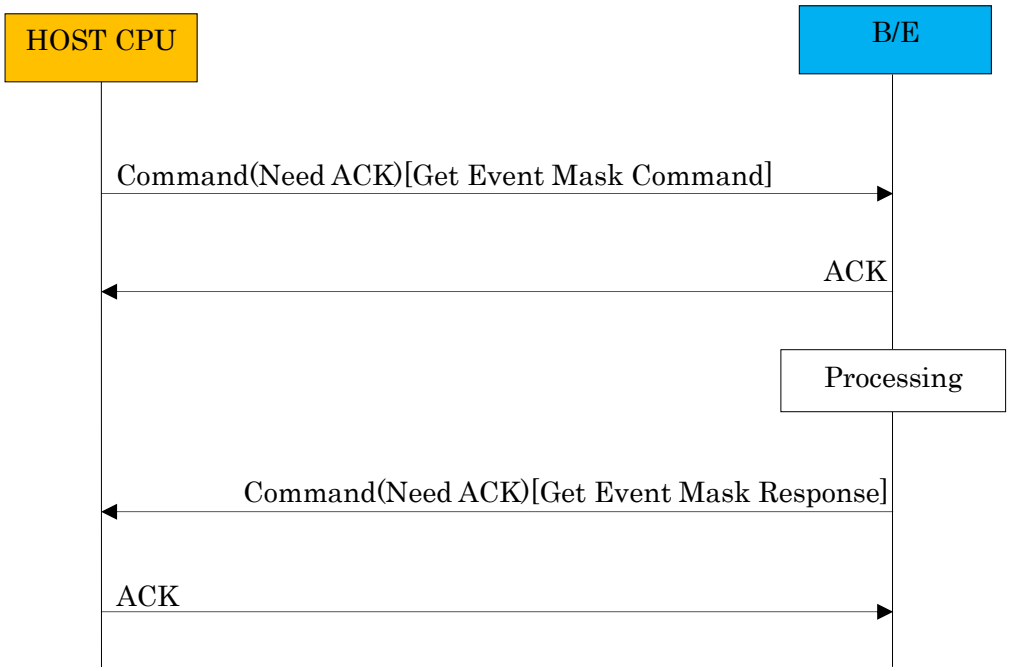
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.4.31. Get Part Number

(1) Get Part Number Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h
Header CRC8 :	--
Data 0 :	01h Get Part Number (Upper)
Data 1 :	43h Get Part Number (Lower)
Data CRC8 :	--

(2) Get Part Number Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : 1Dh Result=NG : 05h
Header CRC8 :	--
Data 0 :	01h Get Part Number (Upper)
Data 1 :	43h Get Part Number (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	BOSCH PN 1
Data 5 :	BOSCH PN 2
Data 6 :	BOSCH PN 3
Data 7 :	BOSCH PN 4
Data 8 :	BOSCH PN 5
Data 9 :	BOSCH PN 6
Data 10 :	BOSCH PN 7
Data 11 :	BOSCH PN 8
Data 12 :	BOSCH PN 9
Data 13 :	BOSCH PN 10
Data 14 :	BOSCH PN 11
Data 15 :	BOSCH PN 12
Data 16 :	Module PN 1
Data 17 :	Module PN 2
Data 18 :	Module PN 3
Data 19 :	Module PN 4
Data 20 :	Module PN 5
Data 21 :	Module PN 6
Data 22 :	Module PN 7
Data 23 :	Module PN 8
Data 24 :	Module PN 9
Data 25 :	Module PN 10
Data 26 :	Module PN 11
Data 27 :	Module PN 12
Data CRC8 :	--

(3) Description

- Get Part Number.
- The format is ASCII.(Ex. "TRH231BS0000")

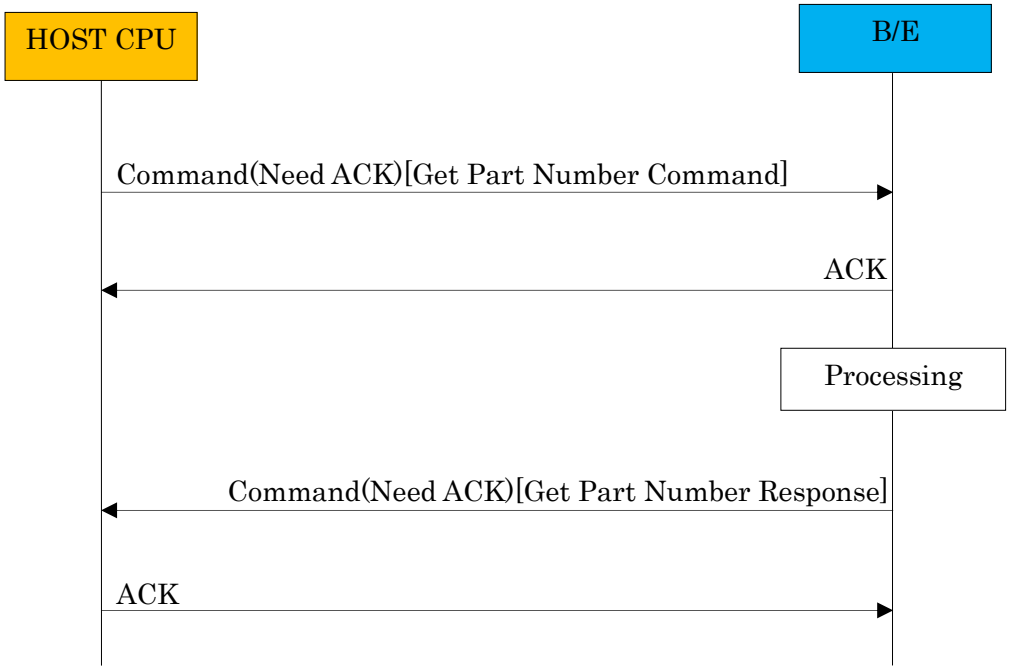
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.4.32. Get Program Name by 3Digit Number

(1) Get Program Name by 3Digit Number Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	07h
Header CRC8 :	--
Data 0 :	01h Get Program Name by 3Digit Number (Upper)
Data 1 :	44h Get Program Name by 3Digit Number (Lower)
Data 2 :	Preset 00h : Home Preset 01h : Drive Preset
Data 3 :	Three Digit Number(Upper)
Data 4 :	Three Digit Number(Lower)
Data 5 :	Branch Identifier Number
Data CRC8 :	--

(2) Get Program Name by 3Digit Number Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h~74h
Header CRC8 :	--
Data 0 :	01h Get Program Name by 3Digit Number (Upper)
Data 1 :	44h Get Program Name by 3Digit Number (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Service ID(Upper)
Data 5 :	Service ID (Lower)
Data 6 :	Event ID(Upper)
Data 7 :	Event ID (Lower)
Data 8 :	Year of Start Time
Data 9 :	Month of Start Time
Data 10 :	Day of Start Time
Data 11 :	Hour of Start Time
Data 12 :	Minute of Start Time
Data 13 :	Year of End Time
Data 14 :	Month of End Time
Data 15 :	Day of End Time
Data 16 :	Hour of End Time
Data 17 :	Minute of End Time
Data 18 :	Program Name Length
Data 19 ~:	Program Name
Data CRC8 :	--

(3) Description

- Get the Current program Name
- "Service ID" identify the service in Broadcaster (0x0000~0xFFFF).
- "Event ID" identify the program (0x0000~0xFFFF).

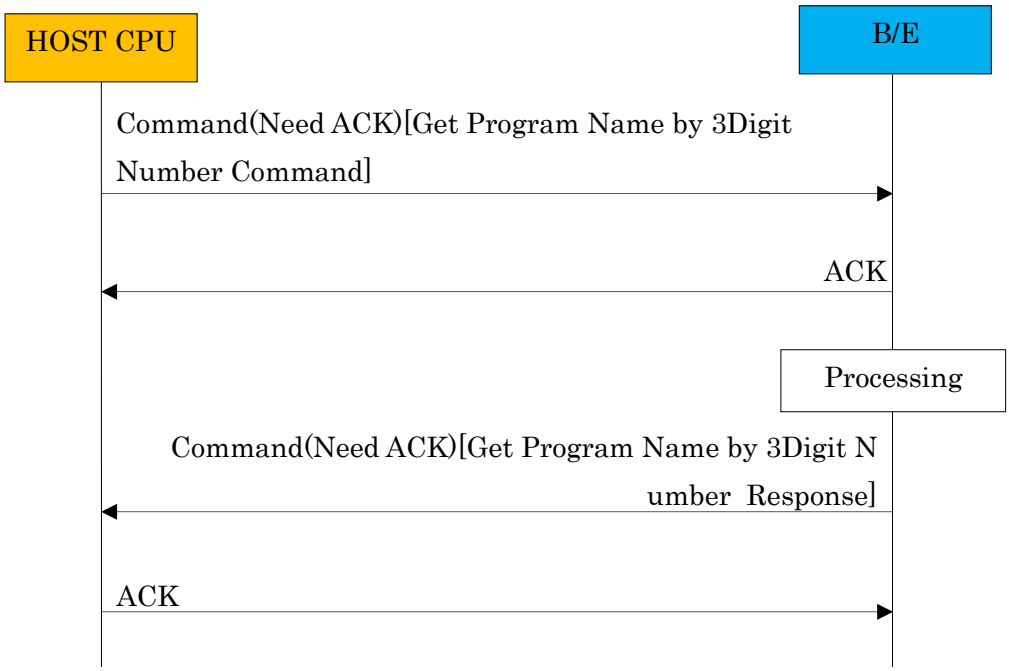
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
02h(Invalid Data)	Event ID is invalid.
03h(Invalid Length)	Data Length is invalid.
04h(No Service)	3Digit Number is invalid
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.4.33. Get Program Name with UTF-8 by 3Digit Number

(1) Get Program Name with UTF-8 by 3Digit Number Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	07h
Header CRC8 :	--
Data 0 :	01h Get Program Name with UTF-8 by 3Digit Number (Upper)
Data 1 :	45h Get Program Name with UTF-8 by 3Digit Number (Lower)
Data 2 :	Preset 00h : Home Preset 01h : Drive Preset
Data 3 :	Three Digit Number(Upper)
Data 4 :	Three Digit Number(Lower)
Data 5 :	Branch Identifier Number
Data CRC8 :	--

(2) Get Program Name with UTF-8 by 3Digit Number Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : 08h~FFh Result=NG : 05h
Header CRC8 :	--
Data 0 :	01h Get Program Name with UTF-8 by 3Digit Number (Upper)
Data 1 :	45h Get Program Name with UTF-8 by 3Digit Number (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Fragment Number
Data 5 :	Last Fragment Number
Data 6 :	Service ID(Upper)
Data 7 :	Service ID (Lower)
Data 8 :	Event ID(Upper)
Data 9 :	Event ID (Lower)
Data 10 :	Year of Start Time
Data 11 :	Month of Start Time
Data 12 :	Day of Start Time
Data 13 :	Hour of Start Time
Data 14 :	Minute of Start Time
Data 15 :	Year of End Time
Data 16 :	Month of End Time
Data 17 :	Day of End Time
Data 18 :	Hour of End Time
Data 19 :	Minute of End Time
Data 20 :	Program Name Length (Upper)
Data 21 :	Program Name Length (Lower)
Data 22 ~:	Program Name

Data CRC8 :	--

(3) Get Program Name with UTF-8 by 3Digit Number Response (BE ⇒ HOST with Header Type 2)

Start Sync :	A5h,59h
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length(2 Bytes) :	Result=OK : 0017h ~ 0197h Result=NG : 0005h
Header CRC8 :	--
Data 0 :	01h Get Program Name with UTF-8 by 3Digit Number (Upper)
Data 1 :	45h Get Program Name with UTF-8 by 3Digit Number (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Fragment Number (00h : Fixed Value)
Data 5 :	Last Fragment Number (00h : Fixed Value)
Data 6 :	Service ID(Upper)
Data 7 :	Service ID (Lower)
Data 8 :	Event ID(Upper)
Data 9 :	Event ID (Lower)
Data 10 :	Year of Start Time
Data 11 :	Month of Start Time
Data 12 :	Day of Start Time
Data 13 :	Hour of Start Time
Data 14 :	Minute of Start Time
Data 15 :	Year of End Time
Data 16 :	Month of End Time
Data 17 :	Day of End Time
Data 18 :	Hour of End Time
Data 19 :	Minute of End Time
Data 20 :	Program Name Length (Upper)
Data 21 :	Program Name Length (Lower)
Data 22 ~:	Program Name
Data CRC8 :	--

(4) Description

- Get the Current program description.
- "Service ID" identify the service in Broadcaster (0x0000~0xFFFF).
- "Event ID" identify the program (0x0000~0xFFFF).
- "Program Name" is UTF-8 character code set Format.
- For Header Type 2, Fragment Numbers - Data4, Data5 are set to 00h.

(5) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:

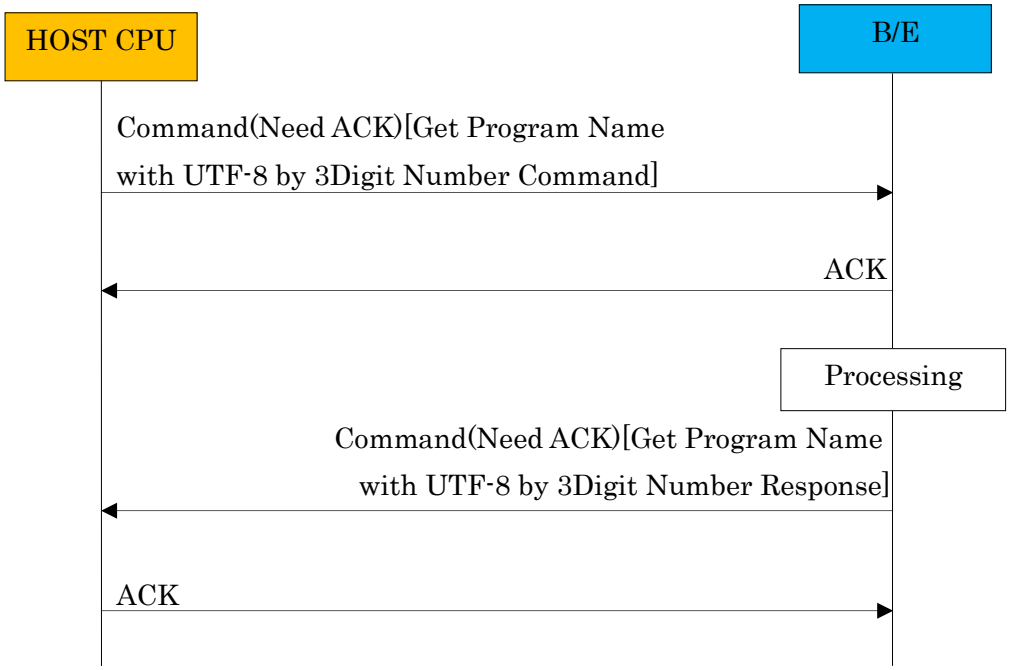
- ✧ Doesn't receive the response within 1sec after HOST transmit the command.
- ✧ The result of response is NG.

・ Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
02h(Invalid Data)	Event ID is invalid.
03h(Invalid Length)	Data Length is invalid.
04h(No Service)	3Digit Number is invalid
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(6) Sequence



5.4.34. Get Logo Data by 3Digit Number

(1) Get Logo Data by 3Digit Number Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	08h
Header CRC8 :	--
Data 0 :	01h Get Logo Data by 3Digit Number (Upper)
Data 1 :	46h Get Logo Data by 3Digit Number (Lower)
Data 2 :	Preset 00h : Home Preset 01h : Drive Preset
Data 3 :	Three Digit Number(Upper)
Data 4 :	Three Digit Number(Lower)
Data 5 :	Branch Identifier Number
Data 6 :	Logo Type
Data CRC8 :	--

(2) Get Logo Data by 3Digit Number Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result=OK : 08h~FFh Result=NG : 05h
Header CRC8 :	--
Data 0 :	01h Get Logo Data by 3Digit Number (Upper)
Data 1 :	46h Get Logo Data by 3Digit Number (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Fragment Number
Data 5 :	Last Fragment Number
Data 6 :	Service ID(Upper)
Data 7 :	Service ID (Lower)
Data 8 ~:	Logo Information Refer to "(4)Description"
Data CRC8 :	--

(3) Get Logo Data by 3Digit Number Response (BE ⇒ HOST with Header Type 2)

Start Sync :	A5h,59h
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length(2 Bytes) :	Result=OK : 0008h~0734h Result=NG : 0005h
Header CRC8 :	--
Data 0 :	01h Get Logo Data by 3Digit Number (Upper)
Data 1 :	46h Get Logo Data by 3Digit Number (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Fragment Number (00h : Fixed Value)
Data 5 :	Last Fragment Number (00h : Fixed Value)
Data 6 :	Service ID(Upper)
Data 7 :	Service ID (Lower)
Data 8 ~:	Logo Information Refer to "(4)Description"
Data CRC8 :	--

(4) Description

- Get Logo Data by 3 Digit number.
- Logo Data refer to "5.4.13. Get Logo Data".
- For Header Type 2, Fragment Numbers – Data4, Data5 are set to 00h.

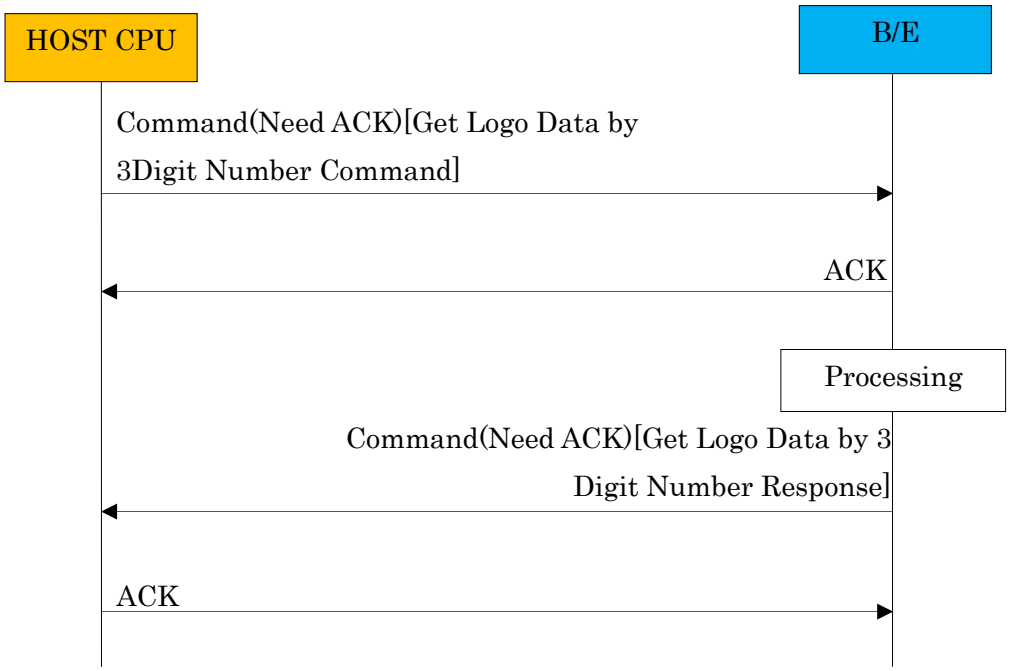
(5) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid. - Logo Type is not 00h ~ 06h.
03h(Invalid Length)	Data Length is invalid.
04h(No Service)	3Digit Number is invalid
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(6) Sequence


HOST CPU
B/E
Command(Need ACK)[Get Logo Data by 3Digit Number Command]
ACK
Processing
Command(Need ACK)[Get Logo Data by 3 Digit Number Response]
ACK

5.5. HOST⇒B/E Event

5.5.1. Key Event

(1) Key Event (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	01h : Command(No Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	02h Key Event(Upper)
Data 1 :	01h Key Event(Lower)
Data 2 :	Key Status 01h : Press 02h : Release
Data 3 :	Key Code 01h : No.1 02h : No.2 03h : No.3 04h : No.4 05h : No.5 06h : No.6 07h : No.7 08h : No.8 09h : No.9 0Ah : No.10/0 0Dh : Up Arrow 0Eh : Down Arrow 0Fh : Left Arrow 10h : Right Arrow 11h : Enter 12h : Back 27h : Data Broadcast 28h : Blue 29h : Red 2Ah : Green 2Bh : Yellow
Data CRC8 :	--

(2) Description

- Notify Key Event to B/E.
- Key Event use only BML Browser.
- When press Remocon button, HOST transmit Key Status=Press
- If release after *N* sec, HOST transmit Key Status=Release.

(3) Error Handling

- HOST doesn't need ACK from B/E.
- If Key Event is CRC Error, B/E doesn't transmit NAK to HOST.

(4) Sequence



5.6. B/E⇒HOST Command

5.6.1. Wakeup

(1) Wakeup Command (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	04h
Header CRC8 :	--
Data 0 :	80h Wakeup(Upper)
Data 1 :	01h Wakeup(Lower)
Data 2 :	Boot Mode 00h : Normal Boot 01h : Software Update
Data CRC8 :	--

(2) Wakeup Response (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	0Bh
Header CRC8 :	--
Data 0 :	80h Wakeup(Upper)
Data 1 :	01h Wakeup(Lower)
Data 2 :	00h : Reserved
Data 3 :	uCom Software Version 0(Upper)
Data 4 :	uCom Software Version 1
Data 5 :	uCom Software Version 2
Data 6 :	uCom Software Version 3(Lower)
Data 7 :	Action Mode 00h : Not tuning after booting(Keep idle status) FFh : Tuning Last Channel after booting
Data 8 :	In case of Normal Boot: 00h : Reserved In case of Software Update 00h : SW Update 01h : HW Test
Data 9 :	00h : Current User 01h to 06h : User ID FFh : Demo Mode
Data CRC8 :	--

(3) Description

- Inform to HOST that B/E finish to boot.
- If HOST doesn't receive this command within 20sec after RESET, HOST should reset B/E again.
- B/E discard all command before receiving Wakeup Response. HOST shouldn't send any commands (except response of Boot Mode) until receiving Wakeup command.
- If Boot Mode = 01h (Software Update), B/E discard all command from HOST (except Supported command (Refer to 5.1. Command List)).
- If there is no response for more than 20 seconds to UART command issued at the time

when boot up should have been completed, B/E should be reset.

- If Action Mode (Data7) is 00h, B/E keeps idle status after booting. If Action Mode (Data7) = FFh, B/E tune the last channel after booting. But if B/E doesn't have the last channel, B/E keeps idle status.

  - If Last Service cannot tune , then Last Channel top Service selected.
  - If Last Channel cannot tune, TV1's smallest preset number station selected.
  - If TV1 was empty, then TV2's smallest preset number station selected.

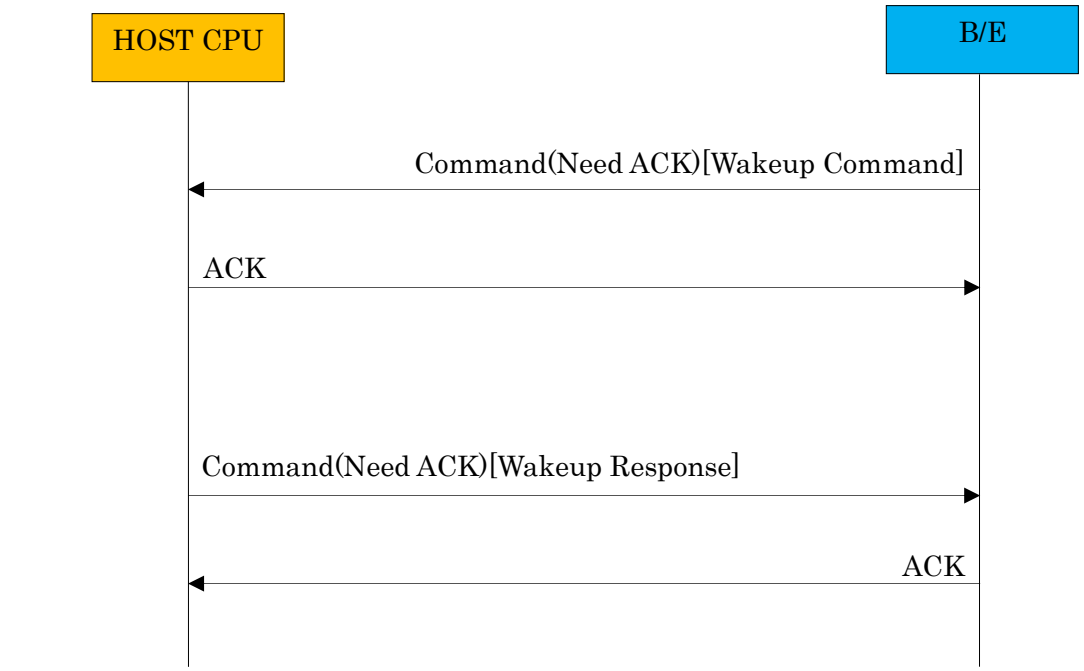
-

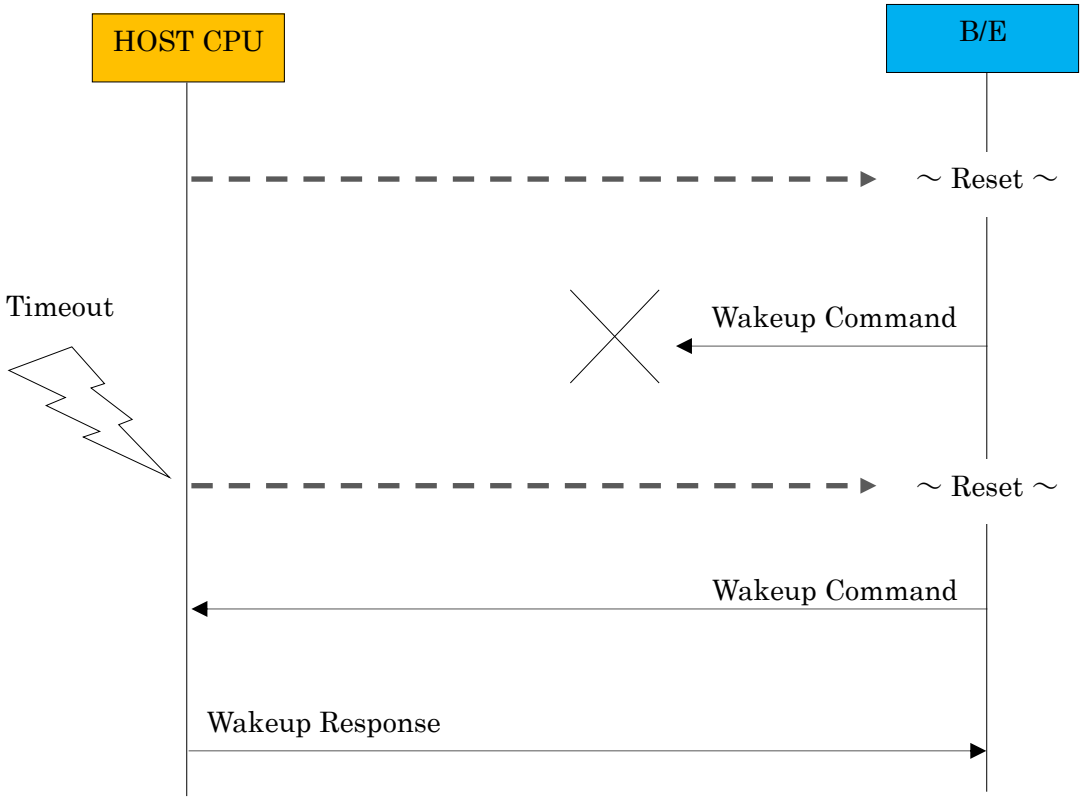
- When the software comes up in DTV mode, it outputs a Wakeup command. After that, output of DTV Status starts regardless of the response from HOST.
  Therefore, When the response from Host is delayed. DVT status may be output earlier than Wakeup response from HOST.

(4) Error Handling

- B/E will continue to wait for Wakeup response.

(5) Sequence


HOST CPU
B/E
Command(Need ACK)[Wakeup Command]
ACK
Command(Need ACK)[Wakeup Response]
ACK


HOST CPU
B/E
～ Reset ～
Timeout
Wakeup Command
～ Reset ～
Wakeup Command
Wakeup Response

5.6.2. Boot Mode

(1) Boot Mode Command (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h
Header CRC8 :	--
Data 0 :	80h Boot Mode(Upper)
Data 1 :	03h Boot Mode(Lower)
Data CRC8 :	--

(2) Boot Mode Response (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Result = OK : 06h Result = NG : 05h
Header CRC8 :	--
Data 0 :	80h Boot Mode(Upper)
Data 1 :	03h Boot Mode(Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	00h : Normal Boot 01h : SW Update
Data CRC8 :	--

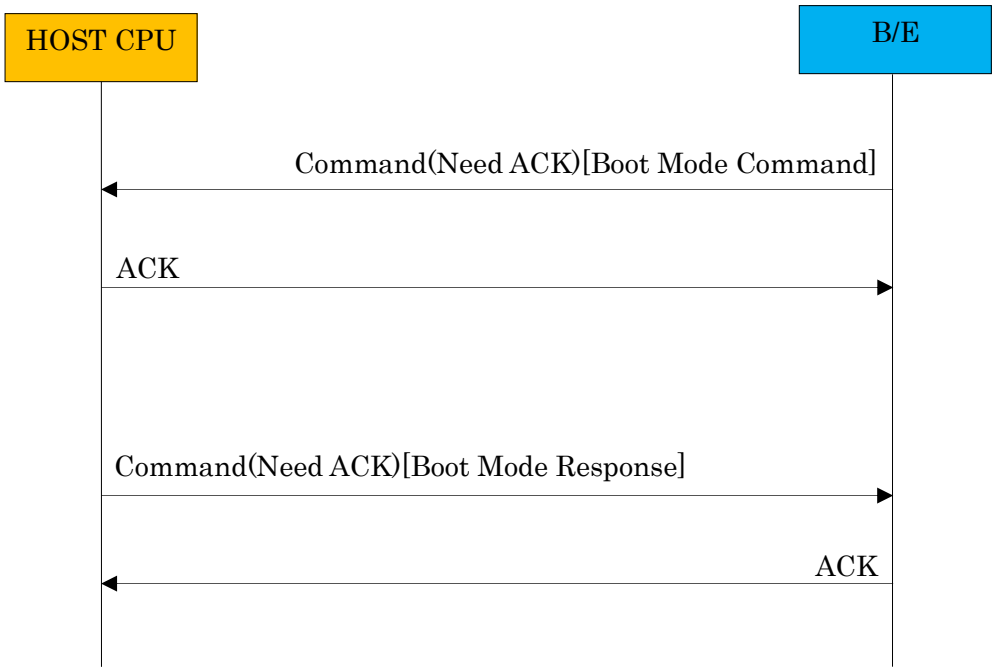
(3) Description

- B/E send this command for confirmation Boot Mode.
- If B/E receive "Data4=00h (Normal Boot)", B/E start DTV Program.
- If B/E receive "Data4=01h (SW Update)", B/E start to SW Update.
- If there is no response for more than 20 seconds to UART command issued at the time when boot up should have been completed, B/E should be reset.

(4) Error Handling

- If B/E Result=NG of response, B/E start DTV Program.
- If B/E didn't receive Boot Mode Response during 200ms after receiving ACK of Wakeup Command, B/E start with Normal Boot.
- If B/E happen to retry over of Boot Mode Command, B/E start with Normal Boot.

(5) Sequence


HOST CPU
B/E
Command(Need ACK)[Boot Mode Command]
ACK
Command(Need ACK)[Boot Mode Response]
ACK

5.7. B/E⇒HOST Event

5.7.1. DTV Status

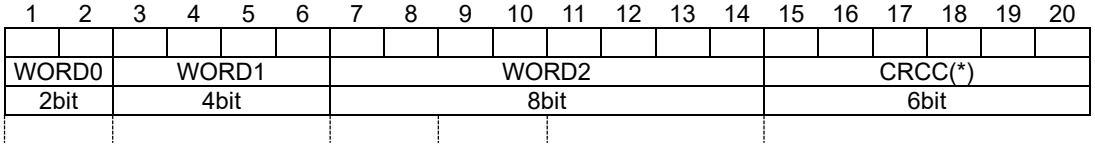
(1) DTV Status Event (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	01h : Command(No Need Ack))
Request No. :	00h ~ FFh
Data Length :	12h
Header CRC8 :	--
Data 0 :	82h DTV Status Event (Upper)
Data 1 :	01h DTV Status Event (Lower)
Data 2 :	DTV Status 00h : Power OFF (Not Support) 01h : Booting (Not Support) 02h : Shutdown (After Shutdown B/E) 03h : In Service (After Booting B/E) 04h : Low Power (Low Power mode) 05h : DTV OFF 06h : DTV OFF with EPG SCAN 07h : Demo mode
Data 3 :	Tuning Status 00h : Idle 01h : Tuning
Data 4 :	Service Type (Now Playing Service Type) 00h : ISDB-T Fullseg 01h : ISDB-T Oneseg FFh : None (Not Tuning)
Data 5 :	Service Status 00h : Invalid 01h : Normal Service 02h : Special Service 03h : Emergency Service 04h : Suspend Service * Refer to "5.4.1(3) Description"
Data 6 :	Video Output 00h : No output (From Tuning to Video Output) 01h : Outputted
Data 7 :	Preset Number 00h : Preset No.1(Home) 01h : Preset No.2(Drive) FFh : None(Not Tuning / Not Stored channel)
Data 8 :	Preset Channel Number 00h : tuned by Search/BCSearch/Physical Channel Set 01h～0Ch : Preset Channel Number FFh : None (Not Tuning)
Data 9 :	Copy Control(Refer to "(2) Description")
Data 10 :	Copy Control(Refer to "(2) Description")
Data 11 :	Event Code Refer to "5.7.4.DTV Event" ※Event code of "0x80" or more will not occur in this data.
Data 12 :	Diver Mode 00h : Dynamic 4Diver System (Fore Br0/1) 01h : Fixed 2x2Diver System (Fore Br0/1) 02h : Fixed 2Diver System(Antenna Br2/3) 03h : Fixed 4Diver System 04h : Fixed 2Diver System(Antenna Br0/1) 05h : Fixed 3x1Diver System (Fore Br0/1/2)

	06h : Dynamic 4Diver System (Fore Br2/3) 07h : Fixed 2x2Diver System (Fore Br2/3) 08h : Fixed 3x1Diver System (Fore Br1/2/3)
Data 13 :	Antenna Level 00h : Antenna Level 0 01h : Antenna Level 1 02h : Antenna Level 2 03h : Antenna Level 3
Data 14 :	Background Status 00h : idle 01h : Background Search 02h : Background scan 03h : Manual Background Search 04h : Manual Background Scan
Data 15 :	EPG/BML Status 00h : No EPG/BML displayed 01h : Updating 02h : EPG List (full seg) 03h : EPG List (one seg) 04h : EPG Information (full seg) 05h : EPG Information (one seg) 06h : BML Display
Data 16 :	Tuner0 Physical Channel Number
Data CRC8 :	--

(2) Description

- B/E notify DTV Status to HOST.
- B/E always transfer to HOST CPU in interval of 1s.
- DTV Status information in Data2 will be set to In Service (03h) immediately after sending "Wakeup" from B/E. B/E will not wait for Wakeup Response for setting this.
- If Data2 = "04h: Low Power mode", tune commands, Display EPG command, etc., are not supported.
- If Data2 = "07h: Demo mode ", Only a few commands are supported. See "5.3.17. Set User ID" for the list of supported commands.
- When Audio/Video are outputted, Data6 becomes 01h almost simultaneously with DTV event code becoming 90h.
- Data 6 Video Output:
  - To make host aware that a video output has started after tune/search.
  - From tuning until the start of playback, this will be 00h.
  - If wrong service is found after broadcast station search, then Audio/Video are muted, and DTV Event code becomes 00h. In this case, DTV status Data6 remains 00h.
  - DTV Status Data6 remains 1 during AV mute after DTV Event 90h.
- Refer to the followings about Copy Control.



b7	b6	b5	b4	b3	b2	b1	b0	b7	b6	b5	b4	b3	b2	b1	b0
Data 9								Data 10							

Data 9:

7~6b: 1~2bit (WORD0)
- 00b: Aspect Ratio 4:3
- 01b: Aspect Ratio 4:3 Letterbox
- 10b: Aspect Ratio 16:9
- 11b: Reserved

5~2b: 3~6bit (WORD1)
- 0000b: CGMS-A Information
- 1111b: Reserved

1~0b: 7~8bit (WORD2 (CGMS-A))
- 00b: Copy Freely
- 01b: Not defined
- 10b: Copy One Generation
- 11b: Copy Never

Data 10:

7~6b: 9~10bit (WORD2 (Analog Output Copy Control Information))
- 00b: Copy Freely
- 01b: A Pseudo-sync pulse is present
- 10b: A Pseudo-sync pulse + 2 line inverted and split burst insertion
- 11b: A Pseudo-sync pulse + 4 line inverted and split burst insertion

5~2b: 11~14bit (WORD2)
- These bits shall be left undefined (Logic 0)

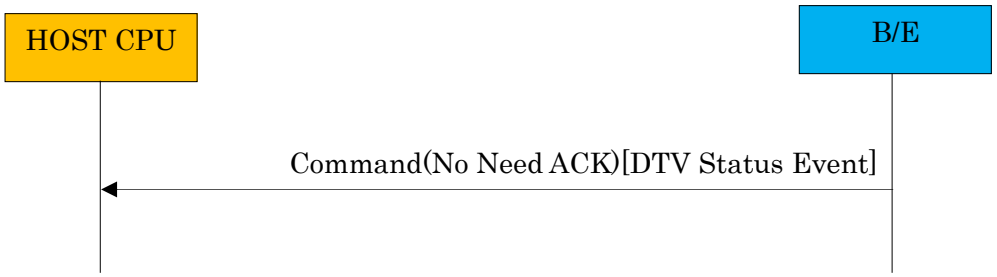
1~0b: Reserved (Logic 0)

*: 15~20bit (CRCC) shall be calculated by HOST CPU

$G(x) = x6+x+1$

- If "Diver Mode" is 00h or 01h, B/E use both tuner0/tuner1.
- Tuner0 is used for Foreground processing, and Tuner1 is used for Background processing.
- "Antenna Level" indicates the number of antenna bars on the display. Antenna Level 0 is bad signal. Antenna Level 3 is good signal. If "Tuning Status (Data3)" is idle, Antenna Level is 0.
- If "EPG Update Status" is 00h, B/E isn't updating EPG of specified channel. But B/E may be updating EPG with back scan. If 01h, B/E is updating forcibly by "5.2.12.Display EPG" command. If finishing to update EPG, "EPG Update Status" change to 00h.
- In the first DTV status event output immediately after booting up, Data2 to Data16 may be filled with initial value (00h). However, B/E is not abnormal state. Please ignore this DTV status event.

(3) Sequence



5.7.2. Scan Status

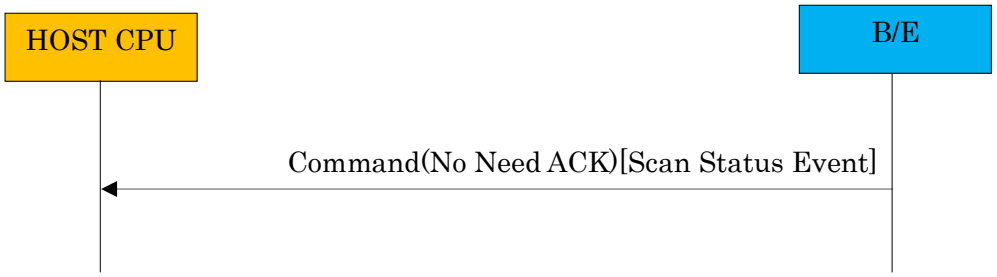
(1) Scan Status (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	01h : Command(No Need Ack)
Request No. :	00h ~ FFh
Data Length :	06h
Header CRC8 :	--
Data 0 :	82h Scan Status(Upper)
Data 1 :	03h Scan Status (Lower)
Data 2 :	Status 01h : notification FFh : end
Data 3 :	Status = 01h 0Dh~34h : Physical Channel Number Status = FFh 00h : Fixed Value
Data 4 :	Status = 01h 00h : No Detection 01h : Detection Status = FFh 00h : No station found or stopped 01h : One or more stations found
Data CRC8 :	--

(2) Description

- B/E notify Scan Status to HOST.
- When changing the physical channel, notify this status (Status=01h).
- When finishing the scan, notify this status (Data2=FFh, Data3=00h, Data4=00h/01h).
- When stopping the scan, notify this status (Data2=FFh, Data3=00h, Data4=00h).

(3) Sequence



5.7.3. Search Status

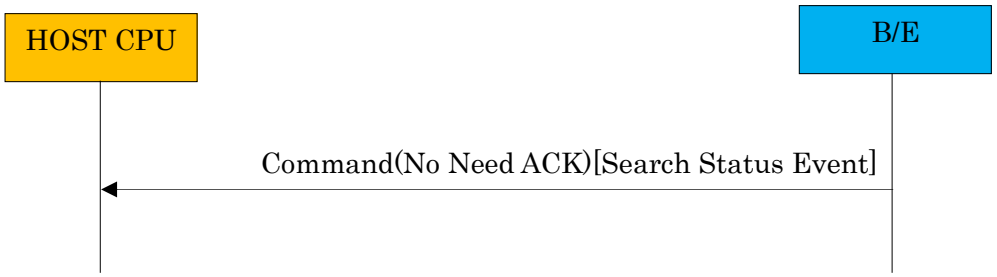
(1) Search Status (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	01h : Command(No Need Ack)
Request No. :	00h ~ FFh
Data Length :	06h
Header CRC8 :	--
Data 0 :	82h Search Status(Upper)
Data 1 :	04h Search Status (Lower)
Data 2 :	Status 01h : notification(Foreground Search) 02h : notification(Broadcast Search – Foreground, Relay station) 03h : notification(Broadcast Search – Foreground, Affiliated station) 04h : notification(Broadcast Search – Foreground, Relay station + Affiliated station) FFh : end (Foreground and Background Search) Background search will be end only when station found successful.
Data 3 :	Status = 01h ~ 04h 0Dh~34h : Physical Channel Number Status = FFh 00h
Data 4 :	Status = 01h ~ 04h 00h : No Detection (Only for foreground search) 01h : Detection Status = FFh 00h : Not find the new channel and return the last channel. 01h : Find the new channel and select new channel. 02h : Select Relay Station(Fore) 03h : Select Affiliated Station(Fore) 04h : Select Relay Station (Back) 05h : Select Affiliated Station (Back)
Data CRC8 :	--

(2) Description

- B/E notify Search/Broadcast Search Status to HOST.
- When changing the physical channel for search, notify this status (Status=01h).
- When changing the physical channel for broadcast search, notify this status (Status=02h).
- When finishing search, notify this status (Data2=FFh, Data3=00h, Data4=00h/01h/02h/03h/04h/05h).
- When stopping search, notify this status (Data2=FFh, Data3=00h, Data4=00h).
- If a background search was performed at the start of the foreground search, the background search will be stopped, but Data 2, FFh will not occur.

(3) Sequence



5.7.4. DTV Event

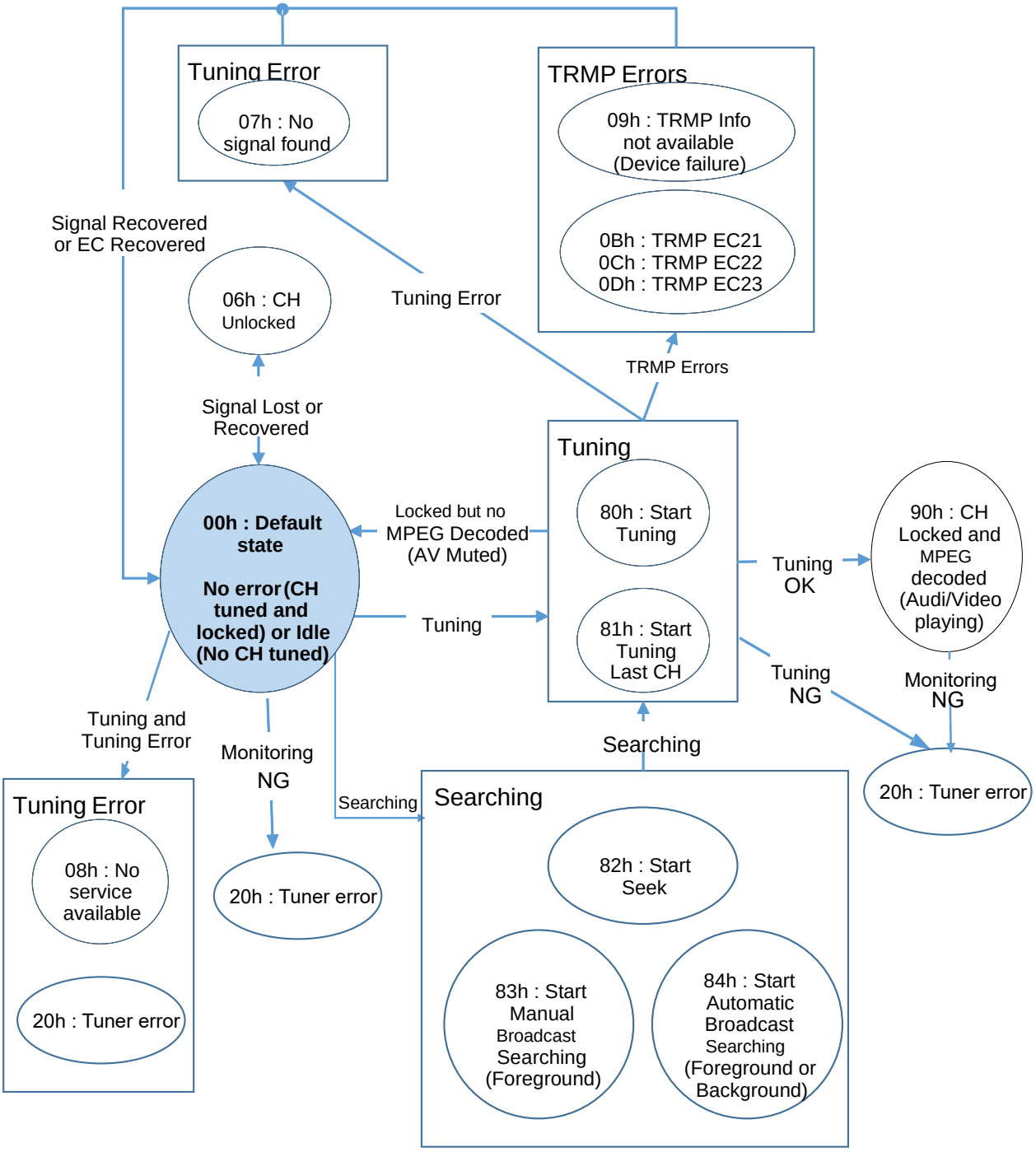
(1) DTV Event Status (BE ⇒ HOST)

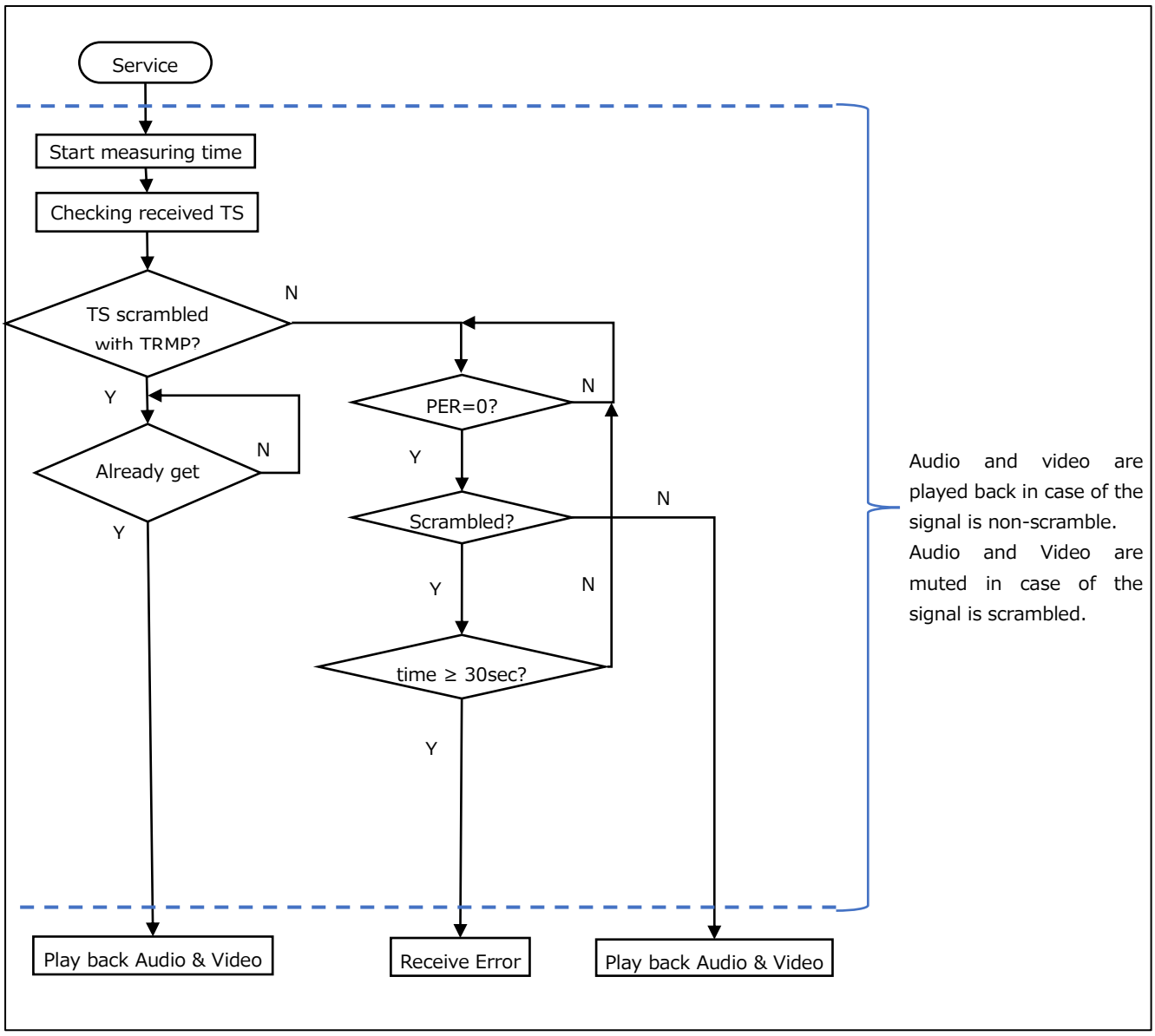
Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	04h
Header CRC8 :	--
Data 0 :	82h DTV Event(Upper)
Data 1 :	06h DTV Event (Lower)
Data 2 :	Event Code Refer to "(2)Description"
Data CRC8 :	--

(2) Description

- B/E notify DTV Event to HOST.
- Refer to the following table and state diagram about Event Code.

Event Code	Content(English)	Content(Japanese)
00h	No Error	エラーなし
06h	Can't receive this channel, because weak signal.	信号レベルが低下している為、このチャンネルは受信できません
07h	Can't receive this channel.	このチャンネルは受信できません
08h	This channel is not being broadcast.	このチャンネルは現在放送されていません
09h	TRMP Information is invalid	受信機情報が不正です
0Bh	RMP Error EC21	受信制御データが設定されていません。しばらくお待ちください（最大で30 秒かかる場合があります）。 コード：EC21
0Ch	RMP Error EC22	このチャンネルは視聴できません。ご覧のチャンネルのカスタマーセンターにお問い合わせください。 コード：EC22
0Dh	RMP Error EC23	この受信機ではご覧になることができません。ご覧のチャンネルのカスタマーセンターへお問い合わせください。 コード：EC23
20h	Tuner error(I2C HW error)	―
80h	Start tuning	―
81h	Start tuning(Last Channel)	―
82h	Start search (Search up/down operation)	―
83h	Start broadcast search(Manually)	―
84h	Start foreground broadcast search (Automatic)	―
90h	Tuning is completed first, and MPEG TS is decoded, then playing audio/video properly.	―

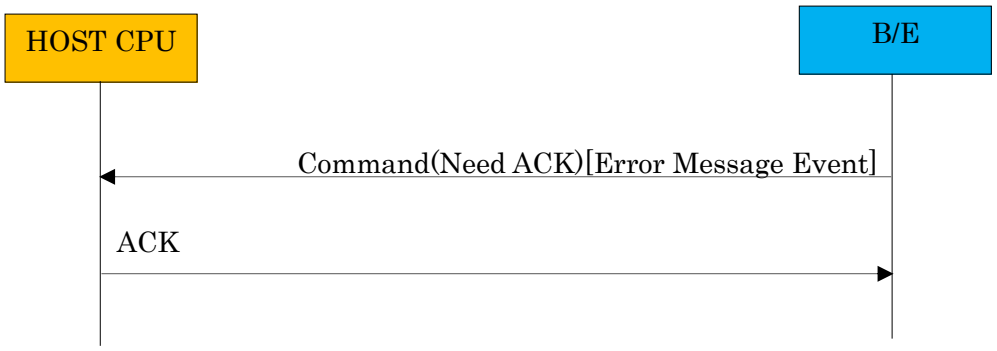

Tuning Error
07h : No signal found
TRMP Errors
09h : TRMP Info not available (Device failure)
0Bh : TRMP EC21
0Ch : TRMP EC22
0Dh : TRMP EC23
Signal Recovered or EC Recovered
06h : CH Unlocked
Tuning Error
TRMP Errors
Signal Lost or Recovered
Tuning
00h : Default state
No error (CH tuned and locked) or Idle (No CH tuned)
Locked but no MPEG Decoded (AV Muted)
80h : Start Tuning
81h : Start Tuning Last CH
Tuning
Tuning OK
90h : CH Locked and MPEG decoded (Audi/Video playing)
Tuning NG
Monitoring NG
Tuning and Tuning Error
Monitoring NG
Searching
Searching
Searching
Tuning Error
08h : No service available
20h : Tuner error
20h : Tuner error
82h : Start Seek
83h : Start Manual Broadcast Searching (Foreground)
84h : Start Automatic Broadcast Searching (Foreground or Background)
20h : Tuner error



Non-TRMP/Non-scrambled' TS flow

- In case of DTV Event code 0b/0c/0dh occurs, B/E change to 1seg if auto segment was used. If B/E receives TRMP key from Transport stream then switch back to 12seg.
- Start tuning code (80h): It occurs at Foreground Tuning.
- Start tuning (Last Channel) code (81h): It occurs at Foreground Tuning.
- Start search (82h): It occurs at Foreground Search.
- Start broadcast search (Manually) (83h): It occurs at Foreground Search.
- Start broadcast search (Automatic) (84h): It occurs at Foreground Search and Background Search.
- Complete tuning (90h): It occurs at Foreground Tuning. In this case, DTV status Data6 becomes 01h almost simultaneously.
- No Error or Idle status (00h): If wrong service is found after broadcast station search, then Audio/Video are muted, and DTV Event code becomes 00h. In this case, DTV status Data6 remains 00h.
- TRMP Information is invalid (09h): It occurs at after Tuning.
    - ⑥ Start tuning (80h) or Start tuning (Last Channel) (81h).
    - ⑦ In case of error: No Error (00h).
    - ⑧ TRMP Information is invalid (09h).
- The case of "Locked but no MPEG Decoded (AV muted)" is only for signal generator use case.
- If an error occurs during I2C communication, it is considered as Monitoring NG.
- When I2C error (20h) happed, Host needs to reset (pin assert) the module.
-

(3) Sequence



5.7.5. EWS Event

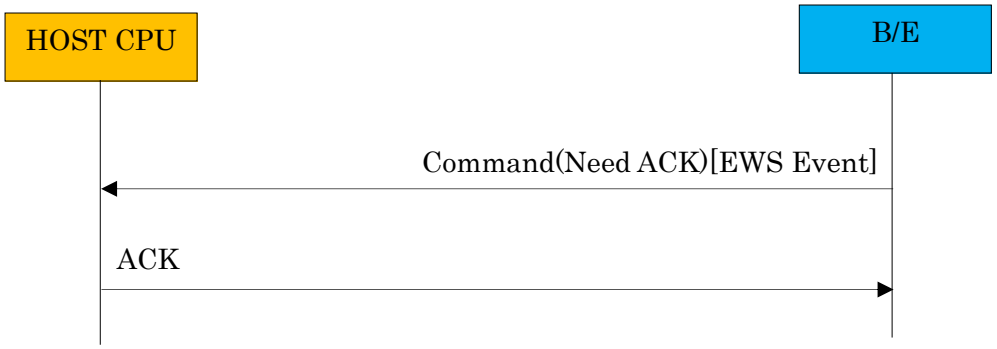
(1) EWS Event (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	06h
Header CRC8 :	--
Data 0 :	82h EWS Event(Upper)
Data 1 :	08h EWS Event (Lower)
Data 2 :	Status 00h : None 01h : Signal Level1 02h : Signal Level2 03h : Test
Data 3 :	Three Digit Number(Upper) Refer to "(2) Description"
Data 4 :	Three Digit Number(Lower) Refer to "(2) Description"
Data CRC8 :	--

(2) Description

- B/E notify EWS Event to HOST.
- If Data 2 is 01h, HU automatically send "5.2.8.Direct Channel Select".
- If Data 2 is 02h or 03h, HU display EWS Confirmation Screen. When user select "yes", HU send "5.2.8.Direct Channel Select".
- Use same branch number that you are currently watching.
- Even when RMP error code is being output in 12-seg fixed mode, 'EWS event' can be sent if EWS event occurs.
- When EWS event ended, Data 2 will be 00h and send to Host.

(3) Sequence



5.7.6. Program Information

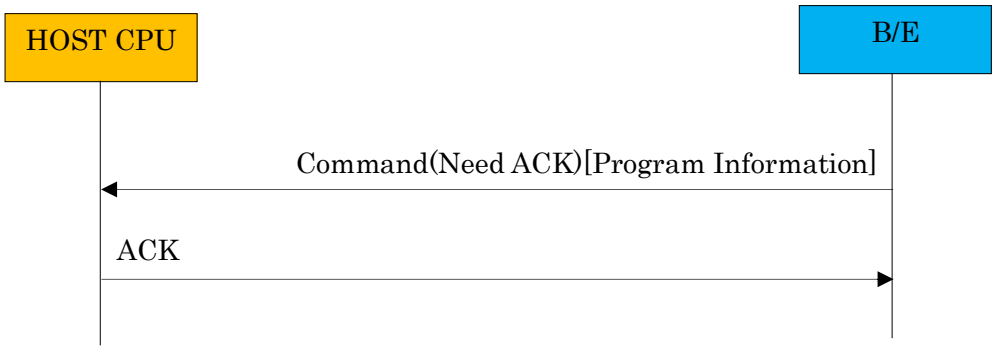
(1) Program Information (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	0Ah~7Eh
Header CRC8 :	--
Data 0 :	82h Program Information (Upper)
Data 1 :	09h Program Information (Lower)
Data 2 :	Preset Number 00h : Preset No.1(Home) 01h : Preset No.2(Drive) FFh：None
Data 3 :	00h : Not stored service 01h~0Ch : Preset Channel Number FFh：None
Data 4 :	Three Digit Number(Upper)
Data 5 :	Three Digit Number(Lower)
Data 6 :	Branch Identifier Number
Data 7 :	Length of Station Name(N1) : 00h~14h
Data 8 ~7+N1 :	Station Name(TS Name)
Data 8+N1 :	Length of Program Name(N2) : 00h~60h
Data 9+N1~8+N1+N2:	Program Name
Data CRC8 :	--

(2) Description

- B/E notify the program information.
- The program information contains Preset Channel Number / Station Name / Program name).
- Preset Channel Number is 01h~0Ch. If current playback channel is not stored, Data3 is 00h and Data2 is FFh.
- Three Digit Number & Branch Identifier Number are current playback service. If current playback channel is not stored, these data are set.
- The length of Station Name is 00h~14h.
- The format of Station Name is 8bit character code set.
- The length of Program Name is 00h~60h.
- The format of Program name is 8bit character code set.
- If B/E doesn't have Program name, Length of Program Name is set to 0.
- When any information is updated, B/E will notify this event.

(3) Sequence



5.7.7. EPG Status

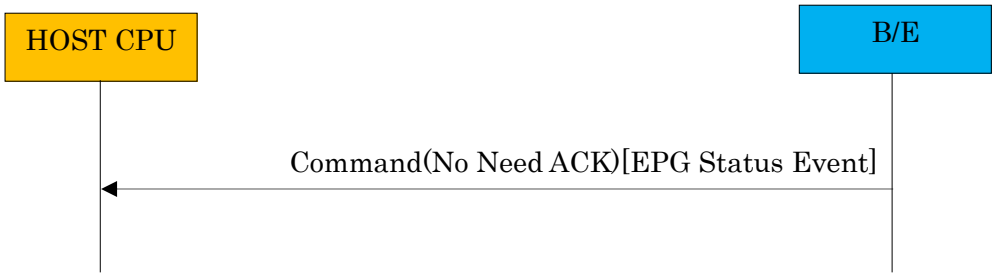
(1) EPG Status (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	01h : Command(No Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	82h EPG Status(Upper)
Data 1 :	0Ah EPG Status (Lower)
Data 2 :	Status 01h : Updating FFh : end
Data 3 :	Status=01h 00h : Fixed Value Status=FFh 00h : Complete to update EPG 01h : Cancel by user
Data CRC8 :	--

(2) Description

- B/E notify EPG Status (Data2=01h) to HOST during EPG Updating.
- When finishing EPG Updating, B/E notify status(Status=FFh)
- If completing to update EPG for all channel in the preset table, Data3 is 00h.
- If cancel by user during updating EPG, Data3 is 01h.
- EPG update will be done at 20s/Ch.

(3) Sequence



5.7.8. Event relay

(1) Event relay (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	09h
Header CRC8 :	--
Data 0 :	82h Event relay(Upper)
Data 1 :	0Bh Event relay (Lower)
Data 2 :	Status 01h – 1Eh (before 1[s] to 30[s]) FFh : end (Event relay done)
Data 3 :	Relay Three Digit Number(Upper)
Data 4 :	Relay Three Digit Number(Lower)
Data 5 :	Relay Branch Identifier Number
Data : 6	Relay Service ID(Upper)
Data : 7	Relay Service ID(Lower)
Data CRC8 :	--

(2) Description

- Data2 : Number of second until switching service.
- Data2 = FFh : Event relay done

(3) Sequence



5.7.9. Number of Program info

(1) Number of Program info(BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	35h
Header CRC8 :	--
Data 0 :	82h Number of Program info (Upper)
Data 1 :	0Ch Number of Program info (Lower)
Data 2 :	Not Stored Channel (Fullseg) : Number of Program info Bit7 : reserved for unknown state Bit6-4 : Notification ID Bit3-0 : Number of Program (0:Not Active) ※FFh unknown
Data 3 :	Not Stored Channel (Oneseg) : Number of Program info
Data 4 :	Home Preset no.1(Fullseg) : Number of Program info
Data 5 :	Home Preset no.2(Fullseg) : Number of Program info
Data 6 :	Home Preset no.3(Fullseg) : Number of Program info
Data 7 :	Home Preset no.4(Fullseg) : Number of Program info
Data 8 :	Home Preset no.5(Fullseg) : Number of Program info
Data 9 :	Home Preset no.6(Fullseg) : Number of Program info
Data 10 :	Home Preset no.7(Fullseg) : Number of Program info
Data 11 :	Home Preset no.8(Fullseg) : Number of Program info
Data 12 :	Home Preset no.9(Fullseg) : Number of Program info
Data 13 :	Home Preset no.10(Fullseg) : Number of Program info
Data 14 :	Home Preset no.11(Fullseg) : Number of Program info
Data 15 :	Home Preset no.12(Fullseg) : Number of Program info
Data 16 :	Home Preset no.1(Oneseg) : Number of Program info
Data 17 :	Home Preset no.2(Oneseg) : Number of Program info
Data 18 :	Home Preset no.3(Oneseg) : Number of Program info
Data 19 :	Home Preset no.4(Oneseg) : Number of Program info
Data 20 :	Home Preset no.5(Oneseg) : Number of Program info
Data 21 :	Home Preset no.6(Oneseg) : Number of Program info
Data 22 :	Home Preset no.7(Oneseg) : Number of Program info
Data 23 :	Home Preset no.8(Oneseg) : Number of Program info
Data 24 :	Home Preset no.9(Oneseg) : Number of Program info
Data 25 :	Home Preset no.10(Oneseg) : Number of Program info
Data 26 :	Home Preset no.11(Oneseg) : Number of Program info
Data 27 :	Home Preset no.12(Oneseg) : Number of Program info
Data 28 :	Travel Preset no.1(Fullseg) : Number of Program info
Data 29 :	Travel Preset no.2(Fullseg) : Number of Program info
Data 30 :	Travel Preset no.3(Fullseg) : Number of Program info
Data 31 :	Travel Preset no.4(Fullseg) : Number of Program info
Data 32 :	Travel Preset no.5(Fullseg) : Number of Program info
Data 33 :	Travel Preset no.6(Fullseg) : Number of Program info
Data 34 :	Travel Preset no.7(Fullseg) : Number of Program info
Data 35 :	Travel Preset no.8(Fullseg) : Number of Program info
Data 36 :	Travel Preset no.9(Fullseg) : Number of Program info
Data 37 :	Travel Preset no.10(Fullseg) : Number of Program info
Data 38 :	Travel Preset no.11(Fullseg) : Number of Program info
Data 39 :	Travel Preset no.12(Fullseg) : Number of Program info
Data 40 :	Travel Preset no.1(Oneseg) : Number of Program info
Data 41 :	Travel Preset no.2(Oneseg) : Number of Program info
Data 42 :	Travel Preset no.3(Oneseg) : Number of Program info
Data 43 :	Travel Preset no.4(Oneseg) : Number of Program info
Data 44 :	Travel Preset no.5(Oneseg) : Number of Program info
Data 45 :	Travel Preset no.6(Oneseg) : Number of Program info

Data 46 :	Travel Preset no.7(Oneseg) : Number of Program info
Data 47 :	Travel Preset no.8(Oneseg) : Number of Program info
Data 48 :	Travel Preset no.9(Oneseg) : Number of Program info
Data 49 :	Travel Preset no.10(Oneseg) : Number of Program info
Data 50 :	Travel Preset no.11(Oneseg) : Number of Program info
Data 51 :	Travel Preset no.12(Oneseg) : Number of Program info
Data CRC8 :	--

(2) Description

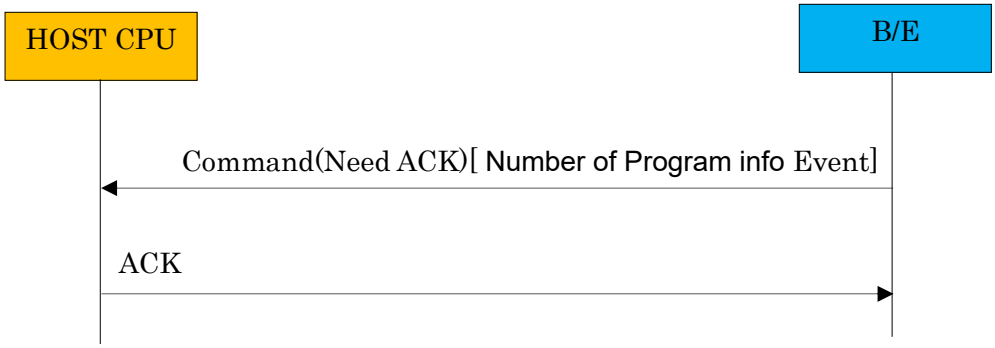
- Get current time Number of programs per Channel.
- Number of Program info = FFh : Haven't got current program info yet.
- Data2 ~ Data51 :

  Bit3-0 – Number of Program Info

  Bit6-4 – Notification ID, incremented (0 to 7) whenever there is an update.

  Bit7 – Set when information is unavailable.
- Send only when there is a change in the data

(3) Sequence



5.7.10. Program Information with UTF-8

(1) Program Information with UTF-8 (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	0Ah~
Header CRC8 :	--
Data 0 :	82h Program Information with UTF-8(Upper)
Data 1 :	0Dh Program Information with UTF-8(Lower)
Data 2 :	Fragment Number
Data 3 :	Last Fragment Number
Data 4 :	Preset Number 00h : Preset No.1(Home) 01h : Preset No.2(Drive) FFh：None
Data 5 :	00h : Not stored service 01h~0Ch : Preset Channel Number FFh：None
Data 6 :	Three Digit Number(Upper)
Data 7 :	Three Digit Number(Lower)
Data 8 :	Branch Identifier Number
Data 9 :	Length of Station Name(N1) : 00h~50h
Data10~9+N1 :	Station Name(TS Name)
Data 10+N1 :	Length of Program Name(N2)(Upper) : 00h~180h
Data 11+N1 :	Length of Program Name(N2)(Lower) : 00h~180h
Data 12+N1~11+N1+N2:	Program Name
Data CRC8 :	--

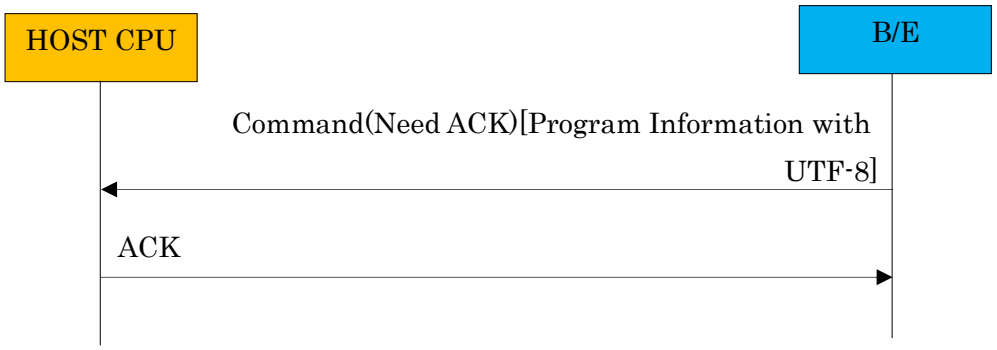
(2) Program Information with UTF-8 (BE ⇒ HOST with Header Type 2)

Start Sync :	A5h,59h
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length(2 Bytes) :	000Ah ~ 01DDh
Header CRC8 :	--
Data 0 :	82h Program Information with UTF-8(Upper)
Data 1 :	0Dh Program Information with UTF-8(Lower)
Data 2 :	Fragment Number (00h : Fixed Value)
Data 3 :	Last Fragment Number (00h : Fixed Value)
Data 4 :	Preset Number 00h : Preset No.1(Home) 01h : Preset No.2(Drive) FFh : None
Data 5 :	00h : Not stored service 01h~0Ch : Preset Channel Number FFh : None
Data 6 :	Three Digit Number(Upper)
Data 7 :	Three Digit Number(Lower)
Data 8 :	Branch Identifier Number
Data 9 :	Length of Station Name(N1) : 00h~50h
Data 10~9+N1 :	Station Name(TS Name)
Data 10+N1 :	Length of Program Name(N2)(Upper) : 00h~180h
Data 11+N1 :	Length of Program Name(N2)(Lower) : 00h~180h
Data 12+N1~11+N1+N2:	Program Name
Data CRC8 :	--

(3) Description

- B/E notify the program information.
- The program information contains Preset Channel Number / Station Name / Program name).
- Preset Channel Number is 01h~0Ch. If current playback channel is not stored, Data3 is 00h and Data2 is FFh.
- Three Digit Number & Branch Identifier Number are current playback service. If current playback channel is not stored, these data are set.
- If B/E doesn't have Program name, Length of Program Name is set to 0.
- When any information is updated, B/E will notify this event.
- For Header Type 2, Fragment Numbers - Data2, Data3 are set to 00h.

(4) Sequence


HOST CPU
B/E
Command(Need ACK)[Program Information with UTF-8]
ACK

5.7.11. TS Name Update

(1) TS Name Update (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h～
Header CRC8 :	--
Data 0 :	82h TS Name Update (Upper)
Data 1 :	0Eh TS Name Update (Lower)
Data 2 :	Fragment Number
Data 3 :	Last Fragment Number
Data 4 :	Preset Number 00h : Preset No.1(Home) 01h : Preset No.2(Drive)
Data 5 ~ 24:	TS Name Not Stored channel
Data 25 ~ :	TS Name Channel 1 to 12 (20byte per channel)
Data CRC8 :	--

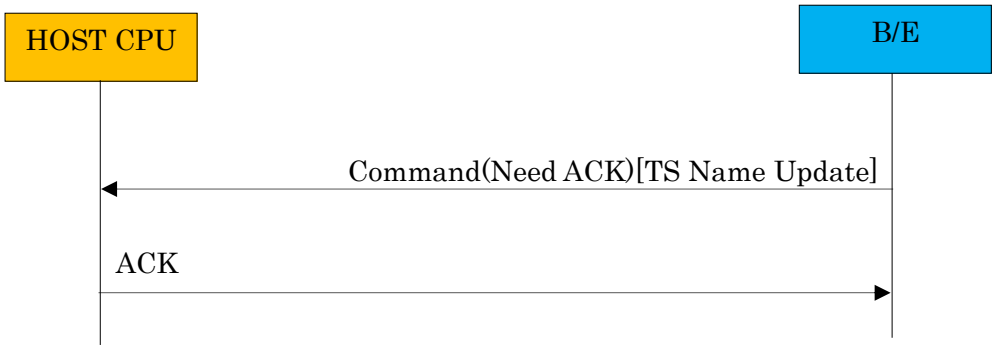
(2) TS Name Update (BE ⇒ HOST with Header Type 2)

Start Sync :	A5h,59h
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length(2 Bytes)	010Ah
Header CRC8 :	--
Data 0 :	82h TS Name Update (Upper)
Data 1 :	0Eh TS Name Update (Lower)
Data 2 :	Fragment Number (00h : Fixed Value)
Data 3 :	Last Fragment Number (00h : Fixed Value)
Data 4 :	Preset Number 00h : Preset No.1(Home) 01h : Preset No.2(Drive)
Data 5 ~ 24:	TS Name Not Stored channel
Data 25 ~ :	TS Name Channel 1 to 12 (20byte per channel)
Data CRC8 :	--

(3) Description

- TS Name with ARIB 8bit character code set.
- TS Name Update
- For Header Type 2, Fragment Numbers - Data2, Data3 are set to 00h.

(4) Sequence



5.7.12. TS Name Update with UTF-8

(1) TS Name Update with UTF-8 (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h～
Header CRC8 :	--
Data 0 :	82h TS Name Update with UTF-8 (Upper)
Data 1 :	0Fh TS Name Update with UTF-8 (Lower)
Data 2 :	Fragment Number
Data 3 :	Last Fragment Number
Data 4 :	Preset Number 00h : Preset No.1(Home) 01h : Preset No.2(Drive)
Data 5 ~ 84:	TS Name Not Stored channel
Data 85 ~ :	TS Name Channel 1 to 12 (80byte per channel)
Data CRC8 :	--

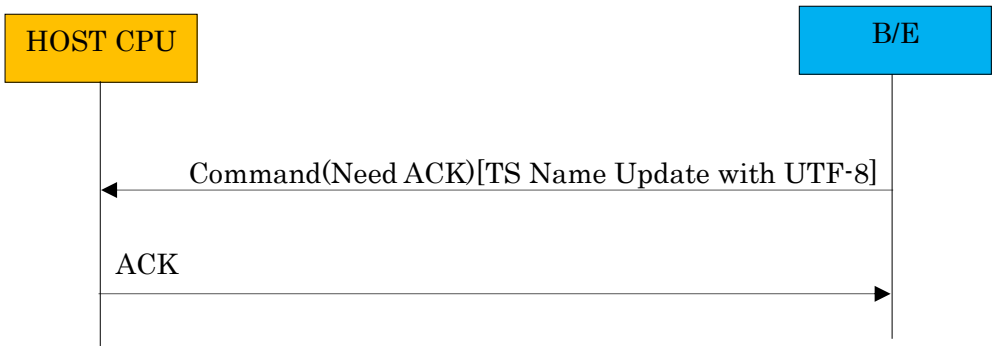
(2) TS Name Update with UTF-8 (BE ⇒ HOST with Header Type 2)

Start Sync :	A5h,59h
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length(2 Bytes) :	0416h
Header CRC8 :	--
Data 0 :	82h TS Name Update with UTF-8 (Upper)
Data 1 :	0Fh TS Name Update with UTF-8 (Lower)
Data 2 :	Fragment Number (00h : Fixed Value)
Data 3 :	Last Fragment Number (00h : Fixed Value)
Data 4 :	Preset Number 00h : Preset No.1(Home) 01h : Preset No.2(Drive)
Data 5 ~ 84:	TS Name Not Stored channel
Data 85 ~ :	TS Name Channel 1 to 12 (80byte per channel)
Data CRC8 :	--

(3) Description

- TS Name with UTF-8.
- TS Name Update
- For Header Type 2, Fragment Numbers - Data2, Data3 are set to 00h.

(4) Sequence



5.7.13. Logo Update

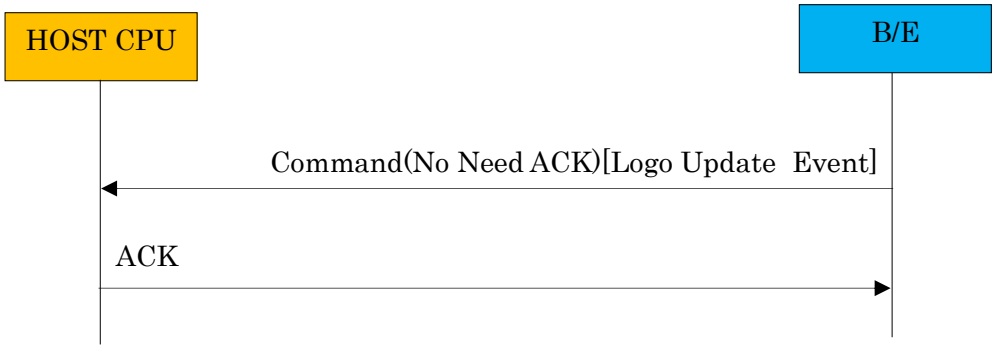
(1) Logo Update (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	08h
Header CRC8 :	--
Data 0 :	82h Logo Update(Upper)
Data 1 :	10h Logo Update (Lower)
Data 2 :	Service ID(Upper)
Data 3 :	Service ID(Lower)
Data 4 :	Logo Type 00h to 05h
Data 5 :	Logo Version (Upper)
Data 6 :	Logo Version (Lower)
Data CRC8 :	--

(2) Description

- Logo Type 6/7 : Not Event Send (Oneseg Logo Include Service Name)
- If the logo data is shared, the event for the shared service will occur.

(3) Sequence



5.7.14. EPG Button Status

(1) EPG Button Status (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	0Fh
Header CRC8 :	--
Data 0 :	82h EPG Button Status (Upper)
Data 1 :	11h EPG Button Status (Lower)
Data 2 :	Up Arrow 0: invalid 1:valid
Data 3 :	Down Arrow 0: invalid 1:valid
Data 4 :	Left Arrow 0: invalid 1:valid
Data 5 :	Right Arrow 0: invalid 1:valid
Data 6 :	Back 0: invalid 1:valid
Data 7 :	Enter 0: invalid 1:valid
Data 8 :	Switch segment 0: invalid 1:valid
Data 9 :	Update 0: invalid 1:valid
Data 10 :	Previous Day(Blue Button) 0: invalid 1:valid
Data 11 :	Next Day(Red Button) 0: invalid 1:valid
Data 12 :	Multi service(Green Button) 0: invalid 1:valid
Data 13 :	Change Scale(Yellow Button) 0: invalid 1:valid
Data CRC8 :	—

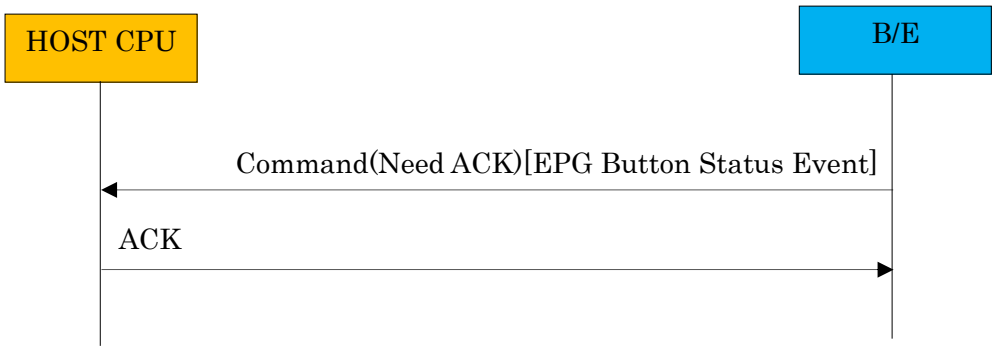
(2) Description

・EPG Button status

・Notifies when EPG is displayed and a change in button status is required.

・This event notification will be started only after B/E receives ACK for Display EPG Response.

(3) Sequence



5.7.15. BML Status

(1) BML Status (BE ⇒ HOST)

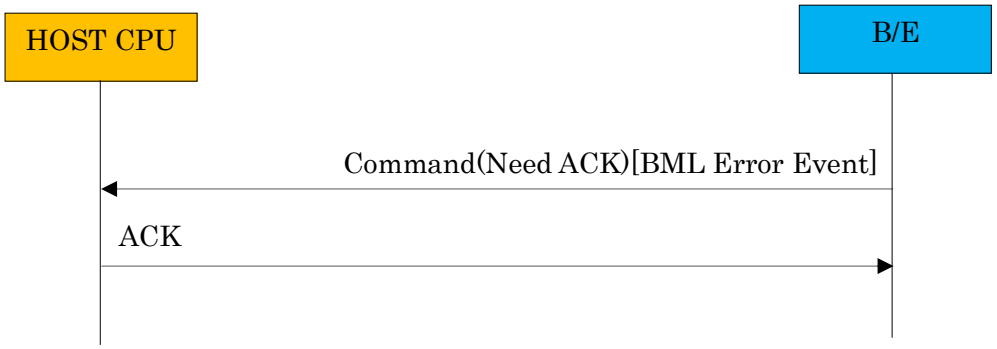
Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	04h
Header CRC8 :	--
Data 0 :	82h BML Status (Upper)
Data 1 :	12h BML Status (Lower)
Data 2 :	BML Status Code
Data CRC8 :	—

(2) Description

・Status Code

Status Code	Content	
B0h	The BML document cannot be obtained.	
B1h	The BML engine of the receiver does not support the version of the BML document obtained.	
B2h	During the display of the contents, an execution error has occurred, and external reference data cannot be obtained.	
B3h	BML failed to start or not supported.	
B4h	BML data not ready. This is raised when the BML cannot be displayed by pressing the 'd' button.	
B5h	BML data ready. This is raised when BML started to be displayed by pressing the 'd' button.	
B6h	BML not available in current service. This is raised when 'd' button presses in case of receiving Oneseg or Fullseg without BML support.	

(3) Sequence



5.7.16. C/N Information

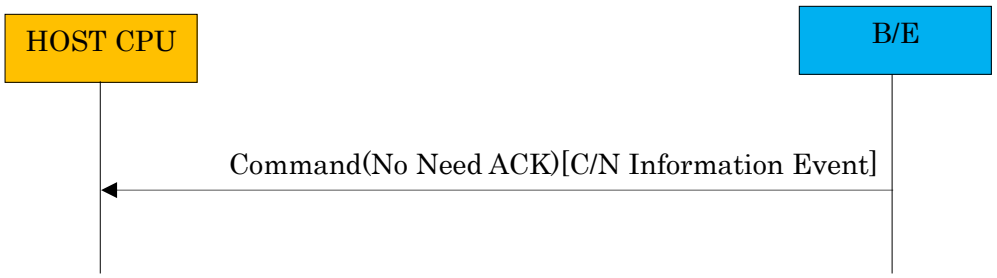
(1) C/N Information (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	01h : Command(No Need Ack)
Request No. :	00h ~ FFh
Data Length :	11h
Header CRC8 :	--
Data 0 :	82h C/N Information (Upper)
Data 1 :	13h C/N Information (Lower)
Data 2 :	Fore physical channel
Data 3-4 :	Fore Diversity C/N
Data 5 :	Back physical channel
Data 6-7 :	Back Diversity C/N
Data 8-9 :	Br0 C/N
Data 10-11 :	Br1 C/N
Data 12-13 :	Br2 C/N
Data 14-15 :	Br3 C/N
Data CRC8 :	—

(2) Description

- B/E notifies the C/N Levels to Host.
- Reports the foreground and background Physical Channel Numbers and C/N levels.
- Notification interval is 1s.
- This event can be enabled/disabled with "Set Event Mask"

(3) Sequence



5.7.17. EPG Scan Result

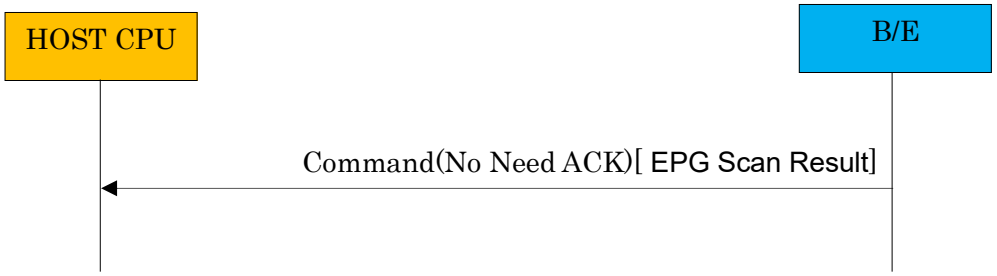
(1) EPG Scan Result (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	01h : Command(No Need Ack)
Request No. :	00h ~ FFh
Data Length :	99h
Header CRC8 :	--
Data 0 :	82h EPG Scan Result (Upper)
Data 1 :	14h EPG Scan Result (Lower)
Data 2 :	Not Stored Channel Fullseg Service 0/1 XYh : X = Service 0 EPG Days Y = Service 1 EPG Days
Data 3 :	Not Stored Channel Fullseg Service 2/3 XYh : X = Service 2 EPG Days Y = Service 3 EPG Days
Data 4 :	Not Stored Channel Fullseg Service 4/5 XYh : X = Service 4 EPG Days Y = Service 5 EPG Days
Data 5 :	Not Stored Channel Fullseg Service 6/7 XYh : X = Service 6 EPG Days Y = Service 7 EPG Days
Data 6 :	Not Stored Channel Oneseg Service 0/1 XYh : X = Service 0 EPG Days Y = Service 1 EPG Days
Data 7 :	Not Stored Channel Oneseg Service 2/Reserve XYh : X = Service 2 EPG Days Y = Reserve
Data 8 :	Home Channel 1 Fullseg Service 0/1
Data 9 :	Home Channel 1 Fullseg Service 2/3
Data 10 :	Home Channel 1 Fullseg Service 4/5
Data 11 :	Home Channel 1 Fullseg Service 6/7
Data 12 :	Home Channel 1 Oneseg Service 0/1
Data 13 :	Home Channel 1 Oneseg Service 2/ Reserve
Data 14 - Data 73 :	・・・(Home Channel 2 —＞ Home Channel 11)
Data 74 :	Home Channel 12 Fullseg Service 0/1
Data 75 :	Home Channel 12 Fullseg Service 2/3
Data 76 :	Home Channel 12 Fullseg Service 4/5
Data 77 :	Home Channel 12 Fullseg Service 6/7
Data 78 :	Home Channel 12 Oneseg Service 0/1
Data 79 :	Home Channel 12 Oneseg Service 2/ Reserve
Data 80 :	Driver Channel 1 Fullseg Service 0/1
Data 81 :	Driver Channel 1 Fullseg Service 2/3
Data 82 :	Driver Channel 1 Fullseg Service 4/5
Data 83 :	Driver Channel 1 Fullseg Service 6/7
Data 84 :	Driver Channel 1 Oneseg Service 0/1
Data 85 :	Driver Channel 1 Oneseg Service 2/ Reserve
Data 86 – Data 145 :	・・・(Driver Channel 2 —＞ Driver Channel 11)
Data 146 :	Driver Channel 12 Fullseg Service 0/1
Data 147 :	Driver Channel 12 Fullseg Service 2/3
Data 148 :	Driver Channel 12 Fullseg Service 4/5
Data 149 :	Driver Channel 12 Fullseg Service 6/7

Data 150 : Driver Channel 12 Oneseg Service 0/1
Data 151 : Driver Channel 12 Oneseg Service 2/ Reserve
Data CRC8 : —

(2) Description

(3) Sequence



5.7.18. DTV Critical Error

(1) DTV Critical Error Event (BE ⇒ HOST)

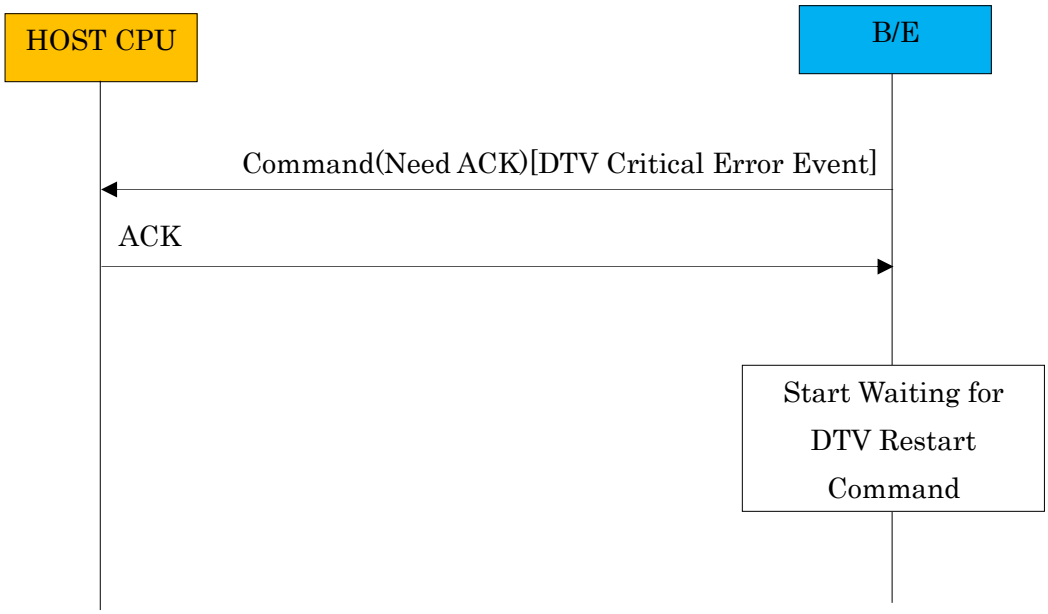
Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	06h
Header CRC8 :	--
Data 0 :	83h DTV Critical Error (Upper)
Data 1 :	01h DTV Critical Error (Lower)
Data 2 :	Error Type 01h : Hangup 02h : Termination
Data 3 :	Module Information Error Type 01h 01h : Audio Decoder 02h : Audio Output 03h : Video Decoder 04h : Video Output 05h : Deinterlacer 06h : Video Scaler 07h : Demux 08h : Timer Error Type 02h 00h : Fixed Value
Data 4 :	Action Request 01h : DTV Restart 02h : DTV Module Reset
Data CRC8 :	—

(2) Description

- BE notifies the host of any Critical Errors.
- Critical Error implies application termination or hangup of any DTV threads.
- Data2 indicates the type of error – Hangup of any threads or Termination.
- Data3 gives a hint about the module which caused the issue.
- Data3 is valid only for Hangup.
- Data4 indicates the action to be taken by the host.
  - If it is 01h (DTV Restart), host may send a command (Refer 5.2.18 DTV Restart) to BE to do an application restart.
  - Alternatively, the host may do the DTV Module Reset.
- For hangup of threads, the notification will be issued within 3 Seconds of occurrence and for unexpected termination, it will be done immediately after the occurrence.
- Upon receiving this event, host should stop sending other commands to BE.
  - After this event BE will accept only the DTV Restart Command (Refer 5.2.18 DTV Restart).
  - No response will be sent for Watch Dog Command.
  - Other commands will be responded with 0Ah(Invalid Status).
- Before sending this event, BE will save the last channel information and after reset,

the information will be used to tune to last channel.

(3) Sequence



5.8. SW Update Command

5.8.1. SW Update

(1) SW Update Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	04h~
Header CRC8 :	--
Data 0 :	A0h SW Update (Upper)
Data 1 :	01h SW Update (Lower)
Data 2 :	Operation 00h : SW Update start
Data CRC8 :	--

(2) SW Update Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	A0h SW Update (Upper)
Data 1 :	01h SW Update (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- B/E start SW Update.
- SW updating transfer update files with YMODEM Protocol.
- B/E wait 100msec after receiving ACK of Response.
- HOST shouldn't send any commands during YMODEM Protocol.
- The new app replaces the older one of two apps in the app partition.
- Non-application data remains unchanged.

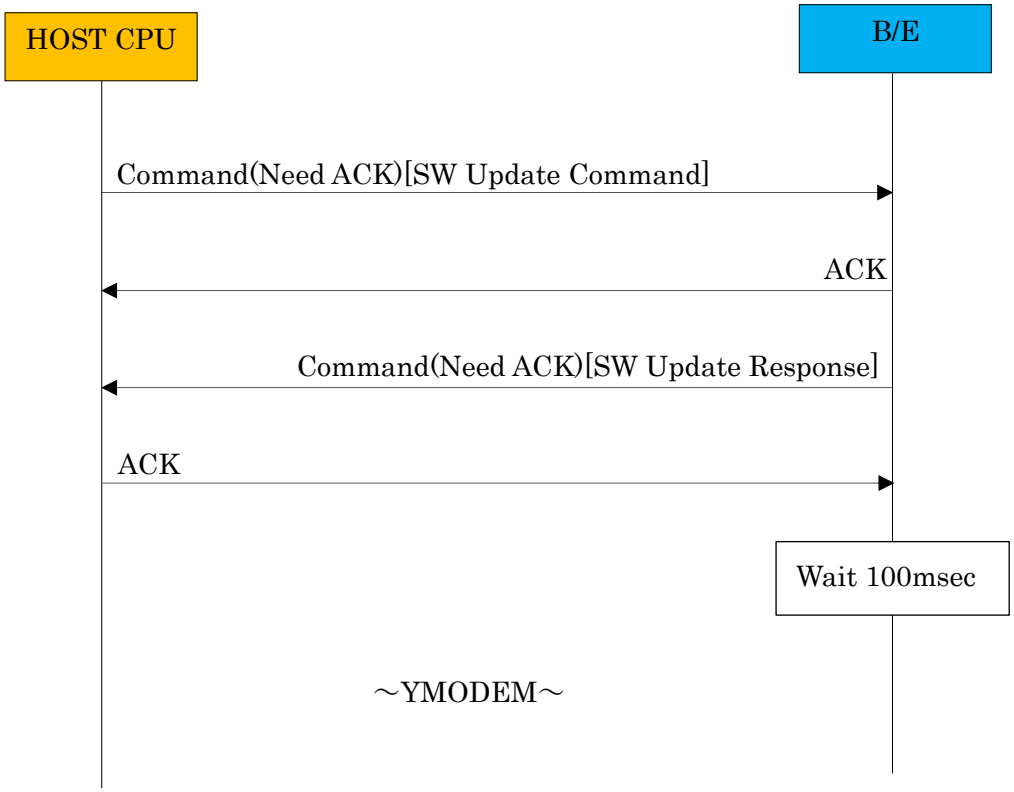
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.

・ Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
21h(Disk Full)	There is not enough space in disk.
FFh(Unknown Error)	unknown error

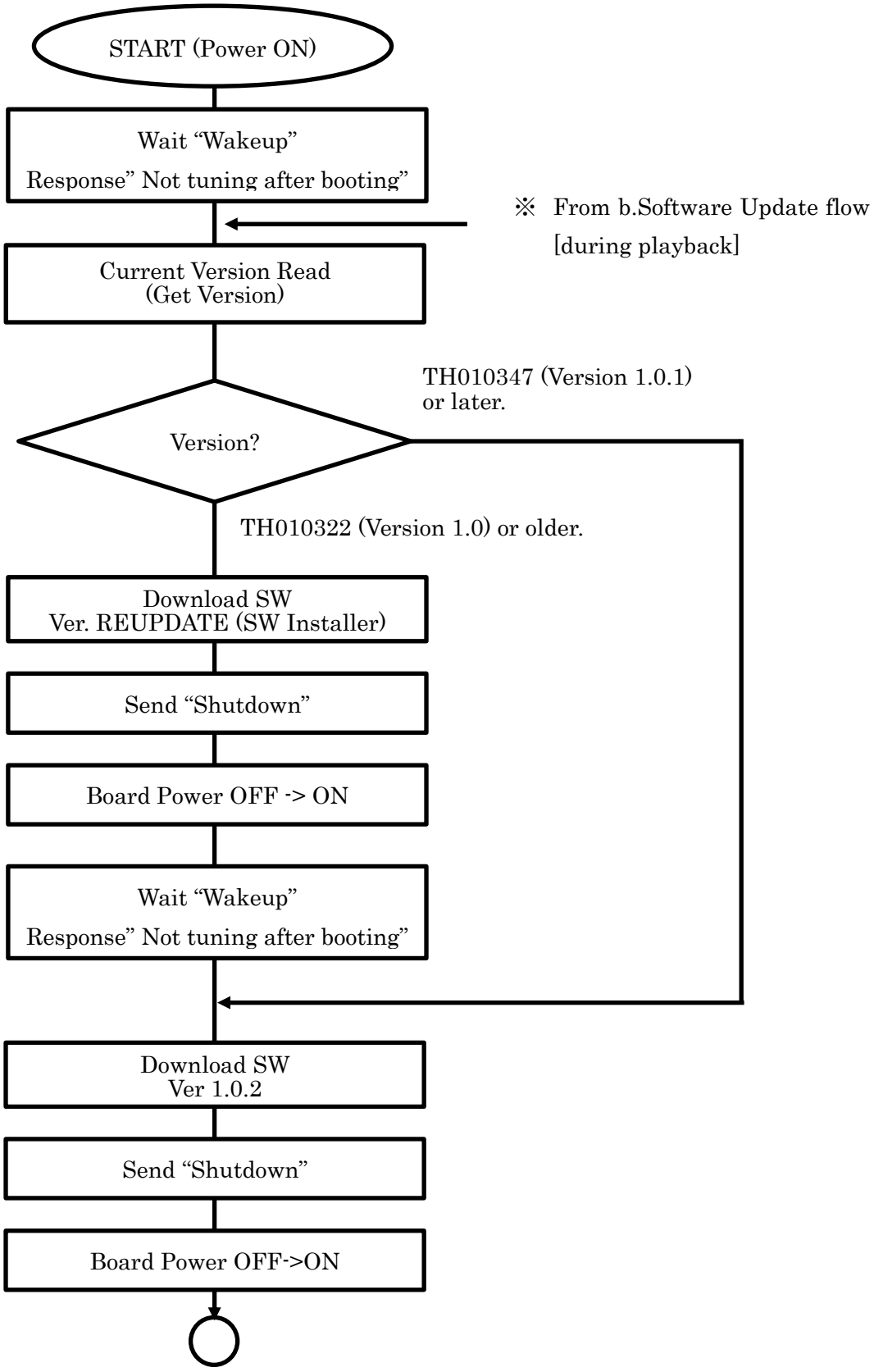
(5) Sequence

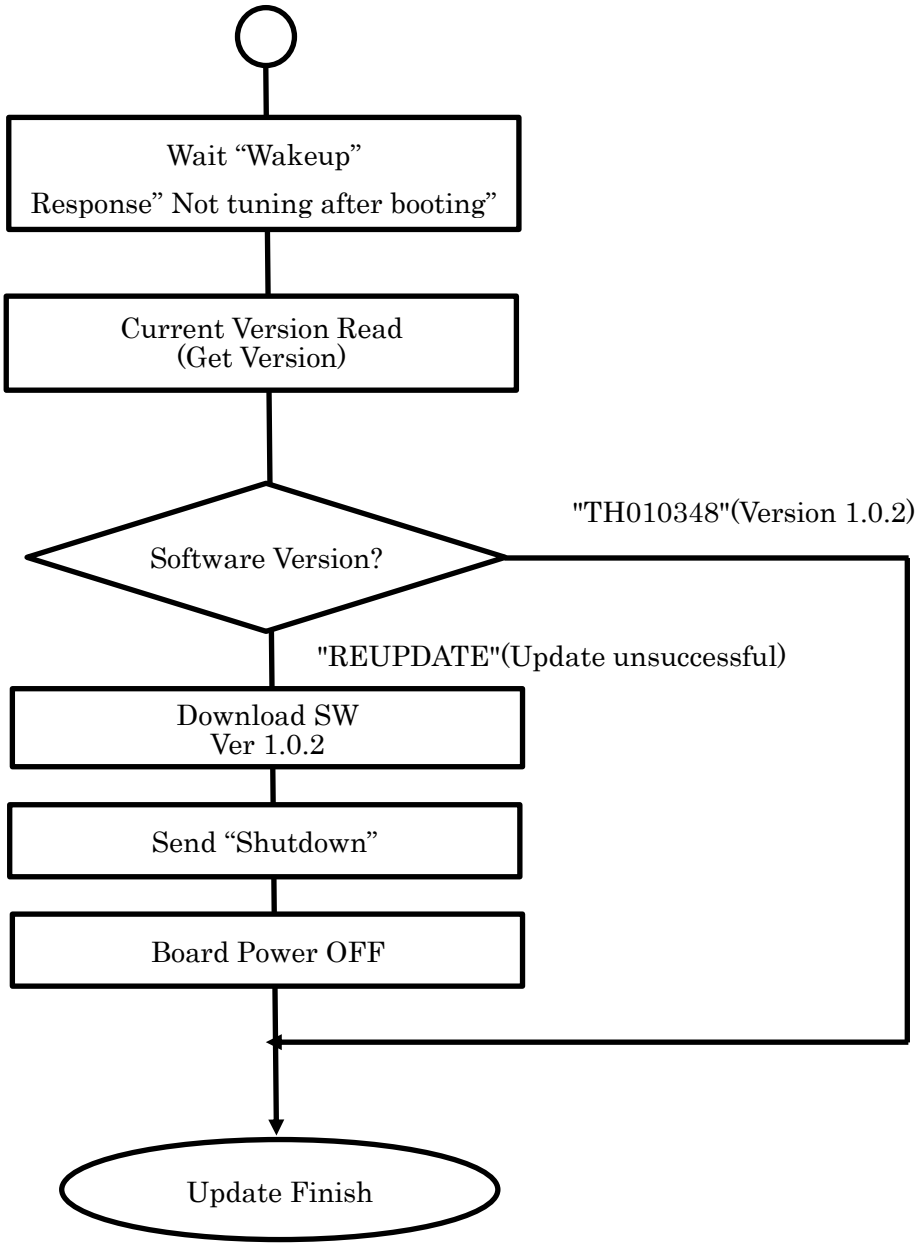


(6) Appendix

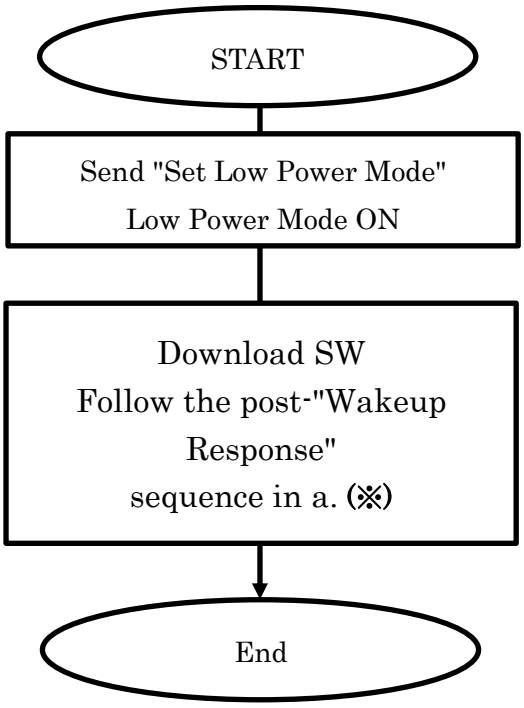
a. Software Update flow

Precondition: Keep the RF connections disconnected during the whole update process.




Wait "Wakeup"
Response" Not tuning after booting"
Current Version Read
(Get Version)
Software Version?
"TH010348"(Version 1.0.2)
"REUPDATE"(Update unsuccessful)
Download SW
Ver 1.0.2
Send "Shutdown"
Board Power OFF
Update Finish

b. Software Update flow [during playback]



5.8.2. SW Update Status

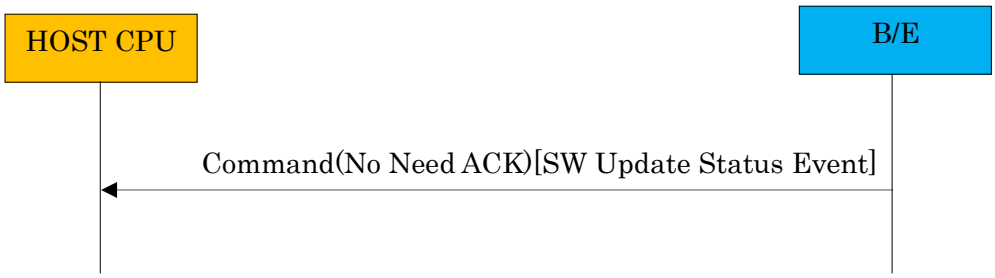
(1) SW Update Status (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	01h : Command(No Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	A0h SW Update Status(Upper)
Data 1 :	02h SW Update Status (Lower)
Data 2 :	Status 00h : Write to NVRAM FFh : Finish SW Update
Data 3 :	Result 00h : Success 01h : Invalid file name 02h : Invalid Data 03h : Invalid file size
Data CRC8 :	--

(2) Description

- B/E notify SW Update Status to HOST.
- B/E always transfers to HOST CPU in interval of 1s during SW Update (except YMODEM).

(3) Sequence



5.8.3. Background SW Update Start

(1) Background SW Update Start Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	27h
Header CRC8 :	--
Data 0 :	A1h Background SW Update Start (Upper)
Data 1 :	01h Background SW Update Start (Lower)
Data 2 ~ 5:	SW Size (4 Bytes Big Endian)
Data 6 ~ 37 :	SW Filename (32 Bytes)
Data CRC8 :	--

(2) Background SW Update Start Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	A1h Background SW Update Start (Upper)
Data 1 :	01h Background SW Update Start (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "4. Error Handling")
Data CRC8 :	--

(3) Description

- B/E prepares itself for receiving the SW Update package.
- The host can send this command even while the B/E is in a playback state.
- After this command, i.e., during the Background SW Update, B/E will continue to send usual events and can accept other commands from the host.
- After receiving the response, the host can start sending the SW Update package.
- (Refer to Section 5.8.5)
- HOST can cancel the Background SW Update at any time. (Refer to Section 5.8.4)

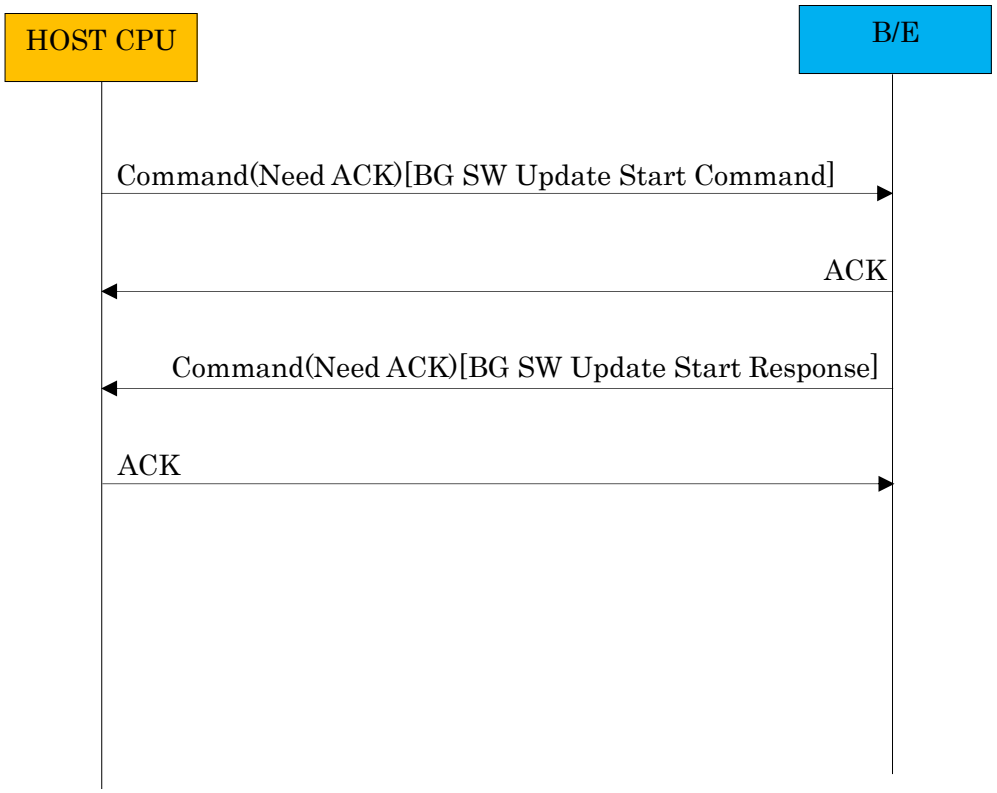
(4) Error Handling

- The host should take necessary action in case of failure.
- The failure can be the following types:
  - ✧ Doesn't receive the response within the timeout.
  - ✧ The result of response is NG.

・ Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter(Size, Filename) is invalid
03h(Invalid Length)	Data Length is not 27h.
0Ah(Invalid Status)	B/E is in invalid status (Ex : Already started BG SW Update)
21h(Disk Full)	There is not enough space in disk.
FFh(Unknown Error)	unknown error

(5) Sequence



5.8.4. Background SW Update Cancel

(1) Background SW Update Cancel Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	03h
Header CRC8 :	--
Data 0 :	A1h Background SW Update Cancel (Upper)
Data 1 :	02h Background SW Update Cancel (Lower)
Data CRC8 :	--

(2) Background SW Update Cancel Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	A1h Background SW Update Cancel (Upper)
Data 1 :	02h Background SW Update Cancel (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "4.Error Handling")
Data CRC8 :	--

(3) Description

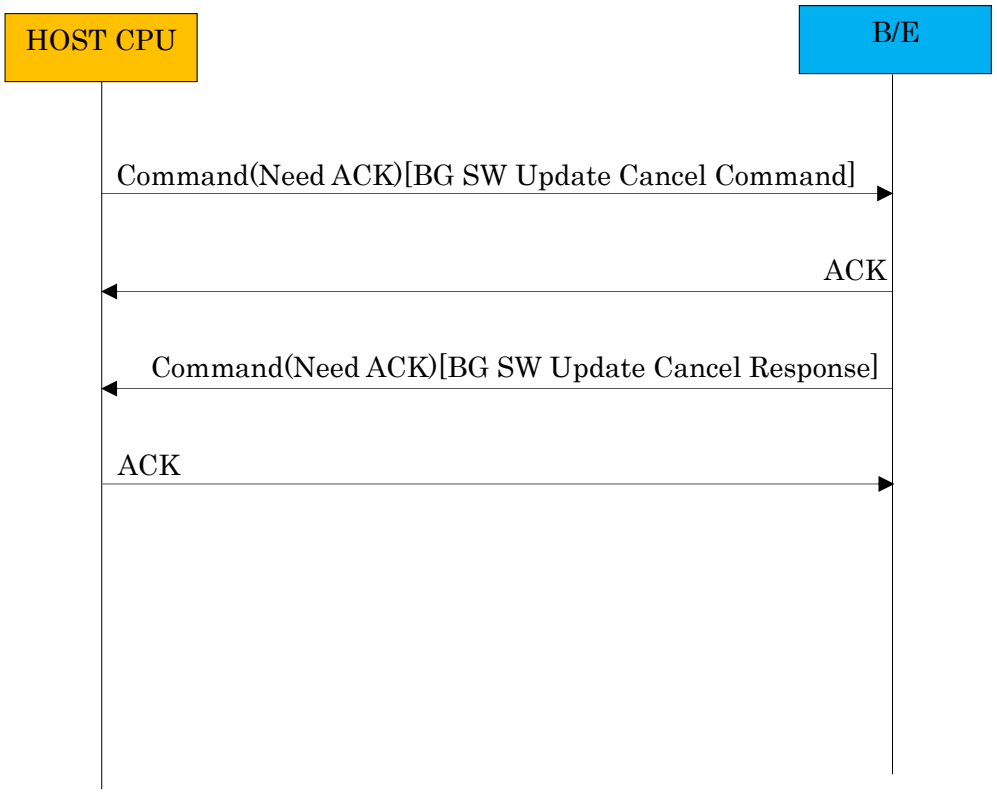
- B/E stops receiving the SW Update package for Background SW Update.
- The host can send this command after Background SW Update Start and before sending the last fragment of the package.

(4) Error Handling

- The host should take necessary action in case of failure.
- The failure can be the following types:
  - ✧ Doesn't receive the response within the timeout.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
03h(Invalid Length)	Data Length is not 03h.
0Ah(Invalid Status)	B/E is in invalid status (BG SW Update not started / completed)
FFh(Unknown Error)	unknown error

(5) Sequence


HOST CPU
B/E
Command(Need ACK)[BG SW Update Cancel Command]
ACK
Command(Need ACK)[BG SW Update Cancel Response]
ACK

5.8.5. Background SW Update

(1) Background SW Update Command (HOST ⇒ BE)

Start Sync :	A5h,59h
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length(2Bytes) :	0008h ~ 0FFFh
Header CRC8 :	--
Data 0 :	A1h Background SW Update (Upper)
Data 1 :	03h Background SW Update (Lower)
Data 2 ~ 3 :	Fragment Number (2 Bytes, Big Endian)
Data 4 ~5 :	Last Fragment Number (2 Bytes, Big Endian)
Data 6 ~ :	Data Bytes
Data CRC8 :	--

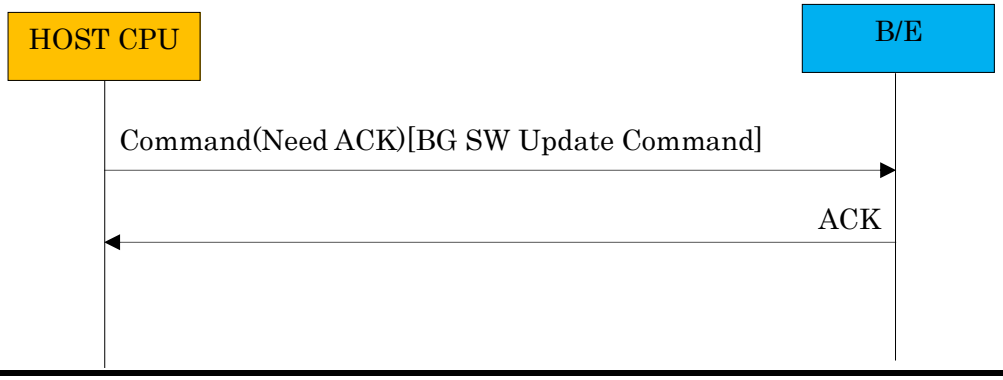
(2) Description

- With this command, the host can send the SW Update Package for Background SW Update.
- The host should divide the SW Update Package into fragments and send the fragments sequentially.
- The host can freely choose the fragment size for each transfer as long as the transfer size is within the maximum(0FFFh).
- The host is free to send other commands between two fragments.
- The host can cancel Background SW Update at any time before sending the last fragment of the SW Update Package.
- When B/E receives the last fragment, it stops receiving the data and proceeds to extract the package and copy it to NAND flash.

(3) Error Handling

- The host should take necessary action in case of failure.
- The failure can be the following types:
  - ✧ Does not receive the ACK or receive NAK(CRC Error) – Host may retry or Cancel SW Update.
  - ✧ Receives Background SW Update Status with status (Data 2) = FFh.

(4) Sequence



5.8.6. Background SW Update Status

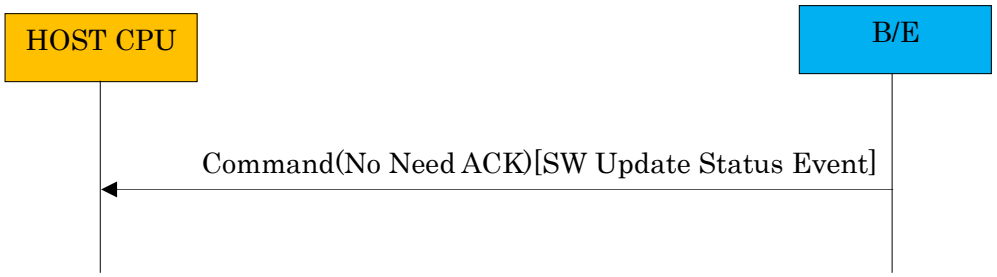
(1) Background SW Update Status (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	01h : Command(No Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	A1h Background SW Update Status(Upper)
Data 1 :	04h Background SW Update Status (Lower)
Data 2 :	Status 00h : Update Completed Successfully 01h : Received First Fragment 02h : Received Last Fragment 03h : Update Canceled FFh : Update Failed (Refer to Data 3 for details)
Data 3 :	Status = 00h ~ 03h 00h : Fixed Value Status = FFh 01h : Invalid Fragment 02h : Invalid Package 03h : Invalid Contents (Individual files NG) 04h : Invalid Status FFh : Unknown Error
Data CRC8 :	--

(2) Description

- B/E will notify the host of the Background SW Update Status.
- B/E notifies the below events
  - 00h : Received all fragments of the SW Update Package and update process completed.
  - 01h : Successfully completed the SW Update
  - FFh : SW Update failed
- In case of SW Update failure, Data 3 will give reason for the failure
  - 01h : Invalid Fragment including wrong fragment number.
  - 02h : Invalid Package
  - 03h : Invalid contents – checksum mismatch, required files not present etc.
  - 04h : Invalid status – BG SW not started, Canceled or Already completed
  - FFh : Unknown Error
- In case of SW Update failure, B/E will internally cancel the update. Host can restart the SW Update.

(3) Sequence



5.9. Test & Debug Command

5.9.1. Test Mode

(1) Test Mode Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	04h~
Header CRC8 :	--
Data 0 :	F0h Test Mode (Upper)
Data 1 :	05h Test Mode (Lower)
Data 2 :	Video Mode 00h : Normal Mode 01h : Test Mode(Output Color Bar)
Data 3 :	Audio Mode 00h : Normal Mode 01h: Test Mode(L/R with 1KHz/16bit-range sine wave) 02h: Test Mode(L with 1KHz/16bit-range sine wave) 03h: Test Mode(R with 1KHz/16bit-range sine wave) 04h: Test Mode(L/R with 1KHz/15bit-range sine wave) 05h: Test Mode(L with 1KHz/15bit-range sine wave) 06h: Test Mode(R with 1KHz/15bit-range sine wave) 07h: Pink Noise(Full-scale with L/R) 08h: Pink Noise(Full-scale with L only) 09h: Pink Noise(Full-scale with R only)
Data CRC8 :	--

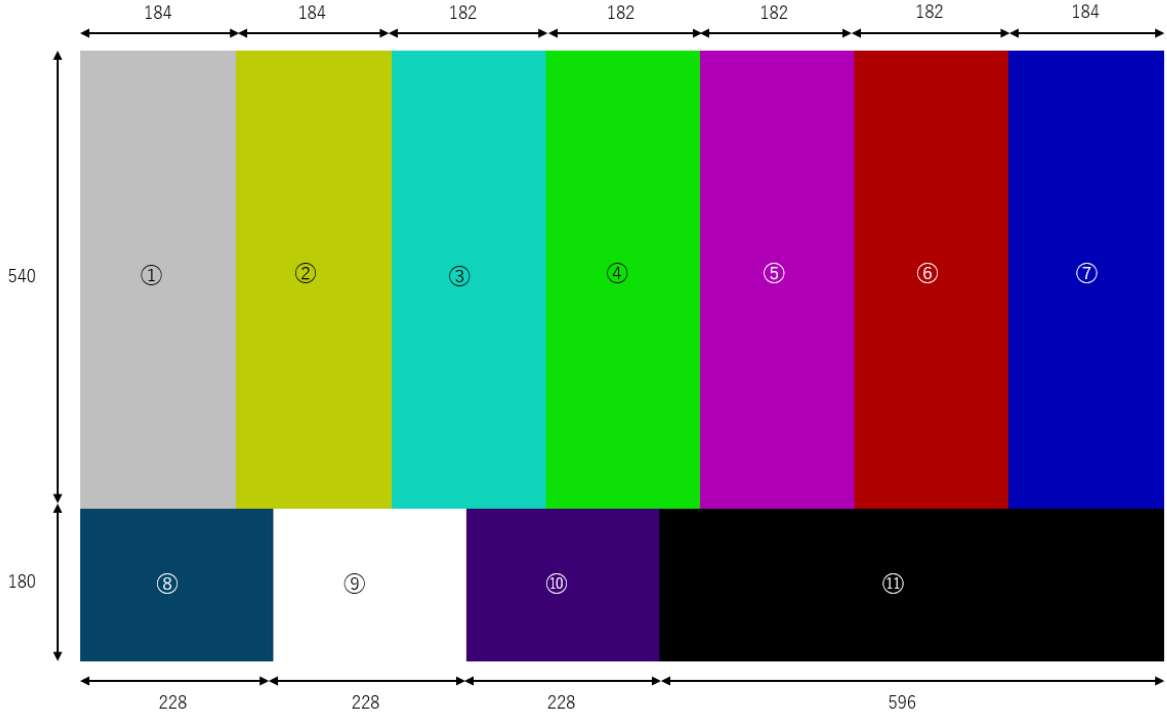
(2) Test Mode Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	F0h Test Mode (Upper)
Data 1 :	05h Test Mode (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- This command is for test mode.
- This command should be used at Idle Status.
- If Video Mode (Data2) is 01h, Video output is color bar as the followings.
- Can use this command for testing purposes only.
- During test mode, only command 5.2.11. Watch Dog is accepted, and other commands are retuned as Invalid.
-

・F039902182【TRS231BSxxxx】, F039902183【TRH231BSxxxx】

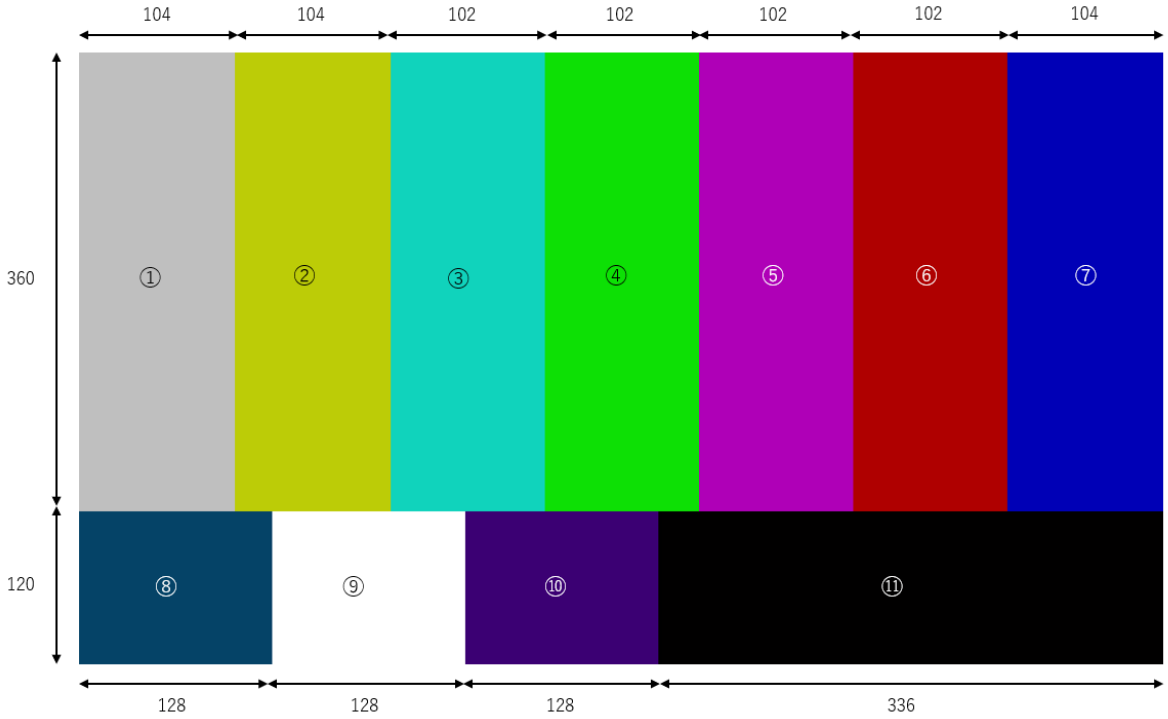


No.	Color	Signal Level *		
		R	G	B
①	Grey	BFh	BFh	BFh
②	Yellow	BCh	CCh	07h
③	Cyan	10h	D3h	BCh
④	Green	0Dh	E0h	05h
⑤	Magenta	AFh	00h	B7h
⑥	Red	AFh	00h	00h
⑦	Blue	00h	00h	B6h
⑧	-I	05h	43h	67h
⑨	White	FFh	FFh	FFh
⑩	+Q	3Bh	00h	73h
⑪	Black	00h	00h	00h

* Comply with ITU-R BT.601 studio encoding parameters.

- If Audio Mode (Data3) is 01h/02h/03h/04h/05h/06h, Audio output is sine wave. It repeats that audio playback for about 20 seconds and stop the short time.
- If Audio Mode (Data3) is 07h/08h/09h, Audio output is pink noise. There will be a short pause every 20seconds.
- Display resolution of Test mode is based on current resolution setting.

・F039990144【FRO232BSyyzz】



No.	Color	Signal Level *		
		Y	Cb	Cr
①	Grey	BFh	BFh	BFh
②	Yellow	BCh	CCh	07h
③	Cyan	10h	D3h	BCh
④	Green	0Dh	E0h	05h
⑤	Magenta	AFh	00h	B7h
⑥	Red	AFh	00h	00h
⑦	Blue	00h	00h	B6h
⑧	-I	05h	43h	67h
⑨	White	FFh	FFh	FFh
⑩	+Q	3Bh	00h	73h
⑪	Black	00h	00h	00h

* Comply with ITU-R BT.601 studio encoding parameters.

- If Audio Mode (Data3) is 01h/02h/03h/04h/05h/06h, Audio output is sine wave. It repeats that audio playback for about 20 seconds and stop the short time.
- If Audio Mode (Data3) is 07h/08h/09h, Audio output is pink noise. There will be a short pause every 20seconds.
- Display resolution of Test mode is based on current resolution setting.

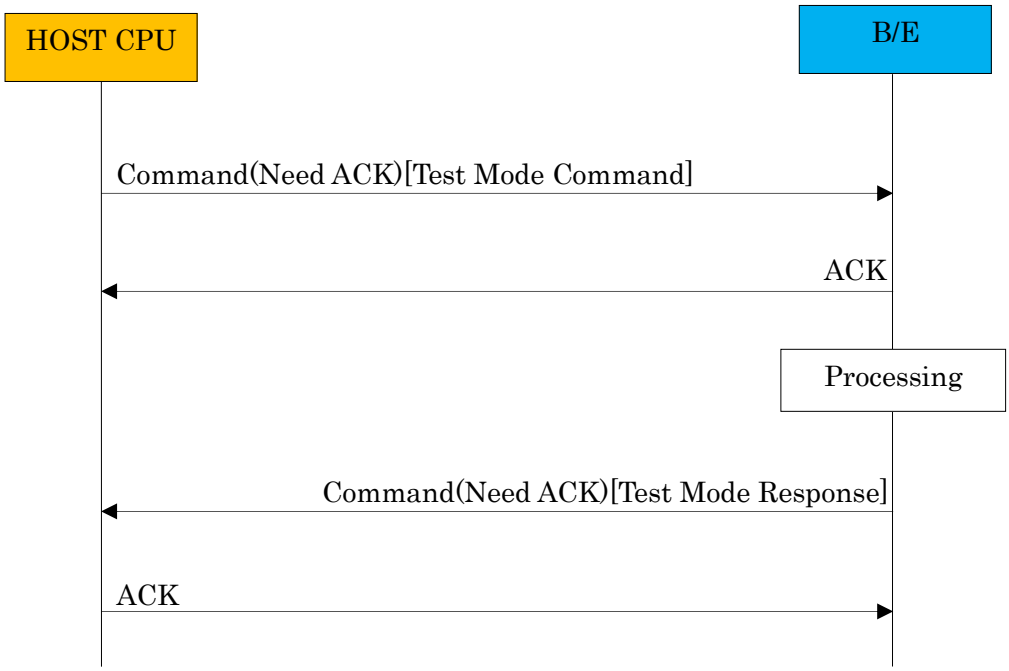
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid. - Data2 is not 00h or 01h. - Data3 is not 00h or 01h or 02h or 03h.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.9.2. Set Debug Parameter

(1) Set Debug Parameter Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	04h
Header CRC8 :	--
Data 0 :	F0h Set Debug Parameter (Upper)
Data 1 :	06h Set Debug Parameter (Lower)
Data 2 :	Monitor Mode Enable 00h : disable(default) 01h : enable FFh : Invalid Value
Data CRC8 :	--

(2) Set Debug Parameter Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	F0h Set Debug Parameter (Upper)
Data 1 :	06h Set Debug Parameter (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- Set Debug Parameter for the DTV Module.
- Data2 is the parameter for enable/disable of "5.9.5.Monitor Status".
- Can use this command for testing purposes only.

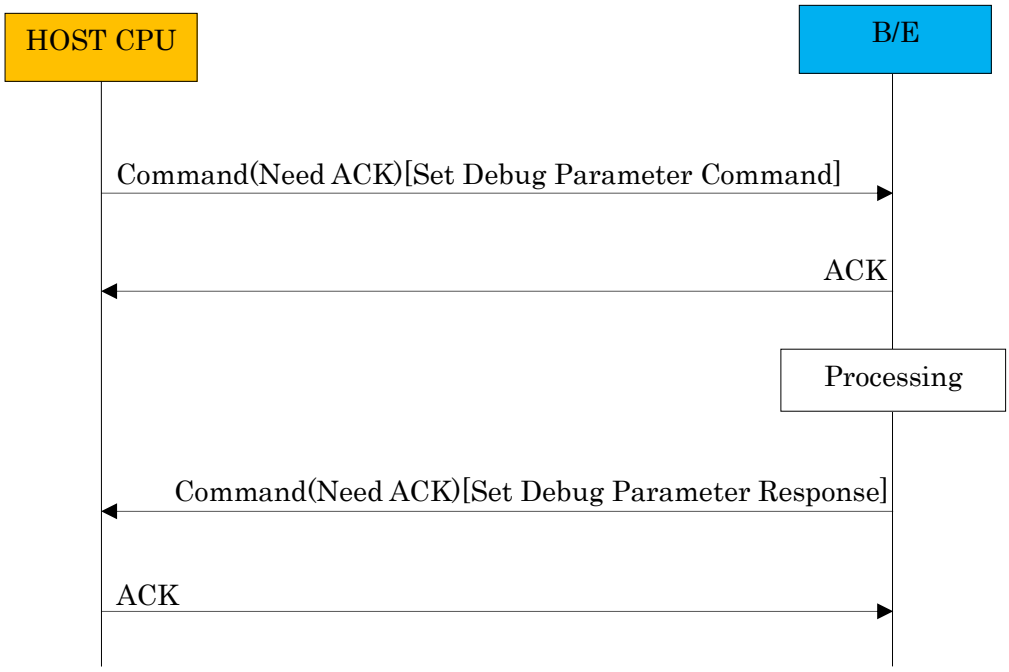
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.9.3. Monitor Status

(1) Monitor Status Event (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	01h : Command(No Need Ack))
Request No. :	00h ~ FFh
Data Length :	2Bh
Header CRC8 :	--
Data 0 :	F0h Monitor Status Event (Upper)
Data 1 :	07h Monitor Status Event (Lower)
Data 2 :	Diver Mode 00h : Dynamic 4Diver System 01h : Fixed 2x2Diver System 02h : Fixed 2Diver System(Antenna Br2/3) 03h : Fixed 4Diver System 04h : Fixed 2Diver System(Antenna Br0/1) 05h : Fixed 3x1Diver System (Fore Br0/1/2) 06h : Dynamic 4Diver System (Fore Br2/3) 07h : Fixed 2x2Diver System (Fore Br2/3) 08h : Fixed 3x1Diver System (Fore Br1/2/3)
Data 3 :	Layer Status Bit0 : Tuner0 A-Layer (0:Invalid 1:Valid) Bit1 : Tuner0 B-Layer (0:Invalid 1:Valid) Bit2 : Tuner0 C-Layer (0:Invalid 1:Valid) Bit3 : Reserved Bit4 : Tuner1 A-Layer (0:Invalid 1:Valid) Bit5 : Tuner1 B-Layer (0:Invalid 1:Valid) Bit6 : Tuner1 C-Layer (0:Invalid 1:Valid) Bit7 : Reserved
Data 4 :	Tuner0 Physical Channel Number(0Ch~34h)
Data 5 :	Tuner0 C/N(Diver) – Upper
Data 6 :	Tuner0 C/N(Diver) – Lower
Data 7 :	Tuner0 A-Layer BER – Upper
Data 8 :	Tuner0 A-Layer BER – Lower
Data 9 :	Tuner0 A-Layer PER – Upper
Data 10 :	Tuner0 A-Layer PER – Lower
Data 11 :	Tuner0 B-Layer BER – Upper
Data 12 :	Tuner0 B-Layer BER – Lower
Data 13 :	Tuner0 B-Layer PER – Upper
Data 14 :	Tuner0 B-Layer PER – Lower
Data 15 :	Tuner0 C-Layer BER – Upper
Data 16 :	Tuner0 C-Layer BER – Lower
Data 17 :	Tuner0 C-Layer PER – Upper
Data 18 :	Tuner0 C-Layer PER – Lower
Data 19 :	Tuner1 Physical Channel Number(0Ch~34h)
Data 20 :	Tuner1 C/N(Diver) – Upper
Data 21 :	Tuner1 C/N(Diver) – Lower
Data 22 :	Tuner1 A-Layer BER – Upper
Data 23 :	Tuner1 A-Layer BER – Lower
Data 24 :	Tuner1 A-Layer PER – Upper
Data 25 :	Tuner1 A-Layer PER – Lower
Data 26 :	Tuner1 B-Layer BER – Upper
Data 27 :	Tuner1 B-Layer BER – Lower
Data 28 :	Tuner1 B-Layer PER – Upper
Data 29 :	Tuner1 B-Layer PER – Lower
Data 30 :	Tuner1 C-Layer BER – Upper
Data 31 :	Tuner1 C-Layer BER – Lower

Data 32 :	Tuner1 C-Layer PER – Upper
Data 33 :	Tuner1 C-Layer PER – Lower
Data 34 :	Antenna0 C/N – Upper
Data 35 :	Antenna0 C/N – Lower
Data 36 :	Antenna1 C/N – Upper
Data 37 :	Antenna1 C/N – Lower
Data 38 :	Antenna2 C/N – Upper
Data 39 :	Antenna2 C/N – Lower
Data 40 :	Antenna3 C/N – Upper
Data 41 :	Antenna3 C/N – Lower
Data CRC8 :	

(2) Description

- B/E notify Tuner Monitor Status to HOST in interval of 1s.
- B/E notify to HOST CPU during Monitor Status Enable of "5.9.4.Set Debug Parameter" is enable.
- If "Diver Mode (Data2)" is 02h/03h/04h, B/E use only tuner0. So, the information of tuner1 (Data 19~33) are all 0.
- Tuner0 is used for Foreground processing, and Tuner1 is used for Background processing.
- "Layer Status" indicates the status of each tuner and each layer by each bit.

bit of Layer Status	Description
Tuner0 A-Layer	0 : Data 7~10 is invalid. HOST can't refer these data. 1 : Data 7~10 is valid. HOST can refer these data.
Tuner0 B-Layer	0 : Data 11~14 is invalid. HOST can't refer these data. 1 : Data 11~14 is valid. HOST can refer these data.
Tuner0 C-Layer	0 : Data 15~18 is invalid. HOST can't refer these data. 1 : Data 15~18 is valid. HOST can refer these data.
Tuner1 A-Layer	0 : Data 22~25 is invalid. HOST can't refer these data. 1 : Data 22~25 is valid. HOST can refer these data.
Tuner1 B-Layer	0 : Data 26~29is invalid. HOST can't refer these data. 1 : Data 26~29 is valid. HOST can refer these data.
Tuner1 C-Layer	0 : Data 30~33 is invalid. HOST can't refer these data. 1 : Data 30~33 is valid. HOST can refer these data.

*:ISDB-T support hierarchical transmission mode include three layers (A/B/C/). Refer to ARIB TR-B14 "General Information of the Technical Report 4.Hierarchical Configuration for Operation and Available Parameters"

・C/N: signed 2byte data (Divide by 100)

・BER : unsigned 2byte data (Divide by 1000000)

・PER : unsigned 2byte data (Divide by 10000)

・C/N range 0dB to 64.0 dB

・BER range 0.0(No error) to 1.0 (Error 100%)

・PER range 0.0(No error) to 1.0 (Error100%)

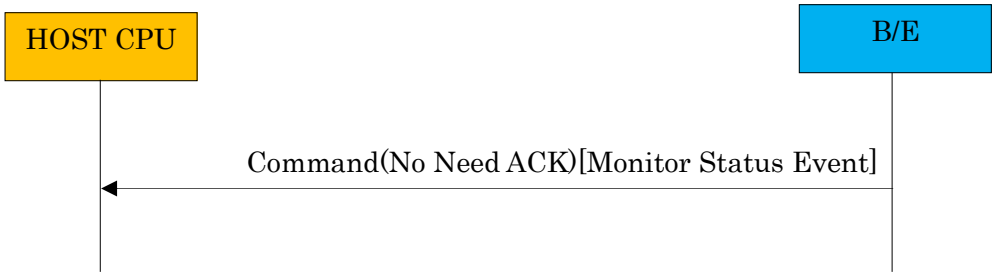
・In case of PER=100%、 Data=2710H

・In case of BER=100%、 Data=F4240H

・but Monitor status display until FFFFH. This means 6.5%.

・F/E IC cannot count BER of 4.41% or more on a structural basis. Therefore, there is no problem until 6.5%.

(3) Sequence



5.9.4. HW Test

(1) HW Test Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	Data2!=03h : 04h Data2=03h : F4h
Header CRC8 :	--
Data 0 :	F0h HW Test (Upper)
Data 1 :	08h HW Test (Lower)
Data 2 :	Operation 00h : HW Test start 01h : DDR Check 02h : Nor Flash Check 03h : UART Check 04h : FONT Check 05h : RMP Check 06h : HDMI EDID Check 07h : Tuner I2C Check FFh : HW Test finish
Data 3~242	Only UART Check Check Data
Data CRC8 :	--

(2) HW Test Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	06h(except UART Check) F6h(UART Check)
Header CRC8 :	--
Data 0 :	F0h HW Test (Upper)
Data 1 :	08h HW Test (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data 4 :	Check Result 00h : NG 01h : OK
Data 5 ~244	Received data by "HW Test Command"
Data CRC8 :	--

(3) Description

- B/E start Hardware Test.
- If checking UART Test, HOST has to set Data3~242. F/E send Data5~244 that are set received data from HOST. HOST can verify check sending data and receiving data
- Can use this command for testing purposes only.

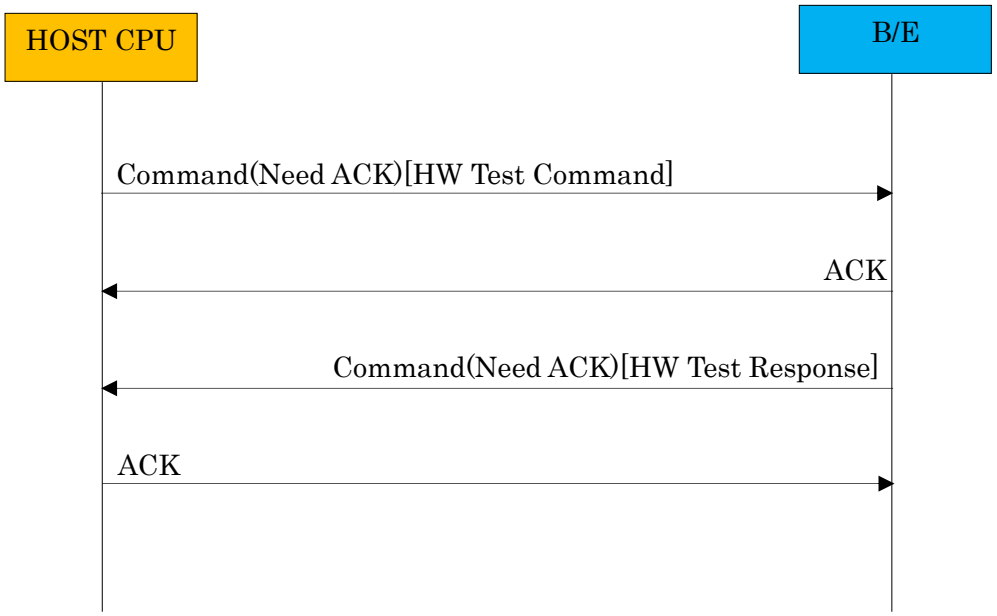
(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.
  - ✧ The result of response is NG.
- Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.9.5. 2nd TS Test

(1) 2nd TS Test Command (HOST ⇒ BE)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	06h
Header CRC8 :	--
Data 0 :	F0h 2nd TS Test (Upper)
Data 1 :	09h 2nd TS Test (Lower)
Data 2 :	Operation 00h : start 01h : stop
Data 3	Physical channel Number 13~52
Data 4	Antenna Config Bit0 : Antenna 1 Bit1 : Antenna 2 Bit2 : Antenna 3 Bit3 : Antenna 4 Bit4~ : Reserved
Data CRC8 :	--

(2) 2nd TS Test Response (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(Need Ack)
Request No. :	00h ~ FFh
Data Length :	05h
Header CRC8 :	--
Data 0 :	F0h 2nd TS Test (Upper)
Data 1 :	09h 2nd TS Test (Lower)
Data 2 :	Result 00h : OK 01h : NG
Data 3 :	Result = OK : 00h : Fixed Value Result = NG : Error Code (Refer to "3.Error Code")
Data CRC8 :	--

(3) Description

- B/E start 2nd TS Test.
- If Data2 = 00h, start "5.9.8.2nd TS Test Status"
- If Data2 = 01h, stop "5.9.8.2nd TS Test Status"
- Can use this command for testing purposes only.

(4) Error Handling

- In HOST, it is solely up to the product planning policy of each manufacture how to handle the failure of response.
- The failure of response means that the followings:
  - ✧ Doesn't receive the response within 1sec after HOST transmit the command.

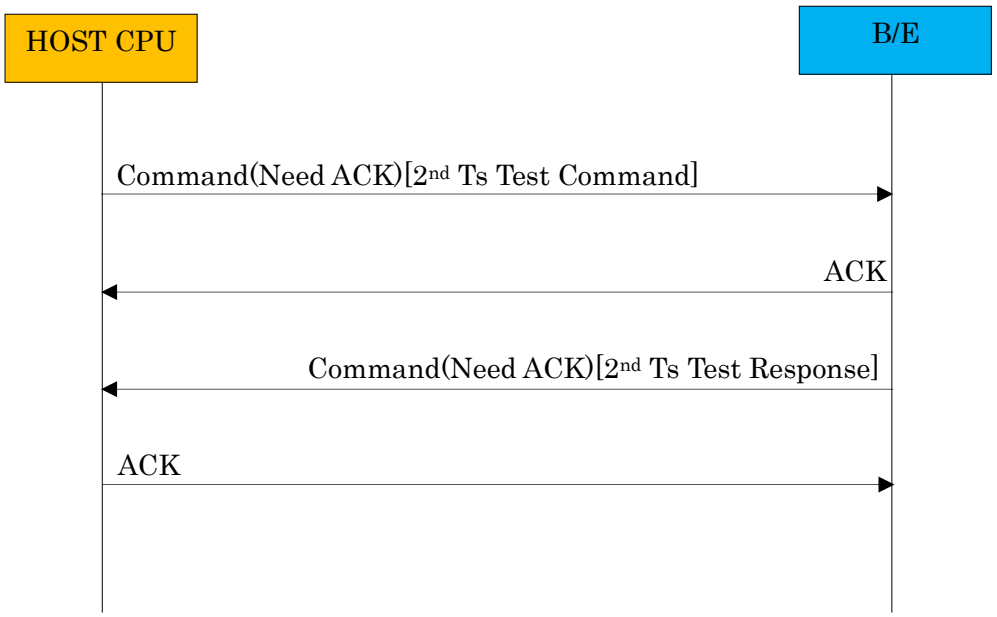
✧ The result of response is NG.

・ Regarding the condition that Result of the response is NG, refer the followings:

Error Code	Condition
01h(Invalid Parameter)	The parameter is invalid.
03h(Invalid Length)	Data Length is invalid.
FFh(Unknown Error)	Unknown error

For other errors, refer to Chapter [3. Error Code]

(5) Sequence



5.9.6. 2nd TS Test Status

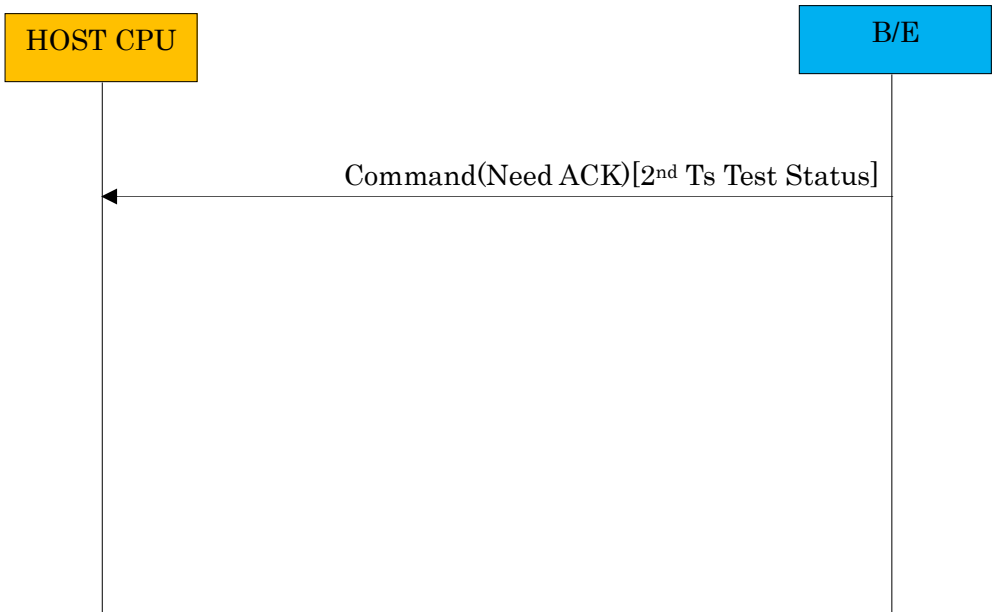
(1) 2nd TS Test Status Command (BE ⇒ HOST)

Start Sync :	A5h,5Ah
Type :	00h : Command(No Need Ack)
Request No. :	00h ~ FFh
Data Length :	14h
Header CRC8 :	--
Data 0 :	F0h 2nd TS Test Status (Upper)
Data 1 :	0Ah 2nd TS Test Status (Lower)
Data 2 :	Layer Status Bit0 : Reserved Bit1 : Reserved Bit2 : Reserved Bit3 : Reserved Bit4 : Tuner1 A-Layer (0:Invalid 1:Valid) Bit5 : Tuner1 B-Layer (0:Invalid 1:Valid) Bit6 : Tuner1 C-Layer (0:Invalid 1:Valid) Bit7 : Reserved
Data 3 :	Physical Channel Number(0Ch~34h)
Data 4 :	C/N(Diver) – Upper
Data 5 :	C/N(Diver) – Lower
Data 6 :	A-Layer BER – Upper
Data 7 :	A-Layer BER – Lower
Data 8 :	A-Layer PER – Upper
Data 9 :	A-Layer PER – Lower
Data 10 :	B-Layer BER – Upper
Data 11 :	B-Layer BER – Lower
Data 12 :	B-Layer PER – Upper
Data 13 :	B-Layer PER – Lower
Data 14 :	C-Layer BER – Upper
Data 15 :	C-Layer BER – Lower
Data 16 :	C-Layer PER – Upper
Data 17 :	C-Layer PER – Lower
Data 18 :	TS Count
Data CRC8 :	--

(2) Description

- B/E notify 2nd TS Status to HOST.
- B/E always transfer to HOST in interval of 1s during 2nd TS Test.

(3) Sequence



6. DTV Information

6.1. Channel List

Information	Channel List
Size	256KByte
Store Time (Erase and Store)	100ms
Store Condition	When receiving "Update preset table". When receiving "DTV Initialize". When updating Channel Information/Service Information in broadcast signal. When receiving "Shutdown" with Shutdown Mode = 01h.
Note	This data is stored for each user ID

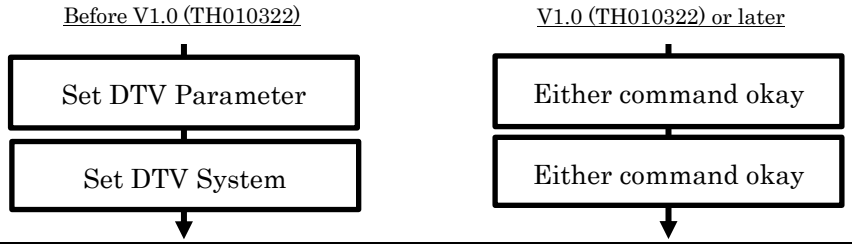
6.2. Last Channel

Information	Last Channel
Size	128Kbyte
Store Time (Erase and Store)	50ms
Store Condition	When receiving "Shutdown" with Shutdown Mode = 01h. When receiving "DTV Initialize"
Note	This data is stored for each user ID

6.3. DTV Configuration

Information	DTV Configuration
Size	128Kbyte
Store Time (Erase and Store)	50ms
Store Condition	When receiving "Set auto tune Mode". When receiving "Set Closed Caption Mode". When receiving "Set Auto Broadcast Search Mode". When receiving "Set Audio Mode". When receiving "Set Residential Area". When receiving "Set Auto Event Relay". When receiving "DTV Initialize". When receiving "Shutdown" with Shutdown Mode = 01h.
Note	This data is stored for each user ID

6.4. DTV System

Information	DTV System
Size	128Kbyte
Store Time (Erase and Store)	50ms
Store Condition	When receiving "Shutdown" with Shutdown Mode = 01h. When receiving "Set DTV System". When receiving "Set DTV Parameter".
Note	Before SW V1.0(TH010322), If both commands are issued in the same power cycle (no reboot in between the commands), the below order should be followed. 

6.5. TRMP Information

Information	TRMP Information
Size	128Kbyte
Store Time (Erase and Store)	50ms
Store Condition	When receiving "Shutdown" with Shutdown Mode = 01h. When receiving broadcast signal at first. When receiving "DTV Initialize".

6.6. Logo

Information	Broadcaster Logo
Size	256KByte
Store Time (Erase and Store)	100ms
Store Condition	When receiving "Shutdown" with Shutdown Mode = 01h. When receiving "DTV Initialize". When there is data update during reception.

6.7. BML Information

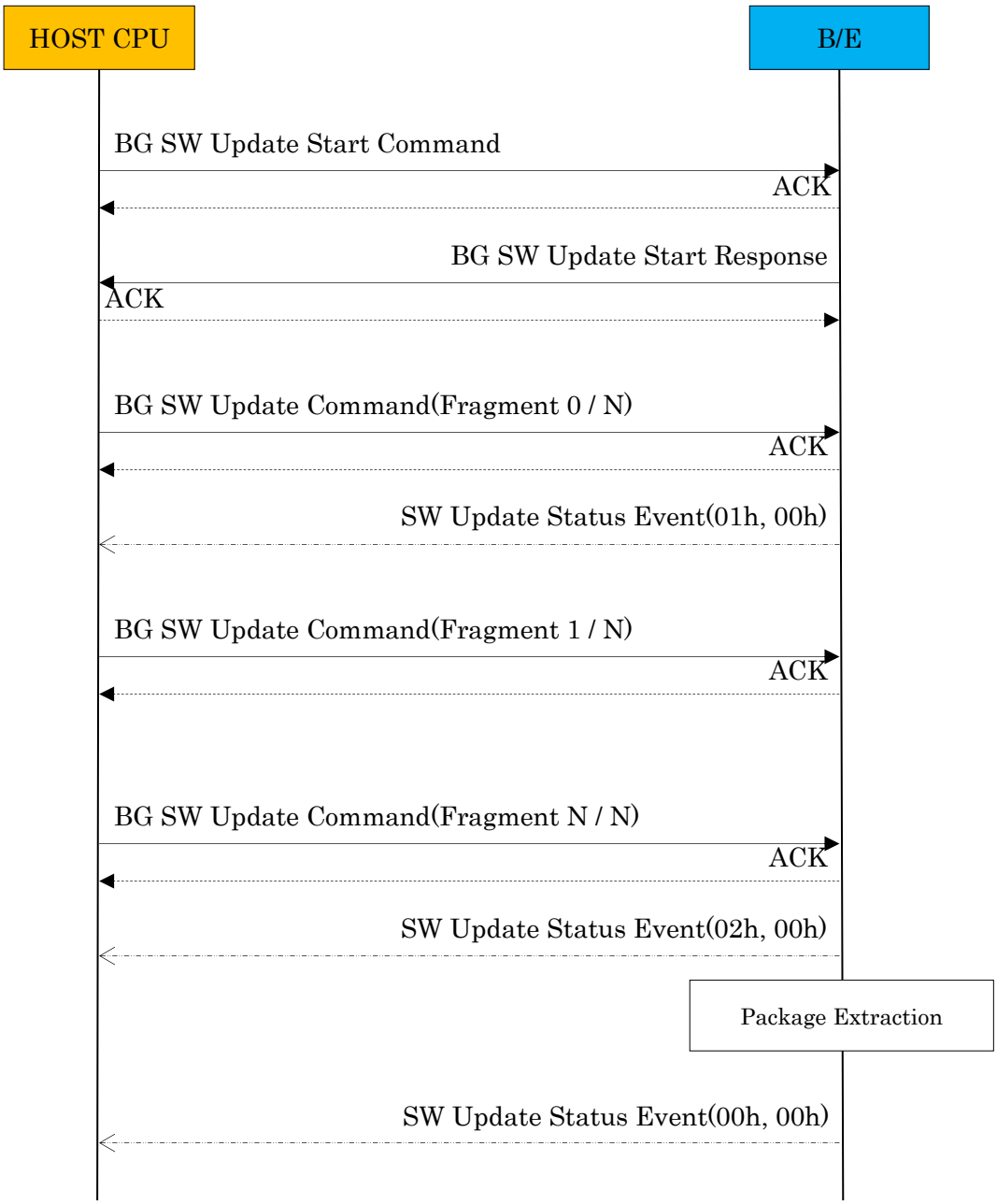
Information	BML Information
Size	512KByte
Store Time (Erase and Store)	200ms
Store Condition	When receiving "Shutdown" with Shutdown Mode = 01h. When receiving "DTV Initialize". When there is data update during reception.

6.8. DTV Parameter2

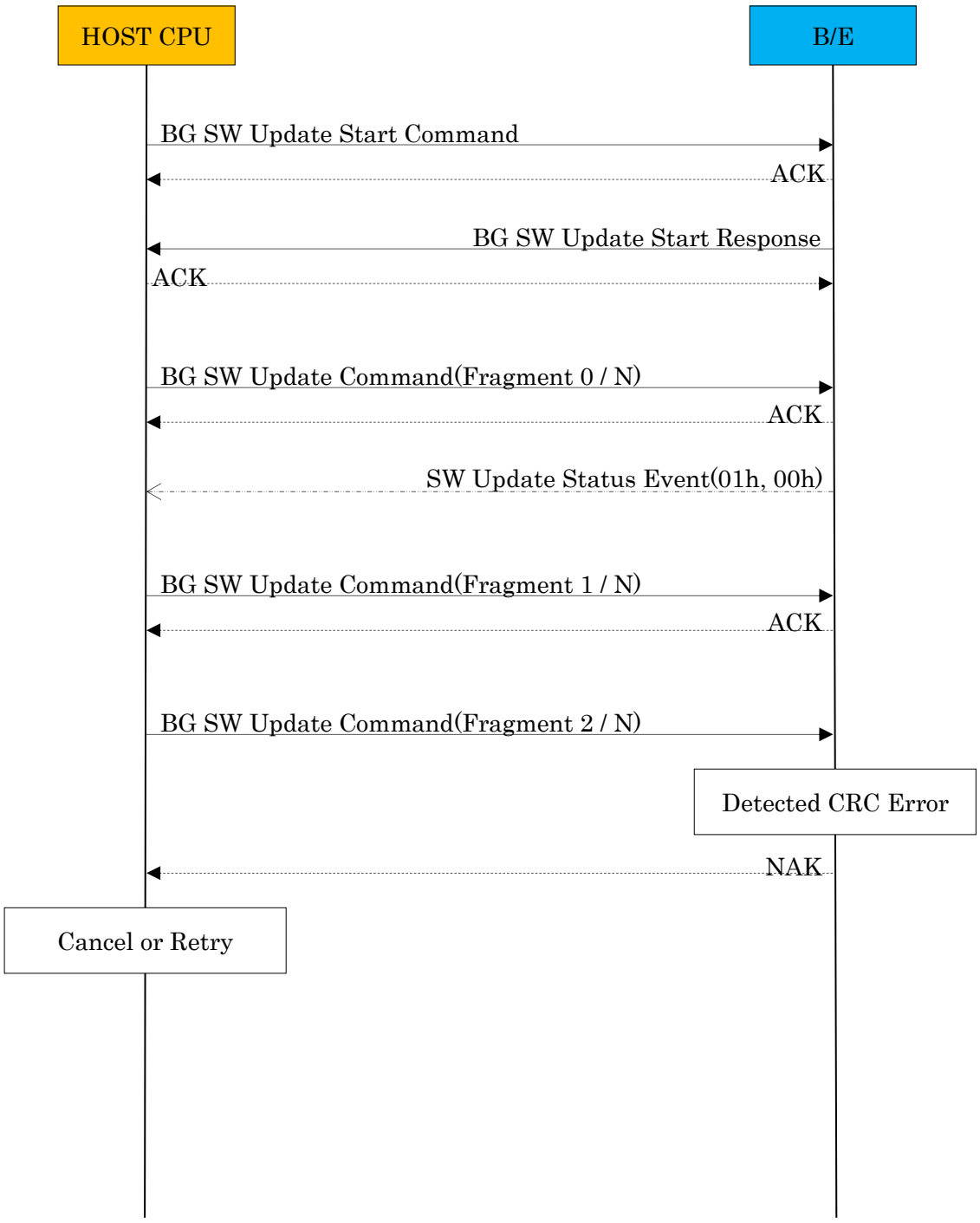
Information	DTV Parameter2
Size	128Kbyte
Store Time (Erase and Store)	50ms
Store Condition	When receiving "Shutdown" with Shutdown Mode = 01h. When receiving "Set DTV Parameter2".

7. Background SW Update Sequence (Examples)

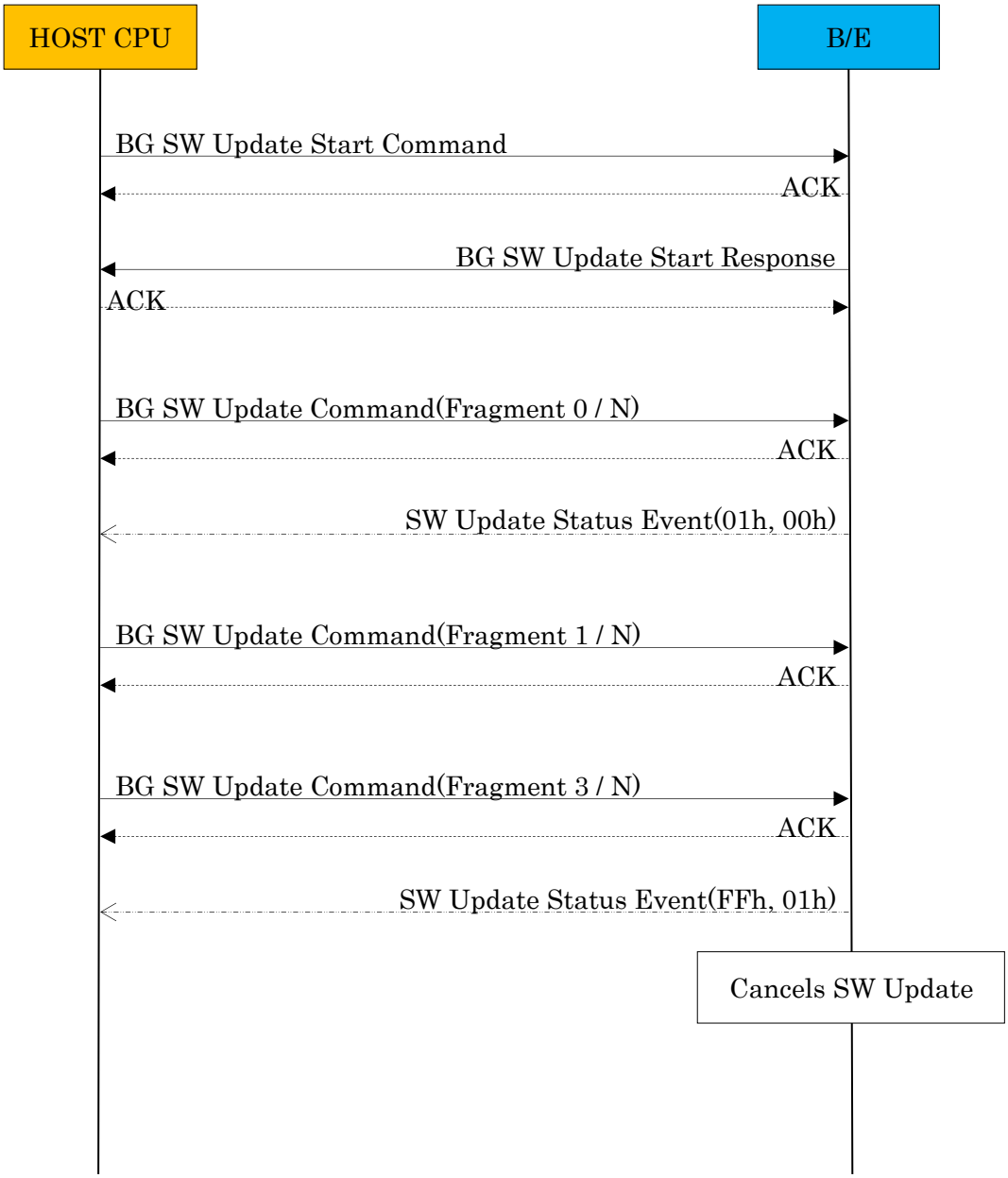
7.1. Update Success



7.2. Update Failed – CRC Error



7.3. Update Failed – Invalid Fragment



8. DTV parameter and DTV parameter 2 data range

8.1. DTV Parameter data range

The DTV Parameter data range is common to following three types.

-F039902182【TRS231BSxxxx】

-F039902183【TRH231BSxxxx】

-F039990144【FRO232BSyyzz】

Data	Data name	Data range
0	Set DTV Parameter (Upper)	01h
1	Set DTV Parameter (Lower	1Dh
2~5	DTV Parameter Enable/Disable(Use default Parameter)	01000000h(Enable) 00000000h(Disable)
6~9	C/N for Tuner Unlock Detection	0 - 4000 [0 - 40dB] HEX : 00000000h - 00000FA0h
10~13	BER for Tuner Unlock Detection	0 - 44100 (0 – 4.41[%]) HEX : 00000000h - 0000AC44h
14~17	PER for Tuner Unlock Detection.	0 - 10000 (0 - 100[%]) HEX : 00000000h - 00002710h
18~21	Time for Tuner Unlock Detection	1 - 10000 (200ms -2000s) HEX : 00000001h - 00002710h
22~25	C/N for change to Fullseg.	0 - 4000 [0 - 40dB] HEX : 00000000h - 00000FA0h
26~29	BER for change to Fullseg	0 - 44100 (0 – 4.41[%]) HEX : 00000000h - 0000AC44h
30~33	PER for change to Fullseg	0 - 10000 (0 - 100[%]) HEX : 00000000h - 00002710h
34~37	Time for change to Fullseg	1 - 10000 (200ms -2000s) HEX : 00000001h - 00002710h
38~41	C/N for change to Oneseg	0 - 4000 [0 - 40dB] HEX : 00000000h - 00000FA0h
42~45	BER for change to Oneseg	0 - 44100 (0 – 4.41[%]) HEX : 00000000h - 0000AC44h
46~49	PER for change to Oneseg	0 - 10000 (0 - 100[%]) HEX : 00000000h - 00002710h
50~53	Time for change to Oneseg	1 - 10000 (200ms -2000s) HEX : 00000001h - 00002710h
54~57	C/N for start Broadcast Search	0 - 4000 [0 - 40dB] HEX : 00000000h - 00000FA0h
58~61	BER for start Broadcast Search	0 - 44100 (0 – 4.41[%]) HEX : 00000000h - 0000AC44h
62~65	PER for start Broadcast Search	0 - 10000 (0 - 100[%]) HEX : 00000000h - 00002710h
66~69	Time for start Broadcast Search	1 - 10000 (200ms -2000s) HEX : 00000001h - 00002710h
70~73	Time (1) for disable Broadcast Search	1 - 10000 (200ms -2000s) HEX : 00000001h - 00002710h
74~77	Time (2) for disable Broadcast Search	1 - 10000 (200ms -2000s) HEX : 00000001h - 00002710h
78~81	Time for Background Tuner Unlock Detection	1 - 10000 (200ms -2000s) HEX : 00000001h - 00002710h
82~85	C/N for start Background Search	0 - 4000 [0 - 40dB] HEX : 00000000h - 00000FA0h
86~89	Time for start Background Search	1 - 10000 (200ms -2000s)

		HEX : 00000001h - 00002710h
90~93	C/N for start Background Scan	0 - 4000 [0 - 40dB] HEX : 00000000h - 00000FA0h
94~97	Time for start Background Scan	1 - 10000 (200ms -2000s) HEX : 00000001h - 00002710h
98~101	Time for change to 4Div mode	1 - 10000 (200ms -2000s) HEX : 00000001h - 00002710h
102~105	C/N difference for switching to background channel	0 - 4000 [0 - 40dB] HEX : 00000000h - 00000FA0h
106~109	BER for switching to background channel	0 - 44100 (0 – 4.41[%]) HEX : 00000000h - 0000AC44h
110~113	PER for switching to background channel	0 - 10000 (0 - 100[%]) HEX : 00000000h - 00002710h
114~117	Time for switching to background channel	1 - 10000 (200ms -2000s) HEX : 00000001h - 00002710h
118~121	C/N for Receive Error Detection	0 - 4000 [0 - 40dB] HEX : 00000000h - 00000FA0h
122~125	BER for Receive Error Detection	0 - 44100 (0 – 4.41[%]) HEX : 00000000h - 0000AC44h
126~129	PER for Receive Error Detection	0 - 10000 (0 - 100[%]) HEX : 00000000h - 00002710h
130~133	Time for Receive Error Detection	1 - 10000 (200ms -2000s) HEX : 00000001h - 00002710h
134~137	C/N for Fullseg Detection at fixed Fullseg search	0 - 4000 [0 - 40dB] HEX : 00000000h - 00000FA0h
138~141	C/N for Oneseg Detection at fixed or auto switch mode search	0 - 44100 (0 – 4.41[%]) HEX : 00000000h - 0000AC44h
142~145	Detect Time (count) for other broadcast (The data value becomes the time as it is)	1 - 2147483647 [s] HEX : 00000001h - 7FFFFFFFh
146~149	Detect During Time for Another broadcast	1 - 10000 (200ms -2000s) HEX : 00000001h - 00002710h
150~153	C/N Offset for Fullseg Detection at fixed Fullseg broadcast search	0 - 4000 [0 - 40dB] HEX : 00000000h - 00000FA0h
154~157	Reserved	
158~161	Reserved	
162~165	Reserved	
166~169	Reserved	
170~173	Reserved	
174~177	Reserved	
178~181	Reserved	
182~185	Reserved	
186~189	Reserved	
190~193	Reserved	
194~197	Reserved	
198~201	Reserved	
202~205	Reserved	
206~209	Reserved	
210~213	Reserved	
214~217	Reserved	
218~221	Reserved	
222~225	Reserved	
226	CRC8	

・C/N: signed 4byte data (Divide by 100)

・BER : unsigned 4byte data (Divide by 1000000)

・PER : unsigned 4byte data (Divide by 10000)

・Time ： unsigned 4byte data (Multiply by 0.2)

・C/N range 0dB to 40.0 dB

・BER range 0.0000 (No error) to 0.0441 (Error 4.41%)

・PER range 0.0(No error) to 1.0 (Error100%)

・Time range 200ms to 2000s (Excluding Data 142 to 145)

8.2. DTV Parameter 2 data range for Thrush

The DTV Parameter 2 data range for Thrush is common to following two types.

-F039902182【TRS231BSxxxx】

-F039902183【TRH231BSxxxx】

Data	**Data name**	**Data range**
0	Set DTV Parameter 2 (Upper)	01h
1	Set DTV Parameter 2 (Lower	35h
2	DTV Parameter 2 Enable/Disable(Use default Parameter)	01h(Enable) 00h(Disable)
3	Auto tune mode default 00h:Auto 01h:Fullseg 02h:Oneseg 03h:Auto with Seamless	0 - 3 HEX : 00h – 03h
4	Default Character Set 00h:ARIB 01h:UTF-8	0 - 1 [%] HEX : 00h - 01h
5	Default Video Resolution 00h:720p 01h:480p 02h:1080p 03h:480i 04h:1080p@30Hz	0 -4 HEX : 00h - 04h
6	Closed Caption display over BML browser Enable/Disable	00h(Enable Display) 01h(Disable Display)
7	BML Error Message Enable/Disable	00h(Enable Display) 01h(Disable Display)
8	Receive Logo Type 0 : Receive, 1 : Not Receive Bit7: 0 (Fixed) Bit6: 0 (Fixed) Bit5: Logo Type 5 Bit4: Logo Type 4 Bit3: Logo Type 3 Bit2: Logo Type 2 Bit1: Logo Type 1 Bit0: Logo Type 0	00XXXXXXb
9	Search Detection Timeout	1 - 10 [s] HEX : 01 – 0Ah
10~11	C/N for Receive Error Detection for Fix Fullseg mode	0 - 4000 [0 - 40dB] HEX : 0000h - 0FA0h
12~13	C/N for Preset Scan Detection for Fix Oneseg mode and Auto mode	0 - 4000 [0 - 40dB] HEX : 0000h - 0FA0h
14~15	C/N for Preset Scan Detection for Fix Fullseg mode	0 - 4000 [0 - 40dB] HEX : 0000h - 0FA0h
16	Timeout for Preset Scan per channel	0 - 10 [s]

		HEX : 00h - 0Ah
17~18	C/N Level 3 threshold for full seg	0 - 4000 [0 - 40dB] HEX : 0000h - 0FA0h
19~20	C/N Level 2 threshold for full seg	0 - 4000 [0 - 40dB] HEX : 0000h - 0FA0h
21~22	C/N Level 1 threshold for full seg	0 - 4000 [0 - 40dB] HEX : 0000h - 0FA0h
23~24	EPG Update scan time / channel	1 - 180 [s] HEX : 0001h – 00B4h
25	EPG Scan Number of Days (Only foreground EPG Scan)	1 - 8 [day] HEX : 01h - 08h
26~27	EPG Scan Signal Threshold (Only foreground EPG Scan)	0 - 4000 [0 - 40dB] HEX : 0000h - 0FA0h
28~29	Audio Const offset for Fullseg	0 - 1000[ms] HEX : 0000h - 03E8h
30~31	Audio Const offset for Oneseg	0 - 1000[ms] HEX : 0000h - 03E8h
32	Event relay : Auto Event relay On/Off *When this configured as off, notification will be skipped	00h : OFF 01h : ON
33	Event relay : Notification interval before switching [s]	0 or 5- 30 [s] HEX : 00h or 05h - 1Eh
34	Event relay : Number of notifications before switching [times]	0 - 6 [times] HEX : 00h - 06h
35~36	Banner : Banner Position X	0 - 1280 HEX : 0000h - 0500h
37~38	Banner : Banner Position Y	0 - 720 HEX : 0000h – 02D0h
39	Banner : Font Size [px]	24 - 40 [px] HEX : 18h - 28h
40	Banner : Transparency [%]	0 - 100 [%] HEX : 00h - 64h
41	Banner : Logo Position Offset [px]	-128 - 127 [px] HEX : 80h - 7Fh
42	Banner : Margin of Top and Bottom [px]	0 – 50 [px] HEX : 00h - 32h
43	Banner : Margin of Left and Right [px]	0 – 50 [px] HEX : 00h - 32h
44	Banner : Line spacing [px]	0 – 50 [px] HEX : 00h - 32h
45	Banner : Max number of characters	1 - 255 HEX : 01h - FFh
46	Banner : Default Timeout [s] 0 : No timeout (Continue to display until Hide Banner) 1 – 255 : Display Time	0 or 1 - 255[s] HEX : 00h or 01h- FFh
47	EPG List: Type Enable 0 : Enable, 1: Disable Bit7: T.B.D Bit6: T.B.D Bit5: T.B.D Bit4: T.B.D Bit3: Type 4 Enable Bit2: Type 3 Enable Bit1: Type 2 Enable Bit0: Type 1 Enable Range : 0 to 14	0 - 15 HEX : 00h – 0Eh

48	EPG List: Type1 Program name Font Size [px]	24 – 40 [px] HEX : 18h - 28h
49	EPG List: Type1 TS Name Font Size [px]	24 – 40 [px] HEX : 18h - 28h
50	EPG List: Type1 EPG Content Font Size [px]	24 – 40 [px] HEX : 18h - 28h
51	EPG List: Type2 Program name Font Size [px]	24 – 40 [px] HEX : 18h - 28h
52	EPG List: Type2 TS Name Font Size [px]	24 – 40 [px] HEX : 18h - 28h
53	EPG List Type2 EPG Content Font Size [px]	24 – 40 [px] HEX : 18h - 28h
54	EPG List: Type3 Program name Font Size [px]	24 – 40 [px] HEX : 18h - 28h
55	EPG List: Type3 TS Name Font Size [px]	24 – 40 [px] HEX : 18h - 28h
56	EPG List: Type3 EPG Content Font Size [px]	24 – 40 [px] HEX : 18h - 28h
57	EPG List: Type4 Program name Font Size [px]	24 – 40 [px] HEX : 18h - 28h
58	EPG List: Type4 TS Name Font Size [px]	24 – 40 [px] HEX : 18h - 28h
59	EPG List: Type4 EPG Content Font Size [px]	24 – 40 [px] HEX : 18h - 28h
60	EPG List: Table Title Color Red	0 - 255 HEX : 00h - FFh
61	EPG List: Table Title Color Green	0 - 255 HEX : 00h - FFh
62	EPG List: Table Title Color Blue	0 - 255 HEX : 00h - FFh
63	EPG List: Table Title Font Color Red	0 - 255 HEX : 00h - FFh
64	EPG List: Table Title Font Color Green	0 - 255 HEX : 00h - FFh
65	EPG List: Table Title Font Color Blue	0 - 255 HEX : 00h - FFh
66	EPG List: Cursor Color Red	0 - 255 HEX : 00h - FFh
67	EPG List: Cursor Color Green	0 - 255 HEX : 00h - FFh
68	EPG List: Cursor Color Blue	0 - 255 HEX : 00h - FFh
69	EPG List: Cursor Font Color Red	0 - 255 HEX : 00h - FFh
70	EPG List: Cursor Font Color Green	0 - 255 HEX : 00h - FFh
71	EPG List: Cursor Font Color Blue	0 - 255 HEX : 00h - FFh
72	EPG List: Current Program Color Red	0 - 255 HEX : 00h - FFh
73	EPG List: Current Program Color Green	0 - 255 HEX : 00h - FFh
74	EPG List: Current Program Color Blue	0 - 255 HEX : 00h - FFh
75	EPG List: Current Program Font Color Red	0 - 255 HEX : 00h - FFh
76	EPG List: Current Program Font Color Green	0 - 255

		HEX : 00h - FFh
77	EPG List: Current Program Font Color Blue	0 - 255 HEX : 00h - FFh
78	EPG List: Balloon color Red	0 - 255 HEX : 00h - FFh
79	EPG List: Balloon color Green	0 - 255 HEX : 00h - FFh
80	EPG List: Balloon color Blue	0 - 255 HEX : 00h - FFh
81	EPG List: Balloon Font color Red	0 - 255 HEX : 00h - FFh
82	EPG List: Balloon Font color Green	0 - 255 HEX : 00h - FFh
83	EPG List: Balloon Font color Blue	0 - 255 HEX : 00h - FFh
84	EPG List: Border line color Red	0 - 255 HEX : 00h - FFh
85	EPG List: Border line color Green	0 - 255 HEX : 00h - FFh
86	EPG List: Border line color Blue	0 - 255 HEX : 00h - FFh
87	EPG Information: Font Size [px]	24 – 40 [px] HEX : 18h - 28h
88	EPG Information: Title Font Color Red	0 - 255 HEX : 00h - FFh
89	EPG Information: Title Font Color Green	0 - 255 HEX : 00h - FFh
90	EPG Information: Title Font Color Blue	0 - 255 HEX : 00h - FFh
91	EPG Information: Line Spacing for Title	0 - 255 HEX : 00h - FFh
92	EPG Information: Line Spacing for Contents	0 - 255 HEX : 00h - FFh
93	EPG Information: Margin	0 - 80 HEX : 00h - 50h
94	Default Auto Broadcast Search mode 0:Automatic Broadcast Search Off 1:Automatic Relay Station Search On 2:Automatic Affiliated Station Search On 3:Automatic Relay & Affiliated Station Search On	0 - 3 HEX : 00h -03h
95~96	C/N Level 3 threshold for one seg	0 - 4000 [0 - 40dB] HEX : 0000h - 0FA0h
97~98	C/N Level 2 threshold for one seg	0 - 4000 [0 - 40dB] HEX : 0000h - 0FA0h
99~100	C/N Level 1 threshold for one seg	0 - 4000 [0 - 40dB] HEX : 0000h - 0FA0h
101	Default event mask 0 bit0 : DTV Status bit1 : Scan Status bit2 : Search Status bit3 : DTV Event bit4 : EWS Event bit5 : EPG Status bit6 : Event relay bit7 : Number of Program info 0 : Not Mask	0 - 255 HEX : 00h - FFh

	1 : Mask - Disables the event notification.	
102	Default event mask 1bit0 : Program Information bit1 : Program Information with UTF-8 bit2 : TS Name Update bit3 : TS Name Update with UTF-8 bit4 : Logo Update bit5 : EGP Button Status bit6 : BML Status bit7 : C/N Information 0 : Not Mask 1 : Mask - Disables the event notification.	0 - 255 HEX : 00h - FFh
103	Default event mask 2 bit0 : EPG Scan Result bit1 - bit7 :Reserved	0 - 255 HEX : 00h - FFh
104	EPG List: Short program border line color Red	0 - 255 HEX : 00h - FFh
105	EPG List: Short program border line color Green	0 - 255 HEX : 00h - FFh
106	EPG List: Short program border line color Blue	0 - 255 HEX : 00h - FFh
107	Banner segment icon Enable/Disable	00h(Enable) 01h(Disable)
108	Number of scrolling line on EPG detail	1 - 255 HEX : 01h - FFh
109	Automatic banner display Enable/Disable	00h(Enable) 01h(Disable)
110	Reserved	
111	Reserved	
112	Reserved	
113	Reserved	
114	Reserved	
115	Reserved	
116	Reserved	
117	Reserved	
118	Reserved	
119	Reserved	
120	Reserved	
121	Reserved	
122	Reserved	
123	Reserved	
124	Reserved	
125	Reserved	
126	Reserved	
127	Reserved	
128	Reserved	
129	Reserved	
130	Reserved	
131	Reserved	
132	Reserved	
133	Reserved	
134	Reserved	
135	Reserved	
136	Reserved	
137	Reserved	
138	Reserved	
139	Reserved	

140	Reserved	
141	Reserved	
142	Reserved	
143	Reserved	
144	Reserved	
145	Reserved	
146	Reserved	
147	Reserved	
148	Reserved	
149	Reserved	
150	Reserved	
151	Reserved	
152	Reserved	
153	Reserved	
154	Reserved	
155	Reserved	
156	Reserved	
157	Reserved	
158	Reserved	
159	Reserved	
160	Reserved	
161	Reserved	
162	Reserved	
163	Reserved	
164	Reserved	
165	Reserved	
166	Reserved	
167	Reserved	
168	Reserved	
169	Reserved	
170	Reserved	
171	Reserved	
172	Reserved	
173	Reserved	
174	Reserved	
175	Reserved	
176	Reserved	
177	Reserved	
178	Reserved	
179	Reserved	
180	Reserved	
181	Reserved	
182	Reserved	
183	Reserved	
184	Reserved	
185	Reserved	
186	Reserved	
187	Reserved	
188	Reserved	
189	Reserved	
190	Reserved	
191	Reserved	
192	Reserved	
193	Reserved	
194	Reserved	

195	Reserved	
196	Reserved	
197	Reserved	
198	Reserved	
199	Reserved	
200	Reserved	
201	Reserved	
202	Reserved	
203	Reserved	
204	Reserved	
205	Reserved	
206	Reserved	
207	Reserved	
208	Reserved	
209	Reserved	
210	Reserved	
211	Reserved	
212	Reserved	
213	Reserved	
214	Reserved	
215	Reserved	
216	Reserved	
217	Reserved	
218	Reserved	
219	Reserved	
220	Reserved	
221	Reserved	
222	Reserved	
223	Reserved	
224	Reserved	
225	Reserved	
226	CRC8	

・C/N: signed 4byte data (Divide by 100)

・C/N range 0dB to 40.0 dB

8.3. DTV Parameter 2 data range for New Frontier

The DTV Parameter 2 data range for New Frontier is adapted to following two type.

-F039990144【FRO232BSyyzz】

Some parameters are not supported.(Not support ones are marked with strikethrough)

Data	**Data name**	**Data range**
0	Set DTV Parameter 2 (Upper)	01h
1	Set DTV Parameter 2 (Lower	35h
2	DTV Parameter 2 Enable/Disable(Use default Parameter)	01h(Enable) 00h(Disable)
3	Auto tune mode default 00h:Auto 01h:Fullseg 02h:Oneseg 03h:Auto with Seamless	0 - 3 HEX : 00h – 03h
4	Default Character Set 00h:ARIB 01h:UTF-8	0 - 1 [%] HEX : 00h - 01h
~~5~~	~~Default Video Resolution~~	~~0 -4~~

	~~00h:720p~~ ~~01h:480p~~ ~~02h:1080p~~ ~~03h:480i~~ ~~04h:1080p@30Hz~~	~~HEX : 00h - 04h~~
~~6~~	~~Closed Caption display over BML browser Enable/Disable~~	~~00h(Enable Display)~~ ~~01h(Disable Display)~~
~~7~~	~~BML Error Message Enable/Disable~~	~~00h(Enable Display)~~ ~~01h(Disable Display)~~
8	Receive Logo Type 0 : Receive, 1 : Not Receive Bit7: 0 (Fixed) Bit6: 0 (Fixed) Bit5: Logo Type 5 Bit4: Logo Type 4 Bit3: Logo Type 3 Bit2: Logo Type 2 Bit1: Logo Type 1 Bit0: Logo Type 0	00XXXXXXb
9	Search Detection Timeout	1 - 10 [s] HEX : 01 – 0Ah
10~11	C/N for Receive Error Detection for Fix Fullseg mode	0 - 4000 [0 - 40dB] HEX : 0000h - 0FA0h
12~13	C/N for Preset Scan Detection for Fix Oneseg mode and Auto mode	0 - 4000 [0 - 40dB] HEX : 0000h - 0FA0h
14~15	C/N for Preset Scan Detection for Fix Fullseg mode	0 - 4000 [0 - 40dB] HEX : 0000h - 0FA0h
16	Timeout for Preset Scan per channel	0 - 10 [s] HEX : 00h - 0Ah
17~18	C/N Level 3 threshold for full seg	0 - 4000 [0 - 40dB] HEX : 0000h - 0FA0h
19~20	C/N Level 2 threshold for full seg	0 - 4000 [0 - 40dB] HEX : 0000h - 0FA0h
21~22	C/N Level 1 threshold for full seg	0 - 4000 [0 - 40dB] HEX : 0000h - 0FA0h
23~24	EPG Update scan time / channel	1 - 180 [s] HEX : 0001h – 00B4h
25	EPG Scan Number of Days (Only foreground EPG Scan)	1 - 8 [day] HEX : 01h - 08h
26~27	EPG Scan Signal Threshold (Only foreground EPG Scan)	0 - 4000 [0 - 40dB] HEX : 0000h - 0FA0h
28~29	Audio Const offset for Fullseg	0 - 1000[ms] HEX : 0000h - 03E8h
30~31	Audio Const offset for Oneseg	0 - 1000[ms] HEX : 0000h - 03E8h
32	Event relay : Auto Event relay On/Off *When this configured as off, notification will be skipped	00h : OFF 01h : ON
33	Event relay : Notification interval before switching [s]	0 or 5- 30 [s] HEX : 00h or 05h - 1Eh
34	Event relay : Number of notifications before switching [times]	0 - 6 [times] HEX : 00h - 06h
35~36	Banner : Banner Position X	0 - 720 HEX : 0000h – 02D0h
37~38	Banner : Banner Position Y	0 - 480 HEX : 0000h – 01E0h
39	Banner : Font Size [px]	18 - 24 [px]

		HEX : 12h - 18h
40	Banner : Transparency [%]	0 - 100 [%] HEX : 00h - 64h
41	Banner : Logo Position Offset [px]	-128 - 127 [px] HEX : 80h - 7Fh
42	Banner : Margin of Top and Bottom	0 – 50 [px] HEX : 00h - 32h
43	Banner : Margin of Left and Right [px]	0 – 50 [px] HEX : 00h - 32h
44	Banner : Line spacing [px]	0 – 50 [px] HEX : 00h - 32h
45	Banner : Max number of characters	1 – 255 HEX : 01h - FFh
46	Banner : Default Timeout [s] 0 : No timeout (Continue to display until Hide Banner) 1 – 255 : Display Time	0 or 1 – 255 [s] HEX : 00h or 01h- FFh
47	EPG List: Type Enable 0 : Enable, 1: Disable Bit7: T.B.D Bit6: T.B.D Bit5: T.B.D Bit4: T.B.D Bit3: Type 4 Enable Bit2: Type 3 Enable Bit1: Type 2 Enable Bit0: Type 1 Enable Range : 0 to 14	0 - 15 HEX : 00h – 0Eh
~~48~~	~~EPG List: Type1 Program name Font Size [px]~~	~~18 – 24 [px]~~ ~~HEX : 12h - 18h~~
~~49~~	~~EPG List: Type1 TS Name Font Size [px]~~	~~18 – 24 [px]~~ ~~HEX : 12h - 18h~~
~~50~~	~~EPG List: Type1 EPG Content Font Size [px]~~	~~18 – 24 [px]~~ ~~HEX : 12h - 18h~~
~~51~~	~~EPG List: Type2 Program name Font Size [px]~~	~~18 – 24 [px]~~ ~~HEX : 12h - 18h~~
~~52~~	~~EPG List: Type2 TS Name Font Size [px]~~	~~18 – 24 [px]~~ ~~HEX : 12h - 18h~~
~~53~~	~~EPG List Type2 EPG Content Font Size [px]~~	~~18 – 24 [px]~~ ~~HEX : 12h - 18h~~
~~54~~	~~EPG List: Type3 Program name Font Size [px]~~	~~18 – 24 [px]~~ ~~HEX : 12h - 18h~~
~~55~~	~~EPG List: Type3 TS Name Font Size [px]~~	~~18 – 24 [px]~~ ~~HEX : 12h - 18h~~
~~56~~	~~EPG List: Type3 EPG Content Font Size [px]~~	~~24 – 40 [px]~~ ~~HEX : 18h - 28h~~
~~57~~	~~EPG List: Type4 Program name Font Size [px]~~	~~18 – 24 [px]~~ ~~HEX : 12h - 18h~~
~~58~~	~~EPG List: Type4 TS Name Font Size [px]~~	~~18 – 24 [px]~~ ~~HEX : 12h - 18h~~
~~59~~	~~EPG List: Type4 EPG Content Font Size [px]~~	~~18 – 24 [px]~~ ~~HEX : 12h - 18h~~
~~60~~	~~EPG List: Table Title Color Red~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~61~~	~~EPG List: Table Title Color Green~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~62~~	~~EPG List: Table Title Color Blue~~	~~0 - 255~~

		~~HEX : 00h - FFh~~
~~63~~	~~EPG List: Table Title Font Color Red~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~64~~	~~EPG List: Table Title Font Color Green~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~65~~	~~EPG List: Table Title Font Color Blue~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~66~~	~~EPG List: Cursor Color Red~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~67~~	~~EPG List: Cursor Color Green~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~68~~	~~EPG List: Cursor Color Blue~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~69~~	~~EPG List: Cursor Font Color Red~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~70~~	~~EPG List: Cursor Font Color Green~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~71~~	~~EPG List: Cursor Font Color Blue~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~72~~	~~EPG List: Current Program Color Red~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~73~~	~~EPG List: Current Program Color Green~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~74~~	~~EPG List: Current Program Color Blue~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~75~~	~~EPG List: Current Program Font Color Red~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~76~~	~~EPG List: Current Program Font Color Green~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~77~~	~~EPG List: Current Program Font Color Blue~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~78~~	~~EPG List: Balloon color Red~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~79~~	~~EPG List: Balloon color Green~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~80~~	~~EPG List: Balloon color Blue~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~81~~	~~EPG List: Balloon Font color Red~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~82~~	~~EPG List: Balloon Font color Green~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~83~~	~~EPG List: Balloon Font color Blue~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~84~~	~~EPG List: Border line color Red~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~85~~	~~EPG List: Border line color Green~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~86~~	~~EPG List: Border line color Blue~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~87~~	~~EPG Information:~~ Font Size [px]	~~18 – 24 [px]~~ ~~HEX : 12h - 18h~~
~~88~~	~~EPG Information: Title Font Color Red~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~89~~	~~EPG Information: Title Font Color Green~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~90~~	~~EPG Information: Title Font Color Blue~~	~~0 - 255~~ ~~HEX : 00h - FFh~~

~~91~~	~~EPG Information: Line Spacing for Title~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~92~~	~~EPG Information: Line Spacing for Contents~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~93~~	~~EPG Information: Margin~~	~~0 - 80~~ ~~HEX : 00h - 50h~~
94	Default Auto Broadcast Search mode 0:Automatic Broadcast Search Off 1:Automatic Relay Station Search On 2:Automatic Affiliated Station Search On 3:Automatic Relay & Affiliated Station Search On	0 - 3 HEX : 00h -03h
95~96	C/N Level 3 threshold for one seg	0 - 4000 [0 - 40dB] HEX : 0000h - 0FA0h
97~98	C/N Level 2 threshold for one seg	0 - 4000 [0 - 40dB] HEX : 0000h - 0FA0h
99~100	C/N Level 1 threshold for one seg	0 - 4000 [0 - 40dB] HEX : 0000h - 0FA0h
101	Default event mask 0 bit0 : DTV Status bit1 : Scan Status bit2 : Search Status bit3 : DTV Event bit4 : EWS Event bit5 : EPG Status bit6 : Event relay bit7 : Number of Program info 0 : Not Mask 1 : Mask - Disables the event notification.	0 - 255 HEX : 00h - FFh
102	Default event mask 1bit0 : Program Information bit1 : Program Information with UTF-8 bit2 : TS Name Update bit3 : TS Name Update with UTF-8 bit4 : Logo Update bit5 : EGP Button Status bit6 : BML Status bit7 : C/N Information 0 : Not Mask 1 : Mask - Disables the event notification.	0 - 255 HEX : 00h - FFh
103	Default event mask 2 bit0 : EPG Scan Result bit1 - bit7 :Reserved	0 - 255 HEX : 00h - FFh
~~104~~	~~EPG List: Short program border line color Red~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~105~~	~~EPG List: Short program border line color Green~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
~~106~~	~~EPG List: Short program border line color Blue~~	~~0 - 255~~ ~~HEX : 00h - FFh~~
107	Banner segment icon Enable/Disable	00h(Enable) 01h(Disable)
108	Number of scrolling line on EPG detail	1 - 255 HEX : 01h - FFh
109	Automatic banner display Enable/Disable	00h(Enable) 01h(Disable)
110	Reserved	
111	Reserved	
112	Reserved	

113	Reserved	
114	Reserved	
115	Reserved	
116	Reserved	
117	Reserved	
118	Reserved	
119	Reserved	
120	Reserved	
121	Reserved	
122	Reserved	
123	Reserved	
124	Reserved	
125	Reserved	
126	Reserved	
127	Reserved	
128	Reserved	
129	Reserved	
130	Reserved	
131	Reserved	
132	Reserved	
133	Reserved	
134	Reserved	
135	Reserved	
136	Reserved	
137	Reserved	
138	Reserved	
139	Reserved	
140	Reserved	
141	Reserved	
142	Reserved	
143	Reserved	
144	Reserved	
145	Reserved	
146	Reserved	
147	Reserved	
148	Reserved	
149	Reserved	
150	Reserved	
151	Reserved	
152	Reserved	
153	Reserved	
154	Reserved	
155	Reserved	
156	Reserved	
157	Reserved	
158	Reserved	
159	Reserved	
160	Reserved	
161	Reserved	
162	Reserved	
163	Reserved	
164	Reserved	
165	Reserved	
166	Reserved	
167	Reserved	

168	Reserved	
169	Reserved	
170	Reserved	
171	Reserved	
172	Reserved	
173	Reserved	
174	Reserved	
175	Reserved	
176	Reserved	
177	Reserved	
178	Reserved	
179	Reserved	
180	Reserved	
181	Reserved	
182	Reserved	
183	Reserved	
184	Reserved	
185	Reserved	
186	Reserved	
187	Reserved	
188	Reserved	
189	Reserved	
190	Reserved	
191	Reserved	
192	Reserved	
193	Reserved	
194	Reserved	
195	Reserved	
196	Reserved	
197	Reserved	
198	Reserved	
199	Reserved	
200	Reserved	
201	Reserved	
202	Reserved	
203	Reserved	
204	Reserved	
205	Reserved	
206	Reserved	
207	Reserved	
208	Reserved	
209	Reserved	
210	Reserved	
211	Reserved	
212	Reserved	
213	Reserved	
214	Reserved	
215	Reserved	
216	Reserved	
217	Reserved	
218	Reserved	
219	Reserved	
220	Reserved	
221	Reserved	
222	Reserved	

223	Reserved	
224	Reserved	
225	Reserved	
226	CRC8	

・C/N: signed 4byte data (Divide by 100)

・C/N range 0dB to 40.0 dB

・Data in gray is where the configuration is not supported.